

INTRODUCTION TO SHIMURA VARIETIES

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ABSTRACT. These are lecture notes for a course on Shimura varieties I am currently teaching at Zhejiang University. Comments are highly welcome and much appreciated.

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1. INTRODUCTION

In this first lecture, we will learn, very roughly, what Shimura varieties are and why they are interesting. Everything brought up today will be covered in much more detail later in the course, and it will be perfectly normal that many terms will be new during a first reading. Our goal today is only to get an overview.

1.1. Why study Shimura varieties? Shimura varieties combine two interesting properties:

- They are varieties defined over number fields which makes them interesting from a number theory perspective. Most importantly, their étale cohomology groups are representations of Galois groups of number fields.
- Their definition is in terms of connected reductive algebraic groups $G/\mathbb{Q}$. They come equipped with an action of the adelic points $G(\mathbb{A}_f)$, which implies that their étale cohomology groups are also $G(\mathbb{A}_f)$ -representations.

Hence, the étale cohomology groups of Shimura varieties are both Galois and $G(\mathbb{A}_f)$ -representations. Conjecturally, this two-fold structure is described by the global Langlands correspondence. Conversely, one can use the cohomology of Shimura varieties to prove important cases of this correspondence. This is the main motivation for our course, and our overall aim is to learn about several important ideas in this context.

Let us mention that Shimura varieties are also interesting for other reasons. For example, the study of heights on the Siegel variety plays an important role in Faltings's proof of the Mordell Conjecture [3]. Another example is the Gross–Zagier formula [7], which states an identity between height pairings of complex multiplication points on the modular curve and derivatives of L -functions. It plays a major role in the proof of cases of the Birch–Swinnerton-Dyer Conjecture. Its higher-dimensional generalizations, the arithmetic Gan–Gross–Prasad Conjectures [5, 25], are an important topic in current arithmetic geometry research. In a related direction, the Kudla program [11] seeks to establish connections between cycles on Shimura varieties and modular forms or Eisenstein series. The proof of the averaged Colmez conjecture [1, 24] has been an application of such ideas.

1.2. This course. The first part of our course will be an introduction to Shimura varieties. We will learn how to define them in terms of moduli spaces of abelian varieties and how to relate this definition to the group-theoretic one of Deligne. One of our goals is to obtain familiarity with the adelic formalism which will become important later.

In the second part of the course, we will study the cohomology of Shimura varieties. We will first get to know Matsushima's formula, which expresses the Betti cohomology of compact Shimura varieties in terms of automorphic representations. We will then learn about point counting in characteristic p (Langlands–Kottwitz method). The aim here is to give an orbital integral expression for the number of $\mathbb{F}_{p^n}$ -points of the reduction mod p of the Shimura variety.

1.3. References. The following two are our main background references.

- The introductory lecture notes by Milne [15]. They focus on the group-theoretic definition of Shimura varieties and the definition of canonical models.
- The first few articles in the lecture notes volume [8]. They provide an introduction to PEL type Shimura varieties. The article of Yihang Zhu [26] is directly related to the material of the second part of the course.

1.4. Prerequisites. We will assume as little as possible. The only necessary background is some familiarity with varieties and algebraic number theory.



In the rest of this introduction, we sketch the definition of Shimura varieties and give an outline of the course contents.

1.5. Shimura data. Shimura varieties are attached to Shimura data. The formalism starts with a connected reductive group G over $\mathbb{Q}$. For example, G might be one of the following.

- $G = \mathrm{GL}_2$
- $G = \mathrm{GSp}_{2g}$, the general symplectic group in $2g$ variables. Let $J = \begin{pmatrix} & 1_g \\ -1_g & \end{pmatrix}$ be the matrix defining the standard symplectic form on $\mathbb{Q}^{2g}$. Then GSp is defined by

$$\mathrm{GSp}_{2g}(\mathbb{Q}) = \{g \in \mathrm{GL}_{2g}(\mathbb{Q}) \mid {}^t g \cdot J \cdot g = c \cdot J \text{ for some } c \in \mathbb{Q}^\times\}. \quad (1.1)$$

It is related to the usual symplectic group Sp_{2g} by the exact sequence

$$1 \longrightarrow \mathrm{Sp}_{2g} \longrightarrow \mathrm{GSp}_{2g} \xrightarrow{c} \mathrm{GL}_1 \longrightarrow 1.$$

The map c is called the *similitude factor*. Note that $\mathrm{GSp}_2 = \mathrm{GL}_2$ and $\mathrm{Sp}_2 = \mathrm{SL}_2$, recovering the previous example.

- $G = \mathrm{U}(V)$, a unitary group. Let $K/\mathbb{Q}$ be an imaginary quadratic extension. (This means that $\mathbb{R} \otimes_{\mathbb{Q}} K \cong \mathbb{C}$.) Let V be an n -dimensional hermitian K -vector space. If V is not positive or negative definite then $\mathrm{U}(V)$ can occur as part of a Shimura datum.

Next, the formalism requires the datum of a homomorphism of real algebraic groups

$$h : \mathbb{C}^\times \longrightarrow G(\mathbb{R}) \quad (1.2)$$

which satisfies certain axioms introduced by Deligne [2]. Such an h is called a *Deligne homomorphism*. If $g \in G(\mathbb{R})$ is a real point of G , then we may conjugate h to define a new Deligne homomorphism,

$$(ghg^{-1})(z) := gh(z)g^{-1}.$$

Let $S_h \subset G(\mathbb{R})$ denote the centralizer of h , meaning the subgroup of elements g with $ghg^{-1} = h$. The quotient $X = G(\mathbb{R})/S_h$ is precisely the set of Deligne homomorphisms that are conjugate to h . An important consequence of Deligne's axioms is that X is a finite union of hermitian symmetric domains for $G(\mathbb{R})$. In particular, it is a complex manifold. The pair (G, X) is called a *Shimura datum*.

Example 1.1. Consider $G = \mathrm{GL}_2$. We can embed $\mathbb{C}$ into $M_2(\mathbb{R})$ as $\mathbb{R}$ -algebra by

$$h(a + bi) := \begin{pmatrix} a & -b \\ b & a \end{pmatrix}. \quad (1.3)$$

If we restrict this embedding to unit groups, then we obtain a Deligne homomorphism $h : \mathbb{C}^\times \rightarrow \mathrm{GL}_2(\mathbb{R})$. Its centralizer is precisely $h(\mathbb{C}^\times)$ and the quotient X is the set of complex structures on $\mathbb{R}^2$. Since $\mathbb{C}^\times$ is connected and since $\mathrm{GL}_2(\mathbb{R})$ has two connected components, X has two connected components. We want to give a more explicit description of X .

Recall that $\mathbb{P}^1(\mathbb{C})$ is the space of complex lines in $\mathbb{C}^2$. Clearly, the Lie group $\mathrm{GL}_2(\mathbb{C})$ acts on it by its natural action on $\mathbb{C}^2$. The subgroup $\mathrm{GL}_2(\mathbb{R})$ preserves the real projective line $\mathbb{P}^1(\mathbb{R})$ and hence acts on the complement,

$$\mathrm{GL}_2(\mathbb{R}) \curvearrowright \mathbb{C} \setminus \mathbb{R}, \quad g \cdot \tau = \frac{a\tau + b}{c\tau + d}. \quad (1.4)$$

The complement $\mathbb{C} \setminus \mathbb{R}$ is the union of the upper and lower half plane which we often denote by $\mathbb{H}^\pm$. As an open subset of $\mathbb{C}$, it is naturally a complex manifold. Let us compute the stabilizer of i :

$$\begin{aligned} i = \frac{ai + b}{ci + d} &\iff -c + di = ai + b \\ &\iff a = d, \quad c = -b. \end{aligned} \quad (1.5)$$

That is, the stabilizer of i is precisely $h(\mathbb{C}^\times)$. Moreover, it is clear that $\mathrm{GL}_2(\mathbb{R})$ acts transitively on $\mathbb{H}^\pm$ because

$$\begin{pmatrix} a & b \\ & 1 \end{pmatrix} \cdot i = ai + b.$$

Hence, we see that

$$X \xrightarrow{\sim} \mathbb{H}^\pm, \quad ghg^{-1} \mapsto g \cdot i \quad (1.6)$$

as smooth manifolds in a $\mathrm{GL}_2(\mathbb{R})$ -equivariant way. We have not defined the complex structure on X , but it is, in fact, given by the complex structure on $\mathbb{H}^\pm$ under (1.6).

Remark 1.2. Some groups, such as GL_n with $n \geq 3$, cannot occur as part of a Shimura datum. For example, the dimension of the symmetric space for $\mathrm{GL}_3(\mathbb{R})$ is

$$\dim \mathrm{SL}_3(\mathbb{R}) - \dim \mathrm{SO}(3) = 8 - 3$$

which is odd and hence cannot be a complex manifold.

1.6. Shimura varieties over $\mathbb{C}$. Given a Shimura datum (G, X) , one next defines a complex variety in the following way. Let $\mathbb{A}$ denote the ring of adeles of $\mathbb{Q}$, and let $\mathbb{A} = \mathbb{A}_f \times \mathbb{R}$ be its factorization into finite and archimedean part. (We will review these definitions later in the course.) Given an open compact subgroup $K \subset G(\mathbb{A}_f)$, the quotient $G(\mathbb{A}_f)/K$ is a discrete countably infinite set with transitive $G(\mathbb{A}_f)$ -action. Hence, the product $X \times G(\mathbb{A}_f)/K$ is a countable union of copies of X . We consider the diagonal action

$$G(\mathbb{Q}) \curvearrowright X \times G(\mathbb{A}_f)/K.$$

The intersection $\Gamma_0 = Z(G)(\mathbb{Q}) \cap K$ of the rational points of the center with K acts trivially. If K is small enough then the $G(\mathbb{Q})/\Gamma_0$ -action is free. (The technical term is “neat” and we will get to know it later in the course.) It is also properly discontinuous, so we can form the quotient complex manifold

$$\mathrm{Sh}_K(G, X)(\mathbb{C}) := G(\mathbb{Q}) \backslash (X \times G(\mathbb{A}_f)/K). \quad (1.7)$$

At this point, we have defined the complex points of the *Shimura variety for Shimura datum (G, X) and level K* as a complex manifold. The theorem of Baily–Borel states that there is a unique way to endow it with an algebraic structure.

Theorem 1.3 (Baily–Borel, see [15, Corollary 3.16]). *There exists a quasi-projective complex variety $\mathrm{Sh}_K(G, X)_{\mathbb{C}}$ such that there exists an isomorphism of complex manifolds $\mathrm{Sh}_K(G, X)_{\mathbb{C}}(\mathbb{C}) \xrightarrow{\sim} \mathrm{Sh}_K(G, X)(\mathbb{C})$. This variety is unique up to isomorphism.*

Remark 1.4. Simple examples of non-unique algebraic structures on complex manifolds can be found in [9].

Example 1.5. Let us again consider the case $G = \mathrm{GL}_2$ and let us give an example of a connected component of (1.7). Let $\widehat{\mathbb{Z}} = \prod_{p < \infty} \mathbb{Z}_p$ be the subring of integral elements of $\mathbb{A}_f$. For $n \geq 1$, consider the kernel

$$K(n) = \ker(\mathrm{GL}_2(\widehat{\mathbb{Z}}) \longrightarrow \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z}))$$

which is an open compact subgroup of $G(\mathbb{A}_f)$. It is small enough (in the above sense) if $n \geq 3$. The intersection

$$\Gamma(n) := \mathrm{GL}_2(\mathbb{Q}) \cap K(n)$$

is the classical congruence subgroup

$$\Gamma(n) = \left\{ \gamma \in \mathrm{GL}_2(\mathbb{Z}) \mid \gamma \equiv \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \pmod{n} \right\}.$$

The quotients $\Gamma(n) \backslash \mathbb{H}^+$ and $\Gamma(n) \backslash \mathbb{H}^-$ will be two of the connected components of the complex manifold $\mathrm{Sh}_{K(n)}(\mathrm{GL}_2, \mathbb{H}^\pm)$.

1.7. Shimura varieties over number fields. Finally, one descends $\mathrm{Sh}_K(G, X)$ to a number field. Starting from a Shimura datum (G, X) , Deligne defines a number field $E \subset \mathbb{C}$ called the *reflex field*. In a suitable sense, it is the smallest field over which the conjugacy class X is defined.

Example 1.6. Consider the three examples from §1.5.

- If $G = \mathrm{GL}_2$ or more generally $G = \mathrm{GSp}_{2g}$, then the reflex field is $\mathbb{Q}$.
- If $G = U(V)$ is a non-definite unitary group for an imaginary-quadratic field $K/\mathbb{Q}$, then the reflex field is the subfield $E \subset \mathbb{C}$ that is isomorphic to K .

Deligne [2] gave a definition of *canonical model* of $\mathrm{Sh}_K(G, X)_{\mathbb{C}}$ over E . It is a variety $\mathrm{Sh}_K(G, X)$ over $\mathrm{Spec}(E)$ together with an isomorphism

$$\mathbb{C} \otimes_E \mathrm{Sh}_K(G, X) \xrightarrow{\sim} \mathrm{Sh}_K(G, X)_{\mathbb{C}}$$

that satisfies a certain reciprocity law for complex multiplication points. Deligne proves that the canonical model $\mathrm{Sh}_K(G, X)$ is unique up to isomorphism if it exists.

Theorem 1.7 (Borovoi, Milne [13]). *For every Shimura datum, the canonical model exists.*

Definition 1.8. Let (G, X) be a Shimura datum with reflex field E and let $K \subset G(\mathbb{A}_f)$ be a sufficiently small level subgroup. The Shimura variety of level K attached to (G, X) is the canonical model $\mathrm{Sh}_K(G, X)$ from Theorem 1.7.

Remark 1.9. Historically, the study of Shimura varieties started with Shimura in the 1960s. He first considered moduli spaces of abelian varieties with **P**olarization, **E**ndomorphisms, and **L**evel structure (PEL). These are the Shimura varieties defined by *PEL type* Shimura data.

Shimura also studied several non-PEL cases and defined the corresponding Shimura varieties as varieties over number fields. Deligne [2] gave a group-theoretic framework for Shimura's work. His definition in terms of a reciprocity law for complex multiplication points is extrapolated from the Shimura–Taniyama reciprocity law for abelian varieties with complex multiplication. Deligne also constructed the canonical model for abelian type Shimura varieties. The proof of existence in the general case was completed by Milne based on ideas of Borovoi. See here for a short summary of the history by Milne [14, §6].

Example 1.10. Consider the two cases from Example 1.6. The unitary group $U(V)$ has no PEL type Shimura data. For the group GSp_{2g} , there exists a PEL type Shimura datum (GSp_{2g}, X) . Consider a principal congruence level subgroup

$$K(n) = \ker(\mathrm{GSp}_{2g}(\widehat{\mathbb{Z}}) \longrightarrow \mathrm{GSp}_{2g}(\mathbb{Z}/n\mathbb{Z}))$$

with $n \geq 3$. Then the canonical model $\mathrm{Sh}_{K(n)}(\mathrm{GSp}_{2g}, X)$ can be described as a moduli space of principally polarized abelian varieties with level- n -structure. For example, if we look at $\mathbb{C}$ -points and specialize to GL_2 , then we obtain

$$\mathrm{Sh}_{K(n)}(\mathrm{GL}_2, X)(\mathbb{C}) \xrightarrow{\sim} \{(E, \eta)/\mathbb{C}\} / \sim \quad (1.8)$$

where the right hand side denotes the set of isomorphism classes of pairs (E, η) with

- E an elliptic curve over $\mathbb{C}$,
- $\eta : (\mathbb{Z}/n\mathbb{Z})^{\oplus 2} \xrightarrow{\sim} E[n]$ a choice of basis for the n -torsion.

The datum η is called a *level structure* for E . Proving (1.8) will be one of our first goals.

1.8. Further topics. We will say more about this when the time comes. For now, let us start looking at Shimura varieties in detail.



Part 1. The Shimura variety of GL_2

2. THE UPPER HALF PLANE

In Example 1.1, we have introduced the action of $\mathrm{GL}_2(\mathbb{Q})$ on the union of upper and lower half plane $\mathbb{H}^\pm = \mathbb{C} \setminus \mathbb{R}$. Recall that it is given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau = \frac{a\tau + b}{c\tau + d}. \quad (2.1)$$

In Example 1.5, we have seen that we are especially interested in actions by subgroups such as $\mathrm{GL}_2(\mathbb{Z})$ and $\Gamma(n)$. Our aim in this section is to give a definition of such *arithmetic subgroups* and to prove properties about their action on $\mathbb{H}^\pm$.

Note that elements of $\mathrm{GL}_2(\mathbb{Z})$ have determinant 1 or -1 , and that the elements of determinant -1 interchange upper and lower half plane. So we will focus on the action of $\mathrm{SL}_2(\mathbb{Q})$ on the upper half plane $\mathbb{H} \subset \mathbb{H}^\pm$.

2.1. The fundamental domain. Let $\mathcal{F}$ be the area defined by

$$\mathcal{F} = \left\{ \tau \in \mathbb{H} \mid |\tau| \geq 1 \text{ and } -\frac{1}{2} \leq \operatorname{Re}(\tau) \leq \frac{1}{2} \right\}. \quad (2.2)$$

Its interior $\mathcal{F}^\circ$ is the open subset where $|\tau| > 1$ and $-1/2 \leq \operatorname{Re}(\tau) \leq 1/2$.

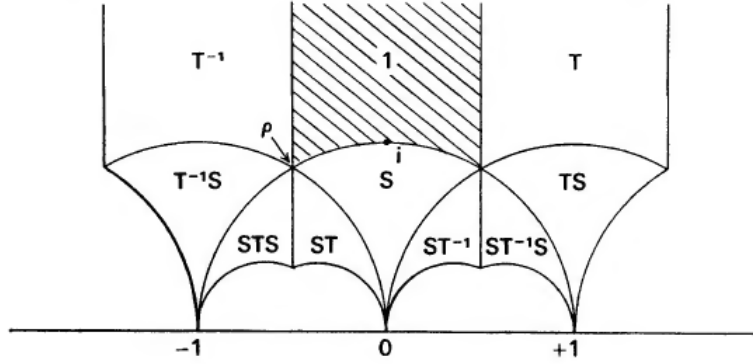


FIGURE 1. The area $\mathcal{F}$ is depicted in grey. The remaining areas show translates of $\mathcal{F}$ under the action of the elements S and T defined in (2.4). By Proposition 2.1 and Remark 2.2, these translates cover all of $\mathbb{H}$. The picture is taken from [21, §VII].

Proposition 2.1. *The set $\mathcal{F}$ is a fundamental domain for the action of $\mathrm{SL}_2(\mathbb{Z})/\{\pm 1\}$ on $\mathbb{H}$. That is, it has the following two properties.*

- (1) *For every $\tau \in \mathbb{H}$, there exists $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that $\gamma\tau \in \mathcal{F}$.*
- (2) *$\mathcal{F}^\circ \cap \gamma\mathcal{F}^\circ = \emptyset$ whenever $\gamma \notin \{\pm 1\}$.*

Proof. Fix $\tau \in \mathbb{H}$ and let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ be any element. By direct computation, we see that

$$\operatorname{Im}(\gamma\tau) = \operatorname{Im}\left(\frac{(a\tau + b)(c\tau - d)}{|c\tau + d|^2}\right) = \frac{(ad - bc)\operatorname{Im}(\tau)}{|c\tau + d|^2} = \frac{\operatorname{Im}(\tau)}{|c\tau + d|^2}. \quad (2.3)$$

The denominator $|c\tau + d|^2$ defines a positive definite quadratic form in $(c, d) \in \mathbb{Z}^2$. It hence takes a minimum on the set of (c, d) that occur as the bottom row of an element of $\mathrm{SL}_2(\mathbb{Z})$. (These are precisely the (c, d) with $\gcd(c, d) = 1$.) So we see that $\{\operatorname{Im}(\gamma\tau) \mid \gamma \in \mathrm{SL}_2(\mathbb{Z})\}$ has a maximum.

Let γ be such that $\text{Im}(\gamma\tau)$ is maximal. Consider the two matrices

$$S = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \quad (2.4)$$

and observe that they act as the very simple transformations

$$S\tau = -\frac{1}{\tau}, \quad T\tau = \tau + 1. \quad (2.5)$$

In particular, acting with a suitable power T^m , $m \in \mathbb{Z}$, we can translate $\gamma\tau$ to assume it lies in the strip $-1/2 \leq \text{Re}(z) \leq 1/2$. Then also $|\gamma\tau| \geq 1$ because otherwise $\text{Im}(S\gamma\tau) > \text{Im}(\gamma\tau)$ would contradict the maximality of $\text{Im}(\gamma\tau)$. This proves statement (1) of the proposition.

We now prove statement (2). Assume that τ and $\gamma\tau$ both lie in $\mathcal{F}^\circ$, our aim being to show that $\gamma \in \{\pm 1\}$. After possibly replacing the pair (γ, τ) by $(\gamma^{-1}, \gamma\tau)$, we can assume that $\text{Im}(\gamma\tau) \geq \text{Im}(\tau)$. Considering again (2.3), this means that $|c\tau + d|^2 \leq 1$.

Clearly, we now have $c = 0$ because $|c\tau + d| > 1$ for every $c \neq 0$ (use $\tau \in \mathcal{F}^\circ$). This means that γ is of the form

$$\gamma = \pm \begin{pmatrix} 1 & m \\ & 1 \end{pmatrix}$$

for some $m \in \mathbb{Z}$. Since both τ and $\gamma\tau$ have real part in $(-1/2, 1/2)$, the only possibility is $m = 0$. This finishes the proof. $\square$

Remark 2.2. One can show that the matrices S and T from (2.4) generate $\text{SL}_2(\mathbb{Z})$. That is, every element of $\text{SL}_2(\mathbb{Z})$ can be written as a product of the three elements S , T and T^{-1} . The proof is not difficult and can be found in [21, §VII.1, Theorem 2].

2.2. Arithmetic subgroups of $\text{SL}_2(\mathbb{Q})$. We now define arithmetic subgroups of $\text{SL}_2(\mathbb{Q})$.

Definition 2.3. (1) For $n \geq 1$, we define the *principal congruence subgroup* $\Gamma(n)$ by

$$\Gamma(n) = \{\gamma \in \text{SL}_2(\mathbb{Z}) \mid \gamma \equiv 1 \pmod{n}\}.$$

(2) We call a subgroup $\Gamma \subset \text{SL}_2(\mathbb{Q})$ *arithmetic* if it contains a principal congruence group $\Gamma(n)$ with finite index.

The group SL_2 has a very interesting property which will come up again later. Namely, for each $n \geq 1$, the projection map

$$\text{SL}_2(\mathbb{Z}) \longrightarrow \text{SL}_2(\mathbb{Z}/n\mathbb{Z}) \quad (2.6)$$

is surjective. This is not hard to show directly, but also follows from Theorem 3.15 (2) below.

Example 2.4. By the surjectivity we just stated for SL_2 , the image of the projection map $\text{GL}_2(\mathbb{Z}) \rightarrow \text{GL}_2(\mathbb{Z}/n\mathbb{Z})$ is the set of matrices with determinant ± 1 . In particular, this projection is not surjective when $n = 5$ or $n \geq 7$.

In the context of Definition 2.3, the surjectivity of (2.6) implies that $\Gamma(n) \trianglelefteq \text{SL}_2(\mathbb{Z})$ is a normal subgroup of index equal to $|\text{SL}_2(\mathbb{Z}/n\mathbb{Z})|$. In particular, if a group Γ contains $\Gamma(n)$ with finite index, then it also contains all $\Gamma(mn)$ with finite index.

Proposition 2.5. *Let Γ be an arithmetic subgroup.*

(1) *There exists a lattice $\Lambda \subset \mathbb{Q}^2$ such that $\Gamma \subseteq \text{SL}(\Lambda)$.*

(2) *More precisely, there exist an integer n and an element $g \in \text{GL}_2(\mathbb{Q})$, $\det(g) > 0$, such that*

$$\Gamma(m) \subseteq g\Gamma g^{-1} \subseteq \text{SL}_2(\mathbb{Z}).$$

Proof. The two statements are proved by very simple and universal arguments. First, by assumption on Γ , there exists an integer n such that $\Gamma(n) \subseteq \Gamma$ with finite index. Let $\gamma_1, \dots, \gamma_r$ be representatives for the cosets $\Gamma/\Gamma(n)$. Then Γ stabilizes the lattice

$$\Lambda := \sum_{i=1}^r \gamma_i \cdot \mathbb{Z}^2.$$

Indeed, since $\gamma\mathbb{Z}^2 = \mathbb{Z}^2$ for every $\gamma \in \Gamma(n)$, we can also write Λ as

$$\Lambda = \sum_{\gamma \in \Gamma} \gamma \cdot \mathbb{Z}^2,$$

and from this second expression the Γ -stability is clear. This means that $\Gamma \subseteq \mathrm{SL}(\Lambda)$ which proves statement (1).

Let $\lambda_1, \lambda_2 \in \Lambda$ be a basis as $\mathbb{Z}$ -module. Viewing λ_1 and λ_2 as column vectors, the base change matrix $g = (\lambda_1 \ \lambda_2)$ lies in $\mathrm{GL}_2(\mathbb{Q})$ and has the property $g\mathbb{Z}^2 = \Lambda$. Changing λ_1 to $-\lambda_1$ if necessary, we may assume $\det(g) > 0$. Then $\mathrm{SL}_2(\mathbb{Z}) = g^{-1}\mathrm{SL}(\Lambda)g$ and hence $g\Gamma g^{-1} \subseteq \mathrm{SL}_2(\mathbb{Z})$.

We still need to show that $g\Gamma g^{-1}$ contains a principal congruence subgroup. This is the content of the next lemma which completes the proof. $\square$

Lemma 2.6. *Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Q})$ be an arithmetic subgroup and $g \in \mathrm{GL}_2(\mathbb{Q})$. Then $g\Gamma g^{-1}$ is again an arithmetic subgroup.*

Proof. Let d be the least common multiple of all the denominators of all the entries of g and g^{-1} . Then, if $A \in d^2m\mathrm{M}_2(\mathbb{Z})$ is an integer matrix divisible by d^2m , we find $g^{-1}Ag \in m\mathrm{M}_2(\mathbb{Z})$. This shows that $g^{-1}\Gamma(d^2m)g \subseteq \Gamma(m)$ which is equivalent to

$$\Gamma(d^2m) \subseteq g\Gamma(m)g^{-1}. \quad (2.7)$$

Now, for the given Γ , choose n with $\Gamma(n) \subseteq \Gamma$. Conjugating this relation by g and using (2.7), we find $\Gamma(d^2n) \subseteq g\Gamma g^{-1}$ which proves that $g\Gamma g^{-1}$ is again arithmetic. $\square$

In other words, Proposition 2.5 shows that the arithmetic subgroups in $\mathrm{SL}_2(\mathbb{Q})$ are precisely the $\mathrm{GL}_2(\mathbb{Q})$ -conjugates of groups between $\mathrm{SL}_2(\mathbb{Z})$ and some $\Gamma(n)$.

2.3. Stabilizers.

Definition 2.7. We say that an arithmetic subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{Q})$ is *neat* if it is torsion free.

Proposition 2.8. *Let Γ be a neat arithmetic subgroup of $\mathrm{SL}_2(\mathbb{Q})$. Then Γ acts with trivial stabilizers on $\mathbb{H}$. That is, if $\gamma\tau = \tau$ for some $\gamma \in \Gamma$ and $\tau \in \mathbb{H}$, then $\gamma = 1$.*

Proof. We have seen in (1.5) that the stabilizer of $i \in \mathbb{H}$ in $\mathrm{GL}_2(\mathbb{R})$ is a copy of $\mathbb{C}^\times$. The unit circle $\mathbb{C}^1 \subset \mathbb{C}^\times$ is compact and equals the intersection $\mathbb{C}^\times \cap \mathrm{SL}_2(\mathbb{R})$. For a general point $\tau \in \mathbb{H}$, we can write $\tau = g \cdot i$ for some $g \in \mathrm{SL}_2(\mathbb{R})$:

$$\begin{pmatrix} 1 & b \\ & 1 \end{pmatrix} \begin{pmatrix} a^{1/2} & \\ & a^{-1/2} \end{pmatrix} \cdot i = ai + b.$$

The stabilizers S_i and S_τ of τ and i in $\mathrm{SL}_2(\mathbb{R})$ are then related by $S_\tau = gS_i g^{-1}$. In this way, we see that for every $\tau \in \mathbb{H}$, the stabilizer $S_\tau \subset \mathrm{SL}_2(\mathbb{R})$ is isomorphic to $\mathbb{C}^1$, in particular compact.

Assume that $\gamma\tau = \tau$, where $\gamma \in \Gamma$ and $\tau \in \mathbb{H}$. This is equivalent to $\gamma \in \Gamma \cap S_\tau$. Since $\Gamma \subset \mathrm{SL}_2(\mathbb{R})$ is a discrete subgroup, the intersection $\Gamma \cap S_\tau$ is a discrete subgroup of S_τ . Since the discrete subgroups of $\mathbb{C}^1$ are all finite cyclic (generated by a root of unity), and since Γ is torsion-free by assumption, we see that $\Gamma \cap S_\tau = \{1\}$. Hence $\gamma = 1$, and the proof is complete. $\square$

Example 2.9. The element $-1 \in \mathrm{SL}_2(\mathbb{Z})$ acts trivially on $\mathbb{H}$ because $(-\tau)/(-1) = \tau$ (substitute in (2.1)). The element $\begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$, which has order 4, stabilizes the point i because $-1/i = i$.

The next proposition provides a simple criterion for detecting neatness.

Proposition 2.10. *For all $n \geq 3$, the principal congruence subgroup $\Gamma(n)$ is neat. In particular, if $\Gamma \subseteq \Gamma(n)$ is an arithmetic subgroup, then Γ is neat.*

Proof. The minimal polynomial $\Phi_d(T)$ of a primitive d -th root of unity has degree $\varphi(d)$ (Euler φ -function). Recall that $\Phi_d(T)$ is called the d -th cyclotomic polynomial and that

$$T^m - 1 = \prod_{d|m} \Phi_d(T)$$

because the roots of $T^m - 1$ are precisely the m -th roots of unity, and each such root of unity is a primitive d -th root of unity for a unique divisor $d \mid m$.

The only values for d such that $\varphi(d) \leq 2$ are 1, 2, 3, 4, and 6. These are precisely the values for d such that $\mathbb{Q}(\zeta_d)$ has degree ≤ 2 over $\mathbb{Q}$.

Let $n \geq 1$ and let $\gamma \in \mathrm{SL}_2(\mathbb{Q})$ be a torsion element, say $\gamma^m = 1$. Then the minimal polynomial of γ divides $T^m - 1$. We know that the minimal polynomial and the characteristic polynomial of a matrix have the same irreducible factors. So the characteristic polynomial $P(T)$ of γ is a product of $\Phi_d(T)$ with $d \mid m$. The only possibilities for $P(T)$ are hence¹

$$(T-1)^2, (T+1)^2, (T-1)(T+1), T^2+1, T^2+T+1, \text{ and } T^2-T+1. \quad (2.8)$$

If $n \geq 3$ and if γ is integral with $\gamma \equiv 1 \pmod{n}$, then also $P(T) \equiv (T-1)^2 \pmod{n}$, leaving $P(T) = (T-1)^2$ as the only possibility. This means that γ is either equal to 1 or $\mathrm{GL}_2(\mathbb{Q})$ -conjugate to $\begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix}$ (Jordan normal form). But γ is also a torsion element by assumption, so $\gamma = 1$ is the only possibility. $\square$

Exercise 2.11. Extend the argument of the previous proof to GL_n . That is, given $n \geq 1$, find an integer $m \geq 1$ such that for $\gamma \in \mathrm{GL}_n(\mathbb{Z})$,

$$\gamma \equiv 1 \pmod{m} \implies \gamma \text{ non-torsion.}$$

Conclusion 2.12. In this lecture, we saw the definition of neat arithmetic subgroups of $\mathrm{SL}_2(\mathbb{Q})$. We have seen in Proposition 2.8 that such groups act freely on $\mathbb{H}$. So the quotient $\Gamma \backslash \mathbb{H}$ will be a Riemann surface and the quotient map

$$\mathbb{H} \longrightarrow \Gamma \backslash \mathbb{H} \quad (2.9)$$

a holomorphic covering map in the sense of topology. We have seen in Proposition 2.5 that, in order to study $\Gamma \backslash \mathbb{H}$, we may always assume $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$. Then we can think of $\Gamma \backslash \mathbb{H}$ as being glued from finitely many $\mathrm{SL}_2(\mathbb{Z})$ -translates of the fundamental domain $\mathcal{F}$ as in Figure 2.1 along their edges.

3. ADELIC DOUBLE QUOTIENTS

In this lecture, we study the adelic double quotients $\mathrm{GL}_2(\mathbb{Q}) \backslash (\mathbb{H}^\pm \times \mathrm{GL}_2(\mathbb{A}_f)/K)$ and relate them to the quotients $\Gamma \backslash \mathbb{H}$ from the previous lecture. We will first revisit the definition of the adeles and explain the definition of $\mathrm{GL}_2(\mathbb{A}_f)$ as a topological group in more detail. In fact, we will use this opportunity to also study groups of the form $G(\mathbb{A}_f)$ more generally.

¹The product $(T-1)(T+1)$ cannot actually occur, of course, because $\det(\gamma) = 1$ for $\gamma \in \mathrm{SL}_2(\mathbb{Q})$. This does not affect the argument, though.

3.1. The adèles. We begin by defining the ring of *integral adèles*. It is the profinite ring given by² $\widehat{\mathbb{Z}} := \varprojlim \mathbb{Z}/n\mathbb{Z}$. The transition maps here are given by the projections $\mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, whenever $m \mid n$. Concretely, we have

$$\widehat{\mathbb{Z}} = \left\{ (x_1, x_2, \dots) \in \prod_{n \geq 1} \mathbb{Z}/n\mathbb{Z} \mid x_{dn} \equiv x_n \pmod{n} \text{ for all } d, n \geq 1 \right\}.$$

Recall that the Chinese remainder theorem identifies $\mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} \prod_p \mathbb{Z}/p^{v_p(n)}\mathbb{Z}$. If we apply this identification to each term of the limit, then we obtain an isomorphism

$$\widehat{\mathbb{Z}} \xrightarrow{\sim} \prod_p \mathbb{Z}_p, \quad (x_1, x_2, \dots) \mapsto ((x_1, x_p, x_{p^2}, \dots))_p. \quad (3.1)$$

We endow each $\mathbb{Z}_p$ with the usual p -adic topology and their product with the product topology. Then (3.1) is an isomorphism of topological rings.

Definition 3.1. The *ring of finite adèles* is defined by $\mathbb{A}_f := \mathbb{Q} \otimes_{\mathbb{Z}} \widehat{\mathbb{Z}}$. Since $\widehat{\mathbb{Z}}$ is torsion-free, we can view it as a subring $\widehat{\mathbb{Z}} \subset \mathbb{A}_f$. We endow $\mathbb{A}_f$ with the topology such that $\widehat{\mathbb{Z}}$ is an open subring.

Let us unravel this definition. First, on the level of rings, $\mathbb{A}_f$ is the ring of fractions x/m with $x \in \widehat{\mathbb{Z}}$ and $m \geq 1$, where the usual rules of arithmetic apply. Using (3.1), we can more explicitly describe it as the subring

$$\mathbb{A}_f = \left\{ (x_p) \in \prod_p \mathbb{Q}_p \mid x_p \in \mathbb{Z}_p \text{ for almost all } p \right\}.$$

Now we describe the topology. In $\widehat{\mathbb{Z}}$, a neighborhood basis of 0 is given by all the kernels of the projections $\widehat{\mathbb{Z}} \rightarrow \mathbb{Z}/n\mathbb{Z}$. These are precisely the ideals $n\widehat{\mathbb{Z}}$. Under the isomorphism (3.1), they are the subsets of the form

$$\prod_{p \in S} p^{m_p} \mathbb{Z}_p \times \prod_{p \notin S} \mathbb{Z}_p$$

where S is a finite set of primes and $(m_p)_{p \in S}$ a tuple of non-negative integers. Such sets forming a neighborhood basis of 0 means that the sets

$$\{x + n\widehat{\mathbb{Z}} \mid x \in \widehat{\mathbb{Z}}, n \geq 1\} \quad (3.2)$$

give a basis of the topology on $\widehat{\mathbb{Z}}$. Declaring $\widehat{\mathbb{Z}} \subset \mathbb{A}_f$ an open subring then simply means that the sets $n\widehat{\mathbb{Z}}$ also form a neighborhood basis of 0 in $\mathbb{A}_f$. Equivalently, the sets

$$\{x + n\widehat{\mathbb{Z}} \mid x \in \mathbb{A}_f, n \geq 1\} \quad (3.3)$$

provide a basis for the topology on $\mathbb{A}_f$.

Definition 3.2. The ring of adèles is defined as the product $\mathbb{A} := \mathbb{A}_f \times \mathbb{R}$ endowed with the product topology.

Proposition 3.3. *The subring $\mathbb{Q} \subset \mathbb{A}$ is discrete.*

Proof. By definitions, the product $U = \widehat{\mathbb{Z}} \times (-1, 1)$ is an open subset of $\mathbb{A}$. The intersection $U \cap \mathbb{Q}$ consists of those rational numbers that lie in $\mathbb{Z} = \widehat{\mathbb{Z}} \cap \mathbb{Q}$ and in the interval $(-1, 1)$. In other words, $U \cap \mathbb{Q} = \{0\}$. Thus, $\{0\} \subset \mathbb{Q}$ is an open subset for the subspace topology. By additive translation invariance of the topology ($\mathbb{A}$ is a topological ring), the same argument applies for all rational numbers. This shows that the subspace topology on $\mathbb{Q}$ is the discrete topology as claimed. $\square$

²We use $\varprojlim$ and $\varinjlim$ to denote the limit and the colimit. In other references, these might be called $\varprojlim$ and $\varinjlim$.

Let $F/\mathbb{Q}$ be a finite extension. The adeles of F can be defined in the same way as for $\mathbb{Q}$. First, we define the integral adeles with profinite topology

$$\widehat{O}_F := \lim_{\mathfrak{a} \subseteq O_F} O_F/\mathfrak{a} \xrightarrow{\sim} \prod_{\mathfrak{p}} O_{F,\mathfrak{p}}. \quad (3.4)$$

Then we tensor by $\mathbb{Q}$ over $\mathbb{Z}$, or equivalently by F over O_F , to define the finite adeles:

$$\begin{aligned} \mathbb{A}_{F,f} &:= \mathbb{Q} \otimes_{\mathbb{Z}} \widehat{O}_F \\ &\xrightarrow{\sim} \left\{ (x_{\mathfrak{p}}) \in \prod_{\mathfrak{p}} F_{\mathfrak{p}} \mid x_{\mathfrak{p}} \in O_{F,\mathfrak{p}} \text{ for almost all } \mathfrak{p} \right\}. \end{aligned} \quad (3.5)$$

Again, the topology on $\mathbb{A}_{F,f}$ is defined by declaring $\widehat{O}_F$ to be an open subring. Finally, we define the adeles as the product

$$\mathbb{A}_F := \mathbb{A}_{F,f} \times (\mathbb{R} \otimes_{\mathbb{Q}} F) \xrightarrow{\sim} \mathbb{A}_{F,f} \times \prod_{\sigma:F \rightarrow \mathbb{R}} \mathbb{R} \times \prod_{\{\sigma, \bar{\sigma}\}: F \rightarrow \mathbb{C}} \mathbb{C}. \quad (3.6)$$

Here, the real factors have their real vector space topology, and the last two products are over the real (resp. complex) places of F .

Recall that O_F is a free abelian group of rank equal to $d = [F : \mathbb{Q}]$. Let $\alpha_1, \dots, \alpha_d$ be a $\mathbb{Z}$ -module basis of O_F . Such a choice provides isomorphisms of $\widehat{\mathbb{Z}}$ -, $\mathbb{A}_f$ -, resp. $\mathbb{A}$ -modules

$$\widehat{\mathbb{Z}}^n \xrightarrow{\sim} O_F \otimes_{\mathbb{Z}} \widehat{\mathbb{Z}}, \quad \mathbb{A}_f^n \xrightarrow{\sim} F \otimes_{\mathbb{Q}} \mathbb{A}_f, \quad \mathbb{A}^n \xrightarrow{\sim} F \otimes_{\mathbb{Q}} \mathbb{A}. \quad (3.7)$$

We endow $\widehat{\mathbb{Z}}^n$, $\mathbb{A}_f^n$ and $\mathbb{A}^n$ with the product topology and use the isomorphisms in (3.7) to define from this the topology on the three tensor products. This topology is independent of the choice of $\alpha_1, \dots, \alpha_d$.

Remark 3.4. The previous definition is a general principle. Let R be a topological ring and let M be a finite free R -module. Any choice of R -basis $\alpha_1, \dots, \alpha_d$ defines an isomorphism $R^d \xrightarrow{\sim} M$ and, in this way, endows M with a topology.

Any two such isomorphisms differ by an element of $\mathrm{GL}_d(R)$. Since the action of every $g \in \mathrm{GL}_d(R)$ on R^d is continuous, the topology is independent of the chosen basis.

Proposition 3.5. *Multiplication defines isomorphisms of topological rings*

$$O_F \otimes_{\mathbb{Z}} \widehat{\mathbb{Z}} \xrightarrow{\sim} \widehat{O}_F, \quad F \otimes_{\mathbb{Q}} \mathbb{A}_f \xrightarrow{\sim} \mathbb{A}_{F,f}, \quad F \otimes_{\mathbb{Q}} \mathbb{A} \xrightarrow{\sim} \mathbb{A}_F.$$

Proof. Every ideal $\mathfrak{a} \subseteq O_F$ contains an ideal nO_F with $n \in \mathbb{Z}_{\geq 1}$. So we can rewrite (3.4) as $\widehat{O}_F = \lim O_F/nO_F$. Having chosen $\alpha_1, \dots, \alpha_d$, we obtain

$$\begin{aligned} \widehat{O}_F &= \lim \left(\bigoplus_{i=1}^d \mathbb{Z}/n\mathbb{Z} \cdot \alpha_i \right) \\ &\xrightarrow{\sim} \bigoplus_{i=1}^d \left(\lim \mathbb{Z}/n\mathbb{Z} \right) \cdot \alpha_i \\ &\xrightarrow{\sim} \bigoplus_{i=1}^d \widehat{\mathbb{Z}} \cdot \alpha_i. \end{aligned}$$

This shows that $O_F \otimes_{\mathbb{Z}} \widehat{\mathbb{Z}} \xrightarrow{\sim} \widehat{O}_F$ as topological rings. The statements for $\mathbb{A}_{F,f}$ and $\mathbb{A}_F$ follow from this. $\square$

Corollary 3.6. *Let $F/\mathbb{Q}$ be a finite extension. Then $F \subset \mathbb{A}_F$ is discrete.*

Proof. Since $\mathbb{Q}$ is discrete in $\mathbb{A}$, we have that $\mathbb{Q}^n$ is discrete in $\mathbb{A}^n$. Choosing a $\mathbb{Q}$ -basis $\alpha = (\alpha_1, \dots, \alpha_d)$ for F , we obtain a commutative square of the form

$$\begin{array}{ccc} \mathbb{Q}^n & \hookrightarrow & \mathbb{A}^n \\ \alpha \parallel & & \parallel \alpha \\ F & \hookrightarrow & \mathbb{A}_F. \end{array}$$

By Proposition 3.5, the right vertical identification is a homeomorphism. Hence we obtain that F is discrete in $\mathbb{A}_F$. $\square$

3.2. Groups of the form $G(\mathbb{A}_f)$. Let us formulate the problem more generally.

Question 3.7. Let X be an affine variety³ over $\mathbb{Q}$ and let R be a topological $\mathbb{Q}$ -algebra. We assume that points of R are closed. For example, R could be $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}_p$, $\mathbb{A}_f$ or $\mathbb{A}$. How to define the topological space $X(R)$ in a natural way?

The answer is very simple. Let us write $\mathcal{A}^N = \text{Spec } \mathbb{Q}[t_1, \dots, t_N]$ for affine N -space over $\mathbb{Q}$ to avoid confusion with the adèle notation. We endow $\mathcal{A}^N(R) = R^N$ with the product topology.

Let $f_1, \dots, f_m \in \mathbb{Q}[t_1, \dots, t_N]$ be polynomials and let $X = V(f_1, \dots, f_m) \subseteq \mathcal{A}^N$ be their vanishing locus. Then $X(R) \subseteq R^N$ is a closed subset because it equals the intersection $\bigcap_{i=1}^m f_i^{-1}(0)$, and we endow it with the subspace topology.

Definition 3.8. Let X be an affine $\mathbb{Q}$ -variety. Choose a presentation $\varphi : X \xrightarrow{\sim} V(f_1, \dots, f_m)$ as above. The topology on $X(R)$ is defined as the subspace topology with respect to $\varphi(R) : X(R) \hookrightarrow R^N$.

Lemma 3.9. *This topology on $X(R)$ is independent of the choices of N , $(f_1, \dots, f_m)$ and φ .*

Proof. Assume that we are given two affine varieties $V(f_1, \dots, f_{m_1}) \subseteq \mathcal{A}^{N_1}$ as well as $V(g_1, \dots, g_{m_2}) \subseteq \mathcal{A}^{N_2}$. Assume that

$$\varphi : V(f_1, \dots, f_{m_1}) \xrightarrow{\sim} V(g_1, \dots, g_{m_2})$$

is an isomorphism of $\mathbb{Q}$ -varieties. Then φ and $\psi = \varphi^{-1}$ lift to morphisms $\Phi : \mathcal{A}^{N_1} \rightarrow \mathcal{A}^{N_2}$ and $\Psi : \mathcal{A}^{N_2} \rightarrow \mathcal{A}^{N_1}$. The induced maps

$$R^{N_1} \xrightleftharpoons[\Psi]{\Phi} R^{N_2}$$

are continuous because they are given by polynomials. Hence their restrictions φ and ψ are continuous as well. Since $\psi = \varphi^{-1}$, this shows that φ is a homeomorphism. $\square$

Example 3.10. Consider the group variety GL_n . One possible presentation as a closed subset of an affine space is given by

$$\begin{aligned} \text{GL}_n &\xrightarrow{\sim} V(1 - t \cdot \det((t_{ij})_{i,j=1}^n)) \subset \mathcal{A} \times_{\text{Spec}(\mathbb{Q})} \mathcal{A}^{n^2} \\ g = (t_{ij})_{i,j=1}^n &\longmapsto (\det(g)^{-1}, g). \end{aligned}$$

For example, if $n = 1$, we recover the closed immersion⁴

$$\mathbb{G}_m \hookrightarrow \mathcal{A}^2, \quad t \longmapsto (t^{-1}, t).$$

According to Definition 3.8, the topology on $\text{GL}_n(\mathbb{A}_f)$ is then given as the subspace topology with respect to

$$\text{GL}_n(\mathbb{A}_f) \hookrightarrow \mathbb{A}_f \times \text{M}_n(\mathbb{A}_f), \quad g \longmapsto (\det(g)^{-1}, g).$$

³More generally, an affine finite type $\mathbb{Q}$ -scheme.

⁴ $\mathbb{G}_m$ is just another notation for GL_1 . The notation symbolizes *multiplicative group*.

The product $\widehat{\mathbb{Z}} \times M_n(\widehat{\mathbb{Z}})$ is an open subset on the right hand side. So the intersection

$$GL_n(\widehat{\mathbb{Z}}) = GL_n(\mathbb{A}_f) \cap (\widehat{\mathbb{Z}} \times M_n(\widehat{\mathbb{Z}}))$$

is an open subset of $GL_n(\mathbb{A}_f)$. (The elements of $GL_n(\widehat{\mathbb{Z}})$ are precisely those elements of $GL_n(\mathbb{A}_f) \cap M_n(\widehat{\mathbb{Z}})$ whose inverse determinant again lies in $\widehat{\mathbb{Z}}$.) As a closed subset of the profinite set $\widehat{\mathbb{Z}} \times M_n(\widehat{\mathbb{Z}})$, $GL_n(\widehat{\mathbb{Z}})$ is again profinite. In fact, we have

$$GL_n(\widehat{\mathbb{Z}}) \xrightarrow{\sim} \lim_{m \geq 1} GL_n(\mathbb{Z}/m\mathbb{Z})$$

as topological group. The principal congruence subgroups

$$K(m) := \ker (GL_n(\widehat{\mathbb{Z}}) \longrightarrow GL_n(\mathbb{Z}/m\mathbb{Z})) \quad (3.8)$$

form a neighborhood basis of 1 in $GL_n(\mathbb{A}_f)$.

Example 3.11. We always view $\mathbb{A}_f^\times$ with the topology coming from $\mathbb{A}_f^\times = \mathbb{G}_m(\mathbb{A}_f)$. Then the inclusion map $\mathbb{A}_f^\times \rightarrow \mathbb{A}_f$ is continuous because it is induced from the morphism of varieties $\mathbb{G}_m \rightarrow \mathcal{A}$, $t \mapsto t$. But it is not an open immersion. For example, $\widehat{\mathbb{Z}}^\times$ is open in $\mathbb{A}_f^\times$, but not in $\mathbb{A}_f$.

Exercise 3.12. Prove the claim in the previous example. That is, show that none of the open subsets $1 + n\widehat{\mathbb{Z}} \subseteq \widehat{\mathbb{Z}}$, which form a neighborhood basis of $1 \in \widehat{\mathbb{Z}}$, is contained in $\widehat{\mathbb{Z}}^\times$.

Example 3.13. Let G be a general linear algebraic group over $\mathbb{Q}$. There always exist some $N \geq 1$ and a closed immersion $G \hookrightarrow GL_N$. Then $G(\mathbb{A}_f) \subseteq GL_N(\mathbb{A}_f)$ has the subspace topology. In particular, the intersections $G(\mathbb{A}_f) \cap K(m)$ with all congruence subgroups form a neighborhood basis of $1 \in G(\mathbb{A}_f)$.

This applies, for example, to the standard representations

$$SL_2 \hookrightarrow GL_2, \quad Sp_{2g} \hookrightarrow GL_{2g}, \quad GSp_{2g} \hookrightarrow GL_{2g}.$$

Let V be a quadratic $\mathbb{Q}$ -vector space. Then it applies to the closed immersions

$$SO(V) \hookrightarrow GL(V), \quad O(V) \hookrightarrow GL(V).$$

Remark 3.14. For local fields k , such as $k \in \{\mathbb{R}, \mathbb{C}, \mathbb{Q}_p\}$, the situation is more straightforward in the following sense. If $X \hookrightarrow Y$ is an open immersion of k -varieties, then $X(k) \rightarrow Y(k)$ is an open immersion with respect to the topologies from Definition 3.8. In particular, the topology on $X(k)$ from Definition 3.8 agrees with the subspace topology in $Y(k)$.

This remark applies, for example, to

$$GL_n(\mathbb{R}) \subset M_n(\mathbb{R}) \quad \text{and} \quad GL_n(\mathbb{Q}_p) \subset M_n(\mathbb{Q}_p).$$

3.3. General adelic double quotients. Let us begin with a general theorem which we will not prove.

Theorem 3.15 ([15, Theorem 4.16]). *(1) Let $G/\mathbb{Q}$ be a connected reductive algebraic group. Then, for every compact open subgroup $K \subset G(\mathbb{A}_f)$, the double quotient $G(\mathbb{Q}) \backslash G(\mathbb{A}_f) / K$ is finite.*

(2, Strong approximation) Let $G/\mathbb{Q}$ be a connected, simply connected semi-simple group of non-compact type. Then $G(\mathbb{Q})$ is dense in $G(\mathbb{A}_f)$. In particular, for every compact open subgroup $K \subset G(\mathbb{A}_f)$,

$$G(\mathbb{A}_f) = \{\gamma \cdot k \mid \gamma \in G(\mathbb{Q}), k \in K\}.$$

As our first application, we obtain a more concrete description of the adelic double quotients that make up the complex points of a Shimura variety (1.7). Let (G, X) be a Shimura datum and let $K \subset G(\mathbb{A}_f)$ be a level subgroup. In particular, G is a connected reductive group over $\mathbb{Q}$, so Theorem 3.15 (1) applies. So we find finitely many double coset representatives $g_1, \dots, g_r \in G(\mathbb{A}_f)$,

$$G(\mathbb{A}_f) = \bigsqcup_{i=1}^r G(\mathbb{Q})g_iK. \quad (3.9)$$

Each of the sets on the right hand side of (3.9) is $G(\mathbb{Q})$ -stable. Moreover, $G(\mathbb{Q})$ acts transitively on the cosets $G(\mathbb{Q})g_iK/K$, and the stabilizer of the coset $g_iK \in G(\mathbb{Q})g_iK/K$ is the subgroup

$$\Gamma_i := G(\mathbb{Q}) \cap g_iKg_i^{-1}.$$

So we obtain

$$\begin{aligned} G(\mathbb{Q}) \backslash (X \times G(\mathbb{A}_f)/K) &= \bigsqcup_{i=1}^r G(\mathbb{Q}) \backslash (X \times G(\mathbb{Q})g_iK/K) \\ &\xrightarrow{\sim} \bigsqcup_{i=1}^r \Gamma_i \backslash X \times \{g_iK\}. \end{aligned} \quad (3.10)$$

Each Γ_i contains the subgroup $\Gamma_0 = K \cap Z(G)(\mathbb{Q})$ and the action of Γ_i on $X \times \{g_iK\}$ is via the quotient $\overline{\Gamma}_i = \Gamma_i/\Gamma_0$. If K is small enough, which we will make precise for GL_2 in a minute, then each $\overline{\Gamma}_i$ is torsion-free and acts without stabilizers on X . So each quotient $\Gamma_i \backslash X$ is then a complex manifold in the same way as we saw before in Conclusion 2.12.

Exercise 3.16. Work out (3.10) for yourself. For example, first prove the following variant. Let H be a group acting on sets X and Y . Let $Y = \sqcup_{i \in I} G \cdot y_i$ be the decomposition of Y into orbits and let Γ_i be the stabilizer of y_i in H . Then

$$H \backslash (X \times Y) \xrightarrow{\sim} \bigsqcup_{i \in I} \Gamma_i \backslash X.$$

Specialize to the situation $H = G(\mathbb{Q})$ and $Y = G(\mathbb{A}_f)/K$.

Exercise 3.17. The group SL_n is connected, simply connected, semi-simple and of non-compact type, so $\mathrm{SL}_n(\mathbb{Q}) \subset \mathrm{SL}_n(\mathbb{A}_f)$ is dense (Strong approximation, see Theorem 3.15 (2)). Using this property, show that

$$\mathrm{SL}_n(\mathbb{Z}) \longrightarrow \mathrm{SL}_n(\mathbb{Z}/m\mathbb{Z})$$

is surjective for all $m \geq 1$. In particular, this shows the surjectivity of (2.6).

3.4. Back to GL_2 . The description in (3.10) is still quite abstract. We now want to make it completely explicit for congruence subgroups of GL_2 . Let us begin by studying $\mathbb{G}_m$.

Proposition 3.18. *Let $K(m) = \ker(\widehat{\mathbb{Z}}^\times \rightarrow (\mathbb{Z}/m\mathbb{Z})^\times)$ be the m -th congruence subgroup of $\mathbb{A}_f^\times$. Then there is an isomorphism*

$$\mathbb{Q}_{>0}^\times \backslash \mathbb{A}_f^\times / K(m) \xrightarrow{\sim} (\mathbb{Z}/m\mathbb{Z})^\times. \quad (3.11)$$

Proof. Let $x = (x_p)_p \in \mathbb{A}_f^\times$ be an element. Here, the component x_p lies in $\mathbb{Q}_p^\times$, and almost all components x_p even lie in $\mathbb{Z}_p^\times$. For each prime p , let $v_p : \mathbb{Q}_p^\times \rightarrow \mathbb{Z}$ denote the valuation normalized by $v_p(p) = 1$. Take the vector of valuations of all the entries of x :

$$(e_p)_p, \quad e_p = v_p(x_p).$$

Only finitely many of the e_p are non-zero. There is a rational number in $\mathbb{Q}_{>0}$ with the same valuations, namely $t = \prod_p p^{e_p}$. So $t^{-1}x$ lies in $\widehat{\mathbb{Z}}^\times$ which shows that every double coset in (3.11) has a representative in $\widehat{\mathbb{Z}}^\times$. Purely formally, we now obtain

$$\mathbb{Q}_{>0}^\times \backslash \mathbb{A}_f^\times / K(m) \xrightarrow{\sim} (\mathbb{Q}_{>0}^\times \cap \widehat{\mathbb{Z}}^\times) \backslash \widehat{\mathbb{Z}}^\times / K(m). \quad (3.12)$$

The rational number t is, in fact, uniquely determined which reflects that $\mathbb{Q}_{>0}^\times \cap \widehat{\mathbb{Z}}^\times = \{1\}$. So (3.12) simplifies to $\widehat{\mathbb{Z}}^\times / K(m)$, which is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^\times$ as claimed. $\square$

We write $\mathrm{GL}_n(\mathbb{Q})_{>0}$ for the subgroup of elements of $\mathrm{GL}_n(\mathbb{Q})$ with positive determinant.

Proposition 3.19. *Let $K \subset \mathrm{GL}_n(\mathbb{A}_f)$ be an open compact subgroup. The determinant map $\det : \mathrm{GL}_n(\mathbb{A}_f) \rightarrow \mathbb{A}_f^\times$ induces a bijection*

$$\det : \mathrm{GL}_n(\mathbb{Q})_{>0} \backslash \mathrm{GL}_n(\mathbb{A}_f) / K \xrightarrow{\sim} \mathbb{Q}_{>0}^\times \backslash \mathbb{A}_f^\times / \det(K). \quad (3.13)$$

Proof. The group SL_n is connected, simply connected, semi-simple and of non-compact type, so $\mathrm{SL}_n(\mathbb{Q}) \subset \mathrm{SL}_n(\mathbb{A}_f)$ is dense (Strong approximation, see Theorem 3.15 (2)). We will use this property freely.

Consider the determinant map in (3.13). It is clearly surjective because already the map $\det : \mathrm{GL}_n(\mathbb{A}_f) \rightarrow \mathbb{A}_f^\times$ is surjective. So our task is to prove that (3.13) is injective.

The source in (3.13) is only a set, so we cannot argue with kernels. Instead, we consider two elements $g_1, g_2 \in \mathrm{GL}_n(\mathbb{A}_f)$ with the same image, meaning that

$$\det(g_1) \in \mathbb{Q}_{>0}^\times \det(g_2) \det(K). \quad (3.14)$$

Our task is to show that $g_1 \in \mathrm{GL}_n(\mathbb{Q})g_2K$.

First, observe that $\det : \mathrm{GL}_n(\mathbb{Q})_{>0} \rightarrow \mathbb{Q}_{>0}^\times$ is surjective. So we find elements $h \in \mathrm{GL}_n(\mathbb{Q})_{>0}$ and $k \in K$ such that $\det(g_1) = \det(hg_2k)$. So after replacing g_2 by hg_2k , we may assume $\det(g_1) = \det(g_2)$.

Next, we consider the conjugate group $g_2Kg_2^{-1}$. Strong approximation for SL_n implies that

$$\mathrm{SL}_n(\mathbb{A}_f) = \mathrm{SL}_n(\mathbb{Q}) \cdot (g_2Kg_2^{-1} \cap \mathrm{SL}_n(\mathbb{A}_f)). \quad (3.15)$$

Hence, there are $h' \in \mathrm{SL}_n(\mathbb{Q})$ and $k' \in K \cap \mathrm{SL}_n(\mathbb{A}_f)$ with

$$g_1g_2^{-1} = h'g_2k'g_2^{-1}.$$

This is equivalent to $g_1 = h'g_2k'$, showing that the double cosets of g_1 and g_2 are equal as claimed. $\square$

Corollary 3.20. *Let $K(m) \subset \mathrm{GL}_2(\mathbb{A}_f)$ be the m -th congruence subgroup. There is a bijection of connected components*

$$\pi_0(\mathrm{GL}_2(\mathbb{Q}) \backslash (\mathbb{H}^\pm \times \mathrm{GL}_2(\mathbb{A}_f) / K(m))) \xrightarrow{\sim} (\mathbb{Z}/m\mathbb{Z})^\times. \quad (3.16)$$

Moreover, the connected components are all of the form $\Gamma \backslash \mathbb{H}$ with $\Gamma = \mathrm{GL}_2(\mathbb{Q})_{>0} \cap gK(m)g^{-1}$ for some element $g \in \mathrm{GL}_2(\mathbb{A}_f)$.

Proof. The two connected components of $\mathbb{H}^\pm$ are interchanged by the elements of negative determinant in $\mathrm{GL}_2(\mathbb{Q})$. Hence, we obtain

$$\begin{aligned} \pi_0(\mathrm{GL}_2(\mathbb{Q}) \backslash (\mathbb{H}^\pm \times \mathrm{GL}_2(\mathbb{A}_f) / K(m))) &\xrightarrow{\sim} \pi_0(\mathrm{GL}_2(\mathbb{Q})_{>0} \backslash (\mathbb{H} \times \mathrm{GL}_2(\mathbb{A}_f) / K(m))) \\ &\xrightarrow{\sim} \mathrm{GL}_2(\mathbb{Q})_{>0} \backslash \mathrm{GL}_2(\mathbb{A}_f) / K(m). \end{aligned} \quad (3.17)$$

Here, the second isomorphism simply used that $\mathbb{H}$ is connected. Next, observe that

$$L := \det(K(m)) = \ker(\widehat{\mathbb{Z}}^\times \rightarrow (\mathbb{Z}/m\mathbb{Z})^\times)$$

is the m -th congruence subgroup in $\mathbb{A}_f^\times$. So, by Proposition 3.19, the determinant allows to rewrite (3.17) as

$$\det : \mathrm{GL}_2(\mathbb{Q})_{>0} \backslash \mathrm{GL}_2(\mathbb{A}_f) / K(m) \xrightarrow{\sim} \mathbb{Q}_{>0}^\times \backslash \mathbb{A}_f^\times / L.$$

By Proposition 3.18, the last expression can be identified with $(\mathbb{Z}/m\mathbb{Z})^\times$ as claimed.

The final statement (each connected component being isomorphic to some $\Gamma \backslash \mathbb{H}$ with Γ of the form $\mathrm{GL}_2(\mathbb{Q})_{>0} \cap gK(m)g^{-1}$) is a special case of the decomposition in (3.10), except that we have already replaced $(\mathbb{H}^\pm, \mathrm{GL}_2(\mathbb{Q}))$ by $(\mathbb{H}, \mathrm{GL}_2(\mathbb{Q})_{>0})$. $\square$

Let us go further and prove a criterion that ensures that all the occurring Γ are torsion free. The arguments will be similar to the ones we saw in §2.3.

Proposition 3.21. *For any $m \geq 3$ and $g \in \mathrm{GL}_2(\mathbb{A}_f)$, the intersection $\Gamma = \mathrm{GL}_2(\mathbb{Q}) \cap gK(m)g^{-1}$ is torsion free.*

Proof. Let γ be an element of $K(m)$. Then, since $K(m) \subseteq \mathrm{GL}_2(\widehat{\mathbb{Z}})$, the characteristic polynomial $P_\gamma(T)$ lies in $\widehat{\mathbb{Z}}[T]$. Since $\gamma \equiv 1 \pmod{m}$, we even know $P_\gamma(T) \equiv (T-1)^2 \pmod{m}$. In general, for every $n \geq 1$ and any ring R , the characteristic polynomial of an element from $\mathrm{GL}_n(R)$ is invariant under conjugation. So, in our setting, the same properties hold for $P_\gamma(T)$ for $\gamma \in gK(m)g^{-1}$.

Assume that $\gamma \in \Gamma = \mathrm{GL}_2(\mathbb{Q}) \cap gK(m)g^{-1}$. Then, the characteristic polynomial of γ has rational coefficients, and hence lies in the intersection

$$\mathbb{Q}[T] \cap ((T-1)^2 + n\widehat{\mathbb{Z}}[T]).$$

This means that $P_\gamma(T) \in \mathbb{Z}[T]$ and $P_\gamma(T) \equiv (T-1)^2 \pmod{m}$.

If γ is a torsion element, then we have already seen during the proof of Proposition 2.10 that $P_\gamma(T)$ comes from the list (2.8). By the congruence condition we just established, the only possibility is $P_\gamma(T) = (T-1)^2$. The matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is not torsion, so cannot be the Jordan normal form of γ . We conclude that $\gamma = 1$, showing that Γ is torsion-free as claimed. $\square$

Conclusion 3.22. Let us come back to the situation of Corollary 3.20. Assume that $m \geq 3$. Then the connected components of

$$\mathrm{GL}_2(\mathbb{Q}) \backslash (\mathbb{H}^\pm \times \mathrm{GL}_2(\mathbb{A}_f))$$

are in natural bijection with $(\mathbb{Z}/m\mathbb{Z})^\times$. Each connected component is of the form $\Gamma \backslash \mathbb{H}$ for a torsion free arithmetic subgroup $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Q})$.

4. GROUP SCHEMES

Our next aim is to endow the complex manifolds $\mathrm{Sh}_K(\mathrm{GL}_2, \mathbb{H}^\pm)(\mathbb{C})$ with an algebraic structure and to even define them over $\mathbb{Q}$ (see 1.7). This relies on their description as moduli spaces of elliptic curves:

Definition 4.1. Let k be a field. An *elliptic curve* over k is a proper, smooth, connected and 1-dimensional k -group scheme.

Later in the course, we will also consider other Shimura varieties and describe them as moduli spaces of abelian varieties:

Definition 4.2. An *abelian variety* over k is a proper, smooth and connected k -group scheme.

In this lecture, we will first discuss some background on group schemes. This will also be useful for talking about groups like GL_n , GSp_{2g} etc. which we have secretly already considered as group schemes over $\mathrm{Spec} \mathbb{Z}$ or $\mathrm{Spec} \mathbb{Q}$ in previous lectures. In general, group schemes are also an interesting topic in itself and come up in many areas of algebra.

Recommended reading closely related to our course: My lecture notes on moduli spaces of elliptic curves [12]. Parts of our discussion here are taken from [12, §2].

General reference on algebraic groups: Milne's book [16], especially §1 about basic definitions.

4.1. Basic definitions. We give the definition over a general base S , but the case to keep in mind is $S = \text{Spec}(k)$ for a field k .

Definition 4.3. Let S be a scheme. A *group scheme* over S is a pair (G, m) that consists of an S -scheme G and an S -scheme morphism (called multiplication morphism)

$$m : G \times_S G \longrightarrow G$$

such that for every S -scheme T , the resulting map on T -valued points

$$m(T) : G(T) \times G(T) \longrightarrow G(T)$$

makes $G(T)$ into a group. We call G *commutative* if $G(T)$ is a commutative group for every T .

Observe that for every morphism $u : T' \rightarrow T$ of S -schemes, the diagram

$$\begin{array}{ccc} G(T) \times G(T) & \xrightarrow{m(T)} & G(T) \\ u^* \times u^* \downarrow & & \downarrow u^* \\ G(T') \times G(T') & \xrightarrow{m(T')} & G(T') \end{array} \quad (4.1)$$

commutes which means that $u^* : G(T) \rightarrow G(T')$ is a group homomorphism. Furthermore, if (G, m) is a group scheme over S , then the Yoneda Lemma implies the existence of two additional S -scheme morphisms:

$$\begin{aligned} e : S &\longrightarrow G, & (\text{neutral element section}) \\ i : G &\longrightarrow G, & (\text{inversion morphism}). \end{aligned} \quad (4.2)$$

The first one is simply the neutral element $e \in G(S)$ of the group $G(S)$. Given $u : T \rightarrow S$, the pullback $u^*(e) = e \circ u \in G(T)$ is the neutral element of $G(T)$. The second one is characterized as the unique morphism that provides the inverse in all the groups $\{G(T)\}_{T \rightarrow S}$:

$$i(T) : G(T) \longrightarrow G(T), \quad g \longmapsto g^{-1}.$$

The datum (G, m, e, i) satisfies the group axioms in a scheme sense, meaning that the three diagrams

$$\begin{array}{ccc} S \times_S G & \xrightarrow{e \times \text{id}} & G \times_S G \\ & \searrow & \swarrow m \\ & G, & \end{array} \quad (4.3)$$

$$\begin{array}{ccc} G \times_S G & \xrightarrow{\text{id} \times i} & G \times_S G \\ \downarrow & & \downarrow m \\ S & \xrightarrow{e} & G, \end{array} \quad \begin{array}{ccc} G \times_S G \times_S G & \xrightarrow{m \times \text{id}} & G \times_S G \\ \text{id} \times m \downarrow & & \downarrow m \\ G \times_S G & \xrightarrow{m} & G. \end{array} \quad (4.4)$$

all commute. In fact, one may also reverse the above logic and obtains the more classical definition of a group scheme over S : It is the same as an S -scheme G together with a morphism $m : G \times_S G \rightarrow G$ such that there exist morphisms $e : S \rightarrow G$ and $i : G \rightarrow G$ such that the diagrams in (4.3) and (4.4) commute. The group scheme (G, m) is commutative

if and only if multiplication interchanges with switching the factors in the sense that also the following diagram commutes:

$$\begin{array}{ccc} G \times_S G & \xrightarrow{(g,h) \mapsto (h,g)} & G \times_S G \\ & \searrow m \quad \swarrow m & \\ & G & \end{array} \quad (4.5)$$

Definition 4.4. Let (G_1, m_1) and (G_2, m_2) be group schemes over S . A group scheme morphism from G_1 to G_2 is a morphism of S -schemes $f : G_1 \rightarrow G_2$ such that $m_2 \circ (f \times f) = f \circ m_1$. Equivalently, it is an S -morphism f such that for all $T \rightarrow S$, the induced map

$$f(T) : G_1(T) \longrightarrow G_2(T)$$

is a group homomorphism.

If (G, m) is a *commutative* S -group scheme, then $\text{End}_{S\text{-Grp.Sch.}}(G, m)$ forms a (possibly non-commutative) ring because endomorphisms can be “added” (meaning multiplied in G) and multiplied (meaning composed). Concretely, sum and product of two elements $f, g \in \text{End}(G)$ are given by

$$f + g := m \circ (f, g), \quad fg := f \circ g.$$

In particular, we can add the identity n times to itself and obtain the n -th power endomorphism $[n] : G \rightarrow G$. On each of the groups $G(T)$, it is given by $[n](g) = g^n$. This is even defined for $n \in \mathbb{Z}$ by $[n] \circ i = [-n]$. In total, these give the ring map

$$[\cdot] : \mathbb{Z} \rightarrow \text{End}(G). \quad (4.6)$$

Coming back to general group schemes, we next define kernels. This is straightforward because fiber products exist in the category of S -schemes. (Defining quotients, on the other hand, is tricky. We refer to [12, §13] for some cases.)

Definition 4.5. Let $f : G_1 \rightarrow G_2$ be a homomorphism of S -group schemes. Let $e_2 : S \rightarrow G_2$ be the neutral element section of G_2 . The kernel of f is defined as the fiber product

$$\begin{array}{ccc} \ker(f) & \longrightarrow & S \\ \downarrow & & \downarrow e_2 \\ G_1 & \xrightarrow{f} & G_2. \end{array} \quad (4.7)$$

It is clear from its definition that $\ker(f)$ has the property

$$\ker(f)(T) = \ker(f(T) : G_1(T) \longrightarrow G_2(T)). \quad (4.8)$$

The multiplication morphism of G_1 restricts to a multiplication on $\ker(f)$ which makes $\ker(f)$ into a group scheme:

$$\begin{array}{ccc} \ker(f) \times_S \ker(f) & \dashrightarrow & \ker(f) \\ \downarrow & & \downarrow \\ G \times_S G & \xrightarrow{m} & G. \end{array} \quad (4.9)$$

Remark 4.6. Recall that if $X \rightarrow S$ is a separated morphism, then every section $\sigma : S \rightarrow X$ is a closed immersion. Thus, if $G \rightarrow S$ is a separated group scheme (e.g. affine or proper), then the neutral element e is a closed immersion. It follows that if in (4.7) $G_2 \rightarrow S$ is separated, then $\ker(f) \rightarrow G_1$ is a closed immersion.

4.2. A commutative example: The multiplicative group. Assume that $S = \operatorname{Spec} R$ is affine. Define $\mathbb{G}_{m,S} = \operatorname{Spec} R[t, t^{-1}]$ which we would like to make into a group scheme over S . Recall that $\operatorname{Spec}(-)$ is an anti-equivalence from R -algebras to affine S -schemes. We define the multiplication map $m : \mathbb{G}_{m,S} \times_S \mathbb{G}_{m,S} \rightarrow \mathbb{G}_{m,S}$ as $\operatorname{Spec}(m^*)$ where m^* is

$$\begin{aligned} m^* : R[t, t^{-1}] &\longrightarrow R[t, t^{-1}] \otimes_R R[t, t^{-1}] \\ t &\longmapsto t \otimes t. \end{aligned} \quad (4.10)$$

We next verify that this makes $\mathbb{G}_{m,S}$ into an S -group scheme. For every S -scheme T , we identify

$$\begin{aligned} \mathbb{G}_{m,S}(T) &\xrightarrow{\sim} \mathcal{O}_T(T)^\times \\ [g : T \rightarrow \mathbb{G}_{m,S}] &\longmapsto g^*(t). \end{aligned} \quad (4.11)$$

Note that this map is obviously defined; the fact that it is an isomorphism is the adjunction $\operatorname{Mor}_S(T, \operatorname{Spec}(A)) \xrightarrow{\sim} \operatorname{Hom}_R(A, \mathcal{O}_T(T))$. Given two morphisms $g_1, g_2 : T \rightarrow \mathbb{G}_{m,S}$, we compute the (dual of the) composition $m \circ (g_1, g_2)$ by

$$\begin{aligned} R[t, t^{-1}] &\xrightarrow{m^*} R[t, t^{-1}] \otimes_R R[t, t^{-1}] \xrightarrow{g_1^* \otimes g_2^*} \mathcal{O}_T(T) \\ t &\longmapsto t \otimes t \longmapsto g_1^*(t)g_2^*(t). \end{aligned}$$

Thus we see that the operation $m(T)$ on $\mathbb{G}_{m,S}(T)$ translates to the usual multiplication under (4.11). In particular, $m(T)$ is a group structure for every T , and hence $(\mathbb{G}_{m,S}, m)$ is a group scheme.

We can next calculate the neutral element e and the inversion map i from (4.2). Under (4.11), the unit element $1 \in R^\times$ corresponds to

$$e^* : R[t, t^{-1}] \longrightarrow R, \quad t \longmapsto 1.$$

Taking $e = \operatorname{Spec}(e^*)$ gives the neutral element section. The inversion map $i = \operatorname{Spec}(i^*)$ is given by

$$i^* : R[t, t^{-1}] \longrightarrow R[t, t^{-1}], \quad t \longmapsto t^{-1}. \quad (4.12)$$

The n -th power maps are given as $[n] = \operatorname{Spec}([n]^*)$ with

$$[n]^* : R[t, t^{-1}] \longrightarrow R[t, t^{-1}], \quad t \longmapsto t^n. \quad (4.13)$$

Note that (4.12) and (4.13) are compatible in the sense that $i = [-1]$, which is always the case for a commutative group scheme. The next proposition, on the other hand, is very specific to $\mathbb{G}_m$.

Proposition 4.7. *Let S be a connected scheme. Then $\operatorname{End}(\mathbb{G}_{m,S}) = \mathbb{Z}$.*

Proof. We only consider the case $S = \operatorname{Spec}(k)$. The extension to general S can be found in [12, Proposition 2.12].

By definition, a group scheme endomorphism f of $\mathbb{G}_{m,k}$ is the same as $f = \operatorname{Spec}(f^*)$ for a unique k -algebra morphism $f^* : k[t, t^{-1}] \longrightarrow k[t, t^{-1}]$ such that

$$(f^* \otimes f^*) \circ m^* = m^* \circ f^* \quad (4.14)$$

where $m^*(t) = t \otimes t$ is as in (4.10). Giving a k -algebra morphism f^* is equivalent to specifying its image $f^*(t) \in k[t, t^{-1}]^\times$. These units are

$$k[t, t^{-1}]^\times = \{\lambda t^n \mid \lambda \in k^\times, n \in \mathbb{Z}\}.$$

If $f^*(t) = \lambda t^n$, then (4.14) evaluated at t becomes

$$\lambda t^n \otimes \lambda t^n \stackrel{?}{=} \lambda(t \otimes t)^n \quad (4.15)$$

which holds if and only if $\lambda^2 = \lambda$, meaning $\lambda = 1$. Note that $f^*(t) = t^n$ precisely defines the multiplication-by- n morphism $[n]$ (meaning taking n -th power in this context) and thus $\operatorname{End}(\mathbb{G}_{m,k}) = \mathbb{Z}$ is proved. $\square$

We next determine the kernel $\mu_{n,S} := \ker([n])$. By definition, see (4.7), we need to compute the fiber product

$$\begin{array}{ccc} \mu_{n,S} & \longrightarrow & S \\ \downarrow & & \downarrow \\ \mathbb{G}_{m,S} & \xrightarrow{[n]} & \mathbb{G}_{m,S}. \end{array}$$

Fiber products of affine schemes are computed by tensor products of rings, so we get

$$\begin{aligned} \mu_{n,S} &= \operatorname{Spec} (R \otimes_{1 \leftarrow t, R[t, t^{-1}], t \mapsto t^n} R[t, t^{-1}]) \\ &= \operatorname{Spec} (R[t]/(t^n - 1)). \end{aligned} \quad (4.16)$$

In terms of (4.8) and (4.11), we see

$$\mu_{n,S}(T) = \{\zeta \in \mathcal{O}_T(T)^\times \mid \zeta^n = 1\}. \quad (4.17)$$

That is, $\mu_{n,S}$ is the *group scheme of n -th roots of unity*. Let us assume that $S = \operatorname{Spec}(k)$. We observe the following interesting phenomenon:

Assume that n is prime to $\operatorname{char}(k)$. Then $t^n - 1 \in k[t]$ is a separable polynomial. Hence, $\mu_{n,k} = \operatorname{Spec} k[t]/(t^n - 1)$ is an étale k -scheme. On the other hand, if $p = \operatorname{char}(k) \mid n$, then $t^n - 1$ is not separable and $k[t]/(t^n - 1)$ is not reduced. For example,

$$\begin{aligned} \mu_{p,k} &= \operatorname{Spec} k[t]/(t^p - 1) \\ &= \operatorname{Spec} k[t]/(t - 1)^p \\ &\xrightarrow{\sim} \operatorname{Spec} k[\varepsilon]/(\varepsilon^p) \end{aligned}$$

is completely infinitesimal. We have the following general results in this direction.

Theorem 4.8 (Cartier, [16, Corollary 8.38]). *Let k be a field of characteristic 0 and let G/k be a finite type group scheme. Then G is smooth.*

Definition 4.9. A morphism $f : X \rightarrow S$ is said to be *finite locally free of rank n* if it is finite and if $f_*(\mathcal{O}_X)$ is locally free of rank n as $\mathcal{O}_S$ -module.

Theorem 4.10. *Let G be a commutative S -group scheme which is finite locally free of rank n .*

(1) *Then G is n -torsion. That is, the multiplication map $[n] : G \rightarrow G$ is the 0-map. (Deligne's Theorem)*

(2) *Assume that $n \in \mathcal{O}_S(S)^\times$. Then G is étale.*

Exercise 4.11. Verify the commutativity of (4.3) and (4.4) for $(\mathbb{G}_m, m, e, i)$.

4.3. A non-commutative example: GL_n . Let $S = \operatorname{Spec}(R)$ be affine as before. The (underlying scheme of the) general linear group in n variables over S is defined as

$$\operatorname{GL}_{n,S} = \operatorname{Spec} R[t_{ij}, 1 \leq i, j \leq n; \det((t_{ij})_{i,j})^{-1}].$$

For every S -scheme T , we can (exercise) identify $\operatorname{GL}_{n,S}(T)$ with $\operatorname{GL}_n(\mathcal{O}_T(T))$ by

$$\Phi : [g : T \rightarrow \operatorname{GL}_{n,S}] \longmapsto (g^*(t_{ij}))_{i,j}. \quad (4.18)$$

We have the usual matrix multiplication on $\operatorname{GL}_n(\mathcal{O}_T(T))$. In terms of (4.18), it comes from the multiplication morphism $m : \operatorname{GL}_{n,S} \times_S \operatorname{GL}_{n,S} \rightarrow \operatorname{GL}_{n,S}$ which is given in coordinates by

$$m^*(t_{ij}) = \sum_{k=1}^n t_{ik} \otimes t_{kj}.$$

The pair $(\mathrm{GL}_{n,S}, m)$ is then an S -group scheme. The identity map $e = \mathrm{Spec}(e^*)$ is given by

$$e^* : R[t_{ij}, \det((t_{ij})_{ij})^{-1}] \longrightarrow R, \quad e^*(t_{ij}) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$$

The inverse of the matrix $(t_{ij})_{ij} \in \mathrm{GL}_n(R[t_{ij}, \det((t_{ij})_{ij})^{-1}])$ has an expression of the form $\det((t_{ij})_{ij})^{-1} \cdot (s_{ij})_{ij}$ where the s_{ij} are polynomials in the variables t_{ij} . (In fact, the s_{ij} are the entries of the adjugate matrix.) Then the inverse morphism $i : \mathrm{GL}_{n,S} \rightarrow \mathrm{GL}_{n,S}$ is given in coordinates by

$$i^*(t_{k\ell}) = \det((t_{ij})_{ij})^{-1} s_{k\ell}.$$

Clearly, $\mathrm{GL}_{1,S}$ is the same as the multiplicative group $\mathbb{G}_{m,S}$. For every S -scheme T , we have a determinant morphism $\mathrm{GL}_n(\mathcal{O}_T(T)) \rightarrow \mathcal{O}_T(T)^\times$. With respect to our identifications (4.11) and (4.18), these come from the group scheme homomorphism

$$\det : \mathrm{GL}_{n,S} \longrightarrow \mathbb{G}_{m,S}, \quad \det^*(t) = \det((t_{ij})_{ij}). \quad (4.19)$$

Its kernel $\ker(\det)$ is the group subscheme $\mathrm{SL}_{n,S} \subset \mathrm{GL}_{n,S}$. Being a closed subscheme of an affine scheme, it is again affine. It can be described explicitly by

$$\mathrm{SL}_{n,S} = \mathrm{Spec}(R[t_{ij}, 1 \leq i, j \leq n] / (\det((t_{ij})_{ij}) - 1)).$$

4.4. Linear algebraic groups. We now specialize to the case of finite type group schemes over a field k . A general classification theorem essentially reduces their study to the affine and the proper case.

Theorem 4.12 (see [16, §8a]). *Let G/k be a connected finite type k -group scheme. Then there exists a unique maximal normal, connected, affine closed group sub-scheme $N \subseteq G$. The quotient G/N is an abelian variety.*

Affine finite type k -group schemes are also called *linear algebraic groups*. The reason for this name is that they can always be realized as a group of linear automorphisms of some vector space. That is, they always embed into some GL_N .

Theorem 4.13 (see [16, Corollary 4.10]). *Let G be an affine finite type k -group scheme. Then there exist an integer n and a closed immersion group scheme morphism $G \rightarrow \mathrm{GL}_n$.*

4.5. Abelian varieties. We have already defined abelian varieties in Definition 4.2. The main point of this definition is that abelian varieties are proper. This implies that they are necessarily commutative which also explains their name.

Theorem 4.14 ([12, Corollary 3.7]). *Let (A, m) be an abelian variety over k . Then (A, m) is a commutative group scheme.*

5. ELLIPTIC CURVES

In the previous section, we defined elliptic curves as proper, smooth, 1-dimensional, connected group schemes and stated that they are always commutative (Definition 4.1 and Theorem 4.14). However, this definition does not shed any light on how to actually write down an example of an elliptic curve. For this reason, we want to next learn about two equivalent definitions:

- An elliptic curve over a field k is a pair (E, e) consisting of a proper smooth connected curve E/k of genus 1 and a rational point $e \in E(k)$.
- An elliptic curve over a field k is a smooth cubic curve $E \subset \mathbb{P}_k^2$ that contains the point $[0 : 1 : 0]$.

Passing between these definitions involves the theory of curves and line bundles. A careful discussion with many details can be found in [12, §4 – §7], but some of these details are tangential for our course. So we will give a shorter and more high-level treatment.

5.1. Cubic curves are elliptic curves. Our first aim is to construct elliptic curves. Let $h(x) = x^3 + ax + b$ be a monic cubic polynomial (without x^2 -term). A polynomial of the form

$$f = y^2 - h(x) \quad (5.1)$$

is called a *simplified Weierstrass equation*. Let

$$F(X, Y, Z) = Y^2 Z - X^3 - aXZ^2 - bZ^3 \quad (5.2)$$

be the homogenization of f , and let $E = V_+(F) \subset \mathbb{P}_k^2$ be its vanishing locus.

Lemma 5.1. *Assume that $\text{char}(k) \neq 2$ and that h is separable. Then E is a smooth curve.*

Proof. First observe by direct substitution in (5.2) that, on the level of sets, $E \cap V_+(Z) = \{[0 : 1 : 0]\}$. We can thus proceed by checking the Jacobi criterion on $E \cap D_+(Z)$ and for the point $[0 : 1 : 0]$.

By definition, we have

$$E \cap D_+(Z) \xrightarrow{\sim} V(y^2 - h(x)) \subset \mathbb{A}_k^2.$$

The Jacobi matrix of the Weierstrass polynomial is the gradient

$$(\partial f / \partial x, \partial f / \partial y) = (-h'(x), 2y). \quad (5.3)$$

Let $e \in E \cap D_+(Z)$ be an arbitrary point. Let $\kappa(e)$ be the residue field of e and let $(e_1, e_2) \in \kappa(e) \times \kappa(e)$ be the coordinates of e .⁵ If $e_2 \neq 0$, then also $2e_2 \neq 0$ by our assumption $\text{char}(k) \neq 2$, meaning $2y$ does not vanish in e . If $e_2 = 0$, however, then $h(e_1) = 0$ since $f(e_1, e_2) = 0$. We have assumed that h is separable, which is equivalent to $h(x)$ and $h'(x)$ being coprime. Thus $h'(e_1) \neq 0$. In summary, we have seen that the gradient (5.3) does not vanish in e .

We now consider the point $[0 : 1 : 0]$. An affine chart is given by

$$E \cap D_+(Y) \xrightarrow{\sim} V(z - x^3 - axz^2 - bz^3) \subset \mathbb{A}_k^2.$$

In these coordinates, $[0 : 1 : 0]$ maps to $(0, 0)$. Moreover, the gradient of that equation is

$$(-3x^2 - az^2, 1 - 2axz - bz^2). \quad (5.4)$$

Its second entry does not vanish in $(0, 0)$, so the Jacobi criterion holds in $(0, 0)$. The proof of the lemma is now complete. $\square$

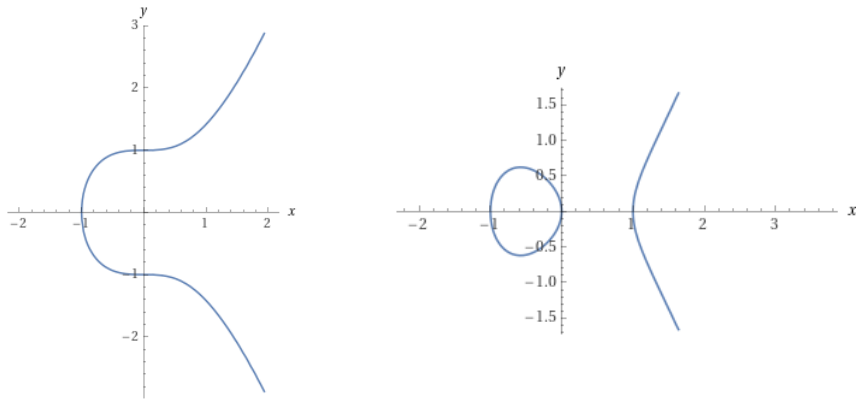


FIGURE 2. The $\mathbb{R}$ -points of the two Weierstrass equations $y^2 = x^3 + 1$ and $y^2 = x^3 - x$. Note that $V(y^2 - (x^3 - x)) \subset \mathbb{A}_{\mathbb{R}}^2$ is a connected scheme. Only its $\mathbb{R}$ -points when endowed with the real topology are disconnected.

⁵Given a scheme X and a point $x \in X$, we use $\kappa(x) = \text{Quot}(\mathcal{O}_{X,x}/\mathfrak{m}_x)$ to denote the residue field in x .

Theorem 5.2. *Let $E = V_+(F) \subset \mathbb{P}_k^2$ be a smooth cubic curve, and let $O \in E(k)$ be a rational point. Then there exists a unique group scheme structure $+: E \times_{\text{Spec}(k)} E \rightarrow E$ on E with neutral element O . By Theorem 4.14, it is necessarily commutative.*

There are two approaches to this theorem. Today, we will explain the more elementary one, which is to give a geometric construction of $+$ in terms of the geometry of $\mathbb{P}^2$. A beautiful aspect of this construction is that it illustrates why *cubic* curves behave so special. Details on some calculations behind this approach may be found in Silverman's book [22, §III.1-3].

The second approach is based on line bundles, the Riemann–Roch Theorem, and the Yoneda Lemma. It is more conceptual, and some of its aspects will be discussed in more detail later in the course. A reference is [12, §7].

Proof of Theorem 5.2. We will admit the uniqueness part of the theorem, which is a general property of abelian varieties [12, Proposition 3.6]. Thus, the main problem is to construct the addition law.

Lemma 5.3. *Let $F \in k[X, Y, Z]$ be homogeneous of degree 3 without linear factor and let $E = V_+(F)$. Let $L \subset \mathbb{P}_k^2$ be any line. Then E intersects L in three points when counted with multiplicities. More precisely, $E \cap L = \text{Spec } A$ for a k -algebra A with $\dim_k(A) = 3$.*

Here, by line we mean a curve of the form $V_+(aX + bY + cZ)$, where $(a, b, c) \neq (0, 0, 0)$.

Proof. After a linear change of coordinates, we may assume that $L = V_+(Z)$. Since F has no linear factor, $Z \nmid F$. Thus $F|_L = F(X, Y, 0)$ is a non-zero homogeneous polynomial of degree 3 and hence has three zeroes (counted with multiplicities) as claimed. $\square$

Construction 5.4. Given $P, Q \in E(k)$, define a line $L \subset \mathbb{P}_k^2$ as follows:

- (1) If $P \neq Q$, then let L be the unique line that passes through P and Q .
- (2) If $P = Q$, then let L be the tangent line to E in that point.

The definition of the tangent uses the smoothness of E . (In a local chart, take the line perpendicular to the gradient of the equation defining E .) The smoothness of E also implies that F has no linear factor. Hence Lemma 5.3 applies and shows that E and L intersect in three points (counting multiplicities). But two of these points are known to be P and Q which lie in $L(k)$! And if a cubic polynomial has two rational roots, then the third root is rational as well. Thus there exists a unique third rational intersection point $R \in (E \cap L)(k)$. Repeating this construction with O, R instead of P, Q , defines a fourth point $S \in E(k)$.

Definition 5.5. The sum of $P, Q \in E(k)$ is defined as $P + Q := S$.

It is true, but not obvious, that this defines a group structure on $E(k)$. The easy part is to show that O is a neutral element and that every element has an inverse (exercise). It is moreover clear that the operation $(P, Q) \mapsto P + Q$ is commutative. Showing associativity is more tricky, however.

So far, we have defined a commutative group $E(k)$. If K/k is a field extension, then we can apply the above construction to $K \otimes_k E \subset \mathbb{P}_K^2$ and obtain a group structure on $E(K) = (K \otimes_k E)(K)$. We know from algebraic geometry that, given reduced varieties (smooth, for example) X and Y over an algebraically closed field K , a morphism $f: X \rightarrow Y$ is uniquely determined by the map $f(K): X(K) \rightarrow Y(K)$ on K -points. So there is at most one morphism $E \times_{\text{Spec}(k)} E \rightarrow E$ that induces the above group structures on all the $E(K)$, K/k . Moreover, if it exists, it will satisfy all group axioms because the sets $E(K)$ do (apply the uniqueness to the diagrams (4.3) and (4.4)).

To complete the proof, one carries out Construction 5.4 in indeterminates and sees that it indeed comes from a morphism of varieties. We refer the curious reader to [22, Theorem 3.6]. $\square$

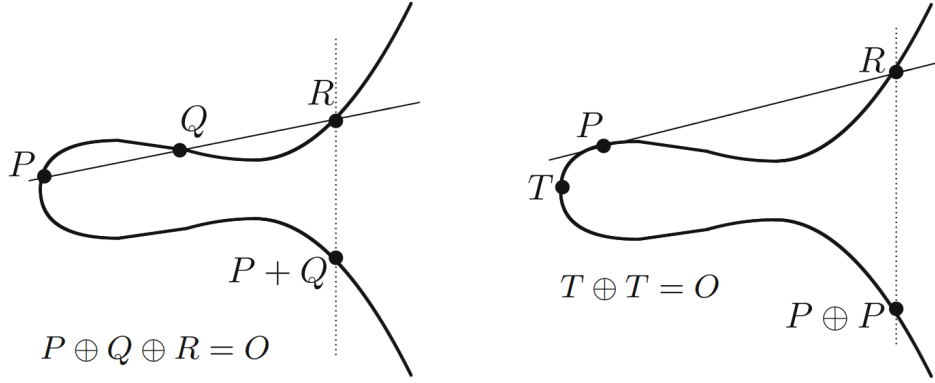


FIGURE 3. The case $P \neq Q$ is shown on the left, the tangent construction when $P = Q$ on the right. The point O here is the point $[0 : 1 : 0]$ at infinity. The vertical dotted lines are the lines through O and R . The picture is taken from [22, §III].

The simplified Weierstrass equations from Lemma 5.1 give simple examples of smooth cubic curves. We will later see that if $\text{char}(k) \neq 2, 3$, then every elliptic curve can be described by $(V_+(F), [0 : 1 : 0])$ for a simplified Weierstrass equation F . In particular, the isomorphism classes of elliptic curves over k can be parametrized by the two coefficients $a, b \in k^2$ of $h(x) = x^3 + ax + b$. (Only those a and b such that h is separable occur, of course.)

5.2. Elliptic curves have genus 1. Our next goal is to show that all elliptic curves come from plane cubic curves. For this, we first need to find a way to extract geometric properties of E from the existence of the group structure. This is done using differential forms. Let us begin by recalling their definition.

Definition 5.6. Let R be a ring, A an R -algebra, and M an A -module. An R -derivation from A to M is an R -linear map $\delta : A \rightarrow M$ such that the Leibniz rule holds: For all $a, b \in A$,

$$\delta(ab) = a\delta(b) + b\delta(a).$$

Lemma 5.7. *There exists a universal R -derivation. That is, there exists an A -module $\Omega_{A/R}^1$ together with an R -derivation $d : A \rightarrow \Omega_{A/R}^1$ such that every R -derivation $\delta : A \rightarrow M$ factors through a unique A -module homomorphism $\varphi : \Omega_{A/R}^1 \rightarrow M$. As diagram,*

$$\begin{array}{ccc} A & \xrightarrow{d} & \Omega_{A/R}^1 \\ & \searrow \forall \delta & \downarrow \exists! \varphi \\ & & M. \end{array} \quad (5.5)$$

The pair $(\Omega_{A/R}^1, d)$ is called the module of Kähler differentials of A over R . It is easy to describe in terms of generators and relations. Let

$$A = R[X_1, \dots, X_n]/(f_1, \dots, f_m)$$

be a presentation of A as a quotient of a polynomial ring over R . Consider the free module $\bigoplus_{i=1}^n A dX_i$ generated by symbols $dX_1, \dots, dX_n$. (This is really A^n ; the symbols dX_i are just the traditional notation for the standard basis here.) For each $f \in R[X_1, \dots, X_n]$, we

can take the gradient vector

$$df := \frac{\partial f}{\partial X_1} \cdot dX_1 + \dots + \frac{\partial f}{\partial X_n} \cdot dX_n. \quad (5.6)$$

Then

$$\begin{aligned} (A dX_1 \oplus \dots \oplus A dX_n) / \langle df_1, \dots, df_m \rangle &\xrightarrow{\sim} \Omega_{A/R}^1 \\ dX_i &\xrightarrow{\sim} d(X_i). \end{aligned} \quad (5.7)$$

The key ideas for proving Lemma 5.7 and (5.7) are as follows:

- Since $d : A \rightarrow \Omega_{A/R}^1$ is supposed to be universal, the module $\Omega_{A/R}^1$ has to be generated by all derivatives $d(a)$ as A -module.
- Since every element of A is a polynomial in the X_i with R -coefficients, the Leibniz rule allows to write every $d(a)$ as an A -linear combination of the $d(X_i)$. Hence, the $d(X_i)$ already generate $\Omega_{A/R}^1$ as A -module.
- Since the $f_j \in A$ are zero, also the $d(f_j)$ in $\Omega_{A/R}^1$ have to be zero. By the Leibniz rule,

$$d(f_j) = (\partial f / \partial X_1) \cdot d(X_1) + \dots + (\partial f / \partial X_n) \cdot d(X_n),$$

which explains the relations $df_1, \dots, df_m$ in (5.7).

Given an element $g \in A$, there is an isomorphism of $A[g^{-1}]$ -modules

$$\Omega_{A/R}^1[g^{-1}] \xrightarrow{\sim} \Omega_{A[g^{-1}]/R}^1 \quad (5.8)$$

which is uniquely characterized by sending $d(a)$ to $d(a)$. In other words, the formation of $\Omega_{A/R}^1$ is compatible with localizations. This means that the construction can be glued from rings to schemes.

Definition 5.8. Let $\pi : X \rightarrow S$ be a morphism of schemes. The quasi-coherent module with derivation $d : \mathcal{O}_X \rightarrow \Omega_{X/S}^1$ is defined as the unique datum (up to isomorphism) that is, locally on affine charts $\text{Spec}(R) \subseteq S$ and $\text{Spec}(A) \subseteq \pi^{-1}(\text{Spec}(R))$, given by $d : A \rightarrow \Omega_{A/R}^1$ glued along (5.8).

Kähler differentials are closely related to smoothness, and we next state one form of this relation.

Theorem 5.9 ([12, Theorem 4.18]). *Let $\pi : X \rightarrow S$ be a morphism that is locally of finite presentation with purely d -dimensional fibers. Then π is smooth if and only if $\Omega_{X/S}^1$ is locally free of rank d as $\mathcal{O}_X$ -module.*

Definition 5.10 (Genus of a curve). (1) By curve over a field k , we mean a proper, smooth, geometrically connected and 1-dimensional k -scheme.

(2) Let $C \rightarrow \text{Spec}(k)$ be a curve. By Theorem 5.9, $\Omega_{C/k}^1$ is a line bundle on C . Being a coherent sheaf on a proper variety, the space of global sections $\Omega_{C/k}^1(C)$ is a finite-dimensional k -vector space. Its dimension is called the genus of C .

Here, recall that a finite type k -scheme X is said to be geometrically reduced, connected, integral, etc. if the base change $\bar{k} \otimes_k X$ is reduced, connected, integral, etc. An equivalent condition is that for all field extensions K/k , the base change $K \otimes_k X$ has the relevant property.

For example, elliptic curves are geometrically connected because they are connected over k (by definition) and have a rational point (the neutral element).

Theorem 5.11. *Let E be an elliptic curve over a field k . Then E has genus 1.*

Sketch of proof. The key point is that the sheaf of differential forms of a group scheme is generated by invariant forms. The proof of this (see [12, Proposition 5.7]) does not concern us here, we will only state and use the result.

Let $\pi : G \rightarrow S$ be a group scheme with neutral element section $e : S \rightarrow G$. Recall that quasi-coherent modules can be pulled back under scheme morphisms. So we may first form $e^*(\Omega_{G/S}^1)$, a quasi-coherent S -module. Then we may again pull back along π . The statement is that there exists an isomorphism

$$\gamma : \pi^* e^* \Omega_{G/S}^1 \xrightarrow{\sim} \Omega_{G/S}^1. \quad (5.9)$$

We now apply (5.9) to our elliptic curve $E \rightarrow \operatorname{Spec}(k)$. The pullback $V = e^*(\Omega_{E/k}^1)$ is a one-dimensional k -vector space because $\Omega_{E/k}^1$ is a line bundle. Then (5.9) states that

$$\gamma : \mathcal{O}_E \otimes_k V \xrightarrow{\sim} \Omega_{E/k}^1.$$

Choosing a basis vector $\omega \in V$, we have thus obtained an isomorphism $\mathcal{O}_E \xrightarrow{\sim} \Omega_{E/k}^1$. The genus of E is hence $\dim_k \mathcal{O}_E(E)$.

Lemma 5.12. *Let $X \rightarrow \operatorname{Spec}(k)$ be a proper k -scheme that is geometrically reduced and geometrically connected. Then $\dim_k \mathcal{O}_X(X) = 1$.*

Proof. The global sections $A = \mathcal{O}_X(X)$ are a finite-dimensional k -algebra. Its formation commutes with base change in the sense that for every field extension K/k , we have

$$K \otimes_k A = \mathcal{O}_{K \otimes_k X}(K \otimes_k X).$$

Hence, if X is geometrically reduced and connected, then $\bar{k} \otimes_k A$ is reduced and has a unique maximal ideal. The residue field is necessarily $\bar{k}$ because $\bar{k}$ is algebraically closed. So $\bar{k} \xrightarrow{\sim} \bar{k} \otimes_k A$. Thus A was one-dimensional to begin with, meaning $k \xrightarrow{\sim} A$. $\square$

Coming back to our elliptic curve $E \rightarrow \operatorname{Spec}(k)$, we see that $\mathcal{O}_E(E) = k$, meaning that E has genus 1 as claimed. $\square$

Remark 5.13. The isomorphism in (5.9) is given by extending the value of a differential form on $e(S)$ in the unique way to a left-translation invariant differential form on G . This concept is also commonly used in differential geometry, where one often identifies the Lie algebra $\mathfrak{g}$ of a Lie group G with the space of translation invariant vector fields on G .

For example, the form dt on $\mathbb{R}$ is translation invariant with respect to addition because $d(t + \lambda) = dt$ for all $\lambda \in \mathbb{R}$. The form $t^{-1}dt$ on $\mathbb{R}^\times$ is translation invariant with respect to multiplication because $(t\lambda)^{-1}d(\lambda t) = t^{-1}dt$ for all $\lambda \in \mathbb{R}^\times$.

5.3. Genus 1 curves as cubics. We have just shown that every elliptic curve has genus 1. In order to complete the circle of equivalent definitions (a triangle, actually), it is left to realize curves of genus 1 as cubic curves in $\mathbb{P}^2$. Let us first briefly recall a bit of general formalism.

Construction 5.14. Let X be a k -scheme. Giving a morphism $f : X \rightarrow \mathbb{P}_k^n$ is the same as giving a line bundle $\mathcal{L}$ on X and a surjection of $\mathcal{O}_X$ -modules

$$\ell : \mathcal{O}_X^{\oplus(n+1)} \twoheadrightarrow \mathcal{L}.$$

Namely, on $\mathbb{P}_k^n$, we have the standard line bundle $\mathcal{O}(1)$. It is generated by the $n+1$ global sections $X_0, \dots, X_n$ corresponding to the $n+1$ coordinates on $\mathbb{P}_k^n$. That is, we have a surjection

$$\mathcal{O}_{\mathbb{P}_k^n}^{\oplus(n+1)} \twoheadrightarrow \mathcal{O}(1), \quad e_i \mapsto X_i.$$

Given $f : X \rightarrow \mathbb{P}_k^n$, we can pull back that surjection and obtain a pair $\mathcal{L} = f^*\mathcal{O}(1)$, $\ell : \mathcal{O}_X^{\oplus(n+1)} \twoheadrightarrow \mathcal{L}$ as desired.

Conversely, assume that $(\mathcal{L}, \ell)$ is given. Let $s_i = \ell(e_i) \in \mathcal{L}(X)$ be the $n+1$ global sections defined by ℓ . Let $U_i = D(s_i) \subseteq X$ be the open subscheme where s_i is a generator. That is, if we locally trivialize $\mathcal{L}$, say

$$\mathcal{O}_U \cdot s \xrightarrow{\sim} \mathcal{L}|_U, \quad s_i = f_i s, \quad f_i \in \mathcal{O}_U(U),$$

then $U_i \cap U = D(f_i)$ is the locus where f_i is invertible.

Over the open subset U_i , every section of $\mathcal{L}$ is a unique multiple of s_i . So we have defined functions $s_j/s_i \in \mathcal{O}_X(U_i)$ by the identity $s_j = (s_j/s_i) \cdot s_i$. This defines a morphism

$$f_i = \left(\frac{s_0}{s_i}, \dots, \frac{\widehat{s_i}}{s_i}, \dots, \frac{s_n}{s_i} \right) : U_i \longrightarrow \mathbb{A}^n.$$

On overlaps $U_i \cap U_j$, we have the (obvious) relation

$$\frac{s_k}{s_i} = \frac{s_j}{s_i} \cdot \frac{s_k}{s_j}.$$

If we spell out how $\mathbb{P}_k^n$ is glued from $n+1$ copies of $\mathbb{A}_k^n$ by the exact same rule of coordinate transformation, then this implies that the f_i glue to a morphism

$$f : X \longrightarrow \mathbb{P}_k^n.$$

A good notation for this morphism is $[s_0 : s_1 : \dots : s_n]$. Namely, if $x \in X$ is a point then we may view $[s_0(x) : \dots : s_n(x)] \in \mathbb{P}^n(\kappa(x))$ as follows. Let $s \in \mathcal{L}_x$ be a generator as $\mathcal{O}_{X,x}$ -module. Then we may write $s_{i,x} = h_i s$ for unique functions $h_i \in \mathcal{O}_{X,x}$. The tuple $[h_0(x) : \dots : h_n(x)]$ is a point of $\mathbb{P}^n(\kappa(x))$. Any other generator of $\mathcal{L}_x$ differs from s by a unit, hence the tuple $(h_0(x), \dots, h_n(x))$ is unique up to $\kappa(x)^\times$, meaning that

$$[s_0(x) : \dots : s_n(x)] := [h_0(x) : \dots : h_n(x)]$$

is well-defined.

Exercise 5.15. Verify that the above two constructions $(\mathcal{L}, \ell) \longleftrightarrow (f : X \rightarrow \mathbb{P}_k^n)$ are inverse to each other.

Example 5.16. We know that every line bundle on $\mathbb{P}_k^1$ is isomorphic to one of the line bundles $\mathcal{O}(d)$. The integer $d \in \mathbb{Z}$ is its degree. We know that

$$\dim_k(\mathcal{O}(d)(\mathbb{P}_k^1)) = \begin{cases} d+1 & \text{if } d \geq 0 \\ 0 & \text{if } d < 0. \end{cases}$$

If $d \geq 0$, then a basis for the global sections $\mathcal{O}(d)(\mathbb{P}_k^1)$ is given by the monomials

$$X_0^d, X_0^{d-1}X_1, \dots, X_0X_1^{d-1}, X_1^d$$

where $X_0, X_1 \in \mathcal{O}(1)(\mathbb{P}_k^1)$ are the coordinates on $\mathbb{P}_k^1$. If $d \geq 0$, then these monomials also generate $\mathcal{O}(d)$ as line bundle. That is, the map

$$\mathcal{O}_{\mathbb{P}_k^1}^{\oplus(d+1)} \longrightarrow \mathcal{O}(d), \quad e_i \longmapsto X_0^{d-i}X_1^i$$

is a surjection of quasi-coherent $\mathcal{O}_{\mathbb{P}_k^1}$ -modules. The corresponding morphism $\mathbb{P}_k^1 \rightarrow \mathbb{P}_k^d$ is called the Veronese map. It is a closed immersion when $d \geq 1$.

Construction 5.14 shows that, if we want to define a morphism $E \rightarrow \mathbb{P}_k^2$ from an elliptic curve to the projective plane, then we need to understand line bundles and their global sections on E . Let us begin with some general observations and definitions.

- Let X be a noetherian scheme and $\mathcal{F}$ a coherent $\mathcal{O}_X$ -module. (This is the same as $\mathcal{F}$ being quasi-coherent and of finite type.) Then $\mathcal{F}$ is locally free (meaning a vector bundle) if and only if for every $x \in X$, the stalk $\mathcal{F}_x$ is a free $\mathcal{O}_{X,x}$ -module.
- Thus, if C is a curve over a field k , then a coherent module $\mathcal{L}$ is a line bundle if and only if for every $x \in X$, the stalk $\mathcal{L}_x$ is free of rank 1 over $\mathcal{O}_{C,x}$.

- By definition, all our curves are smooth, hence normal. So for $x \in C$ closed, the local ring $\mathcal{O}_{C,x}$ is a discrete valuation ring (DVR). By the classification of modules over principal ideal domains (PIDs), a finite type $\mathcal{O}_{C,x}$ -module is free if and only if it is torsion-free.

Conclusion 5.17. Let $0 \neq \mathcal{I} \subseteq \mathcal{O}_C$ be an ideal sheaf in $\mathcal{O}_C$. Then $\mathcal{I}$ is stalk-by-stalk torsion-free because it is a subsheaf of torsion-free sheaf $\mathcal{O}_C$, and hence $\mathcal{I}$ is a line bundle.

Definition 5.18 (Degree of a line bundle). (1) Let $\mathcal{I} \subseteq \mathcal{O}_C$ be a non-zero ideal sheaf. Then $Z = V(\mathcal{I}) \subset C$ is a proper closed subscheme. It has to be 0-dimensional, and hence is a finite k -scheme. As such, it is affine, meaning $Z \cong \operatorname{Spec}(A)$ for a finite dimension k -algebra A . The *degree* of $\mathcal{I}$ is defined as $-\dim_k(A)$. More concretely, because each local ring $\mathcal{O}_{C,x}$ is a DVR, we can write

$$Z = \bigsqcup_{i=1}^r \operatorname{Spec}(\mathcal{O}_{C,x_i}/\mathfrak{m}_{x_i}^{e_i})$$

for uniquely determined pairwise different closed points $x_1, \dots, x_r \in C$ and exponents $e_1, \dots, e_r \geq 1$. Then

$$\deg(\mathcal{I}) = - \sum_{i=1}^r e_i \cdot [\kappa(x_i) : k].$$

(2) Let $\mathcal{L}$ be a line bundle on C . There always exist two ideal sheaves $\mathcal{I}_1, \mathcal{I}_2 \subseteq \mathcal{O}_C$ such that $\mathcal{L} \cong \mathcal{I}_1 \otimes \mathcal{I}_2^{-1}$. We define

$$\deg(\mathcal{L}) := \deg(\mathcal{I}_1) - \deg(\mathcal{I}_2).$$

This does not depend on the choices of $\mathcal{I}_1$ and $\mathcal{I}_2$. In particular, the degree defines a group homomorphism

$$\deg : \operatorname{Pic}(C) \longrightarrow \mathbb{Z}.$$

Motivation 5.19. The degree is a simple numerical invariant of a line bundle on a curve. The following results show that it is extremely helpful when studying global sections of line bundles and hence, by Construction 5.14, maps $C \rightarrow \mathbb{P}_k^n$.

Theorem 5.20 (Riemann–Roch). *Let C be a curve of genus g over a field k . Then, for every line bundle $\mathcal{L}$ on C ,*

$$\dim \mathcal{L}(C) = \deg(\mathcal{L}) + 1 - g + \dim(\Omega_{C/k}^1 \otimes \mathcal{L}^{-1})(C). \quad (5.10)$$

Corollary 5.21. *The degree of $\Omega_{C/k}^1$ is $2g - 2$.*

Proof. Apply the Riemann–Roch Theorem 5.20 to $\Omega_{C/k}^1$. We obtain

$$g = \deg(\Omega_{C/k}^1) + 1 - g + 1$$

which we may rearrange as claimed. $\square$

Corollary 5.22. *Let C/k be a curve of genus 1 and let $\mathcal{L}$ be a line bundle of degree $\deg(\mathcal{L}) \geq 1$ on C . Then*

$$\dim \mathcal{L}(C) = \deg(\mathcal{L}).$$

Proof. By Corollary 5.21, $\deg(\Omega_{C/k}^1) = 0$. Since $\deg(\mathcal{L}) \geq 1$, we then have

$$\deg(\Omega_{C/k}^1 \otimes \mathcal{L}^{-1}) = 0 - \deg(\mathcal{L}) < 0.$$

Line bundles of negative degree cannot have non-zero global sections, so $(\Omega_{C/k}^1 \otimes \mathcal{L}^{-1})(C) = 0$. Evaluating the Riemann–Roch identity (5.10), we find $\dim \mathcal{L}(C) = \deg(\mathcal{L})$ as claimed. $\square$

Theorem 5.23. *Let E be a curve of genus 1 over k such that $E(k) \neq \emptyset$. Then there exists a closed immersion $E \hookrightarrow \mathbb{P}_k^2$ which identifies E with the curve $V_+(F)$ defined by a cubic homogeneous polynomial.*

Proof. Step 1: Construction of a morphism $E \rightarrow \mathbb{P}_k^2$. We have seen in Construction 5.14 that, in order to define a morphism $E \rightarrow \mathbb{P}_k^2$, our task is to find a line bundle $\mathcal{L}$ on E together with a surjection $\ell : \mathcal{O}_E^3 \rightarrow \mathcal{L}$.

We now draw inspiration from the example of $\mathbb{P}_k^1$ above. By assumption, there exists a k -rational point $e \in E(k)$. View $\{e\}$ as a reduced closed subscheme of E , and let $\mathcal{I}_e$ be its ideal sheaf. According to Definition 5.18, its degree is -1 . So the dual line bundle $\mathcal{M} := \mathcal{I}_e^{-1}$ has degree 1.

The degree of $\mathcal{M}^{\otimes d}$ is d .⁶ By Corollary 5.22, this means

$$\dim \mathcal{M}^{\otimes d}(E) = d, \quad d \geq 1.$$

We are mostly interested in $\mathcal{L} = \mathcal{M}^{\otimes 3}$. For every closed point $y \in E$ we have an ideal sheaf $\mathcal{I}_y$ as before. Its degree is $-\kappa(y) : k$, the negative of the residue field extension degree. On the one hand, we may consider $\mathcal{L}$ and $\mathcal{I}_y$ as abstract line bundles. By Riemann–Roch, the dimension of global sections strictly decreases when tensoring with $\mathcal{I}_y$ because the degree goes down:

$$\dim(\mathcal{L} \otimes \mathcal{I}_y)(E) < 3.$$

On the other hand, we can consider the concrete exact sequence

$$0 \longrightarrow \mathcal{I}_y \longrightarrow \mathcal{O}_E \longrightarrow i_*\kappa(y) \longrightarrow 0$$

where $i : \{y\} \rightarrow E$ is the inclusion map. Tensoring by $\mathcal{L}$, which is an exact operation because $\mathcal{L}$ is locally free, we get an exact sequence

$$0 \longrightarrow \mathcal{L} \otimes \mathcal{I}_y \longrightarrow \mathcal{L} \longrightarrow i_*\mathcal{L}(y) \longrightarrow 0.$$

Here, $\mathcal{L}(y) := i^*\mathcal{L}$ is our notation for the 1-dimensional $\kappa(y)$ vector space that forms the fiber of $\mathcal{L}$ in y . Taking global sections, we see that

$$(\mathcal{L} \otimes \mathcal{I}_y)(E) \subseteq \mathcal{L}(E)$$

are precisely those global sections that vanish in y .

We conclude that for every closed point $y \in E$, there exists a global section $s \in \mathcal{L}(E)$ that does not vanish in y . This means that $\mathcal{L}$ is generated by its global sections. That is, after choosing a basis s_0, s_1, s_2 for the three-dimensional vector space $\mathcal{L}(E)$, we obtain a surjection

$$\ell : \mathcal{O}_E^{\oplus 3} \longrightarrow \mathcal{L}, \quad e_i \longmapsto s_i,$$

and hence a morphism $f : E \rightarrow \mathbb{P}_k^2$ as in Construction 5.14.

Step 2: f is a closed immersion. We can prove that f is a closed immersion after base change to $\bar{k}$. So from now on, we assume that k is algebraically closed. This helps, because now every closed point $y \in E$ is k -rational and, in particular, $\deg(\mathcal{I}_y) = -1$. Let $y, y' \in E$ be two (possibly equal) closed points. Corollary 5.22 implies that

$$\begin{aligned} \dim(\mathcal{L} \otimes \mathcal{I}_y)(E) &= 2 \\ \dim(\mathcal{L} \otimes \mathcal{I}_y \otimes \mathcal{I}_{y'})(E) &= 1. \end{aligned}$$

So, after applying a linear change of coordinates on $\mathbb{P}_k^2$, we may assume that our basis $s_0, s_1, s_2 \in \mathcal{L}(E)$ is chosen with

$$\begin{aligned} s_0 &\in \mathcal{L}(E) \setminus (\mathcal{L} \otimes \mathcal{I}_y)(E), \\ s_1 &\in (\mathcal{L} \otimes \mathcal{I}_y)(E) \setminus (\mathcal{L} \otimes \mathcal{I}_y \otimes \mathcal{I}_{y'})(E). \end{aligned} \tag{5.11}$$

⁶All tensor products during the proof are taken over $\mathcal{O}_E$.

If $y \neq y'$, then this means that

$$[s_0(y) : s_1(y) : s_2(y)] \neq [s_0(y') : s_1(y') : s_2(y')]$$

because s_1 vanishes in y while it does not vanish in y' . We conclude that f is injective at the level of topological spaces. Since f is also closed by the properness of E , it is topologically a closed immersion.

Finally, if $y = y'$, then the above choice of s_1 ensures that it vanishes to first order in y , but not to second order. Translating this to local coordinates (omitted), it is possible to deduce that $[s_0 : s_1 : s_2]$ is injective on the tangent space $(\mathfrak{m}_y/\mathfrak{m}_y^2)^\vee$ in y , which means that f is even schematically a closed immersion near y .

Step 3: Its image is defined by a cubic equation. We do not assume anymore that k is algebraically closed. Recall that $e \in E(k)$ is our given rational point and that $\mathcal{M} = \mathcal{I}_e^{-1}$. Dualizing the descending chain

$$\dots \subset \mathcal{I}_e^3 \subset \mathcal{I}_e^2 \subset \mathcal{I}_e \subset \mathcal{O}_E,$$

we obtain an ascending chain

$$\mathcal{O}_E \subset \mathcal{M} \subset \mathcal{M}^2 \subset \mathcal{M}^3 \subset \dots$$

Proceeding with the same logic as in (5.11), we choose elements

$$\begin{aligned} 1 &\in \mathcal{O}_E(E) \\ \mathcal{M}(E) &= \mathcal{O}_E(E) \text{ by Cor. 5.22} \\ x &\in \mathcal{M}^{\otimes 2}(E) \setminus \mathcal{M}(E) \\ y &\in \mathcal{M}^{\otimes 3}(E) \setminus \mathcal{M}^{\otimes 2}(E). \end{aligned} \tag{5.12}$$

View $1, x, y$ as elements of $\mathcal{L}(E) = \mathcal{M}^{\otimes 3}(E)$. Then they form a basis because y generates $\mathcal{L}$ near e , while x vanishes to first order and 1 to third order in e . We consider the morphism

$$[x : y : 1] : E \longrightarrow \mathbb{P}_k^2.$$

Consider the sections

$$1, x, y, x^2, xy, y^2, x^3 \in \mathcal{M}^{\otimes 6}(E). \tag{5.13}$$

These are seven sections of a six-dimensional vector space (use again Corollary 5.22), and hence there exists a non-trivial linear relation

$$a_0 y^2 + b_0 x^3 + a_1 xy + a_2 x^2 + a_3 y + a_4 x + a_6 = 0. \tag{5.14}$$

Claim: Both a_0 and b_0 are non-zero. The section x is a generator of $\mathcal{M}^{\otimes 2}$ near e ; the section y a generator of $\mathcal{M}^{\otimes 3}$ near e . Hence, y^2 and x^3 are both generators of $\mathcal{M}^{\otimes 6}$ near e . Thus either of the set of vectors

$$1, x, y, x^2, xy, y^2, \quad \text{or} \quad 1, x, y, x^2, xy, x^3 \tag{5.15}$$

has the property that the six sections vanish to orders precisely 6, 4, 3, 2, 1, 0 in the stalk $(\mathcal{M}^{\otimes 6})_e$. Thus, either of the two sets forms a basis for $\mathcal{M}^{\otimes 6}(E)$. It follows that $a_0 b_0 \neq 0$ as claimed.

Conclusion: Identity (5.14) means that the morphism $[x : y : 1]$ factors through the cubic curve

$$V_+(F), \quad F = a_0 Y^2 Z + b_0 X^3 + a_1 XYZ + a_2 X^2 Z + a_3 Y Z^2 + a_4 X Z^2 + a_6 Z^3.$$

The linear independence in (5.15) moreover shows that the morphism does not factor through a line or quadric in $\mathbb{P}_k^2$. It follows that F is irreducible and hence $E \xrightarrow{\sim} V_+(F)$ because we already know from Step 2 that $[x : y : 1]$ is a closed immersion. $\square$

Our proof even showed that the affine cubic equation for E may always be chosen in the form (5.14). We can simplify this expression further:

- Scaling y and x by a_0/b_0 , we obtain a relation of the form

$$y^2 + (b_1x + b_3)y = x^3 + b_2x^2 + a_4x + a_6.$$

This kind of cubic equation is called a *general Weierstrass equation*.

- If $\text{char}(k) \neq 2$, then we can change y to $y + (b_1x + b_3)/2$ to simplify further to a relation of the form

$$y^2 = x^3 + c_2x^2 + c_4x + c_6.$$

- If $\text{char}(k) \neq 3$, then we may further replace x by $x + c_2/3$ and arrive at the simplified form

$$y^2 = x^3 + ax + b. \quad (5.16)$$

Ultimately, we conclude that every elliptic curve can be defined by a general Weierstrass equation. Outside of characteristics 2 and 3, we may even restrict to simplified Weierstrass equations.

Corollary 5.24. *Let $E \rightarrow \text{Spec}(k)$ be a curve of genus 1 and let $e \in E(k)$ be a rational point. Then there exists a unique group scheme structure on E with identity element e .*

Proof. Apply Theorem 5.23 to realize E as a cubic in $\mathbb{P}_k^2$. Then use Theorem 5.2 to endow E with a group scheme structure (in a unique way). $\square$

Remark 5.25. Let $F \in k[X, Y, Z]$ be homogeneous of degree d and such that $V_+(F) \subset \mathbb{P}_k^2$ is smooth. Then $V_+(F)$ is a curve of genus $(d-1)(d-2)/2$.

6. ARITHMETIC OF ELLIPTIC CURVES

For every elliptic curve E , we have a multiplication-by- n homomorphism $[n] : E \rightarrow E$ which was defined in (4.6). Let $E[n] := \ker([n])$ be its kernel. Our next goal is to prove that $E[n]$ is always finite of degree n^2 . This is not at all obvious as the following two (also 1-dimensional) examples show.

- The n -torsion $\mathbb{G}_m[n]$ of the multiplicative group is the group scheme μ_n of n -th roots of unity. It is finite of order n .
- Let k be a field and let $\mathbb{G}_{a,k} = \text{Spec } k[t]$ be the additive group over k . Its group scheme structure a (the additional law) is defined by $a^*(t) = t \otimes 1 + 1 \otimes t$. For a k -scheme T , the T -valued points $\mathbb{G}_{a,k}(T)$ are the additive group $(\mathcal{O}_T(T), +)$. In particular,

$$\mathbb{G}_{a,k}[n] = \begin{cases} \{0\} & \text{if } \text{char}(k) \nmid n \\ \mathbb{G}_{a,k} & \text{if } \text{char}(k) \mid n. \end{cases}$$

For example, if $\text{char}(k) = p$, then $[p]$ equals the 0-map $[0]$.

Our proof of $|E[n]| = n^2$ will be in two steps:

Step 1. First, we study elliptic curves over $\mathbb{C}$ where we can use their description by lattices to prove the statement over $\mathbb{C}$. By extension, the statement then even holds over all fields of characteristic 0.

Step 2. We extend the statement from $\mathbb{C}$ to all fields by using the universal Weierstrass family.

6.1. Analytification of complex varieties. Recall from §3.2 that we defined a topology on $X(\mathbb{C})$ for every affine complex variety X . This construction can be upgraded to an analytification functor

$$\begin{aligned} \{\text{Smooth } \mathbb{C}\text{-schemes}\} &\longrightarrow \{\text{Smooth complex manifolds}\} \\ X &\longmapsto X(\mathbb{C}). \end{aligned} \quad (6.1)$$

First, if $X \subseteq \mathbb{A}_{\mathbb{C}}^n$ is a smooth affine variety embedded into affine space, then $X(\mathbb{C}) \subseteq \mathbb{C}^n$ has a unique structure as a smooth complex submanifold. Namely, the Jacobi criterion holds in the algebraic sense for X , and so also holds in the analytic sense for $X(\mathbb{C})$. Hence, $X(\mathbb{C})$ is a complex submanifold by the inverse function theorem.

The construction of this manifold structure is functorial: If $\varphi : X \rightarrow Y$ is a morphism of smooth affine $\mathbb{C}$ -schemes and if $X \subseteq \mathbb{A}_{\mathbb{C}}^n$ and $Y \subseteq \mathbb{A}_{\mathbb{C}}^m$ are embeddings, then there exists an extension of φ to a morphism $\Phi : \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^m$. Passing to $\mathbb{C}$ -points, we obtain a diagram

$$\begin{array}{ccc} X(\mathbb{C}) & \xrightarrow{\varphi(\mathbb{C})} & Y(\mathbb{C}) \\ \downarrow & & \downarrow \\ \mathbb{C}^n & \xrightarrow{\Phi(\mathbb{C})} & \mathbb{C}^m \end{array}$$

where $\Phi(\mathbb{C})$ is holomorphic because it is given by polynomials. It follows that $\varphi(\mathbb{C})$ is holomorphic. If φ is an isomorphism, then the same argument applies to φ^{-1} showing that $\varphi(\mathbb{C})$ is biholomorphic. This shows that the complex manifold structure on $X(\mathbb{C})$ does not depend on the chosen embedding $X \subseteq \mathbb{A}_{\mathbb{C}}^n$. Moreover, the functoriality allows to glue the construction from the affine to the general case.

Analytification has various nice properties of which we mention a few:

- (1) If $X \subseteq \mathbb{P}_{\mathbb{C}}^n$ is a projective variety defined by the vanishing of homogeneous polynomials $F_1, \dots, F_r \in \mathbb{C}[T_0, \dots, T_n]$, then $X(\mathbb{C}) \subseteq \mathbb{P}^n(\mathbb{C})$ is the submanifold defined by the vanishing of the same polynomials.
- (2) X is connected if and only if $X(\mathbb{C})$ is connected.
- (3) X is proper if and only if $X(\mathbb{C})$ is compact.
- (4) Analytification restricts to an equivalence

$$\{\text{Curves over } \mathbb{C}\} \xrightarrow{\sim} \{\text{Compact connected Riemann surfaces}\}. \quad (6.2)$$

This is a non-trivial theorem whose proof requires some functional analysis, see [4, §14]. For curves of genus 1, there is a much simpler proof using the Weierstrass $\wp$ -function.

- (5) Analytification is a faithful functor. It is fully faithful when restricted to proper smooth $\mathbb{C}$ -schemes.

6.2. Application to abelian varieties. Let us now consider analytification in the context of abelian varieties. If $A/\mathbb{C}$ is an abelian variety, then $A(\mathbb{C})$ is a compact connected complex manifold (use (2) and (3) above). Moreover, we can analytify the multiplication morphism and obtain a holomorphic map $A(\mathbb{C}) \times A(\mathbb{C}) \rightarrow A(\mathbb{C})$. Analytification is functorial, so the group axiom diagrams from (4.3) and (4.4) are still commutative. (In fact, this is simply the statement that the set $A(\mathbb{C})$ is a group which we already knew before.) In this way, $A(\mathbb{C})$ is a compact connected complex Lie group.

We have moreover stated that analytification is fully faithful for proper smooth $\mathbb{C}$ -schemes, see (5) above, so for any two abelian varieties A_1, A_2 over $\mathbb{C}$,

$$\text{Hom}_{\mathbb{C}\text{-group scheme}}(A_1, A_2) = \text{Hom}_{\text{complex Lie group}}(A_1(\mathbb{C}), A_2(\mathbb{C})).$$

Theorem 6.1. *Let X be a compact connected complex Lie group of dimension g . Then there exists a lattice $\Lambda \subset \mathbb{C}^g$ and an isomorphism $\mathbb{C}^g/\Lambda \xrightarrow{\sim} X$.*

Proof following [18, p. 1–2]. Consider the action of X on itself by conjugation. It preserves the identity $e \in X$ and hence defines an action of X on the tangent space $V = T_e X$. One can check from the definition of complex Lie group that this defines a holomorphic homomorphism $\text{ad} : X \rightarrow GL_{\mathbb{C}}(V)$. By the maximum principle, a holomorphic function

on a compact complex manifold is compact. Applying this to each of the coordinates of ad proves that this map is trivial, meaning that X is commutative.

Next, consider the exponential map $\exp : T_e X \rightarrow X$. Recall that this map is defined for every complex (or real) Lie group and that it satisfies $\exp(v + w) = \exp(v)\exp(w)$ for all v, w with $[v, w] = 0$. We have already seen that X is commutative, so $[v, w]$ is always 0. It follows that $\exp$ is a group homomorphism.

The exponential map is locally biholomorphic. The image of $\exp$ hence contains an open neighborhood of e . Any such neighborhood generates X as group because X is connected, so $\exp$ is surjective. As $\exp$ is biholomorphic near the identity, we find that $X = V/\Lambda$ for a discrete subgroup $\Lambda \subset V$. Any discrete subgroup of a finite-dimensional real vector space with compact quotient is a lattice, which completes the proof. $\square$

Complex Lie groups of the form $X = \mathbb{C}^g/\Lambda$ are called complex tori. (Here, $\Lambda \subset V$ is a lattice.) They always satisfy $X \cong (\mathbb{R}/\mathbb{Z})^{2g}$ as real Lie group, where $g = \dim_{\mathbb{C}}(V)$, but the complex structure is an additional piece of information.

Corollary 6.2. *There is an equivalence of categories*

$$\{\text{Ell. curves}/\mathbb{C}\} \xrightarrow{\sim} \left\{ \begin{array}{c} \text{Compact complex Lie groups} \\ \text{of the form } \mathbb{C}/\Lambda \end{array} \right\}. \quad (6.3)$$

Proof. We stated above that analytification of proper smooth $\mathbb{C}$ -varieties is a fully faithful functor. So elliptic curves over $\mathbb{C}$ embed fully faithfully into 1-dimensional compact complex Lie groups. These are all of the form $\mathbb{C}/\Lambda$ by Theorem 6.1. The fullness is (6.2). $\square$

Corollary 6.3. *Let A be a g -dimensional abelian variety over a field k of characteristic 0. Then $A[n]$ is a finite k -group scheme of degree n^{2g} .*

Proof. For simplicity, assume that k can be embedded into $\mathbb{C}$ and fix such an embedding. By definition of the kernel, we have $(\mathbb{C} \otimes_k A)[n] \xrightarrow{\sim} \mathbb{C} \otimes_k A[n]$, so $A[n]$ is finite of degree n^{2g} if and only if $(\mathbb{C} \otimes_k A)[n]$ is finite of such degree. We can hence assume from now on that $k = \mathbb{C}$.

Since $\mathbb{C}$ is algebraically closed, $A[n]$ is finite if and only if the $\mathbb{C}$ -points $A[n](\mathbb{C})$ are finite. Moreover, once we know this finiteness, Theorem 4.8 ensures that $A[n]$ is étale. Again since $\mathbb{C}$ is algebraically closed, this is equivalent to $A[n]$ being a disjoint union of copies of $\text{Spec}(\mathbb{C})$. Thus, our proof is complete if we can show that $|A[n](\mathbb{C})| = n^{2g}$.

Recall from (4.8) that the kernel satisfies $A[n](\mathbb{C}) = A(\mathbb{C})[n]$. By Theorem 6.1, $A(\mathbb{C})[n] \xrightarrow{\sim} (n^{-1}\mathbb{Z}/\mathbb{Z})^{\oplus 2g}$, which has order n^{2g} as claimed. $\square$

6.3. The universal Weierstrass family. Let S be a scheme. There is a natural definition of elliptic curve over S which extends the case $S = \text{Spec}(k)$. Sometimes, this is also called a *relative elliptic curve*, or a *family of elliptic curves parametrized by S* .

Definition 6.4. An elliptic curve over S is an S -group schemes $E \rightarrow S$ which is proper and smooth of relative dimension 1 with connected fibers.

Clearly, if $T \rightarrow S$ is a morphism and $E \rightarrow S$ an elliptic curve, then the base change $E_T := T \times_S E$ is an elliptic curve over T . In particular, for every point $s \in S$, the fiber $E_s := \text{Spec}(\kappa(s)) \times_S E$ is an elliptic curve over $\kappa(s)$ in our previous sense.

If $E \rightarrow S$ is an elliptic curve, then E is fiber by fiber a curve of genus 1 (use Theorem 5.11). Moreover, the identity element defines a section $e : S \rightarrow E$. Conversely, we have the following extension of the constructions in §5.

Theorem 6.5. *Let $E \rightarrow S$ be proper and smooth with geometrically connected fibers of dimension 1 and genus 1. Let $e : S \rightarrow E$ be a section. Then there exists a unique S -group scheme structure $E \times_S E \rightarrow E$ with identity element e .*

We apply this theorem to families of cubic equations. Let R be a ring and let $a, b \in R$ be two elements. Consider the homogeneous Weierstrass equation

$$F_{a,b} = Y^2Z - X^3 - aXZ^2 - bZ^3 \in R[X, Y, Z]. \quad (6.4)$$

We take $S = \operatorname{Spec}(R)$ as our base and consider the vanishing locus

$$E_{a,b} := V_+(F_{a,b}) \subset \mathbb{P}_S^2.$$

If k is a field and $s : \operatorname{Spec}(k) \rightarrow S$ a k -valued point of S , then we obtain values $s^*(a), s^*(b) \in k$ by specialization. It is clear from the definition that

$$\operatorname{Spec}(k) \times_S E_{a,b} = E_{s^*(a), s^*(b)}.$$

In particular, the fiber of $E_{a,b}$ in s is a cubic curve in $\mathbb{P}_k^2$. We see that $E_{a,b} \rightarrow S$ is a projective morphism with 1-dimensional fibers.

Consider for a moment an affine curve $V(f) \subseteq \mathbb{A}_S^2$. Recall that $V(f) \rightarrow S$ is smooth if and only if the Jacobi criterion holds, which means that

$$(\partial f / \partial x, \partial f / \partial y) \in R[x, y]dx \oplus R[x, y]dy$$

has rank 1 in each point of $V(f)$. This criterion can be checked fiber by fiber. We conclude that if for all $s \in S$, the specialization

$$E_{a(s), b(s)} \subset \mathbb{P}_{\kappa(s)}^2$$

is a smooth curve, then $E \rightarrow S$ is a smooth morphism. Moreover, by Remark 5.25, all these curves have genus 1.

Assume that $E \rightarrow S$ is smooth. The shape of (6.4) ensures that the section $[0 : 1 : 0] : S \rightarrow \mathbb{P}_S^2$ factors through E . By Theorem 6.5, there is a unique group scheme structure on E with neutral element $[0 : 1 : 0]$. In this way, we have defined an elliptic curve over S .

Construction 6.6 (The universal Weierstrass family). The previous examples all come by specialization from a universal family. Let us, for simplicity, restrict to $\mathbb{Z}[1/6]$ -algebras. The discriminant of a polynomial of the form $x^3 + ax + b$ is $4a^3 + 27b^2$, and

$$x^3 + ax + b \text{ is separable} \iff 4a^3 + 27b^2 \neq 0.$$

Consider the ring

$$R := \mathbb{Z}[1/6][a, b][\Delta^{-1}], \quad \Delta = 4a^3 + 27b^2,$$

set $S = \operatorname{Spec}(R)$, and let $E_{a,b} \rightarrow S$ be as before. We have inverted the discriminant, so for every $s \in S$, the polynomial $x^3 + a(s)x + b(s) \in \kappa(s)[x]$ is separable. By Lemma 5.1, the morphism $E_{a,b} \rightarrow S$ is smooth and hence, as just explained, an elliptic curve with identity section $[0 : 1 : 0]$. It is called the *universal Weierstrass family*.⁷

Why the name “universal”? Let T be any $\mathbb{Z}[1/6]$ -scheme and let $\alpha, \beta \in \mathcal{O}_T(T)$ be functions such that, for all $t \in T$, the polynomial $x^3 + \alpha(t)x + \beta(t)$ is separable. On the one hand, we have previously defined a Weierstrass elliptic curve $E_{\alpha, \beta} \rightarrow T$. On the other hand, (α, β) give rise to a morphism $T \rightarrow S$. We find that

$$E_{\alpha, \beta} = T \times_S E_{a,b}$$

which shows that every Weierstrass family comes by pullback from the universal family.

⁷More precisely, it is the universal *simplified* Weierstrass family. The construction can also be carried out for the general Weierstrass equation (5.14) and then includes residue characteristics 2 and 3.

6.4. Torsion. We can now use the universal Weierstrass family to extend our knowledge about torsion of elliptic curves from characteristic 0 to all cases.

Theorem 6.7. *Let $E \rightarrow S$ be an elliptic curve and let $n \neq 0$. Then $E[n]$ is finite and locally free of rank n^2 over S .*

The proof requires a bit more algebraic geometry which we will leave aside in this course. It suffices for us to have the following summary result.

Proposition 6.8. *Let S be a scheme and let $f : E_1 \rightarrow E_2$ be a homomorphism of elliptic curves over S .*

(1) *Assume that S is connected and that there exists a point $s \in S$ such that the fiber homomorphism $f(s) : E_1(s) \rightarrow E_2(s)$ is 0. Then f is zero.*

(2) *Assume that f is fiberwise non-zero. Then f is finite and locally free.*

Remark 6.9. The properties in Proposition 6.8 are very similar to the ones of $\mathbb{G}_m$ in Proposition 4.7.

Proof of Theorem 6.7. For simplicity, we restrict to $\mathbb{Z}[1/6]$ -schemes.

Step 1: The universal Weierstrass curve. The universal Weierstrass curve is defined over $S = \operatorname{Spec} \mathbb{Z}[1/6][a, b, \Delta^{-1}]$. This ring is an integral domain, so S is connected. Moreover, S has points in characteristic zero; for example, every ring homomorphism

$$\mathbb{Z}[1/6][a, b, \Delta^{-1}] \longrightarrow \mathbb{Q}, \quad a, b \longmapsto \alpha, \beta, \quad \Delta(\alpha, \beta) \neq 0$$

defines such a point. By our results from the complex case (Corollary 6.3), we know that over these points, multiplication by n is non-zero and finite locally free of degree n^2 . By Proposition 6.8, we see that $[n]$ is finite and locally free of degree n^2 for the whole Weierstrass family.

Step 2: Specialization to specific elliptic curves. Let $E \rightarrow T$ be an arbitrary family of elliptic curves with $6 \in \mathcal{O}_T(T)^\times$. For every $t \in T$, the fiber elliptic curve $E(t) \rightarrow \operatorname{Spec}(\kappa(t))$ can be defined by a Weierstrass equation (Theorems 5.11 and 5.23, as well as the discussion up to (5.16)). That is, the fibers $E(t)$ all come by pullback from the universal Weierstrass family. Hence, $[n]$ is fiber by fiber finite of degree n^2 . By Proposition 6.8, $[n]$ itself is finite and locally free of degree n^2 . The kernel $E[n]$ is the pullback of $[n]$ along $S \rightarrow E$, and hence finite and locally free of rank n^2 over S as claimed. $\square$

7. THE MODULAR CURVE (ALGEBRAICALLY)

In the last few lectures, we have

- (1) Defined elliptic curves in terms of group schemes,
- (2) Proved that elliptic curves can always be defined by Weierstrass equations (simplified if $\operatorname{char}(k) \neq 2, 3$),
- (3) Constructed the universal Weierstrass family, and
- (4) Used the universal Weierstrass family to show that $E[n]$ is always of order n^2 .

Today, we want to expand on (3) and (4), and define a space that uniquely classifies elliptic curves together with a trivialization of their n -torsion.

7.1. Moduli spaces. Let us, for simplicity exclude residue characteristics 2 and 3 throughout the lecture. Consider an elliptic curve E over a field k . By (2) above, we may find $\alpha, \beta \in k$ such that E is isomorphic to the (closure in $\mathbb{P}_k^2$ of the) curve defined by

$$y^2 = x^3 + \alpha x + \beta.$$

The parameters α and β are not unique, however. Indeed, let $\lambda \in k^\times$ and consider the curve

$$y^2 = x^3 + (\lambda^{-4}\alpha)x + (\lambda^{-6}\beta). \quad (7.1)$$

It is isomorphic to the previous curve by the substitution $(x, y) \mapsto (\lambda^2 x, \lambda^3 y)$. This shows that our universal Weierstrass family

$$\mathcal{E} \longrightarrow \operatorname{Spec}(\mathbb{Z}[1/6, a, b, \Delta^{-1}]), \quad (7.2)$$

overparametrizes isomorphism classes of elliptic curves. More precisely, for E/k as above, we find a one-dimensional parameter family

$$\mathbb{G}_{m,k} \longrightarrow \operatorname{Spec}(\mathbb{Z}[1/6, a, b, \Delta^{-1}]), \quad \lambda \longmapsto (\lambda^{-4}\alpha, \lambda^{-6}\beta)$$

over which the relative curve $\mathcal{E}$ (7.2) is fiber by fiber isomorphic to E .

Question 7.1. Is it possible to improve on the construction of the universal Weierstrass family, and construct a family in which every elliptic curves occurs exactly once?

The precise mathematical meaning of this question is as follows.

Question 7.2 (Precise form). Do there exist a scheme $\mathcal{M}$ and an elliptic curve $\mathcal{E} \rightarrow \mathcal{M}$ with the following property: For every scheme S and elliptic curve $E \rightarrow S$, there exists a *unique* morphism $u : S \rightarrow \mathcal{M}$ such that $u^*(\mathcal{E}) \cong E$? Here,

$$u^*(\mathcal{E}) := S \times_{u, \mathcal{M}} \mathcal{E}$$

denotes the pullback of $\mathcal{E}$ along u . If $(\mathcal{M}, \mathcal{E})$ exists, then we call $\mathcal{M}$ the *moduli space* of elliptic curves and $\mathcal{E}$ the *universal elliptic curve*.

The pair $(\mathcal{M}, \mathcal{E})$ is uniquely determined up to unique isomorphism. Namely, assume that $\mathcal{E}_1 \rightarrow \mathcal{M}_1$ and $\mathcal{E}_2 \rightarrow \mathcal{M}_2$ are two universal elliptic curves over their respective moduli spaces. By the universal properties, there exist morphisms $u : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ and $v : \mathcal{M}_2 \rightarrow \mathcal{M}_1$ such that $u^*(\mathcal{E}_2) \cong \mathcal{E}_1$ and $v^*(\mathcal{E}_1) \cong \mathcal{E}_2$. The composition $v \circ u : \mathcal{M}_1 \rightarrow \mathcal{M}_1$ satisfies $(v \circ u)^*(\mathcal{E}_1) \cong \mathcal{E}_1$. By the uniqueness part of the universal property of $\mathcal{M}_1$, we find $v \circ u = \operatorname{id}_{\mathcal{M}_1}$. By the same argument, $u \circ v = \operatorname{id}_{\mathcal{M}_2}$. So we see $(\mathcal{M}_1, \mathcal{E}_1) \cong (\mathcal{M}_2, \mathcal{E}_2)$ in a unique way.

Answer 7.3. Assume that $(\mathcal{M}, \mathcal{E})$ exists. Then, in particular, for every field extension $k_0 \subset k$, the map $\mathcal{M}(k_0) \rightarrow \mathcal{M}(k)$ from k_0 -valued points to k -valued points would be injective. (This is simply a property of schemes.) By the universal property, this would mean that for every k_0/k , the map

$$\begin{aligned} \left\{ \begin{array}{l} \text{Isomorphism classes of} \\ \text{ellipt. curves over } k_0 \end{array} \right\} &\longrightarrow \left\{ \begin{array}{l} \text{Isomorphism classes of} \\ \text{ellipt. curves over } k \end{array} \right\} \\ E &\longmapsto k \otimes_{k_0} E \end{aligned} \quad (7.3)$$

would be injective. However, the next example shows that this map is usually not injective, so $(\mathcal{M}, \mathcal{E})$ cannot exist.

Example 7.4 (Quadratic twists). Consider $\alpha, \beta \in \mathbb{Q}$, a non-square integer $D \neq -1$, and the two cubic equations (over $\mathbb{Q}$)

$$y^2 = x^3 + \alpha x + \beta, \quad Dy^2 = x^3 + \alpha x + \beta. \quad (7.4)$$

The second equation can be brought into simplified Weierstrass form by substituting Dx and Dy for x and y , which gives

$$y^2 = x^3 + D^{-2}\alpha x + D^{-3}\beta. \quad (7.5)$$

Let us assume $\Delta(\alpha, \beta) \neq 0$ which also implies $\Delta(D^{-2}\alpha, D^{-3}\beta) = D^{-6}\Delta(\alpha, \beta) \neq 0$, so (7.4) defines two elliptic curves E and E_D over $\mathbb{Q}$. The curve E_D is called a *quadratic twist* of E .

On the one hand, E and E_D are clearly isomorphic over $\mathbb{Q}(\sqrt{D})$, because there we have the substitution $y \mapsto \sqrt{D}y$. On the other hand, one can show that E and E_D are not isomorphic over $\mathbb{Q}$. For example one can prove that two simplified Weierstrass equations over a field k of characteristic $\neq 2, 3$

$$y^2 = x^3 + \alpha_1 x + \beta_1, \quad y^2 = x^3 + \alpha_2 x + \beta_2 \quad (7.6)$$

are isomorphic *if and only if* there exists $\lambda \in k^\times$ with $(\alpha_2, \beta_2) = (\lambda^4 \alpha_1, \lambda^6 \alpha_2)$. Since we have assumed D to be integral, not a square and $\neq -1$, there is no $\lambda \in \mathbb{Q}^\times$ with $(D^{-2}\alpha, D^{-3}\beta) = (\lambda^{-4}\alpha, \lambda^{-6}\beta)$ (unique prime factorization). In terms of (7.4) and (7.5), this means that E and E_D are not isomorphic.

In summary, we have defined two elliptic curves E and E_D over $\mathbb{Q}$ such that

$$\mathbb{Q}(\sqrt{D}) \otimes_{\mathbb{Q}} E \not\cong \mathbb{Q}(\sqrt{D}) \otimes_{\mathbb{Q}} E_D.$$

This shows that (7.3) is not injective.

7.2. Level structure. Heuristically, the reason that there is no moduli space of elliptic curves is that elliptic curves have non-trivial automorphisms. For example, every elliptic curve has multiplication by -1 as automorphism, and this is what underlies the quadratic twist construction from Example 7.4.

In fact, this phenomenon is closely related to $\mathrm{GL}_2(\widehat{\mathbb{Z}}) \subset \mathrm{GL}_2(\mathbb{A}_f)$ not being small enough for the adelic double quotient formalism. We later learned in Proposition 3.21 that a simple family of small enough subgroups are the principal congruence subgroups $K(n)$ with $n \geq 3$. In the same spirit, we now introduce a notion of elliptic curve with level- n -structure. These will then have nice moduli spaces.

Example 7.5. Let us first get some intuition by considering the roots of unity. Let k be a field and let $n \geq 1$ be prime to $\mathrm{char}(k)$. Recall that

$$\mu_{n,k} = \mathrm{Spec} k[t]/(t^n - 1).$$

Consider the factorization of $t^n - 1$ into irreducible polynomials over k ,

$$t^n - 1 = \prod_{\zeta \in \mu_n(k)} (t - \zeta) \cdot \prod_{i=1}^r f_i.$$

Since $t^n - 1$ is separable, the multiplicity of every factor is 1. Moreover, the linear factors correspond to the n -th roots of unity in $k^\times$, denoted by $\mu_n(k)$. The remaining factors $f_1, \dots, f_r$ are of degree ≥ 2 . If we translate this to schemes, we find a disjoint union decomposition

$$\mu_{n,k} = \bigsqcup_{\zeta \in \mu_n(k)} \mathrm{Spec}(k) \sqcup \bigsqcup_{i=1}^r \mathrm{Spec}(K_i) \quad (7.7)$$

where $K_i = k[t]/(f_i)$ is some non-trivial field extension of k . For example, we have

$$\begin{aligned} \mu_{4,\mathbb{Q}} &= \bigsqcup_{\zeta \in \{\pm 1\}} \mathrm{Spec}(\mathbb{Q}) \sqcup \mathrm{Spec}(\mathbb{Q}(i)), \\ \mu_{4,\mathbb{Q}(i)} &= \bigsqcup_{\zeta \in \{\pm 1, \pm i\}} \mathrm{Spec}(\mathbb{Q}(i)), \\ \mu_{19,\mathbb{F}_5} &= \mathrm{Spec}(\mathbb{F}_5) \sqcup \mathrm{Spec}(\mathbb{F}_{5^{18}}). \end{aligned}$$

In general, the union

$$\bigsqcup_{\zeta \in \mu_n(k)} \mathrm{Spec}(k) \subseteq \mu_{n,k}$$

of the connected components corresponding to the k -points itself forms a group scheme. It is a constant group scheme, isomorphic to $\underline{\mu_n(k)}_{\text{Spec}(k)}$.

Definition 7.6. Let S be a scheme and let Γ be a group. The constant group scheme $\underline{\Gamma}_S$ is the S -scheme $\bigsqcup_{\gamma \in \Gamma} S$ together with the S -group scheme structure

$$\underline{\Gamma}_S \times_S \underline{\Gamma}_S = \bigsqcup_{(\gamma_1, \gamma_2) \in \Gamma \times \Gamma} S \longrightarrow \underline{\Gamma}_S$$

that reflects the multiplication of Γ : map the copy of S corresponding to (γ_1, γ_2) with id_S to the copy corresponding to $\gamma_1 \gamma_2$.

Proposition 7.7. *Let E be an elliptic curve over a field k and let $n \geq 1$ be prime to $\text{char}(k)$. Then $E[n](\bar{k})$ is a finite group isomorphic to $(\mathbb{Z}/n\mathbb{Z})^2$.*

Proof. By Theorem 6.7, $E[n]$ is a finite k -group scheme of order n^2 . By Theorem 4.10, it is étale over k . So

$$E[n] \xrightarrow{\sim} \bigsqcup_{i=1}^s \text{Spec}(K_i)$$

for finite separable field extensions K_i/k with $\sum_{i=1}^s [K_i : k] = n^2$. For every i ,

$$\bar{k} \otimes_k K_i \xrightarrow{\sim} \bar{k}^{[K_i : k]},$$

so $E[n](\bar{k})$ is a finite group of order n^2 . For every divisor $d \mid n$, the same argument shows that $E[d](\bar{k})$, which equals the d -torsion in $E[n](\bar{k})$, has order d^2 . By the classification of finite abelian groups, the only possibility is then $E[n](\bar{k}) \cong (\mathbb{Z}/n\mathbb{Z})^2$. $\square$

Example 7.8. Let E be an elliptic curve over a field k and let $n \geq 1$ be prime to $\text{char}(k)$. By the same logic as in Example 7.5, we can decompose $E[n]$ as

$$E[n] = \bigsqcup_{x \in E[n](k)} \text{Spec}(k) \sqcup (\text{Rest}).$$

Where the rest is the union of all connected components $\text{Spec}(K)$ with $[K : k] \geq 2$. The rational part can also be written as the constant group scheme $\underline{E[n](k)}_{\text{Spec}(k)}$. The group $E[n](k)$ is a subgroup of $E[n](\bar{k})$. So we know from Proposition 7.7 that $E[n](k)$ is isomorphic to a subgroup of $(\mathbb{Z}/n\mathbb{Z})^2$. In general, the group $E[n](k)$ will depend on E , n , and k .

Definition 7.9. Let E be an elliptic curve over a $\mathbb{Z}[1/n]$ -scheme S . A *level- n -structure* for E is an isomorphism

$$\eta : \underline{(\mathbb{Z}/n\mathbb{Z})^{\oplus 2}}_S \xrightarrow{\sim} E[n].$$

Equivalently, it is the datum of two sections $\eta_1 = \eta(1, 0)$ and $\eta_2 = \eta(0, 1)$ in $E(S)$ that fiber by fiber induce isomorphisms

$$\eta_s : (\mathbb{Z}/n\mathbb{Z})^{\oplus 2} \xrightarrow{\sim} E[n](\kappa(s)).$$

Remark 7.10. We could have formulated Definition 7.9 for every base scheme S , not just those with $n \in \mathcal{O}_S(S)^\times$. However, constant group schemes are clearly étale. Hence, if a level- n -structure η exists, then $E[n]$ is also étale. One can show that

$$E[n] \text{ is étale} \iff n \in \mathcal{O}_S(S)^\times,$$

so the more general definition would simply be empty for S not over $\mathbb{Z}[1/n]$.

7.3. Back to moduli spaces. Let (E_1, η_1) and (E_2, η_2) be two elliptic curves with level- n -structure over a scheme S . An isomorphism between these pairs is an isomorphism $\gamma : E_1 \rightarrow E_2$ such that $\eta_2 = \gamma \circ \eta_1$. The key point is that adding level structure solves our problem of elliptic curves having automorphisms:

Proposition 7.11 ([12, Proposition 14.8]). *Let n be ≥ 3 and let (E, η) be an elliptic curve with level- n -structure over a scheme S . Then the only automorphism of (E, η) is the identity.*

Assume that $(E, \eta)/S$ is an elliptic curve with level- n -structure and that $u : T \rightarrow S$ is a morphism. Then we may form the pullback

$$u^*(E, \eta) := (T \times_S E, \text{id}_T \times \eta).$$

In terms of the two basis sections $\eta_1 = \eta(1, 0)$ and $\eta_2 = \eta(0, 1)$, we are considering the pullback $E(S) \rightarrow E(T) = (T \times_S E)(T)$.

Theorem 7.12 (The modular curve). *For every integer $n \geq 3$, there exists a moduli space of elliptic curves with level- n -structure.*

That is, there exist a $\mathbb{Z}[1/n]$ -scheme $\mathcal{M}_n$, an elliptic curve $\mathcal{E} \rightarrow \mathcal{M}_n$, and a level- n -structure $\eta \in \mathcal{E}(S)^2$, that together have the following universal property:

For every elliptic curve with level- n -structure (E, η_0) over a scheme S , there exists a unique morphism $u : S \rightarrow \mathcal{M}_n$ such that

$$u^*(\mathcal{E}, \eta) \cong (E, \eta_0).$$

Proof idea. Let us focus on $\mathcal{M}_n[1/6]$. The primes 2 and 3 need to be treated by different arguments. We only sketch some ideas and refer the interested reader to [12, §14] for details.

Step 1. Consider the universal Weierstrass family

$$\mathcal{E} \longrightarrow \mathcal{W} = \text{Spec Spec}[1/6, a, b, \Delta^{-1}].$$

Recall that every elliptic curve already occurs (non-uniquely) in this family; our task is to pass to a quotient that has a uniqueness property.

There is a finite étale morphism

$$\mathcal{W}_n \longrightarrow \mathcal{W}[1/n]$$

that parametrizes the level- n -structures on $\mathcal{E}$. That is, giving a morphism $S \rightarrow \mathcal{W}_n$ is the same as giving a morphism $u : S \rightarrow \mathcal{W}[1/n]$ together with a level- n -structure for $u^*(\mathcal{E})$.

Step 2. Identity (7.1) defines an action of $\mathbb{G}_{m, \mathbb{Z}[1/6]}$ on $\mathcal{W}$. This action can be lifted to an action of $\mathbb{G}_{m, \mathbb{Z}[1/6]}$ on $\mathcal{W}_n$. We define $\mathcal{M}_n$ by taking a quotient

$$\mathcal{M}_n := \mathbb{G}_{m, \mathbb{Z}[1/6n]} \backslash \mathcal{W}_n.$$

Step 3. We have assumed $n \geq 3$, so the action of $\mathbb{G}_{m, \mathbb{Z}[1/6n]}$ on $\mathcal{W}_n$ is without fixed points by Proposition 7.11. This implies that $\mathcal{W}_n \rightarrow \mathcal{M}_n$ is a $\mathbb{G}_m$ -torsor which allows to descend the pair $(\mathcal{E}, \eta)$ from $\mathcal{W}_n$ to $\mathcal{M}_n$. $\square$

8. THE MODULAR CURVE (AS SHIMURA VARIETY)

In the very first lecture, we sketched the general definition of Shimura varieties. Let (G, X) be a Shimura datum. The corresponding Shimura variety for small enough level $K \subset G(\mathbb{A}_f)$ starts life as the complex manifold

$$G(\mathbb{Q}) \backslash (X \times G(\mathbb{A}_f) / K). \quad (8.1)$$

Next, the theorem of Baily–Borel (Theorem 1.3) states that this manifold has a unique structure as complex variety. That is, there exists a complex variety $\text{Sh}_K(G, X)_{\mathbb{C}}$, unique

up to isomorphism, such that the analytification $\mathrm{Sh}_K(G, X)_{\mathbb{C}}(\mathbb{C})$ in the sense of (6.1) is isomorphic to (8.1). Finally, Deligne, Milne, and Borovoi proved that there exists a canonical variety $\mathrm{Sh}_K(G, X)$ over a canonical number field $E \subset \mathbb{C}$ together with an isomorphism

$$\mathbb{C} \otimes_E \mathrm{Sh}_K(G, X) \xrightarrow{\sim} \mathrm{Sh}_K(G, X)_{\mathbb{C}}.$$

The variety $\mathrm{Sh}_K(G, X)$ is then the Shimura variety of level K of (G, X) .

Today, our goal is to carry out this construction for GL_2 . We have already studied the double quotients $\mathrm{GL}_2(\mathbb{Q}) \backslash (\mathbb{H}^{\pm} \times \mathrm{GL}_2(\mathbb{A}_f)/K)$ in §2 and §3. We have proved that K is small enough if it is contained in a principal level subgroup $K(n)$ with $n \geq 3$ (Proposition 3.21). In the more recent lectures, we have then defined the moduli space of elliptic curves with level- n -structure. Taking its generic fiber, we obtain an algebraic curve $\mathcal{M}_{n, \mathbb{Q}}$ over $\mathbb{Q}$.

Theorem 8.1. *For any $n \geq 3$, the rational curve $\mathcal{M}_{n, \mathbb{Q}}$ is the Shimura variety for GL_2 , Deligne homomorphism (1.3), and level subgroup $K(n)$.*

We first explain how to construct the isomorphism

$$\mathcal{M}_n(\mathbb{C}) \xrightarrow{\sim} \mathrm{GL}_2(\mathbb{Q}) \backslash (\mathbb{H}^{\pm} \times \mathrm{GL}_2(\mathbb{A}_f)/K(n))$$

which, essentially, is also the proof of Theorem 8.1. Afterwards, we will look at group actions and more general level subgroups.

8.1. Elliptic curves and the upper half plane. Our first aim is to understand the relation between elliptic curves and the manifold $\mathbb{H}^{\pm}$. Recall from Corollary 6.2 that analytification gives an equivalence

$$\begin{aligned} \{\text{Elliptic curves}/\mathbb{C}\} &\xrightarrow{\sim} \left\{ \begin{array}{c} \text{Compact complex Lie groups} \\ \text{of the form } \mathbb{C}/\Lambda \end{array} \right\} \\ E &\longmapsto E(\mathbb{C}). \end{aligned} \quad (8.2)$$

On the right hand side, we are simply considering $\mathbb{Z}$ -lattices $\Lambda \subset \mathbb{C}$. So, in order to parametrize elliptic curves over $\mathbb{C}$ up to isomorphism, we need to parametrize such lattices and understand when two quotients $\mathbb{C}/\Lambda_1$ and $\mathbb{C}/\Lambda_2$ are isomorphic.

Lemma 8.2. *Two lattices $\Lambda_1, \Lambda_2 \subset \mathbb{C}$ define isomorphic elliptic curves if and only if there exists $\lambda \in \mathbb{C}^{\times}$ such that $\Lambda_2 = \lambda\Lambda_1$.*

More generally, the homomorphisms from $\mathbb{C}/\Lambda_1$ to $\mathbb{C}/\Lambda_2$ are given by

$$\mathrm{Hom}(\mathbb{C}/\Lambda_1, \mathbb{C}/\Lambda_2) = \{\lambda \in \mathbb{C} \mid \lambda\Lambda_1 \subseteq \Lambda_2\}.$$

Proof. Assume $\lambda \in \mathbb{C}$ satisfies $\lambda\Lambda_1 \subseteq \Lambda_2$. Multiplication by λ defines a holomorphic group homomorphism $\mathbb{C} \rightarrow \mathbb{C}$. By the condition $\lambda\Lambda_1 \subseteq \Lambda_2$, it descends to a homomorphism

$$\lambda : \mathbb{C}/\Lambda_1 \longrightarrow \mathbb{C}/\Lambda_2.$$

If even the equality $\lambda\Lambda_1 = \Lambda_2$ holds, then this map is an isomorphism with inverse defined by λ^{-1} .

Conversely, assume we are given a holomorphic group homomorphism $f : \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$. It lifts uniquely to a holomorphic homomorphism on universal covers $\tilde{f} : \mathbb{C} \rightarrow \mathbb{C}$. Writing $\tilde{f}$ as a power series $\tilde{f}(z) = \sum_{i \geq 1} a_i z^i$ centered at 0 (and convergent on all of $\mathbb{C}$), the condition

$$\tilde{f}(z_1 + z_2) = \tilde{f}(z_1) + \tilde{f}(z_2)$$

implies that $\tilde{f}(z) = \lambda z$ for some scalar $\lambda \in \mathbb{C}$. Since $\tilde{f}$ lifts f , this scalar satisfies $\lambda\Lambda_1 \subseteq \Lambda_2$ as claimed. $\square$

Lemma 8.2 states that

$$\begin{aligned} \{\text{Lattices } \Lambda \subset \mathbb{C}\} / \mathbb{C}^\times &\xrightarrow{\sim} \{\text{Elliptic curves} / \mathbb{C}\} / \cong \\ \Lambda &\longmapsto \mathbb{C} / \Lambda. \end{aligned} \quad (8.3)$$

In order find the relation with $\mathbb{H}^\pm$, we now overparametrize all lattices by considering the set

$$\left\{ (\Lambda, \tau_1, \tau_2) \mid \begin{array}{l} \Lambda \subset \mathbb{C} \text{ a lattice} \\ \tau_1, \tau_2 \in \Lambda \text{ a } \mathbb{Z}\text{-basis} \end{array} \right\}. \quad (8.4)$$

It is equipped with an action of $\mathbb{C}^\times \times \text{GL}_2(\mathbb{Z})$ by

$$(\lambda, \gamma) \cdot (\Lambda, \tau_1, \tau_2) := (\lambda\Lambda, \lambda(\tau_1, \tau_2) \cdot \gamma^t). \quad (8.5)$$

Concretely, if $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then

$$(\tau_1, \tau_2) \cdot \gamma^t = (a\tau_1 + b\tau_2, c\tau_1 + d\tau_2).$$

On the one hand, the possible choices of a $\mathbb{Z}$ -basis for a lattice form a simply transitive $\text{GL}_2(\mathbb{Z})$ -orbit. Thus, if we take the quotient by the $\text{GL}_2(\mathbb{Z})$ -action, we precisely recover the set of lattices without extra data:

$$\text{GL}_2(\mathbb{Z}) \backslash \{(\Lambda, \tau_1, \tau_2)\} \xrightarrow{\sim} \{\Lambda \subset \mathbb{C}\}. \quad (8.6)$$

Note that the $\text{GL}_2(\mathbb{Z})$ -action and the action of $\mathbb{C}^\times$ by scaling commute, and that (8.6) is $\mathbb{C}^\times$ -equivariant.

On the other hand, we may also first quotient out the $\mathbb{C}^\times$ -action. Namely, there exists a unique representative in the orbit $\mathbb{C}^\times \cdot (\Lambda, \tau_1, \tau_2)$ of the form $(\Lambda', \tau, 1)$. It is given by

$$\tau_2^{-1} \cdot (\Lambda, \tau_1, \tau_2) = (\tau_2^{-1}\Lambda, \tau_1/\tau_2, 1).$$

Since τ_1, τ_2 are a basis for a lattice in $\mathbb{C}$, they are also an $\mathbb{R}$ -basis of $\mathbb{C}$, and so the ratio τ_1/τ_2 does not lie in $\mathbb{R}$. In this way, we find

$$\begin{aligned} \mathbb{C}^\times \backslash \{(\Lambda, \tau_1, \tau_2)\} &\xrightarrow{\sim} \mathbb{H}^\pm \\ (\Lambda, \tau_1, \tau_2) &\longmapsto \tau_1/\tau_2 \\ (\mathbb{Z}\tau + \mathbb{Z}, \tau, 1) &\longleftarrow \tau. \end{aligned} \quad (8.7)$$

We still have a $\text{GL}_2(\mathbb{Z})$ -action on the left hand side of (8.7). How does it translate to the right hand side? Given $\tau \in \mathbb{H}^\pm$ and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we calculate

$$\begin{array}{ccc} (\mathbb{Z}\tau + \mathbb{Z}, \tau, 1) & \xleftarrow{\quad} & \tau \\ \gamma \downarrow & & \\ (\mathbb{Z}\tau + \mathbb{Z}, a\tau + b, c\tau + d) & \longmapsto & \frac{a\tau + b}{c\tau + d}. \end{array}$$

In other words, we see that the action is given by Moebius transformations. In this way, we have constructed the lower isomorphism in the diagram

$$\begin{array}{ccc} \mathbb{C}^\times \times \text{GL}_2(\mathbb{Z}) \backslash \{(\Lambda, \tau_1, \tau_2)\} & \xrightarrow{\cong} & \mathbb{C}^\times \backslash \{\Lambda \subset \mathbb{C}\} \\ \cong \downarrow (8.7) & & (8.3) \downarrow \cong \\ \text{GL}_2(\mathbb{Z}) \backslash \mathbb{H}^\pm & \xrightarrow{\quad \cong \quad} & \left\{ \text{Isom. classes of} \right. \\ & & \left. \text{ellipt. curves} / \mathbb{C} \right\}. \end{array} \quad (8.8)$$

The lower horizontal map is simply given by

$$\tau \longmapsto \mathbb{C} / (\mathbb{Z}\tau + \mathbb{Z}).$$

8.2. The isomorphism between $\mathcal{M}_n(\mathbb{C})$ and $\mathcal{S}_{K(n)}(\mathbb{C})$. Let us introduce the following shorthand notation: For a level subgroup $K \subset G(\mathbb{A}_f)$, we write

$$\mathcal{S}_K(\mathbb{C}) := \mathrm{GL}_2(\mathbb{Q}) \backslash (\mathbb{H}^\pm \times \mathrm{GL}_2(\mathbb{A}_f)/K).$$

Theorem 8.3. *For ever $n \geq 3$, there is an isomorphism*

$$\mathcal{S}_{K(n)}(\mathbb{C}) \xrightarrow{\sim} \mathcal{M}_n(\mathbb{C}).$$

Proof. Step 1: Parametrizing elliptic curves with level structure by lattices. Our first step is to upgrade the upper and right hand side arrows in (8.8) to include level structure.

Let us, from now on, view a basis $\tau_1, \tau_2 \in \Lambda$ as an isomorphism

$$\alpha : \mathbb{Z}^2 \xrightarrow{\sim} \Lambda, \quad e_i \mapsto \tau_i, \quad i = 1, 2.$$

This helps us keep track of the $\mathrm{GL}_2(\mathbb{Z})$ -action because we now have the cleaner expression

$$\gamma \cdot (\Lambda, \alpha) = (\Lambda, \alpha \circ \gamma^t). \quad (8.9)$$

Next, we observe that the n -torsion of $\mathbb{C}/\Lambda$ is given by the quotient $(n^{-1}\Lambda)/\Lambda$. Using multiplication by n , we identify

$$(n^{-1}\Lambda)/\Lambda \xrightarrow{\sim} \Lambda/n\Lambda, \quad n^{-1}\lambda \mapsto \lambda.$$

Giving a level structure for $\mathbb{C}/\Lambda$ is now the same as giving an isomorphism

$$\eta : (\mathbb{Z}/n\mathbb{Z})^2 \xrightarrow{\sim} \Lambda/n\Lambda.$$

Let us consider the set of triples

$$\left\{ (\Lambda, \alpha, \eta) \mid \begin{array}{l} \Lambda \subset \mathbb{C} \text{ a lattice, } \alpha : \mathbb{Z}^2 \xrightarrow{\sim} \Lambda \text{ a basis} \\ \eta : (\mathbb{Z}/n\mathbb{Z})^2 \xrightarrow{\sim} \Lambda/n\Lambda \text{ a level-}n\text{-structure} \end{array} \right\}.$$

It is equipped by an action of $\mathbb{C}^\times \times \mathrm{GL}_2(\mathbb{Z})$ in the same way as (8.5) before:

$$(\lambda, \gamma) \cdot (\Lambda, \alpha, \eta) := (\lambda\Lambda, \lambda\alpha\gamma^t, \lambda\eta).$$

Taking the quotient by $\mathrm{GL}_2(\mathbb{Z})$ is the same as forgetting α , while scaling by $\mathbb{C}^\times$ is the same as passing to the isomorphism class of elliptic curve. So we find

$$\mathbb{C}^\times \times \mathrm{GL}_2(\mathbb{Z}) \backslash \{(\Lambda, \alpha, \eta)\} \xrightarrow{\sim} \left\{ \begin{array}{l} \text{Isom. classes of} \\ \text{ellipt. curves with} \\ \text{level-}n\text{-str. } (E, \eta)/\mathbb{C} \end{array} \right\}. \quad (8.10)$$

Step 2: Translate to a statement about $\mathbb{H}^\pm$. As in (8.8), we now want to interchange the order and quotient by $\mathbb{C}^\times$ first. To this end, we first simplify our triples (Λ, α, η) . Namely, given such a triple, we may consider the composition

$$\alpha^{-1} \circ \eta := [\alpha^{-1} \bmod n] \circ \eta \in \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z}).$$

Clearly, the two data

$$(\Lambda, \alpha, \eta) \longleftrightarrow (\Lambda, \alpha, \alpha^{-1} \circ \eta) \quad (8.11)$$

are equivalent because we can get one from the other by composing maps. If we act by an element $\gamma \in \mathrm{GL}_2(\mathbb{Z})$ on the left of (8.11), then the matrix on the right transforms by left multiplication with $\gamma^{t,-1} \bmod n$ because

$$(\alpha\gamma^t)^{-1} \circ \eta = \gamma^{t,-1} \circ (\alpha^{-1} \circ \eta). \quad (8.12)$$

This motivates us to renormalize (8.11) as

$$\begin{aligned} \{(\Lambda, \alpha, \eta)\} &\xrightarrow{\sim} \{(\Lambda, \alpha)\} \times \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z}) \\ (\Lambda, \alpha, \eta) &\longmapsto ((\Lambda, \alpha), (\alpha^{-1} \circ \eta)^{t,-1}). \end{aligned} \quad (8.13)$$

Under this normalized isomorphism, the $\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ -action on the right hand side is given by the straightforward formula

$$\gamma \cdot ((\Lambda, \alpha), g) = ((\Lambda, \alpha\gamma^t), \gamma g). \quad (8.14)$$

Moreover, the $\mathbb{C}^\times$ -scaling does not involve the third entry anymore,

$$\lambda \cdot ((\Lambda, \alpha), g) = ((\lambda\Lambda, \lambda\alpha), g).$$

This means that (8.7) applies unchanged and we find

$$\begin{aligned} \mathrm{GL}_2(\mathbb{Z}) \backslash (\mathbb{H}^\pm \times \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})) &\xrightarrow{\sim} \left\{ \begin{array}{c} \text{Isom. classes of ellipt.} \\ \text{curves with level-}n\text{-str. } (E, \eta)/\mathbb{C} \end{array} \right\}. \\ (\tau, g) &\longmapsto (\mathbb{C}/\Lambda, (\tau, 1) \circ g^{t, -1}). \end{aligned} \quad (8.15)$$

Here, $(\tau, 1) \circ g^{t, -1}$ is the map

$$(\mathbb{Z}/n\mathbb{Z})^2 \xrightarrow{\sim} \Lambda/n\Lambda, \quad \begin{pmatrix} e \\ f \end{pmatrix} \longmapsto (\tau, 1) \cdot g^{t, -1} \cdot \begin{pmatrix} e \\ f \end{pmatrix}.$$

Step 3: Relate (8.15) to the definition of $\mathcal{S}_{K(n)}(\mathbb{C})$.

Lemma 8.4. *For every level subgroup $K \subseteq \mathrm{GL}_2(\widehat{\mathbb{Z}})$, the inclusion map induces an isomorphism*

$$\mathrm{GL}_2(\mathbb{Z}) \backslash \mathrm{GL}_2(\widehat{\mathbb{Z}})/K \xrightarrow{\sim} \mathrm{GL}_2(\mathbb{Q}) \backslash \mathrm{GL}_2(\mathbb{A}_f)/K. \quad (8.16)$$

Proof. The inclusion $\mathrm{GL}_2(\widehat{\mathbb{Z}}) \rightarrow \mathrm{GL}_2(\mathbb{A}_f)$ clearly descends to a map on quotients as in (8.16). Assume that for $g_1, g_2 \in \mathrm{GL}_2(\widehat{\mathbb{Z}})$, there exist $h \in \mathrm{GL}_2(\mathbb{Q})$ and $k \in K$ with $g_2 = hg_1k$. Then $h = g_2k^{-1}g_1^{-1}$ lies in $\mathrm{GL}_2(\widehat{\mathbb{Z}}) \cap \mathrm{GL}_2(\mathbb{Q})$ which equals $\mathrm{GL}_2(\mathbb{Z})$. So the map on quotients is injective.

For surjectivity, we recall from Proposition 3.19 that the determinant induces a bijection

$$\det : \mathrm{GL}_2(\mathbb{Q}) \backslash \mathrm{GL}_2(\mathbb{A}_f)/K \xrightarrow{\sim} \mathbb{Q}^\times \backslash \mathbb{A}_f^\times / \det(K).$$

We stated that proposition for $\mathrm{GL}_2(\mathbb{Q})_{>0}$, but it also holds for the full matrix group with the same proof. Since

$$\{\pm 1\} \backslash \widehat{\mathbb{Z}}^\times \xrightarrow{\sim} \mathbb{Q}^\times \backslash \mathbb{A}_f^\times,$$

the determinant map restricted to the left hand side of (8.16) is already surjective. This means that (8.16) is also surjective. $\square$

Lemma 8.4 can be used directly in the Shimura variety definition and provides an isomorphism

$$\mathrm{GL}_2(\mathbb{Z}) \backslash (\mathbb{H}^\pm \times \mathrm{GL}_2(\widehat{\mathbb{Z}})/K) \xrightarrow{\sim} \mathrm{GL}_2(\mathbb{Q}) \backslash (\mathbb{H}^\pm \times \mathrm{GL}_2(\mathbb{A}_f)/K).$$

If we specialize to $K = K(n)$ and use our result from Step 2 (8.15), then we exactly obtain an isomorphism

$$\mathcal{S}_{K(n)}(\mathbb{C}) \xrightarrow{\sim} \mathcal{M}_n(\mathbb{C})$$

as claimed by Theorem 8.3. $\square$

8.3. The $\mathrm{GL}_2(\widehat{\mathbb{Z}})$ -action. Recall that $K(n)$ is defined as a kernel,

$$K(n) = \ker \left(\mathrm{GL}_2(\widehat{\mathbb{Z}}) \longrightarrow \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z}) \right).$$

In particular, it is a normal subgroup of $\mathrm{GL}_2(\widehat{\mathbb{Z}})$ with quotient $\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$. Let us write $[x, h]$ or $[x, hK]$ for the point $\mathrm{GL}_2(\mathbb{Q}) \cdot (x, hK) \in \mathcal{S}_K$. The normality implies that we obtain a right-action of $\mathrm{GL}_2(\widehat{\mathbb{Z}})$ on $\mathcal{S}_{K(n)}$ by

$$\begin{aligned} \mathrm{GL}_2(\widehat{\mathbb{Z}}) \times \mathcal{S}_{K(n)} &\longrightarrow \mathcal{S}_{K(n)} \\ [x, h] \cdot g &= [x, hg]. \end{aligned} \tag{8.17}$$

The point here is that

$$hK(n) \cdot g = hg(g^{-1}K(n)g) = hgK(n)$$

because of the normality. Put differently, the cosets hgK and $hkgK$ are equal for every $k \in K$, which ensures that (8.17) is well-defined. It is clear from the definition that $K(n)$ acts trivially, so this is really a $\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ -action.

We now turn to the moduli space $\mathcal{M}_n$. (Recall that $n \geq 3$ is assumed for $\mathcal{M}_n$ to exist.) Given $g \in \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ and an S -valued point $(E, \eta) \in \mathcal{M}_n(S)$, we can define a new S -valued point $(E, \eta \circ g)$. In this way, $\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ acts on the right of $\mathcal{M}_n$.

Proposition 8.5. *The isomorphism $\iota : \mathcal{S}_{K(n)} \xrightarrow{\sim} \mathcal{M}_n(\mathbb{C})$ constructed during the proof of Theorem 8.3 is $\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ -equivariant in the sense that*

$$\iota([x, h] \cdot g) = \iota([x, h]) \cdot g^{t, -1}.$$

Proof. Start with a point $(E, \eta) \in \mathcal{M}_n(\mathbb{C})$. We choose a lattice $\Lambda \subset \mathbb{C}$ and an isomorphism $\mathbb{C}/\Lambda \xrightarrow{\sim} E(\mathbb{C})$. By composition, we obtain a level structure $\eta : (\mathbb{Z}/n\mathbb{Z})^2 \xrightarrow{\sim} \Lambda/n\Lambda$. We arbitrarily choose a basis $\alpha : \mathbb{Z}^2 \xrightarrow{\sim} \Lambda$. In this way, we have represented (E, η) by a triple (Λ, α, η) as on the right hand side of (8.13). A representative in $\mathbb{H}^\pm \times \mathrm{GL}_2(\mathbb{A}_f)/K(n)$ is given by

$$[\tau := \mathbb{C}^\times(\Lambda, \alpha), hK(n) := (\alpha^{-1} \circ \eta)^{t, -1}]. \tag{8.18}$$

Here, τ is defined by (8.7) as before. If we substitute $\eta \circ g^{t, -1}$ in (8.18), then we change $hK(n)$ to $hgK(n)$. This is precisely what is claimed by Proposition 8.5. $\square$

Remark 8.6. Note that $g \mapsto g^{t, -1}$ defines a group automorphism of GL_2 . The difference between g acting as g or $g^{t, -1}$ in Proposition 8.5 has no mathematical meaning but simply resulted from the choices we made along the way.

8.4. General level groups. We have defined the curve $\mathcal{S}_K$ for every small enough level $K \subset \mathrm{GL}_2(\mathbb{A}_f)$. In fact, one can even work without the “small enough” assumption. First, one defines $\mathcal{S}_K$ as set with quotient topology. Then, one proves that there exists a unique Riemann surface structure on $\mathcal{S}_K$ such that the quotient map

$$\mathbb{H}^\pm \times \mathrm{GL}_2(\mathbb{A}_f)/K \longrightarrow \mathcal{S}_K \tag{8.19}$$

is holomorphic. The only difference with the case K small enough is that (8.19) might not be a covering map in the sense of topology.

By contrast, we have only defined the algebraic modular curve for principal level subgroups. Our final aim in this section is to define an algebraic curve $\mathcal{M}_K$ for every level K , and to state its comparison with $\mathcal{S}_K$.

Proposition 8.7. *For every level $K \subset \mathrm{GL}_2(\mathbb{A}_f)$, there exist a group element $g \in \mathrm{GL}_2(\mathbb{A}_f)$ and $n \geq 1$ such that*

$$K(n) \subseteq g^{-1}Kg \subseteq \mathrm{GL}_2(\widehat{\mathbb{Z}}).$$

We will use the same argument as during the proof of Proposition 2.5. This argument was based on the notion of $\mathbb{Z}$ -lattice in a $\mathbb{Q}$ -vector space, and the fact that for any two lattices $\Lambda_1, \Lambda_2 \subset V$, there exists $g \in \mathrm{GL}(V)$ with $\Lambda_2 = g\Lambda_1$.

This statement is more general. Let R be a PID with fraction field K , and let V be a finite dimensional K -vector space (say of dimension m). An R -lattice in V is simply an R -submodule $\Lambda \subset V$ which is isomorphic to R^m . Then, for any two R -lattices $\Lambda_1, \Lambda_2 \subset V$, there exists $g \in \mathrm{GL}(V)$ with $\Lambda_2 = g\Lambda_1$.

The ring $\mathbb{A}_f$ is not a domain, but there is still a very similar notion of $\widehat{\mathbb{Z}}$ -lattice:

Definition 8.8. Let $\mathbb{V}$ be a finite free $\mathbb{A}_f$ -module. A $\widehat{\mathbb{Z}}$ -submodule $\widehat{\Lambda} \subset \mathbb{V}$ is called a lattice if satisfies the following, equivalent conditions. After choosing a basis $\mathbb{A}_f^m \xrightarrow{\sim} \mathbb{V}$, we have

- (1) $\widehat{\Lambda}$ is finitely generated as $\widehat{\mathbb{Z}}$ -module and $\mathbb{A}_f \cdot \widehat{\Lambda} = \mathbb{V}$.
- (2) There exists an integer $c \geq 1$ such that

$$c \cdot \widehat{\mathbb{Z}}^m \subseteq \widehat{\Lambda} \subseteq c^{-1} \cdot \widehat{\mathbb{Z}}^m.$$

- (3) There exist finitely many primes S and lattices $\Lambda_p \subset \mathbb{Q}_p^m$, $p \in S$, such that

$$\widehat{\Lambda} = \prod_{p \in S} \Lambda_p \times \prod_{p \notin S} \mathbb{Z}_p^m.$$

- (4) There exists an element $g \in \mathrm{GL}_m(\mathbb{A}_f)$ with $\widehat{\Lambda} = g \cdot \widehat{\mathbb{Z}}^m$.

Proof of the equivalence of (1) – (4). Assume statement (1). On the one hand, $\widehat{\Lambda}$ has finitely many generators $\lambda_1, \dots, \lambda_r$, each of which is an m -tuple of elements from $\mathbb{A}_f$. Recall that every $(x_p)_{p \text{ prime}} \in \mathbb{A}_f$ has only finitely many entries x_p not in $\mathbb{Z}_p$. So only finitely many denominators occur among the entries of the λ_j which means there exists $c_1 \geq 1$ with $\widehat{\Lambda} \subseteq c_1^{-1} \cdot \widehat{\mathbb{Z}}^m$. On the other hand, $\mathbb{A}_f \cdot \widehat{\Lambda} = \mathbb{A}_f^m$ means that each standard basis vector e_i can be written as a linear combination

$$e_i = \sum_{j=1}^r a_{ij} \lambda_j, \quad a_{ij} \in \mathbb{A}_f.$$

Only finitely many denominators occur among the entries of the a_{ij} . So there exists $c_2 \geq 1$ with $c_2(a_{ij}) \in M_{m \times r}(\widehat{\mathbb{Z}})$. This means $c_2 \cdot \widehat{\mathbb{Z}}^m \subseteq \widehat{\Lambda}$ and we obtain statement (2) with $c = c_1 c_2$.

Assume statement (2). Let S be the set of all primes dividing c . For $p \in S$, define Λ_p as the lattice generated by the p -components of all elements of $\widehat{\Lambda}$. With these choices, (3) holds.

Assume statement (3). For every $p \in S$, let $g_p \in \mathrm{GL}_m(\mathbb{Q}_p)$ be such that $g_p \cdot \mathbb{Z}_p^m = \Lambda_p$. Define $g = (g_p)_{p \in S} \times (\mathrm{id}_m)_{p \notin S}$ which lies in $\mathrm{GL}_m(\mathbb{A}_f)$. We obtain $\widehat{\Lambda} = g \cdot \widehat{\mathbb{Z}}^m$ as in statement (4).

Finally, assume statement (4). Since multiplication by g defines an isomorphism $\widehat{\mathbb{Z}}^m \xrightarrow{\sim} \widehat{\Lambda}$, we immediately obtain that $\widehat{\mathbb{Z}}$ is finitely generated as $\widehat{\mathbb{Z}}$ -module. Moreover, $g^{-1} \cdot \widehat{\Lambda} = \widehat{\mathbb{Z}}^m$ shows that $\mathbb{A}_f \cdot \widehat{\Lambda}$ equals $\mathbb{A}_f^m$. $\square$

Remark 8.9. Another way to phrase (3) above is to say that a $\widehat{\mathbb{Z}}$ -lattice in $\mathbb{A}_f^m$ is the same as a family of $\mathbb{Z}_p$ -lattices $\Lambda_p \subset \mathbb{Q}_p^m$, almost all of which are equal to $\mathbb{Z}_p^m$.

We can now come back to the proof of Proposition 8.7.

Proof. Let $K \subset \mathrm{GL}_2(\mathbb{A}_f)$ be a level subgroup. The intersection of two open subsets is open, so $K' = \mathrm{GL}_2(\widehat{\mathbb{Z}}) \cap K$ is again an open subgroup. Moreover, K as the disjoint union $K = \bigsqcup_{k \in K/K'} kK'$ of its K' -cosets. By compactness of K , finitely many such cosets suffice to cover K from which we obtain that K/K' is finite. Let $k_1, \dots, k_r$ be a set of coset representatives. Then

$$\widehat{\Lambda} := \sum_{i=1}^r k_i \cdot \widehat{\mathbb{Z}}^2$$

is a lattice in $\mathbb{A}_f^2$; it is clearly finitely generated as $\widehat{\mathbb{Z}}$ -module and satisfies $\mathbb{A}_f \cdot \widehat{\Lambda} = \mathbb{A}_f^2$. It is also clear that K stabilizes $\widehat{\Lambda}$, meaning $K \subseteq \mathrm{GL}(\widehat{\Lambda})$.

By part (4) of Definition 8.8, there exists $g \in \mathrm{GL}_2(\mathbb{A}_f)$ with $\widehat{\Lambda} = g \cdot \widehat{\mathbb{Z}}^2$. This means that $\mathrm{GL}(\widehat{\Lambda}) = g \mathrm{GL}_2(\widehat{\mathbb{Z}}) g^{-1}$ and we find

$$g^{-1}Kg \subseteq \mathrm{GL}_2(\widehat{\mathbb{Z}}).$$

Conjugation by an element in a topological group is a homeomorphism, so $g^{-1}Kg$ is again open. We know that the principal congruence subgroups in $\mathrm{GL}_2(\mathbb{A}_f)$ form a neighborhood basis of the identity. So there exists $n \geq 1$ with $K(n) \subseteq g^{-1}Kg$ and the proof is complete. $\square$

Construction 8.10. (1) In light of Proposition 8.5, we use the new convention

$$(E, \eta) \cdot g := (E, \eta \circ g^{t,-1})$$

for our action of $\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ on $\mathcal{M}_n$. With this renormalization, $\iota : \mathcal{S}_{K(n)} \xrightarrow{\sim} \mathcal{M}_n(\mathbb{C})$ is $\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ -equivariant on the nose.

(2) Let us next come back to the ideas around (8.17). Assume that $K \subset \mathrm{GL}_2(\mathbb{A}_f)$ is a level and $g \in \mathrm{GL}_2(\mathbb{A}_f)$ a group element. Then, multiplication by g on the right induces an isomorphism

$$\begin{aligned} g : \mathcal{S}_K &\xrightarrow{\sim} \mathcal{S}_{g^{-1}Kg} \\ [x, hK] &\longmapsto [x, hKg]. \end{aligned}$$

Note that $hKg = hg(g^{-1}Kg)$ which explains why this is a reasonable definition.

(3) Given K , choose g and $n \geq 3$ such that $K(n) \subseteq g^{-1}Kg \subseteq \mathrm{GL}_2(\widehat{\mathbb{Z}})$. Using the normality of $K(n)$ in $\mathrm{GL}_2(\widehat{\mathbb{Z}})$, we can form the quotient group

$$\mathcal{K} = K(n) \backslash (g^{-1}Kg).$$

Being a subgroup of $\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$, it acts on the right of both $\mathcal{S}_{K(n)}$ and $\mathcal{M}_n$. We define $\mathcal{M}_K$ as the quotient curve $\mathcal{M}_{n,\mathbb{Q}}/\mathcal{K}$ (see below). The isomorphism $\iota : \mathcal{S}_{K(n)} \xrightarrow{\sim} \mathcal{M}_n$ descends to quotients, and we obtain the terms in the following diagram

$$\begin{array}{ccc} \mathcal{S}_{K(n)} & \xrightarrow{\iota} & \mathcal{M}_{n,\mathbb{Q}}(\mathbb{C}) \\ \text{mod } \mathcal{K} \downarrow & & \downarrow \text{mod } \mathcal{K} \\ \mathcal{S}_K & \xrightarrow{g} \mathcal{S}_{g^{-1}Kg} & \dashrightarrow \mathcal{M}_K(\mathbb{C}). \end{array} \tag{8.20}$$

We define the isomorphism $\mathcal{S}_K \xrightarrow{\sim} \mathcal{M}_K(\mathbb{C})$ as the composition of the bottom two arrows.

We still have to explain how to define the quotient $\mathcal{M}_{n,\mathbb{Q}}/\mathcal{K}$. Let A be a ring and let Γ be a finite group. A group action of Γ on $X = \mathrm{Spec}(A)$ is the same as a group homomorphism $\Gamma \rightarrow \mathrm{Aut}_{\mathrm{Scheme}}(X)$. By the equivalence between rings and affine schemes, this is the same as an action of Γ by ring automorphisms on A . Let

$$A^\Gamma = \{a \in A \mid \gamma a = a \text{ for all } \gamma \in \Gamma\}$$

be the ring of invariant elements. The quotient of $\mathrm{Spec}(A)$ by Γ is defined as

$$\Gamma \backslash X := \mathrm{Spec}(A^\Gamma).$$

This construction has the expected universal property in the category of affine schemes. Namely, assume B is a ring and $f : X \rightarrow \mathrm{Spec}(B)$ a Γ -invariant map. This means that $f \circ \gamma = f$ for all $\gamma \in \Gamma$. Translated to rings, it means that $\gamma^* \circ f^* = f^*$ for all γ , which says that $f^* : B \rightarrow A$ has image contained in A^Γ . In other words, there exists a unique factorization over the quotient as in the diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & \mathrm{Spec}(B) \\ \downarrow & \nearrow \bar{f} & \\ \Gamma \backslash X & & \end{array} \quad (8.21)$$

We now need the fact that $\mathcal{M}_n$ is an affine $\mathbb{Z}[1/n]$ -scheme. That is, there exists a finite type $\mathbb{Z}[1/n]$ -algebra A_n with $\mathcal{M}_n = \mathrm{Spec}(A_n)$. The $\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ -action on $\mathcal{M}_n$ translates into a $\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ -action on A_n , and this is how the previous definitions apply.

For a scheme S , we can consider the map $X(S) \rightarrow (\Gamma \backslash X)(S)$ induced by $X \rightarrow \Gamma \backslash X$. It is Γ -invariant, so we obtain a natural map

$$\Gamma \backslash X(S) \longrightarrow (\Gamma \backslash X)(S).$$

In general, this map is neither injective nor surjective. However, we have the following result for affine varieties.

Proposition 8.11. *Let $X = \mathrm{Spec}(A)$ be an affine finite type scheme over a field k and let Γ be a finite group acting on X . For every algebraically closed extension K/k , the quotient map $X \rightarrow \Gamma \backslash X$ defines a bijection*

$$\Gamma \backslash X(K) \xrightarrow{\sim} (\Gamma \backslash X)(K).$$

In the context of (8.20), Proposition 8.11 ensures that

$$\mathcal{M}_n(\mathbb{C})/\mathcal{K} \xrightarrow{\sim} \mathcal{M}_K(\mathbb{C}). \quad (8.22)$$

This defines the bottom dotted arrow in (8.20) and finally completes our construction. One may check that the construction does not depend on n or g up to natural isomorphism.

8.5. The classical moduli problems. We wrap up this section with a description of the classical moduli problems.

Example 8.12 (The j -invariant). Consider $\mathcal{S}_{\mathrm{GL}_2(\widehat{\mathbb{Z}})}$ with its Riemann surface structure from (8.19). By (8.8) (lower arrow), its points are in bijection with the isomorphism classes of elliptic curves over $\mathbb{C}$.

We have now also constructed an algebraic curve $\mathcal{M}_{\mathrm{GL}_2(\widehat{\mathbb{Z}}), \mathbb{Q}}$ over $\mathbb{Q}$ as $\mathcal{M}_n/\mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ for $n \geq 3$ (choose $n = 3$ for example). The j -invariant (see [12, §11.1]) defines an isomorphism

$$j : \mathcal{M}_{\mathrm{GL}_2(\widehat{\mathbb{Z}}), \mathbb{Q}} \xrightarrow{\sim} \mathbb{A}_{\mathbb{Q}}^1.$$

On a Weierstrass elliptic curve

$$E : y^2 = x^3 + \alpha x + \beta,$$

it is given by

$$j(E) = 1728 \frac{4\alpha^3}{4\alpha^3 + 27\beta^2}. \quad (8.23)$$

This value only depends on the isomorphism class of E and not on the choice of α and β . Proposition 8.11 states that for every algebraically closed field $K/\mathbb{Q}$, the j -invariant provides a bijection

$$j : \left\{ \begin{array}{l} \text{Isomorphism classes of} \\ \text{elliptic curves over } K \end{array} \right\} \xrightarrow{\sim} K$$

$$E \longmapsto j(E).$$

In fact, this bijection holds for every algebraically closed field K . Except that in characteristics 2 and 3, one needs to define j by a different formula.

Example 8.13 (Level $K_0(n)$). Consider the level subgroup

$$K_0(n) = \left\{ k \in \mathrm{GL}_2(\widehat{\mathbb{Z}}) \mid k \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{n} \right\}.$$

Note that we always have $-1 \in K_0(n)$, so the intersections $\mathrm{GL}_2(\mathbb{Q}) \cap gK_0(n)g^{-1}$ will never be torsion free. Still, (8.19) defines a Riemann surface $\mathcal{S}_{K_0(n)}$ and Construction 8.10 constructs an algebraic curve $\mathcal{M}_{K_0(n),\mathbb{Q}}$ such that $\mathcal{S}_{K_0(n)} \xrightarrow{\sim} \mathcal{M}_{K_0(n),\mathbb{Q}}(\mathbb{C})$.

The quotient group $K_0(n)/K(n) \subset \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ is the stabilizer of the line spanned by $(1,0)^t$ in $(\mathbb{Z}/n\mathbb{Z})^2$. From (8.22), we obtain a bijection

$$\left\{ \begin{array}{l} \text{Isom. classes of} \\ (E, C) \text{ over } K \end{array} \right\} \xrightarrow{\sim} \mathcal{M}_{K_0(n)}(\mathbb{C})$$

where, on the left hand side, we consider elliptic curves E with a subgroup $C \subset E[n]$ such that C is cyclic of order n .

Example 8.14 (Level $K_1(n)$). Consider the level subgroup

$$K_1(n) = \left\{ k \in \mathrm{GL}_2(\widehat{\mathbb{Z}}) \mid k \equiv \begin{pmatrix} 1 & * \\ 0 & * \end{pmatrix} \pmod{n} \right\}.$$

The quotient $K_1(n)/K(n)$ is the stabilizer of the vector $(1,0)^t \in (\mathbb{Z}/n\mathbb{Z})^2$. One obtains from (8.22) that, for every algebraically closed field $K/\mathbb{Q}$,

$$\left\{ \begin{array}{l} \text{Isom. classes of} \\ (E, P) \text{ over } K \end{array} \right\} \xrightarrow{\sim} \mathcal{M}_{K_1(n)}(\mathbb{C})$$

where, on the left hand side, we consider elliptic curves E with a point $P \in E[n]$ of exact order n .

Observe that the determinant maps $K_0(n), K_1(n) \rightarrow \widehat{\mathbb{Z}}^\times$ are surjective. Proposition 3.19 applies and shows that $\mathcal{S}_{K_0(n)}$ and $\mathcal{S}_{K_1(n)}$ are connected. This allows to define them in terms of SL_2 and the upper half plane. The traditional notation in the literature is

$$\mathcal{Y}_0(n) = \mathcal{S}_{K_0(n)} \quad \text{and} \quad \mathcal{Y}_1(n) = \mathcal{S}_{K_1(n)}.$$

Their traditional definition is

$$\mathcal{Y}_0(n) := \Gamma_0(n) \backslash \mathbb{H}^+, \quad \mathcal{Y}_1(n) := \Gamma_1(n) \backslash \mathbb{H}^+$$

where

$$\Gamma_0(n) = \left\{ \gamma \in \mathrm{SL}_2(\mathbb{Z}) \mid \gamma \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{n} \right\}$$

$$\Gamma_1(n) = \left\{ \gamma \in \mathrm{SL}_2(\mathbb{Z}) \mid \gamma \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod{n} \right\}.$$



Part 2. General Shimura varieties

9. LINEAR ALGEBRAIC GROUPS

Throughout this section, k is a field of characteristic 0 and G a connected linear algebraic group over k . Recall that this means that G is a connected, affine, finite type k -group scheme (Theorem 4.13) and that any such G is automatically smooth over k (Theorem 4.8).

Our goal today is to define reductive groups and to discuss their structure theory. We will also see several important examples like Weil restrictions and unitary groups.

Recommended reading: [16, §19a and §19b] which introduces semi-simple and reductive groups.

Unless indicated otherwise, schemes in this section are over k and products are taken over $\text{Spec}(k)$. By group over k we mean an affine finite type k -group scheme. By subgroup, we mean a closed subgroup scheme. The notation $X_{\bar{k}}$ denotes the base change $\bar{k} \otimes_k X$.

9.1. Quotient groups. Let $H \subseteq G$ be a subgroup. We call H *normal* if the conjugation morphism

$$\begin{aligned} G \times H &\longrightarrow G \\ (g, h) &\longmapsto ghg^{-1} \end{aligned}$$

has image in H . Equivalently, for every point $g \in G(\bar{k})$, we have $g(H_{\bar{k}})g^{-1} = H_{\bar{k}}$ as closed subschemes of $G_{\bar{k}}$. Yet another way to phrase this condition is to say that $H(\bar{k}) \subset G(\bar{k})$ is normal.

Assume that $H \subseteq G$ is normal. Consider the action of H on G by multiplication

$$\begin{aligned} a : H \times G &\longrightarrow G \\ (h, g) &\longmapsto hg. \end{aligned} \tag{9.1}$$

Since G and H are affine, we can write $G = \text{Spec}(A)$ and $H = \text{Spec}(B)$, $B = A/I$, for a finite type k -algebra A and an ideal $I \subseteq A$. Then (9.1) corresponds to a k -algebra homomorphism

$$a^* : B \otimes_k A \longleftarrow A.$$

Definition 9.1 (Quotients, see [16, §5.c]). A function $f \in A$ is *H -invariant* if $a^*(f) = 1 \otimes f$. We denote by $A^H \subseteq A$ the subring of invariant functions. The *quotient* of G by H is the k -scheme

$$G/H := \text{Spec}(A^H).$$

This construction has the following properties:

- A^H is again a finite type k -algebra.
- The comultiplication⁸ map $m^* : A \rightarrow A \otimes_k A$ restricts to a comultiplication on A^H which makes G/H into a k -group.
- The natural map $G \rightarrow G/H$ is a group homomorphism. It is flat and surjective with kernel H .
- Every group scheme homomorphism $\pi : G \rightarrow Q$ with $H \subseteq \ker(\pi)$ factors over G/H .
- The quotient construction defines a bijection

$$\begin{aligned} \{\text{Normal subgroups } H \subseteq G\} &\longleftrightarrow \{\text{Surjective flat homomorphisms } G \rightarrow Q\} \\ H &\longmapsto [G \rightarrow G/H] \\ \ker(\pi) &\longleftarrow [G \xrightarrow{\pi} Q]. \end{aligned} \tag{9.2}$$

⁸ *Comultiplication* is the name for the map of rings dual to the multiplication $m : G \times G \rightarrow G$.

Remark 9.2. An equivalent and more intuitive definition of H -invariant functions is follows. Every $f \in A$ can be viewed as a function on $G(\bar{k})$ with values in $\bar{k}$. Then, f is H -invariant if $f(hg) = f(g)$ for all $h \in H(\bar{k})$ and $g \in G(\bar{k})$.

Example 9.3. Being of finite type over a field, G has finitely many connected components. Let $G^\circ \subseteq G$ be the connected component containing the identity element. Then G° is normal and G/G° is a finite k -group scheme.

Definition 9.4 (Center of G). The *center of G* is the subgroup $Z(G) \subset G$ such that for all k -schemes S , $Z(G)(S)$ is the center of $G(S)$. It can also be characterized as the unique smooth closed subscheme such that $Z(G)(\bar{k})$ is the center of $G(\bar{k})$.

Example 9.5. The center $Z(G) \subseteq G$ is always a normal subgroup. The quotient $G^{\text{ad}} := G/Z(G)$ is called the *adjoint group of G* .

Example 9.6. (1) The center of GL_n is the torus $\mathbb{G}_m$ of diagonal scalar matrices. The center of SL_n is μ_n .

(2) The *projective general linear group* PGL_n is defined as the adjoint group $\text{PGL}_n := \text{GL}_n / \mathbb{G}_m$. If $n = 2$, an explicit description can be obtained as follows. Recall that

$$\text{GL}_2 = \text{Spec}(A), \quad A = k[x_{11}, x_{12}, x_{21}, x_{22}, \delta^{-1}], \quad \delta = x_{11}x_{22} - x_{12}x_{21}.$$

If we write $\mathbb{G}_m = \text{Spec } k[t^{\pm 1}]$, then the multiplication action $a : \mathbb{G}_m \times \text{GL}_2 \rightarrow \text{GL}_2$ is given by

$$\begin{aligned} a^*(x_{ij}) &= t \otimes x_{ij} \\ a^*(\delta^{-1}) &= t^{-2} \otimes \delta. \end{aligned}$$

This means that a monomial $x_{11}^{m_{11}} x_{12}^{m_{12}} x_{21}^{m_{21}} x_{22}^{m_{22}} \delta^{-m}$ is $\mathbb{G}_m$ -invariant if and only if

$$2m = m_{11} + m_{12} + m_{21} + m_{22}.$$

We find that

$$\text{PGL}_2 = \text{Spec}(A^{\mathbb{G}_m}), \quad A^{\mathbb{G}_m} = k \left[\frac{x_{ij}x_{k\ell}}{\delta}, \quad i, j, k, \ell \in \{1, 2\} \right].$$

9.2. Reductive groups. Over an algebraically closed field of characteristic 0, every linear algebraic group is a successive extension of copies of $\mathbb{G}_a$, $\mathbb{G}_m$ and *semi-simple groups*. In this sense, the semi-simple groups are the interesting building blocks of the theory.

Groups that arise as successive extension of copies of $\mathbb{G}_m$ and semi-simple groups (no copies of $\mathbb{G}_a$) are called *reductive*. They have a particularly nice representation theory: every representation of a reductive group is semi-simple (see below).

Definition 9.7. (1) G is said to be *solvable* if there exists a sequence of subgroups

$$\{1\} = G_0 \subset G_1 \subset G_2 \subset \cdots \subset G_r = G \tag{9.3}$$

such that for all $0 \leq i < r$, G_i is normal in G_{i+1} and G_{i+1}/G_i commutative.

(2) G is said to be *unipotent* if there exists such a sequence with the additional condition that $G_{i+1}/G_i \cong \mathbb{G}_a$ for all $0 \leq i < r$.

Example 9.8. The subgroup $B \subset \text{GL}_n$ of upper triangular matrices is solvable. For example, if $n = 3$ then

$$\{1\} \subset \left\{ \begin{pmatrix} 1 & 0 & * \\ & 1 & 0 \\ & & 1 \end{pmatrix} \right\} \subset \left\{ \begin{pmatrix} 1 & * & * \\ & 1 & * \\ & & 1 \end{pmatrix} \right\} \subset B$$

defines a sequence as in (9.3). The quotients are $\mathbb{G}_m^3$, $\mathbb{G}_a^2$, and $\mathbb{G}_a$, respectively. The subgroup U of unipotent upper triangular matrices is unipotent. For $n = 3$, a composition series with successive quotients $\mathbb{G}_a$ is given by

$$\{1\} \subset \left\{ \begin{pmatrix} 1 & 0 & * \\ & 1 & 0 \\ & & 1 \end{pmatrix} \right\} \subset \left\{ \begin{pmatrix} 1 & 0 & * \\ & 1 & * \\ & & 1 \end{pmatrix} \right\} \subset \left\{ \begin{pmatrix} 1 & * & * \\ & 1 & * \\ & & 1 \end{pmatrix} \right\} = U.$$

Definition 9.9 (Reductive groups, see [16, §19]). (1) There exists a maximal connected, normal, solvable subgroup $R(G) \subseteq G$ called its *radical*. Maximality means that every connected, normal, solvable subgroup $N \subseteq G$ is contained in $R(G)$.

(2) Similarly, there exists a maximal connected, normal, unipotent subgroup $R_u(G) \subseteq G$, called its *unipotent radical*. Thus, we have defined subgroups

$$R_u(G) \subseteq R(G) \subseteq G.$$

(3) G is said to be *reductive* if $R_u(G) = \{1\}$. It is said to be *semi-simple* if $R(G) = \{1\}$.

In general, for a connected group G/k , the quotient $G/R_u(G)$ will be reductive and $G/R(G)$ will be semi-simple.

Proposition 9.10 (Centers of reductive groups). *Let G be a connected reductive group over k . Then the radical of G agrees with the identity component of its center. That is, $R(G) = Z(G)^\circ$. In particular, G is semi-simple if and only if its center is finite.*

Example 9.11. Consider the group $G = \mathrm{SL}_2$ over k . One may check that $\mathrm{SL}_2(\bar{k})$ has no proper normal subgroups except for the center $\{\pm 1\}$. So we see that $R_u(G) = R(G) = \{1\}$. Note that all conditions in the definitions of $R_u(G)$ and $R(G)$ are important:

- (1) The diagonal subgroup $\{\pm 1\}$ is normal and commutative, but not connected.
- (2) The subgroup $\left\{ \begin{pmatrix} 1 & b \\ & 1 \end{pmatrix} \right\}$ of upper triangular unipotent matrices is unipotent and connected, but not normal.

Remark 9.12. Under our standing assumption that $\mathrm{char}(k) = 0$, all notions introduced so far are compatible with scalar extension to $\bar{k}$. That is, G is solvable/unipotent/semi-simple/reductive if and only if $G_{\bar{k}}$ has that property. Moreover,

$$R(G_{\bar{k}}) = R(G)_{\bar{k}}, \quad R_u(G_{\bar{k}}) = R_u(G)_{\bar{k}}.$$

Reductive groups can be characterized in terms of their representation theory. Let V be a finite-dimensional k -vector space. The k -group $\mathrm{GL}(V)$ is defined as functor on k -algebras by

$$\mathrm{GL}(V)(R) := \mathrm{GL}_R(R \otimes_k V). \tag{9.4}$$

Any choice of basis for V identifies it $\mathrm{GL}_{n,k}$. A *representation* of a k -group G on V is a homomorphism of k -group schemes

$$\rho : G \longrightarrow \mathrm{GL}(V).$$

A representation is said to be *semi-simple* if, for every G -stable subspace $W \subset V$, there exists a G -stable subspace U with $V = W \oplus U$.

Example 9.13. Consider the inclusion map

$$U := \left\{ \begin{pmatrix} 1 & * \\ & 1 \end{pmatrix} \right\} \hookrightarrow \mathrm{GL}_2$$

viewed as representation of U on k^2 . Clearly, the line $\ell := k \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is U -stable. We claim that it has no complementary U -stable line. Indeed, consider a vector $v = \begin{pmatrix} x \\ y \end{pmatrix}$ with $y \neq 0$. Then v together with

$$\begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \cdot v = \begin{pmatrix} x+y \\ y \end{pmatrix}$$

already generate k^2 . This shows that k^2 is not a semi-simple representation of U .

Proposition 9.14 (see [16, Corollary 19.18] for (2)). *(1) Every representation of a reductive group is semi-simple.*

(2) A connected k -group G is reductive if and only if there exists an injective semi-simple representation $\rho : G \hookrightarrow \mathrm{GL}(V)$.

Example 9.15. The groups GL_n , SL_n , GSp_{2g} and Sp_{2g} are all reductive.

Let $(V, (\ , \))$ be a finite-dimensional k -vector space together with a non-degenerate symmetric bilinear form. If R is a k -algebra, then we obtain a symmetric bilinear form

$$(\ , \)_R : (R \otimes_k V) \times (R \otimes_k V) \longrightarrow R$$

defined on generators by

$$(r_1 \otimes v_1, r_2 \otimes v_2)_R := r_1 r_2 (v_1, v_2).$$

The special orthogonal group is the k -group with functor of points

$$\mathrm{SO}(V)(R) := \{g \in \mathrm{SL}_R(R \otimes_k V) \mid (gx, gy)_R = (x, y)_R \text{ for all } x, y \in R \otimes_k V\}. \quad (9.5)$$

It is a closed subgroup of $\mathrm{GL}(V)$. Just like the groups listed before, it is reductive.

Proof. All listed groups are connected and their standard representation is simple, so Proposition 9.14 applies. $\square$

9.3. Weil restriction. We next explain a construction called Weil restriction. Given a finite extension K/k and a K -group H , it allows to define k -group $\mathrm{Res}_{K/k}(H)$ which is “the same as H but viewed over k ”. It is worth noting that the construction will be very different from just composing the structure map $H \rightarrow \mathrm{Spec}(K)$ with $\mathrm{Spec}(K) \rightarrow \mathrm{Spec}(k)$. In fact, viewing H over k by this naive procedure would not allow to endow it with a k -group structure. For example, the identity element of H is a K -rational point (and not a k -rational one).

Proposition 9.16. *Let K/k be a finite field extension (or product of such) and let X be an affine, finite type K -scheme. Define a functor on k -algebras by*

$$(\mathrm{Res}_{K/k} X)(R) := X(K \otimes_k R). \quad (9.6)$$

Then $\mathrm{Res}_{K/k} X$ is representable by an affine finite type k -scheme which is called the Weil restriction of X . The dimensions are related by

$$\dim(\mathrm{Res}_{K/k} X) = [K : k] \cdot \dim(X).$$

If X is a linear algebraic group over K , then $\mathrm{Res}_{K/k} X$ is naturally a linear algebraic group over k .

Proof. Step 1: The basic construction. Consider first the case of the affine line $X = \mathrm{Spec} K[t]$. To define $\mathrm{Res}_{K/k}(X)$ as affine scheme, we are looking for a k -algebra B together with isomorphisms

$$\mathrm{Hom}_k(B, R) \xrightarrow{\sim} \mathrm{Hom}_K(K[t], K \otimes_k R) \quad (9.7)$$

for all k -algebras R , functorially in R . Let $d = [K : k]$ be the degree of K/k and let $\alpha_1, \dots, \alpha_d$ be a k -basis for K . Giving φ on the right hand side of (9.7) is the same as

giving its value $\varphi(t) = \sum_{j=1}^d \alpha_j \otimes r_j$ in $K \otimes_k R$. Thus, if we define $B = k[y_1, \dots, y_d]$, we can construct (9.7) by

$$\begin{aligned} \mathrm{Hom}_k(k[y_1, \dots, y_d], R) &\xrightarrow{\sim} \mathrm{Hom}_K(K[t], K \otimes_k R) \\ \psi &\mapsto \varphi(t) := \alpha_1 \otimes \psi(y_1) + \dots + \alpha_d \otimes \psi(y_d). \end{aligned} \quad (9.8)$$

This shows that $\mathrm{Res}_{K/k}(\mathbb{A}_K^1)$ is representable by $\mathbb{A}_k^d$. By taking products, we constructed an isomorphism

$$\mathbb{A}_k^{d \cdot n} \xrightarrow{\sim} \mathrm{Res}_{K/k}(\mathbb{A}_K^n). \quad (9.9)$$

Step 2: Closed subschemes. Assume now that $X = V(f_1, \dots, f_m)$ is a closed subscheme of $\mathbb{A}_K^n$. We claim that $\mathrm{Res}_{K/k}(X)$ is a closed subscheme of $\mathrm{Res}_{K/k}(\mathbb{A}_K^n)$, and we are going to exhibit the corresponding equations in $\mathbb{A}_k^{d \cdot n}$ under (9.9).

Let $t_1, \dots, t_n$ be the coordinates on $\mathbb{A}_K^n$ and let y_{ij} , $1 \leq i \leq n$, $1 \leq j \leq d$ denote the corresponding coordinates on $\mathbb{A}_k^{d \cdot n}$. For each $1 \leq k \leq m$, we substitute $\alpha_1 \otimes y_{i1} + \dots + \alpha_d \otimes y_{id}$ for t_i to obtain an element in $K \otimes_k k[y_{11}, \dots, y_{nd}]$:

$$f_k \left(\sum_{j=1}^d \alpha_j \otimes y_{1j}, \sum_{j=1}^d \alpha_j \otimes y_{2j}, \dots, \sum_{j=1}^d \alpha_j \otimes y_{nj} \right) = \sum_{j=1}^d \alpha_j \otimes g_{kj}(y_{11}, \dots, y_{nd}).$$

Then, if we look at the higher-dimensional variant of (9.8),

$$\begin{aligned} \mathrm{Hom}_k(k[y_{11}, \dots, y_{nd}], R) &\xrightarrow{\sim} \mathrm{Hom}_K(K[t_1, \dots, t_n], K \otimes_k R) \\ \psi &\mapsto \varphi(t_i) := \alpha_1 \otimes \psi(y_{i1}) + \dots + \alpha_d \otimes \psi(y_{id}), \end{aligned}$$

we find that $\varphi(f_k) = 0$ if and only if $\psi(g_{k1}) = \dots = \psi(g_{kd}) = 0$. This proves that

$$V(g_{11}, \dots, g_{md}) \xrightarrow{\sim} \mathrm{Res}_{K/k}(V(f_1, \dots, f_m))$$

which concludes the proof of representability for $\mathrm{Res}_{K/k}(X)$.

Wrap-up: Group structures. It is clear from the functor of points description (9.6) that Weil restriction is functorial and compatible with products. In particular, if we are given a K -group structure $m : X \times_{\mathrm{Spec}(K)} X \rightarrow X$, then we obtain a multiplication

$$\mathrm{Res}_{K/k}(X) \times_{\mathrm{Spec}(k)} \mathrm{Res}_{K/k}(X) \longrightarrow \mathrm{Res}_{K/k}(X).$$

The claim about dimensions (at least when k is of characteristic 0) follows from the next proposition. $\square$

Let K/k be a finite separable extension or product of such extensions. For every field extension L/k , we have a natural map

$$K \otimes_k L \longrightarrow \prod_{\varphi \in \mathrm{Hom}_k(K, L)} L$$

which is $a \otimes b \mapsto \varphi(a)b$ in the φ -component. We say that L is a splitting field for K if this map is an isomorphism. This is equivalent to $\mathrm{Hom}_k(K, L)$ having cardinality $[K : k]$. For example, K is a splitting field for K if and only if K is Galois.

Proposition 9.17. *Assume that K/k is a product of finite separable field extensions and L/k a splitting field for k . Then, for X/K as before,*

$$L \otimes_k \mathrm{Res}_{K/k}(X) \xrightarrow{\sim} \prod_{\varphi \in \mathrm{Hom}_k(K, L)} L \otimes_{\varphi, K} X.$$

Proof. Let R be an L -algebra. We write $R|_k$ or $R|_{\varphi,K}$ if we view R as k algebra or as K algebra via $\varphi : K \rightarrow L$. Observe that

$$\begin{aligned} K \otimes_k R &= (K \otimes_k L) \otimes_L R \\ &\xrightarrow{\sim} \prod_{\varphi: K \rightarrow L} R|_{\varphi,K} \end{aligned}$$

as K -algebras. By definition of the Weil restriction, we have

$$\begin{aligned} (L \otimes_k \operatorname{Res}_{K/k} X)(R) &= (\operatorname{Res}_{K/k} X)X(R|_k) \\ &= X(K \otimes_k R|_k) \\ &\xrightarrow{\sim} \prod_{\varphi: K \rightarrow L} X(R|_{\varphi,K}) \\ &= \prod_{\varphi: K \rightarrow L} (L \otimes_{\varphi,K} X)(R). \end{aligned}$$

□

Example 9.18 (Deligne torus). The Weil restriction $\mathbb{S} := \operatorname{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_m)$ is called the *Deligne torus*. The proof of Proposition 9.16 explains that we can explicitly describe it as follows. First, we have the presentation

$$\mathbb{G}_{m,\mathbb{C}} = \operatorname{Spec} \mathbb{C}[x, y]/(f), \quad f = xy - 1.$$

Now we introduce new variables x_1, x_2, y_1, y_2 and substitute

$$x = x_1 + x_2 \cdot i, \quad y = y_1 + y_2 \cdot i$$

into f which yields

$$\underbrace{(x_1 y_1 - x_2 y_2 - 1)}_{g_1} + \underbrace{(x_1 y_2 + x_2 y_1)}_{g_2} \cdot i$$

and hence gives

$$\mathbb{S} \cong \operatorname{Spec} A, \quad A = \mathbb{R}[x_1, x_2, y_1, y_2]/(g_1, g_2). \quad (9.10)$$

The group law is given by the multiplication rule for complex numbers of the form $x_1 + x_2 \cdot i$ and $y_1 + y_2 \cdot i$. That is,

$$m^*(x_1) = x_1 \otimes x_1 - x_2 \otimes x_2$$

$$m^*(x_2) = x_1 \otimes x_2 + x_2 \otimes x_1$$

and analogously for y_1 and y_2 . The equations g_1 and g_2 precisely encode that

$$(x_1 + x_2 \cdot i)(y_1 + y_2 \cdot i) = 1. \quad (9.11)$$

In the ring A , we check that (exercise)

$$(x_1^2 + x_2^2)(y_1^2 + y_2^2) = 1$$

as we would expect from identity (9.11) for actual complex numbers. We also check

$$\begin{aligned} x_1 &= y_1(x_1^2 + x_2^2) \\ x_2 &= -y_2(x_1^2 + x_2^2). \end{aligned}$$

In this way, we find

$$\begin{aligned} \mathbb{R}[x_1, x_2, (x_1^2 + x_2^2)^{-1}] &\xrightarrow{\sim} A \\ x_i &\longleftrightarrow x_i \\ (-1)^{i+1} x_i (x_1^2 + x_2^2)^{-1} &\longleftarrow y_i. \end{aligned}$$

In other words, for every $\mathbb{R}$ -algebra R , we have

$$\mathbb{S}(R) = \{x_1 + x_2 \cdot i \mid x_1, x_2 \in R, x_1^2 + x_2^2 \in R^\times\} \quad (9.12)$$

where i is a symbol with $i^2 = -1$. The multiplication rules are the same as in $\mathbb{C} = \mathbb{R} \oplus \mathbb{R} \cdot i$.

9.4. Unit groups of algebras. A more compact way of writing (9.12) is as

$$\mathbb{S}(R) = (\mathbb{C} \otimes_{\mathbb{R}} R)^{\times}.$$

In this interpretation, it becomes an example of an interesting more general construction. Let $\mathcal{A}$ be a (not necessarily commutative) finite-dimensional k -algebra. Then we define a functor on k -algebras by

$$\underline{\mathcal{A}}^{\times}(R) := (R \otimes_k \mathcal{A})^{\times}. \quad (9.13)$$

This is a group valued functor by multiplication in $R \otimes_k \mathcal{A}$.

Proposition 9.19. *The functor $\underline{\mathcal{A}}^{\times}$ is representable by an affine finite type k -scheme.*

Proof. Choose a k -basis $\alpha_1, \dots, \alpha_d$ for $\mathcal{A}$. The functor $\underline{\mathcal{A}}(R) = R \otimes_k \mathcal{A}$ is representable by $\mathbb{A}_k^d$ via

$$\begin{aligned} \mathbb{A}_k^d(R) &\xrightarrow{\sim} \underline{\mathcal{A}}(R) \\ (r_1, \dots, r_d) &\mapsto \sum_{i=1}^d r_i \otimes \alpha_i. \end{aligned} \quad (9.14)$$

Our aim is to describe the subfunctor $\underline{\mathcal{A}}^{\times} \subset \underline{\mathcal{A}}$. We recall the Cayley–Hamilton theorem over a general ring.

Proposition 9.20 (Cayley–Hamilton). *Let R be a ring and let $f \in \text{End}_R(R^d)$ be an endomorphism with characteristic polynomial $P(T) \in R[T]$. Then $P(f) = 0$.*

Corollary 9.21. *Let $T^d + a_{d-1}T^{d-1} + \dots + a_0$ be the characteristic polynomial of $f \in \text{End}_R(R^d)$. Then f is invertible if and only if $a_0 = \det(f) \in R^{\times}$, in which case*

$$f^{-1} = -a_0^{-1}(f^{d-1} + a_{d-1}f^{d-2} + \dots + a_1).$$

Let us come back to our proof. An element $f \in R \otimes_k \mathcal{A}$ acts by left multiplication on $R \otimes_k \mathcal{A}$ which is a free R -module of rank d . So we may consider its determinant $\det_R(f) \in R$. For example, we may use the basis $1 \otimes \alpha_1, \dots, 1 \otimes \alpha_d$ to first express f as a $(d \times d)$ -matrix and then take the determinant. This does not depend on the choice of basis. Coming back to our proof, we have the following application.

Corollary 9.22. *An element $f \in R \otimes_k \mathcal{A}$ lies in $(R \otimes_k \mathcal{A})^{\times}$ if and only if $\det_R(f) \in R^{\times}$.*

Proof. If $\det_R(f) \in R^{\times}$, then $f \in \text{GL}_R(R \otimes_k \mathcal{A})$. By Corollary 9.21, f^{-1} is a polynomial in f , hence again lies in $R \otimes_k \mathcal{A}$. $\square$

So far, we have established that $\underline{\mathcal{A}}^{\times}(R) \subseteq \underline{\mathcal{A}}(R)$ is the subset of elements whose determinant lies in $R^{\times}$. We claim that this condition is defined by a principal open subset $D(\det) \subset \underline{\mathcal{A}}$ for a morphism $\det : \underline{\mathcal{A}} \rightarrow \mathbb{A}_k^1$.

In the basis $\alpha_1, \dots, \alpha_d$, multiplication by α_i is given by a $(d \times d)$ -matrix $A_i \in M_d(k)$. Then

$$\begin{aligned} \det_R\left(\sum_{i=1}^d r_i \otimes \alpha_i\right) &= \det\left(\sum_{i=1}^d r_i \cdot A_i\right) \\ &= \delta(r_1, \dots, r_d) \end{aligned}$$

for the (degree d , homogeneous) polynomial

$$\delta(x_1, \dots, x_d) = \det(x_1 A_1 + \dots + x_d A_d) \in k[x_1, \dots, x_d].$$

This polynomial defines the morphism $\det : \underline{\mathcal{A}} \rightarrow \mathbb{A}_k^1$ in terms of the coordinates from (9.14). The proof of Proposition 9.19 is now complete. $\square$

Example 9.23. (1) Let K/k be a finite-dimensional commutative k -algebra (e.g. a field extension). For every k -algebra R , we have

$$\begin{aligned} \operatorname{Res}_{K/k}(\mathbb{G}_{m,K})(R) &= \mathbb{G}_{m,K}(K \otimes_k R) \\ &= (K \otimes_k R)^\times \\ &= \underline{K}^\times(R). \end{aligned}$$

So we see that $\underline{K}^\times = \operatorname{Res}_{K/k}(\mathbb{G}_{m,K})$.

(2) Let us come back to the Deligne torus. We choose the $\mathbb{R}$ -basis $\alpha_1 = 1, \alpha_2 = i$ of $\mathbb{C}$. In this basis, left multiplication with α_1 and α_2 is given by

$$A_1 = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} & -1 \\ 1 & \end{pmatrix}$$

and

$$\det(x_1 A_1 + x_2 A_2) = \det \begin{pmatrix} x_1 & -x_2 \\ x_2 & x_1 \end{pmatrix} = x_1^2 + x_2^2.$$

The description of $\underline{\mathcal{A}}^\times$ during the proof of Proposition 9.19 now directly gives

$$\mathbb{S} = \operatorname{Spec} \mathbb{R}[x_1, x_2, (x_1^2 + x_2^2)^{-1}].$$

(3) Let V be a finite-dimensional k -vector space and consider $\mathcal{A} = \operatorname{End}_k(V)$. Then

$$\begin{aligned} \underline{\mathcal{A}}^\times(R) &= (R \otimes_k \operatorname{End}_k(V))^\times \\ &= \operatorname{End}_R(R \otimes_k V)^\times \\ &= \operatorname{GL}_R(R \otimes_k V). \end{aligned}$$

This group scheme is called $\operatorname{GL}(V)$. It is a coordinate independent variant of $\operatorname{GL}_{\dim(V)}$.

(4) Let K/k be a field extension and V a finite-dimensional K -vector space. Then, by definitions,

$$\operatorname{Res}_{K/k} \operatorname{GL}(V) = \underline{\operatorname{End}_K(V)}^\times$$

where $\operatorname{End}_K(V)$ is viewed as k -algebra. In general, if L/k is a field extension, then we immediately see

$$\underline{L \otimes_k \mathcal{A}}^\times = L \otimes_k \underline{\mathcal{A}}^\times.$$

So, if K/k is separable and if L splits K , then we obtain

$$\begin{aligned} L \otimes_k \operatorname{Res}_{K/k} \operatorname{GL}(V) &= \underline{L \otimes_k \operatorname{End}_K(V)}^\times \\ &= \prod_{\varphi: K \rightarrow L} \underline{\operatorname{End}_L(L \otimes_{\varphi, K} V)}^\times \\ &\xrightarrow{\sim} (\operatorname{GL}_{\dim(V)})^{[K:k]} \end{aligned}$$

which is in line with Proposition 9.17.

10. SHIMURA DATA

Our aim in this section is to explain the following definition and to give several examples.

Definition 10.1 (Shimura datum, [15, Definition 5.5]). A Shimura datum is a pair (G, X) consisting of a connected reductive group $G/\mathbb{Q}$ and a $G(\mathbb{R})$ -conjugacy class X of homomorphisms $\mathbb{S} \rightarrow G_{\mathbb{R}}$ satisfying the following three axioms. For all $h \in X$:

(SV 1) The image $h(G_{m,\mathbb{R}})$ is contained in $Z(G_{\mathbb{R}})$. In particular, $h \bmod Z(G_{\mathbb{R}})$ factors over a homomorphism $\bar{h} : U(1) \rightarrow G_{\mathbb{R}}^{\text{ad}}$,

$$\begin{array}{ccc} \mathbb{S} & \xrightarrow{h} & G_{\mathbb{R}} \\ \downarrow z \mapsto z\bar{z} & & \downarrow \\ U(1) & \xrightarrow{\bar{h}} & G_{\mathbb{R}}^{\text{ad}}. \end{array}$$

We require that only the weights $-1, 0$ and 1 occur in the representation $\text{ad} \circ \bar{h}$ of $U(1)$ on $\text{Lie}(G_{\mathbb{R}})$.

(SV 2) Conjugation by $h(i)$ defines a Cartan involution θ on G^{ad} . That is, the Lie group

$$G^{\text{ad},(\theta)}(\mathbb{R}) := \{g \in G^{\text{ad}}(\mathbb{C}) \mid g = h(i) \cdot \bar{g} \cdot h(i)^{-1}\}$$

is compact.

(SV 3) There exists no product decomposition $G^{\text{ad}} = G_1 \times G_2$ with both $G_i \neq \{1\}$ such that one of the components of $\text{ad} \circ h$ is trivial.

It is worth pointing out that if one element $h \in X$ satisfies (SV 1)–(SV 3), then all such elements do. So one could have phrased the definition in terms of a single homomorphism h , and then taken X as its $G(\mathbb{R})$ -conjugacy class. As in Milne [15], the formulation here avoided endowing X with a distinguished point.

Axiom (SV 3) is a non-triviality condition. Namely, if $G(\mathbb{R})$ is compact, then any homomorphism $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ satisfying (SV 2) is central. So we would always get $X = \{\text{pt}\}$. Excluding factors of this form with (SV 3) matters in Deligne's definition of canonical models. However, we will not discuss this aspect any further.

10.1. The adjoint representation and $\text{ad} \circ h$. We begin with a discussion of Lie algebras and the adjoint representation. Any finite type group scheme G/k has a Lie algebra defined by

$$\text{Lie}(G) := \ker \left(G(k[\varepsilon]/(\varepsilon^2)) \longrightarrow G(k) \right). \quad (10.1)$$

This is an abelian group even when G is non-commutative.

Example 10.2. (1) The Lie algebra of GL_n is the additive group of $(n \times n)$ -matrices,

$$\begin{aligned} \text{Lie}(\text{GL}_n) &= (1 + \varepsilon \cdot \text{M}_n(k), \text{mult}) \\ &\xrightarrow{\sim} (\text{M}_n(k), \text{add}). \end{aligned}$$

The point is that

$$(1 + \varepsilon X)(1 + \varepsilon Y) \equiv 1 + \varepsilon(X + Y) \pmod{(\varepsilon)},$$

so multiplication translates to addition.

(2) The determinant of $1 + \varepsilon X \in \text{M}_n(k[\varepsilon]/(\varepsilon^2))$ is $1 + \varepsilon \text{tr}(X)$. So we obtain

$$\text{Lie}(\text{SL}_n) = \text{M}_n(k)^{\text{tr}=0}.$$

(3) For a general linear algebraic group G , we can always find some n and an embedding $\rho : G \rightarrow \text{GL}_n$ (Theorem 4.13) and obtain

$$\text{Lie}(G) = \text{Lie}(\text{GL}_n) \cap \rho(G(k[\varepsilon]/(\varepsilon^2))).$$

For example, consider $\text{SO}(k^n, 1_n)$ for a field k with $\text{char}(k) \neq 2$. For vectors $v, w \in (k[\varepsilon]/(\varepsilon^2))^n$ and $X \in \text{M}_n(k)$, we have

$$\begin{aligned} ((1 + \varepsilon X)v, (1 + \varepsilon X)w) &= v^t(1 + \varepsilon X)^t(1 + \varepsilon X)w \\ &= v^t w + \varepsilon \cdot v^t(X^t + X)w. \end{aligned}$$

This equals $(v, w) = v^t w$ for all v and w if and only if $X^t + X = 0$. So we see that $\text{Lie}(SO(k^n, 1_n)) \subset M_n(k)$ is the subset of skew-symmetric matrices $X^t = -X$.

For every $\lambda \in k$, there is a scaling map

$$k[\varepsilon]/(\varepsilon^2) \xrightarrow{\sim} k[\varepsilon]/(\varepsilon^2), \quad \varepsilon \mapsto \lambda\varepsilon.$$

Composition with these maps defines an action of k on $\text{Lie}(G)$ making it into a k -vector space. In the above examples, this is simply the usual k -vector space structure on $M_n(k)$.

The Lie algebra is also the same as the tangent space to G at 1. In our upcoming applications, we will be in the case $\text{char}(k) = 0$. Then G is necessarily smooth and we have $\dim_k \text{Lie}(G) = \dim(G)$. If $k = \mathbb{R}$ or $\mathbb{C}$, then this algebraic definition agrees with the definition in terms of the tangent space at 1 of the Lie group $G(\mathbb{R})$ (resp. $G(\mathbb{C})$).

Definition 10.3. In general, G acts on $\text{Lie}(G)$ by conjugation. Conjugation by central elements is trivial, so this action factors over the adjoint group. That is, we obtain a representation

$$\text{ad} : G \longrightarrow G^{\text{ad}} \hookrightarrow \text{GL}(\text{Lie}(G))$$

called the *adjoint representation* of G .

Example 10.4. The adjoint representation of GL_n on $\text{Lie}(\text{GL}_n)$ is given by the usual conjugation action of GL_n on $M_n(k)$. For a general linear algebraic group G , we may always choose an embedding $\rho : G \rightarrow \text{GL}_n$ to realize $\text{Lie}(G)$ as a subspace of $M_n(k)$. Then G acts by conjugation via ρ .

We next explain the representation $\text{ad} \circ \bar{h}$ that occurs in (SV 1). The determinant construction during the proof of Proposition 9.19 restricts to a group homomorphism $\underline{A}^\times \rightarrow \mathbb{G}_m$. Applying this to the Deligne torus $\mathbb{S} = \underline{\mathbb{C}}^\times$ defines an exact sequence

$$1 \longrightarrow U(1) \longrightarrow \mathbb{S} \xrightarrow{N_{\mathbb{C}/\mathbb{R}}} \mathbb{G}_{m,\mathbb{R}} \longrightarrow 1 \quad (10.2)$$

where

$$N_{\mathbb{C}/\mathbb{R}}(x_1 + x_2 \cdot i) = x_1^2 + x_2^2$$

is the norm map from $\mathbb{C}$ to $\mathbb{R}$, but viewed as algebraic morphism $\mathbb{A}_{\mathbb{R}}^2 \rightarrow \mathbb{A}_{\mathbb{R}}^1$. The kernel $U(1)$ is the $\mathbb{R}$ -group scheme unit circle

$$U(1) = \text{Spec } \mathbb{R}[x_1, x_2]/(x_1^2 + x_2^2 - 1).$$

Its $\mathbb{R}$ -points $U(1)(\mathbb{R})$ are literally the unit circle in $\mathbb{C}^\times$. In addition to the exact sequence (10.2), we also have the sequence

$$\begin{aligned} 1 \longrightarrow \mathbb{G}_m \longrightarrow \mathbb{S} \longrightarrow U(1) \longrightarrow 1 \\ z \longmapsto z/\bar{z}. \end{aligned} \quad (10.3)$$

Note that the composition

$$U(1) \hookrightarrow \mathbb{S} \xrightarrow{N_{\mathbb{C}/\mathbb{R}}} U(1)$$

is $z \mapsto z^2$. The two exact sequences express that $\mathbb{S}$ is almost the product of $\mathbb{G}_{m,\mathbb{R}}$ and $U(1)$, namely

$$(\mathbb{G}_{m,\mathbb{R}} \times U(1))/\Delta(\pm 1) \xrightarrow{\sim} \mathbb{S}, \quad (t, z) \mapsto t \cdot z.$$

Consider now a homomorphism $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ as in Definition 10.1. If $h(\mathbb{G}_m) \subset Z(G_{\mathbb{R}})$, then the composition $h : \mathbb{S} \rightarrow G \rightarrow G^{\text{ad}}$ factors through the quotient $U(1)$ in (10.3). If we compose with the adjoint representation, we obtain a representation

$$\text{ad} \circ \bar{h} : U(1) \longrightarrow \text{GL}(\text{Lie}(G_{\mathbb{R}})), \quad z \longmapsto \text{ad}(h(\sqrt{z})). \quad (10.4)$$

Here, the notation $\sqrt{z}$ means pick any inverse image of z under $U(1) \rightarrow U(1)$, $z \mapsto z^2$. The second part of (SV 1) is a condition on (10.4) which we explain next.

10.2. Representation theory of $U(1)$. We need a classification of the representations of $U(1)$. Let us first note that there is no difference between the representation theory of $U(1)$ as algebraic group and that of $\mathcal{U} := U(1)(\mathbb{R})$ as real Lie group. Every finite-dimensional algebraic representation $\rho : U(1) \rightarrow \mathrm{GL}(V)$ gives rise (pass to $\mathbb{R}$ -points) to a representations as real Lie group

$$\rho(\mathbb{R}) : \mathcal{U} \longrightarrow \mathrm{GL}(V)(\mathbb{R}).$$

This functor defines an equivalence of categories

$$\mathrm{Rep}^{\mathrm{alg}}(U(1)) \xrightarrow{\sim} \mathrm{Rep}^{\mathrm{Lie group}}(\mathcal{U}).$$

We will work with $\mathcal{U}$ to prove our classification theorem.

Let k be a non-zero integer. Define a two-dimensional real representation (W_k, ρ_k) of $\mathcal{U}$ as $W_k = \mathbb{C}$ and

$$\rho_k : \mathcal{U} \rightarrow \mathrm{GL}_{\mathbb{R}}(W_k), \quad \rho_k(z)(v) = z^k v.$$

It is clear that W_k has no $\mathcal{U}$ -stable lines and is hence irreducible. It is also clear that the complex conjugation defines an isomorphism

$$W_k \xrightarrow{\sim} W_{-k}, \quad x_1 + x_2 \cdot i \longmapsto x_1 - x_2 \cdot i.$$

Moreover, $W_k \not\cong W_{k'}$ if $k \neq \pm k'$ because the trace of $\rho_k(z)$ is $\mathrm{Re}(z^k)$ and

$$\mathrm{Re}(z^k) = \mathrm{Re}(z^{k'}) \text{ for all } z \in \mathcal{U} \iff k = \pm k'.$$

Finally, we also define $W_0 = \mathbb{R}$ (trivial representation).

Proposition 10.5. (1) *Every finite-dimensional representation of $U(1)$ is a direct sum of irreducible representations.*

(2) *Every irreducible such representation is isomorphic to precisely one of the W_k , $k \geq 0$.*

Proof. Decomposition into irreducibles. The following argument works more generally for compact Lie groups (Peter–Weyl Theorem). Let V be a finite-dimensional continuous representation of $\mathcal{U}$. If V is not already irreducible, then there exists a proper $\mathcal{U}$ -stable subspace $0 \neq W \subset V$. Choose a projection $f_0 : V \twoheadrightarrow W$, meaning a surjection such that $f_0|_W = \mathrm{id}_W$. Define $f : V \rightarrow W$ by

$$f(v) = \int_{\mathcal{U}} z^{-1} \cdot f_0(z \cdot v) \, dz$$

where dz is the translation invariant measure on $\mathcal{U}$ of total volume 1. We still have $f|_W = \mathrm{id}$, because

$$\begin{aligned} f(w) &= \int_{\mathcal{U}} z^{-1} \cdot f_0(z \cdot w) \, dz \\ &= \int_{\mathcal{U}} z^{-1} \cdot z \cdot w \, dz \\ &= w. \end{aligned}$$

This shows that f is still a projection to W and hence that $V = W \oplus \ker(f)$. The translation invariance of the measure moreover implies that

$$\begin{aligned} f(gv) &= \int_{\mathcal{U}} z^{-1} f_0(z \cdot gv) \, dz \\ &= \int_{\mathcal{U}} (zg^{-1})^{-1} \cdot f_0((zg^{-1}) \cdot gv) \, dz \\ &= \int_{\mathcal{U}} g \cdot z^{-1} f_0(zv) \, dz \\ &= gf(v), \end{aligned}$$

which shows that f is $\mathcal{U}$ -equivariant. So $\ker(f)$ is a $\mathcal{U}$ -stable subspace and $V = W \oplus \ker(f)$ is an $\mathcal{U}$ -stable direct sum decomposition. By induction on $\dim(V)$, we have proved that V is a direct sum of irreducible representations.

Classification of irreducibles over $\mathbb{C}$. We first consider finite-dimensional complex representations of $\mathcal{U}$. The first part of the proof still applies and shows that every such representation is a direct sum of irreducible ones. Let $\rho : \mathcal{U} \rightarrow \mathrm{GL}_{\mathbb{C}}(V)$ be irreducible. The operators $\{\rho(z)\}_{z \in \mathcal{U}}$ all pairwise commute because $\mathcal{U}$ is commutative. So there exists a joint eigenvector $0 \neq v \in V$,

$$\rho(z)(v) = \chi(z)v,$$

with eigenvalues $\chi(z) \in \mathbb{C}$. In particular, $\mathbb{C} \cdot v \subset V$ is $\mathcal{U}$ -stable. Since V is irreducible by assumption, $V = \mathbb{C} \cdot v$. In other words, V is defined by a character $\chi : \mathcal{U} \rightarrow \mathbb{C}^\times$.

Recall that we are always considering continuous representations. So $\chi(\mathcal{U}) \subset \mathbb{C}^\times$ is a compact subgroup. (Images of compact sets under continuous maps are compact.) Composing with the absolute value map $|\cdot| : \mathbb{C}^\times \rightarrow \mathbb{R}_{>0}$, we similarly see that $|\chi(\mathcal{U})| \subset \mathbb{R}_{>0}$ is compact. But the only bounded subgroup of $\mathbb{R}_{>0}$ is $\{1\}$, meaning that $\chi(\mathcal{U}) \subseteq \mathcal{U}$. The characters $\mathcal{U} \rightarrow \mathcal{U}$ are precisely the group homomorphisms $\chi_k(z) = z^k$, $k \in \mathbb{Z}$. Taking all these arguments together, we have shown that every complex finite-dimensional representation (V, ρ) has a direct sum decomposition

$$V = \bigoplus_{k \in \mathbb{Z}} V_k \tag{10.5}$$

where V_k is the subspace of v with $\rho(z)(v) = \chi_k(z)v$, and each V_k is isomorphic to $\chi_k^{\oplus \dim(V_k)}$.

Classification of irreducibles over $\mathbb{R}$. Let V be an irreducible $\mathcal{U}$ -representation over $\mathbb{R}$ and let $V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$ viewed as representation over $\mathbb{C}$. There is a complex conjugation on $V_{\mathbb{C}}$ by $\sigma(x \otimes v) = \bar{x} \otimes v$, and $V = (V_{\mathbb{C}})^{\sigma=\mathrm{id}}$. The ($\mathcal{U}$ -stable) real subspaces $W \subseteq V$ are in bijection with the ($\mathcal{U}$ -stable and) σ -stable complex subspaces $W' \subseteq V_{\mathbb{C}}$, with bijection given by

$$W \longmapsto \mathbb{C} \otimes_{\mathbb{R}} W, \quad (W')^{\sigma=\mathrm{id}} \longleftarrow W'.$$

Consider now the decomposition

$$V_{\mathbb{C}} = \bigoplus_{k \in \mathbb{Z}} V_k \tag{10.6}$$

obtained before for complex representations. Because of the simple relation

$$\sigma(z^k v) = \bar{z}^k \sigma(v),$$

we have $\sigma(V_k) = V_{-k}$. Thus $(V_k \oplus V_{-k})^{\sigma=\mathrm{id}} \subseteq V$ is a $\mathcal{U}$ -stable subspace. Since V was assumed irreducible, there is a unique $k \geq 0$ with $V = V_k \oplus V_{-k}$.

First case: $V_{\mathbb{C}} = V_0$. Then $\mathcal{U}$ acts trivial on $V_{\mathbb{C}}$. Hence V was a trivial 1-dimensional representation by irreducibility, meaning $V \cong W_0$.

Second case: $V_{\mathbb{C}} = V_k \oplus V_{-k}$ with $k > 0$. Pick any $0 \neq v \in V_k$. Then $0 \neq \sigma(v) \in V_{-k}$ and $\mathbb{C}v + \mathbb{C}\sigma(v)$ is $\mathcal{U}$ -stable and σ -stable. By irreducibility of V ,

$$V = (\mathbb{C}v + \mathbb{C}\sigma(v))^{\sigma=\mathrm{id}}$$

which is isomorphic to W_k . □

Definition 10.6. For a complex representation V as in (10.5), the integers k with $V_k \neq 0$ are called the *weights* of V , and (10.5) is called the *weight decomposition*. For a real representation V we define the weights of V to be those of $V_{\mathbb{C}}$.

We have now discussed all notions that go into (SV 1): We consider homomorphisms $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ such that $\text{ad} \circ h$ factors over the quotient $\mathbb{S} \rightarrow U(1)$. We require that the weights of the representation of $U(1)$ on $\text{Lie}(G_{\mathbb{R}})$ are all contained in $\{-1, 0, 1\}$.

10.3. Cartan involutions. We call a real linear algebraic group G *compact* if $G(\mathbb{R})$ is a compact Lie group. For example, the special orthogonal group $\text{SO}(V, Q)$ of a vector space V with positive or negative definite quadratic form Q is compact. An example is $U(1) = \text{SO}(2)$.

Definition 10.7. (1) In general, an *involution* of an object is an automorphism τ such that $\tau^2 = \text{id}$.

(2) Let G be a connected linear algebra group over $\mathbb{R}$. A *Cartan involution* is an involution $\theta : G \rightarrow G$ (automorphism as algebraic group over $\mathbb{R}$) such that

$$G^{(\theta)}(\mathbb{R}) := \{g \in G(\mathbb{C}) \mid \theta(\bar{g}) = g\}$$

is a compact.

Theorem 10.8 ([15, Theorem 1.16]). *Let G be a connected algebraic group over $\mathbb{R}$. Cartan involutions exist if and only if G is reductive. Any two Cartan involutions are conjugate by an element of $G(\mathbb{R})$.*

Example 10.9. (1) On $\mathbb{G}_m$, consider the involution $\theta(t) = t^{-1}$. Then

$$\mathbb{G}_m^{(\theta)}(\mathbb{R}) = \{z \in \mathbb{C}^\times \mid \bar{z}^{-1} = z\} = \mathcal{U}$$

which is compact.

(2) More generally, on GL_n , consider the involution $\theta(g) = g^{t,-1}$. Note that $(gh)^t = h^t g^t$ and $(gh)^{-1} = h^{-1} g^{-1}$, so the composition of both is really a group automorphism. We find that

$$\text{GL}_n^{(\theta)}(\mathbb{R}) = \{g \in \text{GL}_n(\mathbb{C}) \mid \bar{g}^t = g^{-1}\} = U(n)(\mathbb{R})$$

is the unitary group of the standard positive definite hermitian space $(\mathbb{C}^n, 1_n)$ which is compact.

We can now clarify (SV 2). Given $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$, define $\theta : G_{\mathbb{R}} \rightarrow G_{\mathbb{R}}$ by

$$\theta(g) := h(i)gh(i)^{-1}.$$

The first part of axiom (SV 1) implies that $h(-1)$ is central. So θ is an involution because

$$\theta(\theta(g)) = h(-1)gh(-1) \stackrel{h(-1) \text{ central}}{=} h(-1)h(-1)g = g.$$

10.4. Example: GL_2 . We continue from Example 1.1. That is, we embed $\mathbb{C}$ into $\text{M}_2(\mathbb{R})$ by

$$h(a + bi) := \begin{pmatrix} a & -b \\ b & a \end{pmatrix}. \quad (10.7)$$

Applying the formalism of unit group schemes from §9.4, we obtain a Deligne homomorphism

$$h : \mathbb{S} \longrightarrow \text{GL}_{2,\mathbb{R}}.$$

It is clear that (SV 3) holds; our task is to verify (SV 1) and (SV 2).

Verification of (SV 1). The image $h(\mathbb{G}_{m,\mathbb{R}})$ is the subgroup of scalar matrices in $\text{GL}_{2,\mathbb{R}}$. These agree with $Z(\text{GL}_{2,\mathbb{R}})$ and there is nothing further to check for the first part. For the second part, let $\sigma \in \text{M}_2(\mathbb{R})$ be an element that satisfies

$$\sigma \cdot h(a + bi) = h(a - bi) \cdot \sigma$$

for all $a + bi \in \mathbb{C}$. For example, we could choose

$$\sigma = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix},$$

but every multiple $h(z) \cdot \sigma$, $z \neq 0$ will work as well because $h(z)$ commutes with $h(\mathbb{C})$. Then we have the vector space decomposition

$$M_2(\mathbb{R}) = h(\mathbb{C}) \oplus h(\mathbb{C}) \cdot \sigma.$$

Conjugation by an element $h(z)$ is given by

$$\begin{aligned} h(z) \cdot (h(w_1) + h(w_2)\sigma) \cdot h(z)^{-1} &= h(z)h(w_1)h(z)^{-1} + h(z)h(w_2)\sigma h(z)^{-1} \\ &= h(w_1) + h(z)h(w_2)h(\bar{z})^{-1}\sigma \\ &= h(w_1) + h(z/\bar{z})h(w_2)\sigma. \end{aligned}$$

In particular, we see that $\mathcal{U} \subset \mathbb{S}(\mathbb{R})$ acts as

$$h(z) \cdot (h(w_1) + h(w_2)\sigma) \cdot h(z)^{-1} = h(w_1) + h(z^2 w_2)\sigma, \quad z \in \mathcal{U}.$$

Thus, in terms of the classification from Proposition 10.5, we have that

$$\mathrm{Lie}(\mathrm{GL}_{2,\mathbb{R}}) \xrightarrow{\sim} W_0^{\oplus 2} \oplus W_2$$

as representation $\mathrm{ad} \circ (h|_{\mathcal{U}})$. We see that $\{\pm 1\} \subset \mathcal{U}$ acts trivially, which we already knew from $h(\mathbb{G}_{m,\mathbb{R}}) \subset Z(G_{\mathbb{R}})$. If we descend $\mathrm{ad} \circ (h|_{\mathcal{U}})$ along the map $\mathcal{U} \rightarrow \mathcal{U}$, $z \mapsto z^2$, then we find that

$$\mathrm{ad} \circ \bar{h} \cong W_0^{\oplus 2} \oplus W_1.$$

That is, all occurring weights are $-1, 0$ and 1 , as we needed to show.

Verification of (SV 2): Conjugation by $h(i)$ defines the involution

$$\theta \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}. \quad (10.8)$$

The matrices $g \in \mathrm{GL}_2(\mathbb{C})$ with $\theta(g) = \bar{g}$ are precisely the ones of the form

$$\mathrm{GL}_2^{(\theta)}(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \mid a, b \in \mathbb{C} \right\}.$$

This group is not compact; for example, $\{\mathrm{diag}(a, \bar{a})\}$ is a closed subspace isomorphic to $\mathbb{C}^\times$ which is not compact. In particular, θ is not a Cartan involution on $\mathrm{GL}_{2,\mathbb{R}}$. However, axiom (SV 2) only required θ to be a Cartan involution for $\mathrm{PGL}_{2,\mathbb{R}}$. Equivalently, because the map $\mathrm{SL}_{2,\mathbb{R}} \rightarrow \mathrm{PGL}_{2,\mathbb{R}}$ is finite flat of degree 2 (its kernel is μ_2), we may check that θ is a Cartan involution of $\mathrm{SL}_{2,\mathbb{R}}$. We see

$$\mathrm{SL}_2^{(\theta)}(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \in \mathrm{GL}_2^{(\theta)}(\mathbb{R}) \mid |a|^2 + |b|^2 = 1 \right\}.$$

The condition $|a|^2 + |b|^2 = 1$ defines a compact subset of $\mathbb{C}^2$, so $\mathrm{SL}_2^{(\theta)}(\mathbb{R})$ is compact as was to be shown.

In conclusion, we have shown that (GL_2, X) with $X = \mathrm{GL}_2(\mathbb{R}) \cdot h$ is a Shimura datum.

10.5. The complex structure on X . Given $h \in X$, we have a stabilizer

$$K_h := \{g \in G(\mathbb{R}) \mid gh(z)g^{-1} = h(z) \text{ for all } z \in \mathbb{C}\}.$$

This is a closed subgroup of $G(\mathbb{R})$ because of each the conditions $gh(z)g^{-1} = h(z)$ is closed. In particular, K_h is a Lie group (closed subgroup theorem). Recall that the action of $G(\mathbb{R})$ on X is transitive by definition, so we may make X into a smooth manifold by identifying it with the quotient

$$G(\mathbb{R})/K_h \xrightarrow{\sim} X, \quad gK_h \mapsto g \cdot h.$$

By the following theorem, X has only finitely many connected components.

Theorem 10.10 ([15, Corollary 5.3]). *Let V be a variety over $\mathbb{R}$. Then $V(\mathbb{R})$ has only finitely many connected components.*

Recall that a smooth manifold M has a tangent bundle TM . This is the vector bundle $\pi : TM \rightarrow M$ whose fiber over $x \in M$ is the tangent space $T_x M$ of M at x . Equivalently, one may take a Grothendieck perspective and consider the sheaf of smooth sections

$$\mathcal{T}_M : U \mapsto TM(U).$$

That is, for every open $U \subset M$, $TM(U)$ denotes the set of smooth maps $s : U \rightarrow TM$ such that $s \circ \pi = \text{id}_U$. Let $\mathcal{C}^\infty(-, \mathbb{R})$ denote the sheaf of smooth real valued maps on M . Then $\mathcal{T}_M$ is a locally free $\mathcal{C}^\infty(-, \mathbb{R})$ -module of rank equal to $\dim(M)$. It does not matter which of the two perspectives one takes in the following discussion.

Definition 10.11. An *almost complex structure* on a smooth manifold M is a vector bundle endomorphism $J \in \text{End}_{\mathcal{C}^\infty(-, \mathbb{R})}(\mathcal{T}_M)$ such that $J^2 = -1$.

On any smooth manifold, we also have the sheaf $\mathcal{C}^\infty(-, \mathbb{C})$ of smooth maps to the complex plane. Assume that M is even a complex manifold. We define $\mathcal{T}_M$ as before by viewing M as a smooth manifold. But the complex structure makes $\mathcal{T}_M$ into a locally free $\mathcal{C}^\infty(-, \mathbb{C})$ -module of rank equal to $\dim_{\mathbb{C}}(M)$. Multiplication by the constant section $i \in \mathcal{C}^\infty(M, \mathbb{C})$ defines an almost complex structure $J \in \text{End}_{\mathcal{C}^\infty(-, \mathbb{R})}(\mathcal{T}_M)$. In this way, one obtains a fully faithful functor

$$\{\text{Complex manifolds}\} \hookrightarrow \left\{ \begin{array}{l} \text{Smooth manifolds with} \\ \text{almost complex structure} \end{array} \right\}$$

where morphisms in the target are smooth maps that are compatible with almost complex structures. An almost complex structure is said to be *integrable* if it comes from a complex structure. Fully faithfulness means that this structure is unique if it exists.

Our next goal is to endow X with an almost complex structure. Let $Z = Z(G(\mathbb{R}))$ be the center of $G(\mathbb{R})$. It is contained in K_h , and we denote by $\mathfrak{z} \subseteq \mathfrak{k}_h \subseteq \mathfrak{g}$ the Lie algebras of $Z \subseteq K_h \subseteq G(\mathbb{R})$. Using axiom (SV 1), we have a weight decomposition

$$\mathfrak{g} = \mathfrak{g}_h^0 \oplus \mathfrak{g}_h^{\pm 1} \tag{10.9}$$

with respect to $\text{ad} \circ \bar{h}$ where at most the weights 0 and ± 1 occur. For every $g \in G(\mathbb{R})$, we have

$$\mathfrak{g}_{g \cdot h}^0 = \text{ad}(g)(\mathfrak{g}_h^0), \quad \mathfrak{g}_{g \cdot h}^{\pm 1} = \text{ad}(g)(\mathfrak{g}_h^{\pm 1}).$$

Lemma 10.12. *There is the equality $\mathfrak{g}_h^0 = \mathfrak{k}_h$.*

Proof. The inclusion $\mathfrak{k}_h \subseteq \mathfrak{g}_h^0$ is clear because $h(z)$ centralizes K_h for every $z \in \mathbb{C}^\times$. For the converse, we recall that for every $\gamma \in \mathfrak{g}$, there is a unique group homomorphism

$$\exp_\gamma : \mathbb{R} \rightarrow G(\mathbb{R}), \quad t \mapsto \exp_\gamma(t)$$

such that

$$\left. \frac{d}{dt} \right|_{t=0} \exp_\gamma(t) = \gamma.$$

It satisfies $\exp_{\text{ad}(g)(\gamma)}(t) = g \exp_\gamma(t) g^{-1}$ by uniqueness. By axiom (SV 1), we have $h(\mathbb{C}^\times) \subseteq Z \cdot h(\mathcal{U})$. So if $\gamma \in \mathfrak{g}_h^0$, then $\text{ad}(h(z))(\gamma) = \gamma$ for every $z \in \mathbb{C}^\times$ and hence

$$\begin{aligned} h(z) \cdot \exp_\gamma(t) h(z)^{-1} &= \exp_{\text{ad}(h(z))(\gamma)}(t) \\ &= \exp_\gamma(t). \end{aligned}$$

This means that $\exp_\gamma$ has image in K_h , and thus $\gamma \in \mathfrak{k}$. $\square$

For every $h \in X$, this constructs an identification

$$\mathfrak{g}_h^{\pm 1} \xrightarrow{\sim} \mathfrak{g}/\mathfrak{k}_h \xrightarrow{\sim} T_h X.$$

By definition, the $\mathcal{U}$ -representation $\mathfrak{g}_h^{\pm 1}$ is isomorphic to a direct sum of copies of $W_1 = (\mathbb{C}, \rho_1(z)(v) = zv)$. That is, $J := \text{ad} \circ \bar{h}(i)$ is an endomorphism of $\mathfrak{g}_h^{\pm 1}$ with $J^2 = -1$. This construction, which we have only described for an individual $h \in X$, is compatible as h varies and endows TX with an almost complex structure. This structure is integrable, i.e. comes from a (unique) complex manifold structure on X .

10.6. The hermitian structure on X . So far, we have only used axiom (SV 1). Axiom (SV 2) is used to endow X with a hermitian structure. We note upfront that we have a subgroup

$$G(\mathbb{R})/Z \hookrightarrow G^{\text{ad}}(\mathbb{R}) \quad (10.10)$$

which is a union of connected components. Indeed, the quotient map $G \rightarrow G^{\text{ad}}$ is smooth and surjective, so (10.10) is submersive near the identity (surjective on tangent spaces), hence a diffeomorphism near the identity because both have the same dimension. So the image of (10.10) contains the identity connected component $G^{\text{ad}}(\mathbb{R})^\circ$. But in general, the map $G(\mathbb{R}) \rightarrow G^{\text{ad}}(\mathbb{R})$ is not surjective:

Example 10.13. The adjoint group of SL_n is PGL_n , but the natural map

$$\text{SL}_n(\mathbb{R}) \longrightarrow \text{PGL}_n(\mathbb{R}) = \text{GL}_n(\mathbb{R})/\mathbb{R}^\times$$

is surjective if and only if n is odd. If n is even, then the target has two connected components distinguished by the sign of the determinant. The image consists of the elements $\mathbb{R}^\times \cdot g$ with positive determinant.

Let $\theta : G_{\mathbb{R}} \rightarrow G_{\mathbb{R}}$ denote conjugation by $h(i)$ (an involution). The centralizer K_h equals

$$G(\mathbb{R}) \cap G^{(\theta)}(\mathbb{R}) = \{g \in G(\mathbb{R}) \mid \theta(g) = g\}.$$

The image K_h/Z in $G^{\text{ad}}(\mathbb{R})$ is then a closed subgroup of

$$G^{\text{ad},(\theta)}(\mathbb{R}) = \{g \in G^{\text{ad}}(\mathbb{C}) \mid \theta(g) = \bar{g}\}$$

which, by axiom (SV 2), is compact. So K_h/Z is a compact subgroup of $G(\mathbb{R})/Z$.

Definition 10.14. A *hermitian pairing* on a complex vector bundle $E \rightarrow M$ is a smooth map

$$(\ , \) : E \times_M E \longrightarrow \mathbb{C}$$

that is conjugate $\mathbb{C}$ -linear in the first coordinate, $\mathbb{C}$ -linear in the second, satisfies $(s_1, s_2) = \overline{(s_2, s_1)}$ for all $s_1, s_2 \in E(M)$, and is positive definite in the sense that $(s, s)(x) > 0$ for all $s \in E(M)$ and $x \in M$ with $s(x) \neq 0$. A *hermitian structure* on a complex manifold is a hermitian pairing on its tangent bundle.

Using (SV 2), we can define a hermitian structure on X as follows. Fixing $h \in X$, choose a positive definite hermitian form

$$(\ , \)_{h,0} : T_h X \times_X T_h X \longrightarrow \mathbb{C}.$$

Let dk denote a translation invariant measure on K_h/Z ; such a measure exists and is unique up to scaling by $\mathbb{R}_{>0}$. Using that K_h/Z is compact, we may *average* the hermitian from $(\cdot, \cdot)_{h,0}$ to define

$$(v, w)_h := \int_{K_h/Z} (kv, kw)_{h,0} dk. \quad (10.11)$$

This form is still hermitian and positive definite. It is additionally K_h -invariant because the measure dk is translation invariant. For every $h' \in X$, there exists an element $g \in G(\mathbb{R})$ with $gh = h'$. Translation by g induces an isomorphism

$$dg : T_h X \xrightarrow{\sim} T_{gh} X.$$

We define a hermitian form $(\cdot, \cdot)_{gh}$ on $T_{gh} X$ by

$$(v, w)_{gh} := ((dg)^{-1}v, (dg)^{-1}w).$$

This is well-defined, because g is unique up to $k \in K_h$ and $(\cdot, \cdot)_h$ is K_h -invariant. One may check (omitted) that this fiberwise construction defines a hermitian pairing on TX . Moreover, this hermitian structure is $G(\mathbb{R})$ -invariant: For every pair of vector fields $s_1, s_2 \in TX(X)$ and every $g \in G(\mathbb{R})$,

$$((dg)(s_1), (dg)(s_2)) = g_*(s_1, s_2).$$

By general classification theorems, one can show that every connected component $X^\circ \subseteq X$ is a hermitian symmetric domain. That is, X° is a complex manifold with hermitian structure such that

- For any two points $p, q \in X^\circ$, there exists a holomorphic isometry $f \in \text{Aut}(X^\circ)$ such that $f(p) = q$.
- For every point p , there exists a holomorphic isometry $f \in \text{Aut}(X^\circ)$ with $f(p) = p$ and such that $df : T_p X^\circ \rightarrow T_p X^\circ$ is multiplication by -1 .

In this context, we also have the following property.

- For every $h \in X$, the subgroup $K_h^\circ := K_h/Z \cap G^{\text{ad}}(\mathbb{R})^\circ$ is a maximal compact subgroup of $G^{\text{ad}}(\mathbb{R})^\circ$. That is, if K' is a compact subgroup with $K_h^\circ \subseteq K'$, then $K_h^\circ = K'$.

We will further develop these topics as needed. For now, we refer the curious reader to [15, §1 – §5].

Example 10.15. Let $h : \mathbb{S} \rightarrow \text{GL}_{2,\mathbb{R}}$ be as in (10.7). Then K_h° is the image of the circle $\text{SO}(2) \subset \text{SL}_2(\mathbb{R})$ in $\text{PGL}_2(\mathbb{R})^\circ$. We claim that this is a maximal compact subgroup and that every other maximal compact subgroup is conjugate to it. Since the projection map $\text{SL}_2(\mathbb{R}) \rightarrow \text{PGL}_2(\mathbb{R})^\circ$ is 2-to-1, it suffices to show that $\text{SO}(2) \subset \text{SL}_2(\mathbb{R})$ is maximal compact and has the uniqueness up to conjugation property.

Step 1: Every compact subgroup $K \subset \text{SL}_2(\mathbb{R})$ is contained in $\text{SO}(\lambda)$ for some positive definite quadratic form λ on $\mathbb{R}^2$. First note that K is a Lie group by the closed subgroup theorem. Start with any positive definite quadratic form λ_0 on $\mathbb{R}^2$ and average it as in (10.11),

$$\lambda(v, w) = \int_K \lambda_0(v, w) dk.$$

Then λ is K -invariant, meaning $K \subseteq \text{SO}(\lambda)$.

Step 2: All $\text{SO}(\lambda)$ are conjugate. Since all positive definite quadratic forms on $\mathbb{R}^2$ are isometric, there exists $g \in \text{GL}_2(\mathbb{R})$ with $g\text{SO}(\lambda)g^{-1} = \text{SO}(2)$. If $\det(g) > 0$, then we may scale g by $\sqrt{\det(g)}^{-1}$ to assume $g \in \text{SL}_2(\mathbb{R})$. If $\det(g) < 0$, then we can scale after modifying g with an element from $\text{O}(2) \setminus \text{SO}(2)$. In this way, we see that $\text{SO}(\lambda)$ and $\text{SO}(2)$ are conjugate in $\text{SL}_2(\mathbb{R})$.

Step 3: $\mathrm{SO}(2)$ is maximal among compact subgroups of $\mathrm{SL}_2(\mathbb{R})$. Let $\mathrm{SO}(2) \subseteq K$ be a compact subgroup containing $\mathrm{SO}(2)$. By Steps 1 and 2, $K \subseteq g\mathrm{SO}(2)g^{-1}$ for some $g \in \mathrm{SL}_2(\mathbb{R})$. We necessarily have $\mathrm{SO}(2) = g\mathrm{SO}(2)g^{-1}$ because both are connected Lie subgroups and hence uniquely determined by their Lie algebras as subspaces of $\mathrm{Lie}(\mathrm{SL}_2(\mathbb{R}))$. This shows $\mathrm{SO}(2) = K$.

10.7. General examples and remarks. We have now discussed Shimura data in some detail. Let us collect a few easy remarks and examples.

Definition 10.16 (Tori). An algebraic *torus* over a field k is an algebraic group T/k such that $\bar{k} \otimes_k T \cong \mathbb{G}_{m,\bar{k}}^d$ for some $d \geq 0$. In particular, tori are affine, connected, and of finite type.

Tori in characteristic 0 are the same as connected *commutative* reductive groups.

Example 10.17. If K/k is a finite separable field extension, then $T = \mathrm{Res}_{K/k}(\mathbb{G}_{m,K})$, which agrees with $K^\times$, is a torus over k by Proposition 9.17. In general, Weil restrictions of tori over K are tori over k .

To give a concrete example, let $\mathbb{Q}(i)/\mathbb{Q}$, set $T = \mathbb{Q}(i)^\times$ and consider the exact sequence

$$1 \longrightarrow T^1 \longrightarrow T \xrightarrow{N_{\mathbb{Q}(i)/\mathbb{Q}}} \mathbb{G}_{m,\mathbb{Q}} \longrightarrow 1. \quad (10.12)$$

Both T^1 is the group scheme

$$T^1 = \mathrm{Spec} \mathbb{Q}[x, y]/(x^2 + y^2 - 1);$$

it provides one of the many possibilities of defining $U(1)/\mathbb{R}$ over $\mathbb{Q}$. Then T^1 and T are tori over $\mathbb{Q}$. After base change to $\mathbb{Q}(i)$, (10.12) becomes isomorphic to

$$1 \longrightarrow \mathbb{G}_m \xrightarrow{t \mapsto (t, t^{-1})} \mathbb{G}_m \times_k \mathbb{G}_m \xrightarrow{(x, y) \mapsto xy} \mathbb{G}_m \longrightarrow 1.$$

Example 10.18 (Shimura data for tori). Let $T/\mathbb{Q}$ be a torus and let $h : \mathbb{S} \rightarrow T_{\mathbb{R}}$ be any morphism. Then $(T, \{h\})$ is a Shimura datum. Indeed (SV 1)–(SV 3) essentially only concern the adjoint group $T^{\mathrm{ad}} = \{1\}$ and are trivially satisfied. Note that (SV 3) explicitly requires factors to be $\neq \{1\}$.

Example 10.19 (Twisting a Shimura datum by central homomorphisms). Let (G, X) be any Shimura datum, and let $h_0 : \mathbb{S} \rightarrow Z(G)_{\mathbb{R}}$ be any homomorphism to the center of $G_{\mathbb{R}}$. For any $h \in X$, the product $(h_0 h)(z) := h_0(z)h(z)$ defines a new Deligne homomorphism. Since the axioms (SV 1)–(SV 3) essentially only concern the adjoint group, the pair $(G, h_0 \cdot X)$ defines a new Shimura datum.

In particular, this example shows that a Shimura datum for G is more information than just a X viewed as complex manifold with $G(\mathbb{R})$ -action. For example, let (GL_2, X) be our usual Shimura datum for GL_2 (§10.4). Let h_0 be the homomorphism

$$h_0 : \mathbb{S} \xrightarrow{N_{\mathbb{C}/\mathbb{R}}} \mathbb{G}_{m,\mathbb{R}} = Z(\mathrm{GL}_{2,\mathbb{R}}).$$

Then $(\mathrm{GL}_2, h_0 \cdot X)$ is a new Shimura datum, but the resulting complex manifold datum is still $\mathrm{GL}_2(\mathbb{R})$ acting on $\mathbb{H}^\pm$. In this sense, calling $(\mathrm{GL}_2, \mathbb{H}^\pm)$ a Shimura datum in the first part of the lecture was an abuse of notation.

Example 10.20 (Passing to the adjoint group). If (G, X) is a Shimura datum and if $T \subseteq Z(G)$ is a subgroup of the center, then $(G/T, (G/T)(\mathbb{R}) \cdot \bar{h})$ for any $h \in X$ is a new Shimura datum. Here $\bar{h} = [G_{\mathbb{R}} \rightarrow (G/T)_{\mathbb{R}}] \circ h$ is the composition of h with the quotient map. For example, we always have the adjoint Shimura datum $(G^{\mathrm{ad}}, G^{\mathrm{ad}}(\mathbb{R}) \cdot \bar{h})$. The reason this construction works is that the Shimura variety axioms essentially only depend on the composition $\mathbb{S} \rightarrow G_{\mathbb{R}} \rightarrow G_{\mathbb{R}}^{\mathrm{ad}}$.

Example 10.21 (SL_2 does not admit a Shimura datum). By the previous example, we obtain a Shimura datum (PGL_2, X) for PGL_2 from our datum for GL_2 . But there is no Shimura datum for SL_2 , even though the two groups are closely related by $\mathrm{PGL}_2 = \mathrm{SL}_2 / \{\pm 1\}$.

Sketch: Indeed, we have $Z(\mathrm{SL}_2) = \{\pm 1\}$. So any homomorphism $h : \mathbb{S} \rightarrow \mathrm{SL}_{2,\mathbb{R}}$ with $h(\mathbb{G}_{m,\mathbb{R}}) \subseteq Z(\mathrm{SL}_{2,\mathbb{R}})$ has to be trivial on $\mathbb{G}_m$ and hence factor through the quotient $q : \mathbb{S} \rightarrow U(1)$. But for any $h_0 : U(1) \rightarrow \mathrm{SL}_{2,\mathbb{R}}$, if we look at the composition $\mathrm{ad} \circ h_0 \circ q|_{U(1)}$, we will find only weights $\in 4\mathbb{Z}$ in $\mathrm{Lie}(\mathrm{SL}_{2,\mathbb{R}})$. This means that after descending along $U(1) \rightarrow U(1)$, $z \mapsto z^2$, we can have only even weights. So (SV 1) can never be satisfied. (Exercise: Fill in the details of this argument and compare with 10.4.)

Example 10.22 (Dimension of symmetric space). Assume that $G = \mathrm{GL}_n$ admits some Shimura datum X . By what was said at the end of §10.6, for every $h \in X$, the group $K_h^\circ := (K_h/\mathbb{R}^\times)^\circ \subset \mathrm{PGL}_n(\mathbb{R})^\circ$ would be a maximal compact subgroup. By the same argument as in Example 10.15, the maximal compact subgroups in $\mathrm{PGL}_n(\mathbb{R})^\circ$ are the conjugates of the image of $\mathrm{SO}(n)$. (Exercise: Check this.) So we see that

$$\dim_{\mathbb{R}}(X) = (n^2 - 1) - \dim_{\mathbb{R}} \mathrm{SO}(n).$$

In Example 10.2, we saw that $\mathrm{Lie}(\mathrm{SO}(n))$ is the vector space of skew-symmetric $(n \times n)$ -matrices. So its dimension is $n(n-1)/2$. Thus

$$\dim_{\mathbb{R}}(X) = n(n+1)/2 - 1.$$

If $n \equiv 3, 0 \pmod{4}$, then this dimension is odd, so X cannot be a complex manifold. In fact, GL_n with $n \geq 3$ can never define a Shimura datum because there is no Cartan involution on PGL_n which is given by conjugation with a group element (compare with Example 10.9).

Example 10.23 (Forms of groups). Let (G, X) be a Shimura datum and let $G'/\mathbb{Q}$ be a connected reductive group such that $G'_{\mathbb{R}} \cong G_{\mathbb{R}}$. Choosing any such isomorphism, we may view X as a $G'(\mathbb{R})$ -conjugacy class of Deligne homomorphisms $\mathbb{S} \rightarrow G'_{\mathbb{R}}$ and obtain a pair (G', X) . Axioms (SV 1) and (SV 2) are immediately satisfied because they only concern the situation over $\mathbb{R}$. If (SV 3) holds as well, then we have constructed a Shimura datum for G' .

10.8. Shimura varieties. Let (G, X) be a Shimura datum. We have sketched in §10.5 and §10.6 that X is naturally a union of hermitian symmetric domains on which $G(\mathbb{R})$ acts transitively by isometries. For every level subgroup $K \subset G(\mathbb{A}_f)$, we can now define

$$\mathrm{Sh}_K(G, X)(\mathbb{C}) := G(\mathbb{Q}) \backslash (X \times G(\mathbb{A}_f)/K).$$

As explained in §3.3, this is a finite union of quotients of the form

$$\Gamma_g \backslash X, \quad \Gamma_g = G(\mathbb{Q}) \cap gKg^{-1}, \quad g \in G(\mathbb{A}_f).$$

If K is small enough,⁹ then each Γ_g is torsion free and the projection

$$X \times G(\mathbb{A}_f)/K \longrightarrow \mathrm{Sh}_K(G, X)(\mathbb{C})$$

is a topological covering. In this way, $\mathrm{Sh}_K(G, X)(\mathbb{C})$ becomes a complex manifold. This defines the Shimura variety for Shimura datum (G, X) and level K over $\mathbb{C}$.

So far, we have only discussed the case of the modular curve ($G = \mathrm{GL}_2$). Our goal for the next few lectures is to get to know several important examples of the theory. We will first look at Hilbert modular varieties and their quaternionic twist. We will then look at the Siegel case and PEL moduli problems.

⁹We will not use the precise definition, but see [15, §3] and the references there for details.

11. HILBERT MODULAR VARIETIES

Let $h_0 : \mathbb{S} \rightarrow \mathrm{GL}_{2,\mathbb{R}}$ be the Deligne homomorphism from (10.7) and let $X_0 = \mathrm{GL}_2(\mathbb{R}) \cdot h_0$ be its conjugacy class. The pair (GL_2, X_0) is our (by now very familiar) Shimura datum for GL_2 . For an integer $d \geq 1$, we can take its d -fold self product $(\mathrm{GL}_2^d, X = X_0^d)$ to obtain a Shimura datum for GL_2^d . Hilbert modular varieties arise by considering forms of GL_2^d as in Example 10.23.

11.1. Definition. Let $F/\mathbb{Q}$ be a totally real field of degree d . Totally real means that all archimedean places of F are required to be real. In particular, if we denote these places by $\beta_1, \dots, \beta_d : F \rightarrow \mathbb{R}$, then

$$\begin{aligned} \beta : \mathbb{R} \otimes_{\mathbb{Q}} F &\xrightarrow{\sim} \mathbb{R} \times \dots \times \mathbb{R} \\ a \otimes x &\longmapsto (a\beta_1(x), \dots, a\beta_d(x)). \end{aligned} \quad (11.1)$$

Consider the group $G = \mathrm{Res}_{F/\mathbb{Q}}(\mathrm{GL}_2)$. Recall that this means that for every $\mathbb{Q}$ -algebra R ,

$$G(R) = \mathrm{GL}_2(R \otimes_{\mathbb{Q}} F).$$

By Proposition 9.16, G is a linear algebra group over $\mathbb{Q}$ of dimension $4d$. In general, being connected reductive can be checked after base change to the algebraic closure. Applying this to G , we find by Proposition 9.17 that

$$\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} G \xrightarrow{\sim} \mathrm{GL}_{2,\overline{\mathbb{Q}}}^d$$

is connected reductive, so G is connected reductive.

Let us define $\mathbb{Q}_{\infty} = \mathbb{R}$ to have a uniform notation for $\mathbb{Q}_p$, where $p \leq \infty$ is a place of $\mathbb{Q}$. Given p , let $v_1, \dots, v_r$ be the places of F above p . Generalizing (11.1), we have

$$\mathbb{Q}_p \otimes_{\mathbb{Q}} F \xrightarrow{\sim} F_{v_1} \times \dots \times F_{v_r}.$$

So for every $\mathbb{Q}_p$ -algebra R , we have

$$\begin{aligned} G(R) &= \mathrm{GL}_2(R \otimes_{\mathbb{Q}} F) \\ &= \mathrm{GL}_2(R \otimes_{\mathbb{Q}_p} \mathbb{Q}_p \otimes_{\mathbb{Q}} F) \\ &\xrightarrow{\sim} \mathrm{GL}_2(R \otimes_{\mathbb{Q}_p} F_{v_1}) \times \dots \times \mathrm{GL}_2(R \otimes_{\mathbb{Q}_p} F_{v_r}). \end{aligned}$$

Denoting by $G_{\mathbb{Q}_p}$ the base change $\mathbb{Q}_p \otimes_{\mathbb{Q}} G$, this shows that

$$G_{\mathbb{Q}_p} \xrightarrow{\sim} \prod_{i=1}^r \mathrm{Res}_{F_{v_i}/\mathbb{Q}_p}(\mathrm{GL}_{2,F_{v_i}}).$$

In particular, (11.1) constructs an isomorphism

$$\beta : G_{\mathbb{R}} \xrightarrow{\sim} (\mathrm{GL}_{2,\mathbb{R}})^d.$$

We also see that

$$G(\mathbb{Q}_p) \xrightarrow{\sim} \prod_{i=1}^n \mathrm{GL}_2(F_{v_i}).$$

Definition 11.1. Define a Shimura datum (G, X) by taking X as the $G(\mathbb{R})$ -conjugacy class of the composition

$$h : \mathbb{S} \xrightarrow{(h_0, \dots, h_0)} (\mathrm{GL}_{2,\mathbb{R}})^d \xrightarrow{\beta^{-1}} G_{\mathbb{R}}.$$

For level $K \subset \mathrm{GL}_2(\mathbb{A}_{F,f})$, the corresponding double quotient

$$\mathcal{S}_K = \mathrm{GL}_2(F) \backslash (X \times \mathrm{GL}_2(\mathbb{A}_{F,f}) / K)$$

is the *Hilbert modular variety* for field F and level K .

Let us give a more concrete description of $\mathcal{S}_K$. Each component β_i of β provides a projection map $\beta_i : G(\mathbb{R}) \rightarrow \mathrm{GL}_2(\mathbb{R})$ which comes from the map $M_2(\mathbb{R} \otimes_{\mathbb{Q}} F) \rightarrow M_2(\mathbb{R})$,

$$\beta_i \left(a \otimes \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} \right) = \begin{pmatrix} a\beta_i(x_{11}) & a\beta_i(x_{12}) \\ a\beta_i(x_{21}) & a\beta_i(x_{22}) \end{pmatrix}.$$

So each β_i defines an action of $G(\mathbb{R})$ on $\mathbb{H}^{\pm}$ by Moebius transformations,

$$(g, \tau) \mapsto \beta_i(g) \cdot \tau. \quad (11.2)$$

Taking the product of all these actions, we obtain an identification

$$\begin{aligned} X &\xrightarrow{\sim} \mathbb{H}^{\pm} \times \dots \times \mathbb{H}^{\pm} \\ g \cdot h &\mapsto (\beta_1(g) \cdot i, \dots, \beta_d(g) \cdot i). \end{aligned} \quad (11.3)$$

With respect to this action, we can rewrite the definition of $\mathcal{S}_K$ as

$$\mathcal{S}_K = \mathrm{GL}_2(F) \backslash (\mathbb{H}^{\pm} \times \dots \times \mathbb{H}^{\pm} \times \mathrm{GL}_2(\mathbb{A}_{F,f}) / K).$$

11.2. Connected components. We next determine the connected components of $\mathcal{S}_K$. In general, a connected reductive group G has a derived subgroup G^{der} which is the smallest subgroup containing all commutators $[g_1, g_2]$, $g_1, g_2 \in G$. It is contained in the kernel of every homomorphism $\varphi : G \rightarrow T$ to a commutative group scheme T . So every such φ factors over the quotient $G^{\mathrm{ab}} = G/G^{\mathrm{der}}$ which is hence called the *maximal abelian quotient* of G . By definition, there is an exact sequence

$$1 \longrightarrow G^{\mathrm{der}} \longrightarrow G \xrightarrow{\nu} G^{\mathrm{ab}} \longrightarrow 1.$$

In many situations of interest, G^{der} is simply connected and a general theorem allows to describe the connected components of $\mathrm{Sh}_K(G, X)(\mathbb{C})$ in terms of G^{ab} , see [15, Theorem 5.17]. We will now discuss this for Hilbert modular varieties.

The derived subgroup of GL_2 is SL_2 , and the maximal abelian quotient is the determinant map $\mathrm{GL}_2 \rightarrow \mathbb{G}_m$. These notions are directly preserved under Weil restrictions, so

$$\mathrm{Res}_{F/\mathbb{Q}}(\mathrm{GL}_2)^{\mathrm{der}} = \mathrm{Res}_{F/\mathbb{Q}}(\mathrm{SL}_2), \quad \mathrm{Res}_{F/\mathbb{Q}}(\mathrm{GL}_2)^{\mathrm{ab}} = \mathrm{Res}_{F/\mathbb{Q}}(\mathbb{G}_m).$$

Let $F_{>0} \subset F^{\times}$ be the subgroup of totally positive elements. That is,

$$F_{>0} := \beta^{-1}(\underbrace{\mathbb{R}_{>0} \times \dots \times \mathbb{R}_{>0}}_{\text{a subgroup of } (\mathbb{R}^{\times})^d}).$$

Since $\gamma(F)$ is dense in $\mathbb{R}^d$, we have

$$\begin{aligned} F^{\times} / F_{>0} &\xrightarrow{\sim} \{\pm 1\}^d \\ x &\mapsto (\mathrm{sgn}(\beta_1(x)), \dots, \mathrm{sgn}(\beta_d(x))). \end{aligned}$$

Proposition 11.2 (extending Corollary 3.20). *The determinant map induces a bijection*

$$\begin{aligned} \pi_0(\mathcal{S}_K) &\longrightarrow F_{>0} \backslash \mathbb{A}_{F,f}^{\times} / \det(K) \\ [x, gK] &\longmapsto \det(g), \end{aligned}$$

where the representative $[x, gK]$ is chosen with $x \in \mathbb{H}^+ \times \dots \times \mathbb{H}^+$.

Proof. The product $(\mathbb{H}^{\pm})^d$ has 2^d connected components on which $\mathrm{GL}_2(F)$ acts transitively. Concretely, an element $\gamma \in \mathrm{GL}_2(F)$ acts with signs

$$(\mathrm{sgn}(\beta_1(\det(\gamma))), \dots, \mathrm{sgn}(\beta_d(\det(\gamma))))$$

on $\pi_0(\mathbb{H}^{\pm})^d \xrightarrow{\sim} \{\pm 1\}^d$. In particular, the stabilizer of the connected component $(\mathbb{H}^+)^d \subset (\mathbb{H}^{\pm})^d$ is the subgroup $\mathrm{GL}_2(F)_{>0}$ of matrices γ with $\det(\gamma) \in F_{>0}$. In particular, every

$\mathrm{GL}_2(F)$ -orbit $[x, gK]$ has a representative $(x', g'K)$ with $x' \in (\mathbb{H}^+)^d$, and this representative is unique up to $\mathrm{GL}_2(F)_{>0}$. In this way, we obtain an isomorphism

$$\mathcal{S}_K \xrightarrow{\sim} \mathrm{GL}_2(F)_{>0} \backslash (\mathbb{H}^+ \times \dots \times \mathbb{H}^+ \times \mathrm{GL}_2(\mathbb{A}_{F,f})/K).$$

Taking connected components, since $\mathbb{H}^+$ is connected, we deduce

$$\pi_0(\mathcal{S}_K) \xrightarrow{\sim} \mathrm{GL}_2(F)_{>0} \backslash \mathrm{GL}_2(\mathbb{A}_{F,f})/K.$$

Next, we use that $\mathrm{Res}_{F/\mathbb{Q}}(\mathrm{SL}_2)$ is simply connected (Weil restrictions of simply connected groups are simply connected). So $\mathrm{SL}_2(F) \subset \mathrm{SL}_2(\mathbb{A}_{F,f})$ is dense by Theorem 3.15. By the same arguments as around (3.15), we get that

$$\det : \mathrm{GL}_2(F)_{>0} \backslash \mathrm{GL}_2(\mathbb{A}_{F,f})/K \xrightarrow{\sim} F_{>0} \backslash \mathbb{A}_{F,f}^\times / \det(K) \quad (11.4)$$

is an isomorphism. (Exercise: Recall the argument from (3.15) and fill in the details here.) $\square$

Recall that the class group $\mathcal{C}\ell_K$ of a number field K is the group of fractional ideals $\mathcal{I}_K$ modulo the group of principal fractional ideals $\mathcal{P}_K$. We have

$$\begin{aligned} K^\times \backslash \mathbb{A}_{K,f}^\times / \widehat{O}_K^\times &\xrightarrow{\sim} \mathcal{I}_K / \mathcal{P}_K \\ (x_v)_v &\longmapsto \prod_{v < \infty} \mathfrak{p}_v^{v(x_v)}, \end{aligned}$$

where we have written non-archimedean valuations additively and normalized them by $v(\mathfrak{p}_v) = 1$. We can also describe

$$\mathcal{C}\ell_K \xrightarrow{\sim} \mathrm{Pic}(O_K)$$

by viewing a fractional ideal as a projective O_K -module of rank 1. In the context of totally real fields, one often uses the following refinement.

Definition 11.3. The *narrow* class group $\mathcal{C}\ell_F^+$ of the totally real field F is the group of fractional ideals $\mathcal{I}_K$ modulo the principal ideals $\mathcal{P}_{F,>0}$ generated by a totally positive element. That is,

$$F_{>0} \backslash \mathbb{A}_{F,f}^\times / \widehat{O}_F^\times \xrightarrow{\sim} \mathcal{C}\ell_F^+. \quad (11.5)$$

Let $\mathfrak{a} \subseteq O_F$ be an ideal. The classical level subgroups as in §8.5 are defined by

$$\begin{aligned} K_0(\mathfrak{a}) &= \left\{ k \in \mathrm{GL}_2(\widehat{O}_F) \mid k \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{\widehat{\mathfrak{a}}} \right\} \\ K_1(\mathfrak{a}) &= \left\{ k \in \mathrm{GL}_2(\widehat{O}_F) \mid k \equiv \begin{pmatrix} 1 & * \\ 0 & * \end{pmatrix} \pmod{\widehat{\mathfrak{a}}} \right\}. \end{aligned} \quad (11.6)$$

If $K = K_0(\mathfrak{a})$ or $K = K_1(\mathfrak{a})$, then $\det(K) = \widehat{O}_F^\times$. In such cases, the narrow class group gives a description of the connected components of $\mathfrak{S}_K$:

Corollary 11.4. Assume that $\det(K) = \widehat{O}_F^\times$. Then Proposition 11.2 and (11.5) provide a bijection

$$\pi_0(\mathcal{S}_K) \xrightarrow{\sim} \mathcal{C}\ell_F^+.$$

An *orientation* of a projective O_F -module $\mathfrak{a}$ of rank 1 is the choice of an orientation for all the real lines $\mathbb{R} \otimes_{\beta_i, O_F} \mathfrak{a}$. Then $\mathcal{C}\ell_F^+$ is the group of isomorphism classes of oriented projective O_F -modules of rank 1.

In general, there are 2^d choices of orientation on a given $\mathfrak{a}$. The roots of unity in a totally real field are ± 1 , and -1 acts by flipping the orientation on each $\mathbb{R} \otimes_{\beta_i, O_F} \mathfrak{a}$. Using this, one can show that there is an exact sequence

$$1 \longrightarrow \{\pm 1\}^{d-1} \longrightarrow \mathcal{C}\ell_F^+ \longrightarrow \mathcal{C}\ell_F \longrightarrow 1. \quad (11.7)$$

Exercise 11.5. Fill in the details in the construction of (11.7).

11.3. Moduli description. In the case $F = \mathbb{Q}$, we have given a description of $\mathcal{S}_K$ as a moduli space of elliptic curves with level structure (see §8). It is desirable to have an analogous description of Hilbert modular varieties. Let us consider pairs (A, κ) where

- A is an abelian variety over $\mathbb{C}$ of dimension d ,
- $\kappa : O_F \rightarrow \text{End}(A)$ is an O_F -action on A .

An isomorphism $(A, \kappa) \xrightarrow{\sim} (A', \kappa')$ is an isomorphism of abelian varieties $f : A \rightarrow A'$ with $f \circ \kappa(x) = \kappa'(x) \circ f$ for all $x \in O_F$.

Proposition 11.6. *There exists a natural construction of a bijection*

$$\mathcal{S}_{\text{GL}_2(\widehat{O}_F)} \xrightarrow{\sim} \{\text{Isom. classes of } (A, \kappa)\}.$$

Proof. We give a refined construction that explains how the connected components on the left hand side reflect on the right hand side.

Step 1: The main construction. Let $\mathfrak{a}_1, \dots, \mathfrak{a}_r$ be representatives for the isomorphism classes of oriented projective O_F -modules of rank 1. For each $j = 1, \dots, r$, form the direct sum $\mathfrak{A}_j = O_F \oplus \mathfrak{a}_j$. For each $i = 1, \dots, d$, choose a positive generator $\alpha_{ij} \in \mathbb{R} \otimes_{\beta_i, O_F} \mathfrak{a}_j$. We orient

$$V_{ij} := \mathbb{R} \otimes_{\beta_i, O_F} \mathfrak{A}_j$$

by declaring $(1, 0), (0, \alpha_{ij})$ to be an oriented basis. This basis also provides us with an isomorphism

$$1, \alpha_{ij} : \mathbb{R}^2 \xrightarrow{\sim} V_{ij}. \quad (11.8)$$

Observe that the group of O_F -automorphisms of $\mathfrak{A}_j$ is given by

$$\Gamma_j = \left\{ \begin{pmatrix} O_F & \mathfrak{a}_j^{-1} \\ \mathfrak{a}_j & O_F \end{pmatrix} \right\}^\times.$$

Under (11.8), an element $\gamma \in \Gamma_j$ acts by the matrix $\beta_i(\gamma) \in \text{GL}_2(\mathbb{R})$. In particular,

$$\gamma \curvearrowright V_{1j} \oplus \dots \oplus V_{dj} \text{ as } (\beta_1(\gamma), \dots, \beta_j(\gamma)). \quad (11.9)$$

Consider the element $J = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix}$. It defines a complex structure on V_{ij} under (11.8). The tuple $(J, \dots, J)$ lies in

$$\text{End}_{\mathbb{R} \otimes_{\mathbb{Q}} F}(\mathbb{R} \otimes_{\mathbb{Z}} \mathfrak{A}_j) = \prod_{i=1}^d \text{End}_{\mathbb{R}}(\mathbb{R} \otimes_{\beta_i, O_F} \mathfrak{A}_j).$$

We can view $\mathfrak{A}_j$ as an O_F -stable lattice in $\mathbb{R} \otimes_{\mathbb{Z}} \mathfrak{A}_j$. So we can define a d -dimensional complex torus with O_F -action by

$$\left((\mathbb{R} \otimes_{\mathbb{Z}} \mathfrak{A}_j) / \mathfrak{A}_j, J, \text{ natural } O_F\text{-action} \right). \quad (11.10)$$

It will be explained in the next lecture (see Proposition 12.6) that this torus is, in fact, an abelian variety.

Let $\tau_1, \dots, \tau_d \in (\mathbb{H}^+)^d$ be a d -tuple of points on the upper half plane. There is $g = (g_1, \dots, g_d)$, $g_i \in \text{GL}_2(\mathbb{R})_{\det > 0}$ with $\tau_i = g_i \cdot J$. Then, in a slight abuse of notation,

$$J_\tau := (g_1 J g_1^{-1}, \dots, g_d J g_d^{-1})$$

defines a new complex structure on $\mathbb{R} \otimes_{\mathbb{Z}} \mathfrak{A}_j$ via (11.8). In this way, we upgraded the definition in (11.10) to a map

$$\begin{aligned} (\mathbb{H}^+)^d &\longrightarrow \{\text{Isom. classes of } (A, \kappa)\} \\ \tau &\longmapsto \left((\mathbb{R} \otimes_{\mathbb{Z}} \mathfrak{A}_j) / \mathfrak{A}_j, J_\tau, \text{ natural } O_F\text{-action} \right). \end{aligned} \quad (11.11)$$

Assume that τ and τ' define isomorphic pairs under (11.11). This means there exists an O_F -linear isomorphism on lattices $\gamma : \mathfrak{A}_j \xrightarrow{\sim} \mathfrak{A}_j$ which extends to a complex linear map $(\mathbb{R} \otimes_{\mathbb{Z}} \mathfrak{A}_j, J_\tau) \xrightarrow{\sim} (\mathbb{R} \otimes_{\mathbb{Z}} \mathfrak{A}_j, J_{\tau'})$. That is, $\gamma \in \Gamma_j$ such that

$$\gamma \circ J_\tau = J_{\tau'} \circ \gamma \iff J_\tau = \gamma J_{\tau'} \gamma^{-1}.$$

By (11.9), this precisely translates to $\tau = \gamma \tau'$ where γ acts by the Moebius transformation described in (11.2) and (11.3). Moreover, note that γ necessarily lies in the subgroup $\Gamma_{j,>0}$ of matrices with totally positive determinant because this is the stabilizer of $(\mathbb{H}^+)^d$ in $(\mathbb{H}^\pm)^d$. In this way, we have constructed an injective map

$$\Gamma_{j,>0} \backslash (\mathbb{H}^+)^d \hookrightarrow \{\text{Isom. classes of } (A, \kappa)\}. \quad (11.12)$$

Taking their disjoint union, we get a map

$$\bigsqcup_{j=1}^r \Gamma_{j,>0} \backslash (\mathbb{H}^+)^d \longrightarrow \{\text{Isom. classes of } (A, \kappa)\}. \quad (11.13)$$

Step 2: Identifying $\Gamma_{j,>0} \backslash (\mathbb{H}^+)^d$ with a piece of $\mathcal{S}_{\text{GL}_2(\widehat{O}_F)}$. For each $j = 1, \dots, r$, choose an idele $x^{(j)} = (x_v^{(j)})_v \in \mathbb{A}_{F,f}^\times$ representing $\mathfrak{a}_j$. A lift under the determinant map is

$$g^{(j)} := \begin{pmatrix} x^{(j)} & \\ & 1 \end{pmatrix} \in \text{GL}_2(\mathbb{A}_{F,f}).$$

The bijection in (11.4) concretely says that

$$\text{GL}_2(\mathbb{A}_{F,f}) = \bigsqcup_{j=1}^r \text{GL}_2(F)_{>0} \cdot g^{(j)} \cdot \text{GL}_2(\widehat{O}_F).$$

The module $\widehat{O}_F \cdot x^{(j)}$ is isomorphic to $\widehat{\mathfrak{a}}_j$, so we get

$$g^{(j)} \cdot \text{GL}_2(\widehat{O}_F) \cdot (g^{(j)})^{-1} = \text{GL}(\widehat{O}_F \oplus \widehat{\mathfrak{a}}_j).$$

Taking the intersection with $\text{GL}_2(F)_{>0}$, we recover

$$\Gamma_{j,>0} = \text{GL}_2(F)_{>0} \cap \text{GL}(\widehat{O}_F \oplus \widehat{\mathfrak{a}}_j). \quad (11.14)$$

Following the logic from (3.9) and (3.10), we find

$$\mathcal{S}_K \xrightarrow{\sim} \bigsqcup_{j=1}^r \Gamma_{j,>0} \backslash (\mathbb{H}^+)^d. \quad (11.15)$$

So the left hand side of (11.13) can be identified with $\mathcal{S}_{\text{GL}_2(\widehat{O}_F)}$.

Step 3: Proving bijectivity of (11.13). Let (A, κ) be a d -dimensional abelian variety with O_F -action over $\mathbb{C}$. By Theorem 6.1, we can write $A(\mathbb{C}) = V/\Lambda$ for a d -dimensional $\mathbb{C}$ -vector space and a lattice $\Lambda \subset V$. The O_F -action makes Λ into a torsion-free O_F -module which is necessarily projective of rank 2. Then V can be identified with

$$\mathbb{R} \otimes_{\mathbb{Z}} \Lambda \xrightarrow{\sim} \prod_{i=1}^d \mathbb{R} \otimes_{\beta_i, O_F} \Lambda.$$

Moreover, since O_F has to act $\mathbb{C}$ -linearly on V , each piece $\mathbb{R} \otimes_{\beta_i, O_F} \Lambda$ in that decomposition is a 1-dimensional complex subspace of V . As such, it is naturally oriented by orienting $\mathbb{C}$ such that $1, J$ is an oriented basis. In this way, Λ is naturally an oriented projective O_F -module of rank 2.

By the classification of projective modules over Dedekind domains (essentially a reformulation of (11.4) in our current setting), there exists an oriented projective O_F -module $\mathfrak{a}$ of rank 1 such that

$$\Lambda \xrightarrow{\sim} O_F \oplus \mathfrak{a}.$$

Note that $\mathfrak{a}$ is uniquely determined because we may recover it by taking the exterior square:

$$\begin{aligned} \bigwedge_{O_F}^2(O_F \oplus \mathfrak{a}) &\xrightarrow{\sim} \mathfrak{a} \\ 1 \wedge x &\longmapsto x. \end{aligned}$$

There is a unique $j = 1, \dots, r$ such that $\mathfrak{a} \cong \mathfrak{a}_j$ as oriented projective module. Since every F -linear complex structure on $\mathbb{R} \otimes_{\mathbb{Z}} (O_F \oplus \mathfrak{a}_j)$ that is compatible with orientations occurs in the family constructed in (11.12), we find that (A, κ) lies in the image of the j -component under (11.13). This shows surjectivity. Since j was uniquely determined by A , we also have injectivity, completing the proof. $\square$

Remark 11.7. Recall that in the modular curve case, we were able to construct a moduli space with universal elliptic curve after adding a bit of level structure (Theorem 7.12). After the above discussion, it seems natural to hope for something similar for the pairs (A, κ) that come up in Proposition 11.6, but this hope is dashed by the following observation: By Dirichlet's Unit Theorem, the structure of $O_F^\times$ as abelian group is given by

$$O_F^\times \xrightarrow{\sim} \{\pm 1\} \times \mathbb{Z}^{d-1}.$$

So if $F \neq \mathbb{Q}$, then pairs (A, κ) will always have infinitely many automorphisms, namely at least all element in $\kappa(O_F^\times)$. So even after adding level structures

$$\eta : (O_F/(n))^{\oplus 2} \xrightarrow{\sim} A[n] \quad (O_F\text{-linear}),$$

triples (A, κ, η) still have infinitely many automorphisms. This is a real difference between the cases $F = \mathbb{Q}$ and $F \neq \mathbb{Q}$ which Milne captures by his axiom (SV 5), see [15, §5].

12. THE SIEGEL MODULAR VARIETY

12.1. Parametrizing complex tori. Let us begin very leisurely by extending the ideas in §8.1 from elliptic curves to complex tori of dimension n . That is, our aim is to parametrize all complex tori of dimension n up to isomorphism. Recall that these are the complex Lie groups of the form V/Λ where V is a n -dimensional $\mathbb{C}$ -vector space and $\Lambda \backslash V$ a lattice.

There are two approaches to this problem: One is to fix V and vary Λ , the other to fix Λ and vary V . In §8.1, we have chosen the first approach. But this turns out to be more tricky in the higher-dimensional setting, so we now present the second.

All $\mathbb{Z}$ -lattices of rank $2n$ are isomorphic. So we may assume that $\Lambda = \mathbb{Z}^{2n}$. This also identifies $V = \mathbb{R} \otimes_{\mathbb{Z}} \Lambda$ with $\mathbb{R}^{2n}$. In order to make $\mathbb{R}^{2n}$ into a complex vector space, we need to provide an endomorphism $J \in M_{2n}(\mathbb{R})$ with $J^2 = -1$ (a *complex structure*). The function of J is to describe multiplication by $i \in \mathbb{C}$ on $\mathbb{R}^{2n}$. For every such J , we obtain a complex torus as

$$T_J := (\mathbb{R}^{2n}, J) / \mathbb{Z}^{2n}$$

and all complex tori are of this form. We find that $T_J \cong T_{J'}$ if and only if there exists an element $\gamma \in \mathrm{GL}_{2n}(\mathbb{Z})$ such that

$$\gamma \circ J = J' \circ \gamma, \quad \text{equivalently} \quad J' = \gamma J \gamma^{-1}.$$

In this way, we see that

$$\mathrm{GL}_{2n}(\mathbb{Z}) \backslash \left\{ \begin{array}{c} \text{Complex structures} \\ J \in M_{2n}(\mathbb{R}) \end{array} \right\} \xrightarrow{\sim} \left\{ \begin{array}{c} n\text{-Dimensional} \\ \text{complex tori} \end{array} \right\}.$$

In fact, all of $\mathrm{GL}_{2n}(\mathbb{R})$ acts on the set of complex structures by

$$g \cdot J := gJg^{-1}$$

and the action of $\mathrm{GL}_{2n}(\mathbb{Z})$ was just the restriction to a subgroup. We know that all complex structures are $\mathrm{GL}_{2n}(\mathbb{R})$ -conjugate because there is only one isomorphism class of n -dimensional complex vector space. Choose one complex structure $\mathbb{C}^n \xrightarrow{\sim} \mathbb{R}^{2n}$. Its stabilizer is $\mathrm{GL}_n(\mathbb{C})$. So the set of complex structures is in bijection with $\mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$. We have proved the following proposition.

Proposition 12.1. *This constructs a natural bijection*

$$\mathrm{GL}_{2n}(\mathbb{Z}) \backslash \mathrm{GL}_{2n}(\mathbb{R}) / \mathrm{GL}_n(\mathbb{C}) \xrightarrow{\sim} \left\{ \begin{array}{c} n\text{-Dimensional} \\ \text{complex tori} \end{array} \right\}. \quad (12.1)$$

12.2. Abelian varieties vs. complex tori. The quotient $\mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ can be viewed as the conjugacy classe of a Deligne homomorphism $h : \mathbb{S} \rightarrow \mathrm{GL}_{2n,\mathbb{R}}$ that satisfies (SV 1). But the stabilizer $\mathrm{GL}_n(\mathbb{C})$ is not compact modulo center as in (SV 2), so this does not give a Shimura datum. This is related to the fact that the right hand side only classifies certain complex Lie groups, and not abelian varieties.

Recall that we have an analytification functor giving rise to fully faithful embeddings

$$\left\{ \begin{array}{c} \text{Smooth projective} \\ \text{complex varieties} \end{array} \right\} \hookrightarrow \left\{ \begin{array}{c} \text{Smooth proper} \\ \text{complex varieties} \end{array} \right\} \hookrightarrow \left\{ \begin{array}{c} \text{Compact complex} \\ \text{manifolds} \end{array} \right\}. \quad (12.2)$$

A non-trivial theorem, see [18, §6], states that every abelian variety is projective. So, if we restrict attention to proper group varieties, the first arrow becomes an equality:

$$\left\{ \begin{array}{c} \text{Projective abelian} \\ \text{varieties} \end{array} \right\} = \{ \text{Abelian varieties} \} \hookrightarrow \{ \text{Complex tori} \}. \quad (12.3)$$

The point now is that the second arrow is not essentially surjective. Only the *projective* tori are algebraic, i.e. the tori $T = V/\Lambda$ such that there exists a closed immersion (as complex manifold) into some $\mathbb{P}^N(\mathbb{C})$.

12.3. Projective complex tori. Let $T = V/\Lambda$ be a complex torus. Recall that in order to define a closed embedding into some $\mathbb{P}^N(\mathbb{C})$, we need to find a very ample holomorphic line bundle L on T and then the embedding is given by $[s_0 : \dots : s_N]$ for a global generating set $s_0, \dots, s_N \in H^0(T, L)$. Recall that L is said to be ample if some tensor power $L^{\otimes r}$ is very ample. So the question of whether T is algebraic is all about classifying (ample) line bundles on T .

The line bundles on T can be completely classified with the Appell–Humbert theorem. We give a sketch of this classification; a more complete summary can be found in [6, §1.1]; full details are provided in [18, §1–3].

Let $\pi : V \rightarrow V/\Lambda$ be the projection map and let L be a holomorphic line bundle on V/Λ . Then $\pi^*(L)$ is a holomorphic line bundle on V . For every $u \in \Lambda$, we have a canonical isomorphism $u^*(\pi^*(L)) \xrightarrow{\sim} \pi^*(L)$ because $\pi \circ (\text{translate by } u) = \pi$. Holomorphic line bundles on $\mathbb{C}^n$ are all trivial, so we may choose an isomorphism $\beta : \mathcal{O}_V \xrightarrow{\sim} \pi^*(L)$. For every u , we can then consider the composition

$$\begin{aligned} \mathcal{O}_V &= u^*\mathcal{O}_V \xrightarrow{u^*(\beta)} u^*(\pi^*(L)) \xrightarrow{\sim} \pi^*(L) \xrightarrow{\beta^{-1}} \pi^*(L) \\ 1 &\longmapsto 1. \end{aligned} \quad (12.4)$$

It is of the form $e_u \cdot \beta$ for a nowhere vanishing holomorphic function $e_u : V \rightarrow \mathbb{C}^\times$. A small calculation shows that for all $u_1, u_2 \in \Lambda$, we have¹⁰

$$e_{u_1+u_2}(z) = e_{u_1}(z+u_2)e_{u_2}(z). \quad (12.5)$$

¹⁰This identity says that e_u is a cocycle of Λ with values in $\mathcal{O}_V(V)^\times$.

Since V is simply connected, we may choose logarithms of all the e_u :

$$\begin{array}{ccc} & & \mathbb{C} \\ & \nearrow \exists f_u & \downarrow z \mapsto \exp(2\pi iz) \\ V & \xrightarrow{e_u} & \mathbb{C}^\times. \end{array}$$

The f_u are only unique up to adding a constant in $\mathbb{Z}$. So (12.5) states that

$$F(u_1, u_2) := f_{u_1+u_2}(z) - f_{u_1}(z+u_2) - f_{u_2}(z) \in \mathbb{Z}.$$

This difference depends on how we chose the $\{f_u\}_{u \in \Lambda}$, of course. But the symmetrization

$$E(u_1, u_2) := F(u_1, u_2) - F(u_2, u_1) \quad (12.6)$$

does not. Namely, assume for example that we replace f_{u_1} by $f_{u_1} + c$. Then this constant cancels in the difference

$$f_{u_1}(z+u_2) - f_{u_1}(z)$$

which occurs when writing out $E(u_1, u_2)$. In this way, we have constructed an alternating pairing

$$E : \Lambda \times \Lambda \longrightarrow \mathbb{Z}$$

that only depends on L . Extending E in the $\mathbb{R}$ -bilinear way, we obtain an alternating pairing $E : V \times V \longrightarrow \mathbb{R}$. At this point, an additional argument (see [6, §1.1]) shows that E is compatible with the complex structure in the sense that

$$E(ix, iy) = E(x, y) \quad \text{for all } x, y \in V.$$

Then the pairing

$$\lambda(x, y) = E(ix, y) + iE(x, y)$$

is a hermitian pairing $\lambda : V \times V \longrightarrow \mathbb{C}$. The last step is just a reformulation: we may recover E as the imaginary part of λ . More precisely,

$$\begin{aligned} \left\{ \begin{array}{l} \text{Alternating } E : V \times V \rightarrow \mathbb{R} \\ \text{s.t. } E(ix, iy) = E(x, y) \end{array} \right\} &\xrightarrow{\sim} \{ \text{Hermitian } \lambda : V \times V \rightarrow \mathbb{C} \} \\ E &\longmapsto \lambda(x, y) := E(ix, y) + iE(x, y) \\ \text{Im}(\lambda) &\longleftarrow \lambda. \end{aligned} \quad (12.7)$$

Theorem 12.2 (Appell–Humbert, see [6, Theorem 1.1.3]). *The holomorphic line bundles on V/Λ correspond bijectively to pairs (λ, α) where*

- $\lambda : V \times V \rightarrow \mathbb{C}$ is a hermitian form such that $E = \text{Im}(\lambda)|_{\Lambda \times \Lambda}$ takes values in $\mathbb{Z}$.
- $\alpha : \Lambda \rightarrow S^1$ is a map to the unit circle such that

$$\alpha(u_1 + u_2) = e^{i\pi E(u_1, u_2)} \alpha(u_1) \alpha(u_2). \quad (12.8)$$

The line bundle $L(\lambda, \alpha)$ corresponding to (λ, α) is given by the cocycle $\{e_u\}_{u \in \Lambda}$ defined by

$$e_u(z) = \alpha(u) e^{\pi \lambda(z, u) + \pi \lambda(u, u)/2}.$$

In particular, the parametrization satisfies

$$L(\lambda_1, \alpha_1) \otimes L(\lambda_2, \alpha_2) \cong L(\lambda_1 + \lambda_2, \alpha_1 \alpha_2).$$

Moreover, $L(\lambda, \alpha)$ is ample if and only if λ is positive definite.

In this way, we have singled out the abelian varieties among the complex tori V/Λ . They are precisely those quotients such that there exists a non-degenerate alternating pairing $E : V \times V \rightarrow \mathbb{R}$ such that

- (1) $E|_{\Lambda \times \Lambda}$ takes values in $\mathbb{Z}$.
- (2) $E(ix, iy) = E(x, y)$ for all $x, y \in V$.

(3) The hermitian form $\lambda(x, y) = E(ix, y) + iE(x, y)$ is positive definite.

Indeed, for every alternating pairing E satisfying (1) and (2), there exist maps α satisfying (12.8). If E also satisfies (3), then we have defined an ample line bundle by the theorem.

Remark 12.3. The forms λ and E in (12.7) and Theorem 12.2 are allowed to be degenerate. For example, if L is the trivial line bundle on V/Λ , then the $\{e_u\}$ in (12.5) will all be constantly 1 and the alternating form E in (12.7) will be 0.

Definition 12.4. The *Néron–Severi* group $\text{NS}(T)$ of $T = V/\Lambda$ is the abelian group of alternating forms $E : \Lambda \times \Lambda \rightarrow \mathbb{Z}$ such that (1) and (2) hold. By Theorem 12.2, there is an exact sequence

$$0 \longrightarrow \text{Hom}(\Lambda, S^1) \longrightarrow \text{Pic}(T) \xrightarrow{(12.6)} \text{NS}(T) \longrightarrow 0. \quad (12.9)$$

A *Riemann form* for T is an element $E \in \text{NS}(T)$ such that also (3) holds. Riemann forms exist if and only if T is isomorphic to the analytification $A(\mathbb{C})$ of an abelian variety $A/\mathbb{C}$.

For example, we may apply Theorem 12.2 to prove that all 1-dimensional complex tori are algebraic.

Proposition 12.5. *Let $T = \mathbb{C}/\Lambda$ be a 1-dimensional complex torus. Then Riemann forms exist. In particular, all one-dimensional complex tori are algebraic (elliptic curves).*

Proof. Let $e_1, e_2 \in \Lambda$ be a $\mathbb{Z}$ -basis. The alternating pairings $E : \Lambda \times \Lambda \rightarrow \mathbb{Z}$ are precisely those bilinear pairings that satisfy

$$E(e_1, e_1) = E(e_2, e_2) = 0 \quad \text{and} \quad E(e_1, e_2) = -E(e_2, e_1).$$

They are hence the pairings represented by the matrices of the form

$$\begin{pmatrix} 0 & c \\ -c & 0 \end{pmatrix}, \quad c \in \mathbb{Z}. \quad (12.10)$$

Let E be any non-degenerate such pairing (meaning $c \neq 0$), and extend it $\mathbb{R}$ -linearly to the ambient $\mathbb{C}$. For an automorphism $g \in \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(\mathbb{R})$, we calculate

$$\begin{aligned} (ge_1) \wedge (ge_2) &= (ae_1 + be_2) \wedge (ce_1 + de_2) \\ &= (ae_1) \wedge (de_2) + (be_2) \wedge (ce_1) \\ &= (ad - bc) \cdot (e_1 \wedge e_2). \end{aligned} \quad (12.11)$$

This shows that the $\text{GL}_2(\mathbb{R})$ -action scales E with the determinant. In particular, E automatically satisfies

$$E(ix, iy) = \det(i)E(x, y) = E(x, y)$$

and hence lies in $\text{NS}(\mathbb{C}/\Lambda)$. Let $\lambda : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ be the hermitian pairing obtained from E by (12.7). Then λ can be of signature $(1, 0)$ or $(0, 1)$. So one out of $\pm\lambda$ will be positive definite, which shows by Theorem 12.2 that ample line bundles exist on $\mathbb{C}/\Lambda$. $\square$

A beautiful observation is that this argument can be extended to the complex tori that were constructed in (11.11).

Proposition 12.6. *Let $(T = V/\Lambda, \kappa)$ be a d -dimensional complex torus with an action $\kappa : O_F \rightarrow \text{End}(T)$ of the ring of integers in a totally real field of degree $[F : \mathbb{Q}] = d$. Then T is the analytification of an abelian variety.*

Proof. Via κ , the lattice Λ is a projective O_F -module of rank 2. Let e_1, e_2 be an F -basis for $W = F \otimes_{O_F} \Lambda$. Any $c \in F$ defines an F -bilinear alternating pairing

$$\begin{pmatrix} 0 & c \\ -c & 0 \end{pmatrix} : W \times W \longrightarrow F.$$

Let us call this pairing E_c ; it does not need to satisfy $E_c(\Lambda \times \Lambda) \subset O_F$ of course. Let again $\beta_1, \dots, \beta_d : F \rightarrow \mathbb{R}$ denote the d different embeddings and consider the decomposition

$$V = \bigoplus_{i=1}^d W_i, \quad W_i = \mathbb{R} \otimes_{\beta_i, F} W.$$

By base change, each summand W_i carries the $\mathbb{R}$ -linear alternating pairing

$$E_{c,i} = \mathbb{R} \otimes_{\beta_i, F} E_c : W_i \times W_i \longrightarrow \mathbb{R}.$$

Each W_i is 2-dimensional as $\mathbb{R}$ -vector space, so the same argument as during the proof of Proposition 12.5 shows that $g_i \in \mathrm{GL}_{\mathbb{R}}(W_i)$ satisfies

$$E_{c,i}(g_i x, g_i y) = \det(g_i) \cdot E_{c,i}(x, y).$$

An element $g \in \mathrm{GL}_{\mathbb{R} \otimes_{\mathbb{Q}} F}(V)$ hence scales the $E_{c,i}$ as

$$g \cdot (E_{c,1}, \dots, E_{c,d}) = (\beta_1(\det(g))E_{c,1}, \dots, \beta_d(\det(g))E_{c,d}).$$

In particular, the given complex structure $J \in \mathrm{GL}_{\mathbb{R} \otimes_{\mathbb{Q}} F}(V)$ satisfies

$$E_c(Jx, Jy) = E_c(x, y).$$

The associated hermitian pairing $\lambda_c : V \times V \rightarrow \mathbb{C} \otimes_{\mathbb{R}} F$ is the tuple of the hermitian pairings

$$\lambda_{c,i} : W_i \times W_i \longrightarrow \mathbb{C}, \quad \lambda_{c,i}(x, y) = E_{c,i}(Jx, y) + JE_{c,i}(x, y).$$

(We again denoted by $J \in \mathbb{C}$ the imaginary element i to avoid confusion with the index i .) Let $\delta \in F^\times$ be an element with

$$\mathrm{sign}(\beta_i(\delta)) = \begin{cases} +1 & \text{if the signature of } \lambda_{c,i} \text{ is } (1, 0) \\ -1 & \text{if the signature of } \lambda_{c,i} \text{ is } (0, 1). \end{cases}$$

Then $\lambda_{\delta c, i}$ is positive definite for all $i = 1, \dots, d$. Define

$$E' : \Lambda \times \Lambda \longrightarrow \mathbb{Q}$$

as the composition of

$$\Lambda \times \Lambda \xrightarrow{E_{\delta c}} F \xrightarrow{\mathrm{tr}_{F/\mathbb{Q}}} \mathbb{Q}.$$

Let $\lambda'(x, y) = E'(Jx, y) + JE'(x, y)$ be the corresponding hermitian pairing. It is the sum of all the $\lambda_{\delta c, i}$, and hence positive definite. Finally, pick an integer $m \geq 1$ such that $E'(\Lambda \times \Lambda) \subseteq m^{-1}\mathbb{Z}$. Then $(m \cdot E')(\Lambda \times \Lambda) \subseteq \mathbb{Z}$, meaning mE' is a Riemann form for V/Λ . $\square$

12.4. The Siegel modular variety. Consider the $2n$ -dimensional standard symplectic pairing $\langle \cdot, \cdot \rangle$ on $\mathbb{Z}^{2n}$. That is, denoting the basis of $\mathbb{Z}^{2n}$ by $e_1, \dots, e_n, f_1, \dots, f_n$, the pairing is given by

$$\begin{aligned} \langle e_i, e_j \rangle &= \langle f_i, f_j \rangle = 0 \\ \langle e_i, f_j \rangle &= -\langle f_j, e_i \rangle = \delta_{ij}. \end{aligned}$$

The general symplectic group GSp_{2n} is defined by

$$\mathrm{GSp}_{2n}(R) = \left\{ (g, \nu) \in \mathrm{GL}_{2n}(R) \times R^\times \mid \begin{array}{l} \langle gx, gy \rangle = \nu \langle x, y \rangle \\ \text{for all } x, y \in R^{2n} \end{array} \right\}.$$

That is, elements of GSp_{2n} are required to preserve $\langle \cdot, \cdot \rangle$ up to scalar; this factor ν is called the similitude factor. Projection to ν defines an exact sequence

$$1 \longrightarrow \mathrm{Sp}_{2n} \longrightarrow \mathrm{GSp}_{2n} \xrightarrow{\nu} \mathbb{G}_m \longrightarrow 1.$$

We have seen in (12.10) and (12.11) that $\mathrm{GSp}_2 = \mathrm{GL}_2$ with $\nu = \det$.

Definition 12.7. The *Siegel Shimura datum* is the datum (GSp_{2n}, X) where X is the $\mathrm{GSp}_{2n}(\mathbb{R})$ -conjugacy class of

$$h(a + ib) := \begin{pmatrix} a \cdot 1_n & -b \cdot 1_n \\ b \cdot 1_n & a \cdot 1_n \end{pmatrix}. \quad (12.12)$$

Here, the matrix form is with respect $e_1, \dots, e_n, f_1, \dots, f_n$. Let

$$\mathcal{S}_K := \mathrm{GSp}_{2n}(\mathbb{Q}) \backslash (X \times \mathrm{GSp}_{2n}(\mathbb{A}_f) / K)$$

denote the attached Shimura variety, which is called the *Siegel modular variety* of level K .

We introduce one last piece of terminology. A *polarization* for a complex torus V/Λ is a Riemann form for Λ . A *principal polarization* is a perfect Riemann form E . That is, E gives an isomorphism

$$E : \Lambda \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{Z}}(\Lambda, \mathbb{Z}).$$

A principally polarized abelian variety over $\mathbb{C}$ is a pair (A, E) where A is an abelian variety over $\mathbb{C}$, say $A(\mathbb{C}) = V/\Lambda$, and E a principal polarization for V/Λ . An isomorphism of such pairs is an isomorphism $f : V_1/\Lambda_1 \xrightarrow{\sim} V_2/\Lambda_2$ of abelian varieties such that $E_2(f(x), f(y)) = E_1(x, y)$ for all $x, y \in \Lambda_1$.

Theorem 12.8 (Siegel moduli description). *There is an isomorphism*

$$\mathcal{S}_{\mathrm{GSp}_{2n}(\widehat{\mathbb{Z}})} \xrightarrow{\sim} \left\{ \text{Isom. classes of principally polarized } n\text{-dimensional abelian varieties over } \mathbb{C} \right\}. \quad (12.13)$$

Proof. Step 1: Structure of $\mathcal{S}_{\mathrm{GSp}_{2n}(\widehat{\mathbb{Z}})}$. The group Sp_{2n} is simply connected. Thus, by strong approximation (Theorem 3.15), for every level subgroup K , we obtain that

$$\nu : \mathrm{GSp}_{2n}(\mathbb{Q}) \backslash \mathrm{GSp}_{2n}(\mathbb{A}_f) / K \xrightarrow{\sim} \mathbb{Q}^\times \backslash \mathbb{A}_f^\times / \nu(K).$$

For $K = \mathrm{GSp}_{2n}(\widehat{\mathbb{Z}})$, we have $\nu(K) = \widehat{\mathbb{Z}}^\times$ and hence find that

$$\mathrm{GSp}_{2n}(\mathbb{A}_f) = \mathrm{GSp}_{2n}(\mathbb{Q}) \cdot 1 \cdot \mathrm{GSp}_{2n}(\widehat{\mathbb{Z}}).$$

Thus $\mathcal{S}_{\mathrm{GSp}_{2n}(\widehat{\mathbb{Z}})} = \mathrm{GSp}_{2n}(\mathbb{Z}) \backslash X$ because

$$\mathrm{GSp}_{2n}(\mathbb{Z}) = \mathrm{GSp}_{2n}(\mathbb{Q}) \cap \mathrm{GSp}_{2n}(\widehat{\mathbb{Z}}).$$

Step 2: Analyzing h from (12.12). The image $J := h(i)$ defines a complex structure on the vector space $\mathbb{R}^{2n}$. Concretely, we have the isomorphism

$$(\mathbb{R}^{2n}, J) \xrightarrow{\sim} \mathbb{C}^n \quad (12.14)$$

$$\sum_{k=1}^n a_k e_k + b_k f_k \mapsto (a_1 + b_1 i, \dots, a_n + b_n i).$$

Let us define $E := -\langle \cdot, \cdot \rangle$, which is a perfect symplectic pairing on $\mathbb{Z}^{2n}$. Substitution shows that

$$E(Jx, Jy) = E(x, y) \quad \text{for all } x, y \in \mathbb{R}^{2n}.$$

That is, J is compatible with E in the sense of (12.7), so we obtain the hermitian form $\lambda(x, y) := E(Jx, y) + iE(x, y)$ on $(\mathbb{R}^{2n}, J)$. By definitions, we have

$$E(Je_k, e_k) = -\langle f_k, e_k \rangle = 1, \quad (12.15)$$

which shows that λ is the standard positive definite hermitian pairing on $\mathbb{C}^n$ under the isomorphism (12.14). In particular, E is a principal polarization of the torus $(\mathbb{R}^{2n}/\mathbb{Z}^{2n}, J)$, and we have hence defined a point

$$((\mathbb{R}^{2n}/\mathbb{Z}^{2n}, J), E) \in \text{RHS of (12.13)}. \quad (12.16)$$

Step 3: Extending construction (12.16) to all of X . For every $g \in \mathrm{GSp}_{2n}(\mathbb{R})$, we can consider the conjugate complex structure $J_g = gJg^{-1}$. We have

$$\begin{aligned} E(gJg^{-1}x, gJg^{-1}y) &= \nu(g)E(Jg^{-1}x, Jg^{-1}y) \\ &\stackrel{(12.15)}{=} \nu(g)E(g^{-1}x, g^{-1}y) \\ &= E(x, y). \end{aligned}$$

So J_g is still compatible with E in the sense of (12.7). Let $\lambda_g(x, y) = E(J_g x, y) + iE(x, y)$ be the hermitian pairing associated to g . The group $\mathrm{GSp}_{2n}(\mathbb{R})$ has two connected components which are distinguished by the sign of $\nu(g)$. The form λ_g is always non-degenerate and varies continuously in g . So for all g with $\nu(g) > 0$, λ_g is positive definite and E defines a polarization for $(\mathbb{R}^{2n}/\mathbb{Z}^{2n}, J_g)$. If $\nu(g) < 0$, then λ_g is negative definite instead and $-E$ defines a polarization. So we constructed a map

$$\begin{aligned} X &\longrightarrow \left\{ \begin{array}{l} \text{Isom. classes of principally polarized} \\ n\text{-dimensional abelian varieties over } \mathbb{C} \end{array} \right\} \\ ghg^{-1} &\longmapsto ((\mathbb{R}^{2n}/\mathbb{Z}^{2n}, J_g), \text{sign}(\nu(g))E). \end{aligned} \quad (12.17)$$

It is immediately clear (apply definitions) that an element $\gamma \in \mathrm{GSp}_{2n}(\mathbb{Z})$ defines an isomorphism

$$\gamma : ((\mathbb{R}^{2n}/\mathbb{Z}^{2n}, J_g), \text{sign}(\nu(g))E) \xrightarrow{\sim} ((\mathbb{R}^{2n}/\mathbb{Z}^{2n}, J_{\gamma g}), \nu(\gamma)\text{sign}(\nu(g))E). \quad (12.18)$$

Since $\nu(\gamma) \in \mathbb{Z}^\times = \{\pm 1\}$ is already a sign, the RHS here is the same as

$$((\mathbb{R}^{2n}/\mathbb{Z}^{2n}, J_{\gamma g}), \text{sign}(\nu(\gamma g))E).$$

In this way, we see that (12.17) factors over the quotient

$$X \longrightarrow \mathrm{GSp}_{2n}(\mathbb{Z}) \backslash X.$$

Conversely an isomorphism between two polarized tori of the form $(\mathbb{R}^{2n}/\mathbb{Z}^{2n}, J_g, \pm E)$ has to come from an element of $\gamma \in \mathrm{GSp}_{2n}(\mathbb{Z})$. In this way, we constructed an injection

$$\mathrm{GSp}_{2n}(\mathbb{Z}) \backslash X \hookrightarrow \text{RHS of (12.17)}. \quad (12.19)$$

Step 4: Surjectivity of (12.19). Let $(V/\Lambda, E')$ be a principally polarized abelian variety of dimension n . There is only one isomorphism class of perfect alternating pairing on $\mathbb{Z}^{2n}$, so (Λ, E') is isomorphic to $(\mathbb{Z}^{2n}, E)$. After fixing such an isomorphism, our given pair is of the form

$$((\mathbb{R}^{2n}/\mathbb{Z}^{2n}, J'), E)$$

for some complex structure J' on $\mathbb{R}^{2n}$ that is compatible with E in the sense of (12.7). It is left to show that $J' = gJg^{-1}$ for some $g \in \mathrm{GSp}_{2n}(\mathbb{R})$ with $\nu(g) > 0$. Let $\lambda = \lambda(E, J)$ and $\lambda' = \lambda(E, J')$ be the hermitian forms attached to J and J' via (12.7). Both are positive definite because this is part of the definition of polarization. There is only one isomorphism class of hermitian form of signature $(n, 0)$, so there exists $g \in \mathrm{GL}_{2n}(\mathbb{R})$ defining an isometry

$$g : (\mathbb{R}^{2n}, J, \lambda) \xrightarrow{\sim} (\mathbb{R}^{2n}, J', \lambda').$$

That is, $\lambda(x, y) = \lambda'(gx, gy)$. The imaginary parts of λ and λ' are equal to E , so we obtain $E(x, y) = E(gx, gy)$. This means $g \in \mathrm{Sp}_{2n}(\mathbb{R})$, and the proof is complete. $\square$



Part 3. Counting points mod p on $\mathcal{M}_n$

13. MOTIVATION AND BACKGROUND

The problem we want to study in the remainder of our lecture is the following question.

Question 13.1. Given $n \geq 3$ and $q = p^d$, what is the cardinality of $\mathcal{M}_n(\mathbb{F}_q)$?

It would be worthwhile to study this question without any further motivation. After all, elliptic curves play an important role in number theory, and determining their isomorphism classes over finite fields is an interesting task. However, this question is also part of a much larger circle of ideas. Our aim today is to sketch this background. In the lectures after that, we will again focus on $\mathcal{M}_n$.

13.1. Overview. Every smooth projective variety $X/\mathbb{Q}$ defines a holomorphic function $\zeta(X, s)$ on the right half plane $\operatorname{Re}(s) > 1 + \dim(X)$, called its *Hasse–Weil ζ -function*. This definition generalizes the Riemann ζ -function or the Dedekind ζ -functions known from number theory. One can always express $\zeta(X, s)$ as a Dirichlet series

$$\zeta(X, s) = \sum_{n \geq 1} \frac{a_n}{n^s}, \quad \operatorname{Re}(s) > 1 + \dim(X), \quad (13.1)$$

where the coefficients $a_n \in \mathbb{Z}$ capture a lot of information about the arithmetic of X . We will see a definition of $\zeta(X, s)$ and some examples below.

Extrapolating from known cases such as the Riemann ζ -function, it is conjectured that the Hasse–Weil ζ -function admits a meromorphic continuation to all $s \in \mathbb{C}$ and that it satisfies a functional equation for $s \longleftrightarrow \dim(X) + 1 - s$. These properties (if true) express a very strong form of coherence among all the a_n in (13.1).

The definition of $\zeta(X, s)$ is purely algebraic, so it is not at all clear how to establish such analytic properties. The roundabout way in known cases is to relate the étale cohomology of X to automorphic representations (Langlands correspondence). This allows to write $\zeta(X, s)$ in terms of automorphic L -functions. For these, analytic properties are naturally easier to establish.

Shimura varieties play a special role in this context because their definition is so closely related to group theory. A precise conjecture (Kottwitz conjecture) describes their étale cohomology in terms of automorphic representations, and this makes it possible to access their ζ -functions systematically. We very briefly sketch the strategy for this at the end of this section (Langlands–Kottwitz method).

The ζ -function $\zeta(X, s)$ bundles the arithmetic information of all the étale cohomology groups of X into a single object. The Kottwitz Conjecture involves finer information. It describes the cohomology of Shimura varieties in terms of summands built from the (conjectural) Langlands correspondence. In this way, studying the ζ -function of Shimura varieties goes hand in hand with constructing instances of the Langlands correspondence in their cohomology.

13.2. The Hasse–Weil ζ -function. The Hasse–Weil ζ -function $\zeta(X, s)$ is defined as a product $\prod_p \zeta_p(X, s)$ of factors for all prime numbers. The factors for primes of good reduction can be described in terms of smooth integral models of X . The definition at the “bad” primes is more subtle; see [20] which also defines the archimedean factors for ζ . A general reference is [10].

Lemma 13.2. *Let X be a smooth projective variety over $\mathbb{Q}$. Then there exists a finite set of primes S and a smooth projective $\mathbb{Z}[S^{-1}]$ -scheme $\mathcal{X}$ such that $\mathcal{X}_{\mathbb{Q}} \cong X$.*

Proof. Since X is projective by assumption, we may write $X = V_+(F_1, \dots, F_m) \subseteq \mathbb{P}_{\mathbb{Q}}^N$ for some integer N and some homogeneous polynomials $F_1, \dots, F_m$. Scaling the F_i by the denominators of their coefficients, we may assume $F_i \in \mathbb{Z}[T_0, \dots, T_N]$. Then $\mathcal{X}' = V_+(F_1, \dots, F_m) \subseteq \mathbb{P}_{\mathbb{Z}}^m$ is a projective $\mathbb{Z}$ -scheme with $\mathcal{X}'_{\mathbb{Q}} \cong X$.

Let $\mathcal{X}'_{\text{sm}} \subset \mathcal{X}'$ be the subset of all points in which $f : \mathcal{X}' \rightarrow \text{Spec}(\mathbb{Z})$ is smooth. This subset is open: if the Jacobi criterion holds in a point $x \in \mathcal{X}'$, then it also holds on an open neighborhood of x . Since f is projective, hence proper, the image $Z = f(\mathcal{X}' \setminus \mathcal{X}'_{\text{sm}})$ is a closed subset of $\text{Spec}(\mathbb{Z})$. The generic fiber $\mathcal{X}'_{\mathbb{Q}}$ is smooth, so $(0) \notin Z$. We conclude that $S := Z$ is a finite set of primes, and that $\mathcal{X} := \mathcal{X}'[S^{-1}]$ is a smooth projective model for X over $\mathbb{Z}[S^{-1}]$. $\square$

Definition 13.3. Let X be a smooth projective over $\mathbb{Q}$. Let S be a finite set of primes such that for each $p \notin S$, X has a smooth projective model $\mathcal{X}_p$ over $\mathbb{Z}_{(p)}$. For $p \notin S$, define the p -factor

$$\zeta_p(X, s) := \exp \left(\sum_{n=1}^{\infty} |\mathcal{X}_p(\mathbb{F}_{p^n})| \cdot \frac{p^{-ns}}{n} \right). \quad (13.2)$$

This expression converges and defines a holomorphic function on the right half plane $\text{Re}(s) > \dim(X)$. The partial *Hasse–Weil ζ -function* of X is defined as the product

$$\zeta^S(X, s) := \prod_{p \notin S} \zeta_p(X, s). \quad (13.3)$$

This product converges on the right half plane $\text{Re}(s) > \dim(X) + 1$.

Example 13.4. (1) Consider $X = \text{Spec}(\mathbb{Q})$. Then $\mathcal{X} = \text{Spec}(\mathbb{Z})$ is a smooth projective model over all of $\text{Spec}(\mathbb{Z})$, so we can take $S = \emptyset$. We find $|\mathcal{X}(\mathbb{F}_{p^n})| = 1$ for all p and n , so

$$\zeta_p(\text{Spec}(\mathbb{Q}), s) = \exp \left(\sum_{n=1}^{\infty} \frac{p^{-ns}}{n} \right).$$

Recall that $\log(1+x) = \sum_{n=1}^{\infty} (-x)^n/n$, so we see that

$$\begin{aligned} \zeta_p(\text{Spec}(\mathbb{Q}), s) &= \exp(-\log(1 - p^{-s})) \\ &= (1 - p^{-s})^{-1}. \end{aligned}$$

We obtain that

$$\zeta(\text{Spec}(\mathbb{Q}), s) = \zeta(s) \quad (\text{Riemann } \zeta\text{-function}). \quad (13.4)$$

(2) Consider $X = \mathbb{P}_{\mathbb{Q}}^m$. We have the smooth integral model $\mathbb{P}_{\mathbb{Z}}^m$, so we may again take $S = \emptyset$. Recall that we may set-theoretically decompose $\mathbb{P}^m$ as

$$\mathbb{P}^m = \mathbb{A}^m \sqcup \mathbb{A}^{m-1} \sqcup \dots \sqcup \mathbb{A}^0.$$

Hence, the number of $\mathbb{F}_q$ -points of $\mathbb{P}^m$ is given by

$$\mathbb{P}^m(\mathbb{F}_q) = 1 + q + q^2 + \dots + q^m.$$

Substituting this into the definition, we find

$$\begin{aligned} \zeta_p(\mathbb{P}^m, s) &= \exp \left(\sum_{j=0}^m \sum_{n=1}^{\infty} p^{nj} \cdot \frac{p^{-ns}}{n} \right) \\ &= \prod_{j=0}^m \exp \left(\sum_{n=1}^{\infty} \frac{p^{-n(s-j)}}{n} \right) \\ &= \zeta_p(s) \zeta_p(s-1) \cdots \zeta_p(s-m). \end{aligned}$$

We see that the Hasse–Weil ζ -function of projective space is a product of shifts of Riemann ζ -functions,

$$\zeta(\mathbb{P}_{\mathbb{Q}}^m, s) = \zeta(s)\zeta(s-1)\cdots\zeta(s-m). \quad (13.5)$$

There is a slightly different expression for the ζ -function which is more similar to the definition of the Dedekind ζ -function. Let $\mathcal{X}_p$ be a smooth model of X over $\operatorname{Spec} \mathbb{Z}_{(p)}$ and let $x \in \mathcal{X}_p$ be a closed point. Its residue field $\kappa(x)$ is a finite extension of $\mathbb{F}_p$; let $N(x) = |\kappa(x)|$ be its order (the “norm” of x). For every $q = p^n$, we have

$$|\operatorname{Spec}(\kappa(x))(\mathbb{F}_q)| = \begin{cases} [\kappa(x) : \mathbb{F}_p] & \text{if } N(x) \mid q \\ 0 & \text{otherwise.} \end{cases} \quad (13.6)$$

It follows that the contribution of x to (13.2) is

$$\exp\left(\sum_{\ell=1}^{\infty} [\kappa(x) : \mathbb{F}_p] \frac{p^{-[\kappa(x) : \mathbb{F}_p] \cdot \ell s}}{[\kappa(x) : \mathbb{F}_p] \cdot \ell}\right) = (1 - N(x)^{-s})^{-1}.$$

Let $|\mathcal{X}_p|_{\text{cl}}$ denote the set of closed points of $\mathcal{X}_p$. We find that

$$\zeta_p(X, s) = \prod_{x \in |\mathcal{X}_p|_{\text{cl}}} (1 - N(x)^{-s})^{-1}. \quad (13.7)$$

Example 13.5. Consider $X = \operatorname{Spec}(K)$ where $K/\mathbb{Q}$ is a finite extension. Let S be the set of primes ramified in K . Then $\operatorname{Spec} O_K[S^{-1}]$ is a smooth projective model for X over $\mathbb{Z}[S^{-1}]$. With (13.7), we see that

$$\zeta^S(\operatorname{Spec}(K), s) = \prod_{\mathfrak{p} \subset O_K, \mathfrak{p} \nmid S} (1 - N(\mathfrak{p})^{-s})^{-1} \quad (13.8)$$

is the Dedekind ζ -function of K , up to the finitely many Euler factors coming from ramified primes.

Example 13.6 (L -function of an elliptic curve). Finally, let us give an example that does not reduce to Dedekind ζ -functions. Let $E/\mathbb{Q}$ be an elliptic curve and let S be as in Definition 13.3. A concrete way to find S is as follows. Choose a Weierstrass equation $y^2 = x^3 + ax + b$ for E . Its discriminant is $\Delta = 4a^3 + 27b^2$. Let S be the set of all primes dividing the denominators of a and b or the numerator of 2Δ . Then

$$a, b \in \mathbb{Z}[S^{-1}] \quad \text{and} \quad \Delta \in \mathbb{Z}[S^{-1}]^{\times},$$

so $y^2 = x^3 + ax + b$ defines an elliptic curve $\mathcal{E}$ over $\mathbb{Z}[S^{-1}]$ (see Theorem 6.5 and the ensuing discussion). For each $p \notin S$ and every $q = p^n$, the finite abelian group $\mathcal{E}(\mathbb{F}_q)$ has cardinality estimated by

$$|\#\mathcal{E}(\mathbb{F}_q) - q - 1| \leq 2\sqrt{p} \quad (13.9)$$

which is a theorem of Hasse, see [22, Theorem V.1.1]. Define $\{a_p\}_{p \notin S}$ by

$$a_p := \#\mathcal{E}(\mathbb{F}_p) - p - 1$$

and set

$$L_p(E/\mathbb{Q}, s) = \frac{1}{1 - a_p p^{-s} + p^{-2s+1}}, \quad L^S(E/\mathbb{Q}, s) = \prod_{p \notin S} L_p(E/\mathbb{Q}, s).$$

This is the L -function of E , except that we have not defined the factors for the primes $p \in S$. The Hasse bound (13.9) ensures that $L^S(E/\mathbb{Q}, s)$ converges on the right half plane $\operatorname{Re}(s) > 3/2$. Then

$$\zeta^S(E, s) = \frac{\zeta^S(s) \cdot \zeta^S(s-1)}{L^S(E/\mathbb{Q}, s)}. \quad (13.10)$$

The Riemann ζ -function has a meromorphic continuation to the complex plane and satisfies a functional equation with respect to $s \longleftrightarrow 1 - s$. It is conjectured that this is a general property of Hasse–Weil ζ -functions:

Conjecture 13.7 (Hasse, Weil, see [20]). For every smooth projective variety $X/\mathbb{Q}$, the Hasse–Weil ζ -function $\zeta^S(X, s)$ admits a meromorphic continuation to all $s \in \mathbb{C}$ and satisfies a functional equation with symmetry $s \longleftrightarrow \dim(X) + 1 - s$.

13.3. Link with étale cohomology. A surprising property of the local factor $\zeta_p(X, s)$ is that it is independent of the choice of integral model at p . That is, for any two choices of smooth projective model $\mathcal{X}_1$ and $\mathcal{X}_2$ of X over $\mathbb{Z}_{(p)}$, we have $|\mathcal{X}_1(\mathbb{F}_q)| = |\mathcal{X}_2(\mathbb{F}_q)|$ for all $q = p^n$. This is explained by the relation of $\zeta_p(X, s)$ with étale cohomology.

First, we have a very general definition of q -Frobenius that applies to all $\mathbb{F}_q$ -schemes.

Definition 13.8. Let Y be an $\mathbb{F}_q$ -scheme. The q -Frobenius of Y is the $\mathbb{F}_q$ -scheme morphism $F_q : Y \rightarrow Y$ defined by the following two properties:

- (1) The map $|F_q| : |Y| \rightarrow |Y|$ on underlying topological spaces is the identity.
- (2) For every open $U \subseteq Y$, the pullback map on functions $F_q^* : \mathcal{O}_Y(U) \rightarrow \mathcal{O}_Y(U)$ is $f \mapsto f^q$.

In general, if $\phi : Y \rightarrow Y$ is a morphism of S -schemes, then we can define its *fixed points* subscheme $\text{Fix}(\phi) \subseteq Y$ by the Cartesian diagram

$$\begin{array}{ccc} \text{Fix}(\phi) & \longrightarrow & Y \\ \downarrow & & \downarrow \Delta_{Y/S} \\ Y & \xrightarrow{\Gamma_\phi} & Y \times_S Y. \end{array} \quad (13.11)$$

Here, $\Gamma_\phi = (\text{id}, \phi)$ is the graph of ϕ . By definition, for every S -scheme T , we have

$$\text{Fix}(\phi)(T) = \{y \in Y(T) \mid \phi(y) = y\}.$$

If $Y \rightarrow S$ is separated, then $\Delta_{Y/S}$ is a closed immersion, and hence $\text{Fix}(\phi) \rightarrow Y$ a closed immersion by pullback.

Coming back to $Y/\mathbb{F}_q$, let us also assume that Y is a smooth projective variety. Then

$$\text{Fix}(F_q) = Y(\mathbb{F}_q) \quad (13.12)$$

as the finite, reduced, discrete subschemes of Y that is given by all the $\mathbb{F}_q$ -points. Thus, counting the points $Y(\mathbb{F}_q)$ is the same problem as determining the number of fixed points of the q -Frobenius F_q . Recall next a result from topology, which we state for smooth manifolds for simplicity.

Theorem 13.9 (Lefschetz fixed point formula). *Let M be a smooth compact manifold and let $\phi : M \rightarrow M$ be a map with isolated fixed point in the sense that the intersection*

$$\Gamma_\phi \cap \Delta_M \subseteq M \times M$$

is transversal. Then each fixed point $\phi(x) = x$ has a well-defined intersection multiplicity $\text{mult}(\phi, x) \in \{\pm 1\}$, and

$$\sum_{x \in M, \phi(x)=x} \text{mult}(\phi, x) = \sum_{i=0}^{\dim(M)} (-1)^i \text{tr}(\phi^* \mid H^i(M, \mathbb{Q})).$$

That is, the number of fixed points of ϕ counted with signs agrees with the alternating sum of the traces of ϕ^* on the (finite-dimensional) cohomology groups $H^i(M, \mathbb{Q})$. An analogous result holds for étale cohomology groups. Applying it to the Frobenius map in (13.12) gives the following result.

Theorem 13.10 (Grothendieck–Lefschetz fixed point formula for $Y(\mathbb{F}_q)$). *Let $Y/\mathbb{F}_q$ be a smooth projective variety. Then*

$$|Y(\mathbb{F}_q)| = \sum_{i=0}^{2\dim(Y)} (-1)^i \operatorname{tr}\left(F_q^* \mid H_{\text{ét}}^i(Y_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)\right).$$

Finally, let us again consider a smooth projective variety X and a prime p such that X has a smooth projective model $\mathcal{X}_p/\mathbb{Z}_{(p)}$. In this situation, the étale cohomology of the special fiber $Y = \mathbb{F}_p \otimes_{\mathbb{Z}} \mathcal{X}_p$ agrees with the étale cohomology of X (proper smooth base change theorem). More precisely, there is an isomorphism of $\mathbb{Q}_\ell$ -vector spaces

$$\begin{array}{ccc} H_{\text{ét}}^i(X_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell) & \xrightarrow{\sim} & H_{\text{ét}}^i(\mathcal{X}_{p, \overline{\mathbb{F}}_p}, \mathbb{Q}_\ell) \\ \circlearrowleft & & \circlearrowleft \\ \operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \supset \operatorname{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) & \twoheadrightarrow & \operatorname{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \end{array} \quad (13.13)$$

which is equivariant for the action of any decomposition group $\operatorname{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$. Thus, if $\sigma_p \mapsto F_p$ denotes a Frobenius element in $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, then we obtain

$$|\mathcal{X}_p(\mathbb{F}_{p^n})| = \sum_{i=0}^{2\dim(X)} (-1)^i \operatorname{tr}\left((\sigma_p^n)^* \mid H_{\text{ét}}^i(X_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)\right). \quad (13.14)$$

Using elementary manipulations of power series, one can finally deduce the following expression for the ζ -function of X :

$$\zeta_p(X, s) = \prod_{i=0}^{2\dim(X)} \det\left(1 - \sigma_p \cdot t \mid H_{\text{ét}}^i(X_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)\right)^{(-1)^{i+1}} \Big|_{t=p^{-s}}. \quad (13.15)$$

This expression is purely in terms of X , showing independence of $\zeta^S(X, s)$ from the choice of integral models!

13.4. The Langlands–Kottwitz method. Finally, let us consider the Hasse–Weil ζ -function and (13.15) in the context of Shimura varieties. That is, $X = \operatorname{Sh}_K(G, X)$ is the canonical model of a Shimura variety over the reflex field $E = E(G, X)$. We may pretend $E = \mathbb{Q}$ at our current altitude of discussion, but see [15, §12] for a definition. Let us also assume that X is projective for simplicity. Then, there is a precise conjectural description of the alternating sum of the $H_{\text{ét}}^i(X_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)$ as Galois representation due to Kottwitz. An introduction to this conjecture can be found in [17]. For our current discussion, the only relevant point is that this description is in terms of summands of the form

$$\pi_{\text{fin}}^K \otimes [r_\mu \circ \operatorname{LC}(\pi)] \quad (13.16)$$

where π_{fin}^K denotes the K -invariants of an automorphic representation π of G , where $\operatorname{LC}(\pi)$ is the conjectural Langlands parameter of π (a Galois representation), and where r_μ is a representation constructed from the Shimura datum. In this way, via (13.15), there is a precise expectation for the ζ -function in terms of automorphic L -functions. This makes it possible to attack Conjecture 13.7 for Shimura varieties, the strategy for which goes back to Langlands and Kottwitz. Let us mention the main ideas going into this *Langlands–Kottwitz method*:

(1) Let us assume that the Shimura variety in question is of PEL type. That is, X has a concrete description as moduli space of abelian varieties with polarizations and endomorphisms, similar to what we saw in §11 and §12. Then, at almost all primes p , the integral model $\mathcal{X}_p$ can be constructed by simply extending this moduli description. The quantities $|\mathcal{X}_p(\mathbb{F}_q)|$ now acquire a very concrete meaning: they count the number of certain polarized abelian varieties with endomorphisms and level structure over $\mathbb{F}_q$.

(2) Next, one partitions the set $\mathcal{X}_p(\overline{\mathbb{F}_p})$ into isogeny classes. This is an equivalence relation which is weaker than isomorphism classes. The point is that the set of isogeny classes of abelian varieties over $\mathbb{F}_q$ has a group-theoretic description by Honda–Tate theory.

(3) Within each isogeny class, the number of $\mathbb{F}_{p^n}$ -points is counted by a linear combination of expressions of the form

$$\text{Orb}(\gamma, \mathbb{1}_{K^p}) \cdot \text{TOrb}(\delta, \phi_p) \quad (13.17)$$

where the first factor is a $G(\mathbb{A}_f^p)$ -orbital integral and the second factor a Frobenius-twisted $G(\mathbb{Q}_{p^n})$ -orbital integral.

(4) Finally, one hopes to compare the expressions of the form (13.17) with the orbital integral side of the Arthur–Selberg trace formula. This involves, among other things, a comparison of twisted orbital integrals with actual orbital integrals, which is the content of the Fundamental Lemma.

Let us come back to the case $G = \text{GL}_2$. We have already completed Step (1) by introducing the moduli space $\mathcal{M}_n$ of elliptic curves with level structure over $\mathbb{Z}[n^{-1}]$. The final goal of our course will be to carry out Steps (2) and (3) for $\mathcal{M}_n$.

14. ENDOMORPHISM RINGS

Recall that our overarching problem is to count the number of isomorphism classes of elliptic curves over $\mathbb{F}_q$ with level structure. To this end, we first need to gain a better understanding of the endomorphism rings of elliptic curves. The main result in this direction is that there are only three possibilities for $\text{End}(E)$:

- (1) $\text{End}(E) = \mathbb{Z}$, or
- (2) $\text{End}(E)$ is an order in an imaginary-quadratic extension $K/\mathbb{Q}$, or
- (3) $\text{End}(E)$ is an order in a definite quaternion algebra $B/\mathbb{Q}$. This can only happen in positive characteristic (Corollary 14.17).

14.1. Isogenies. Let us first explain why $\text{End}(E)$ has no zero-divisors.

Lemma 14.1. *Let $\phi : E_1 \rightarrow E_2$ be a homomorphism of elliptic curves over some field k . Then either $\phi = 0$, or ϕ is finite locally free (Definition 4.9).*

Proof. Morphisms between proper schemes are proper, so ϕ is proper. Hence $\phi(E_1)$ is closed. It is also connected since E_1 is connected. Since E_2 is irreducible and 1-dimensional, we find that either $\phi(E_1) = \{0\}$ or $\phi(E_1) = E_2$. In the first case $\phi = 0$ ($\star$). In the second case, ϕ is surjective.

Assume ϕ is surjective and let $x \in E_2$ be any closed point. Then $\phi^{-1}(x)$ is a proper closed subset of E_1 . It is hence a finite set of closed points. Moreover, we necessarily have $\phi^{-1}(\eta_2) = \{\eta_1\}$, where $\eta_i \in E_i$ denotes the generic point ($\dagger$). This shows that ϕ has finite fibers. In this situation, the following extremely useful statement applies and yields that ϕ is finite.

Proposition 14.2 ([23, Tag 02LS]). *Let $f : X \rightarrow S$ be a morphism of schemes. Then f is finite if and only if it is proper and has finite fibers.*

Thus $\phi_*(\mathcal{O}_{E_1})$ is a coherent $\mathcal{O}_{E_2}$ -module. It is torsion free because both E_1 and E_2 are integral and because the map $\mathcal{O}_{E_2, \eta_2} \rightarrow \mathcal{O}_{E_1, \eta_1}$ is injective. (This map is the field extension $\kappa(\eta_2) \rightarrow \kappa(\eta_1)$ defined by ϕ .) By the structure theorem for finitely generated modules over DVRs or Dedekind domains, $\phi_*(\mathcal{O}_{E_1})$ is locally free as $\mathcal{O}_{E_2}$ -module. $\square$

Exercise 14.3. Add details to the claims made in ($\star$) and ($\dagger$).

Corollary 14.4. *The ring $\text{End}(E)$ has no zero divisors. That is, if $\phi \circ \psi = 0$, then $\phi = 0$ or $\psi = 0$.*

Variant: Let $\phi : E_1 \rightarrow E_2$ and $\psi_1, \psi_2 : E_2 \rightarrow E_3$ be homomorphisms of elliptic curves. If $\psi_1 \circ \phi = \psi_2 \circ \phi$, then $\psi_1 = \psi_2$.

Proof. A composition of surjective maps is surjective. So if $\phi \neq 0$ and $\psi \neq 0$, then also $\phi \circ \psi \neq 0$. The variant is shown with the same argument. $\square$

Definition 14.5. (1) A non-zero homomorphism $\phi : E_1 \rightarrow E_2$ of elliptic curves is also called an *isogeny*. The *degree* of a homomorphism $\phi : E_1 \rightarrow E_2$ is defined as

$$\deg(\phi) = \begin{cases} 0 & \text{if } \phi = 0 \\ \text{rank of } \phi_*(\mathcal{O}_{E_1}) \text{ as } \mathcal{O}_{E_2}\text{-module} & \text{if } \phi \neq 0. \end{cases}$$

It is clear that $\deg(\phi \circ \psi) = \deg(\phi) \deg(\psi)$. By Corollary 6.3, $[n]$ is an isogeny with $\deg([n]) = n^2$.

(2) The space of *quasi-homomorphisms* from E_1 to E_2 , and the ring of *quasi-endomorphisms* of E are defined as the localizations

$$\text{Hom}^0(E_1, E_2) = \mathbb{Q} \otimes_{\mathbb{Z}} \text{Hom}(E_1, E_2), \quad \text{End}^0(E) = \mathbb{Q} \otimes_{\mathbb{Z}} (E). \quad (14.1)$$

That is, quasi-homomorphisms are formal fractions of the form $\phi = \frac{\phi_0}{n}$ with $n \in \mathbb{Z}_{\geq 1}$ and $\phi_0 \in \text{Hom}(E_1, E_2)$. By Corollary 14.4, we have¹¹

$$\frac{\phi_0}{n} = \frac{\psi_0}{m} \iff m\phi_0 = n\psi_0.$$

Quasi-homomorphism can be added and multiplied with all usual rules of fractional arithmetic. Moreover, the definition of degree extends multiplicatively to a map

$$\deg : \text{Hom}^0(E_1, E_2) \longrightarrow \mathbb{Q}, \quad \deg\left(\frac{\phi_0}{n}\right) = \frac{\deg(\phi_0)}{n^2}. \quad (14.2)$$

Non-zero quasi-homomorphisms are also called *quasi-isogenies*.

14.2. The dual isogeny. Consider a diagram of abelian groups with exact upper row

$$\begin{array}{ccccccc} 0 & \longrightarrow & K & \longrightarrow & M & \xrightarrow{\phi_1} & Q \longrightarrow 0 \\ & & & \searrow & \downarrow \phi_2 & \nearrow \psi & \\ & & & & N & & \end{array}$$

If ϕ_2 is a homomorphism with $\phi_2|_K = 0$, then there exists a unique map $\psi : Q \rightarrow N$ with $\phi_2 = \psi\phi_1$. We next study the same kind of factorization property for isogenies between elliptic curves. This is another example of a quotient property in algebraic geometry, similar to what we briefly saw around (8.21) and in §9.1. We also have an existence statement for said quotients.

Proposition 14.6 (see Proposition [12, Proposition 14.6]). (1) *Let E/S be an elliptic curve and let $K \subset E$ be an S -subgroup scheme that is finite locally free over S . Then there exists an elliptic curve E/K (the quotient) together with an isogeny $\phi : E \rightarrow E/K$ such that $\ker(\phi) = K$ (the quotient map).*

¹¹Note that $m\phi_0$ is the same as $[m] \circ \phi_0$. Both notations will be used interchangeably.

(2) Let E , E_1 and E_2 be elliptic curves over S . Let $\phi_1 : E \rightarrow E_1$ and $\phi_2 : E \rightarrow E_2$ be isogenies that fit into the diagram

$$\begin{array}{ccccc} \ker(\phi_1) & \longrightarrow & E & \xrightarrow{\phi_1} & E_1 \\ & \searrow 0 & \downarrow \phi_2 & \swarrow \psi & \\ & & E_2 & & \end{array}$$

In other words, assume $\ker(\phi_1) \subseteq \ker(\phi_2)$. Then there exists a unique isogeny $\psi : E_1 \rightarrow E_2$ such that $\phi_2 = \psi\phi_1$.

Corollary 14.7. *Let $\phi : E_1 \rightarrow E_2$ be an isogeny of degree n . Then there exists a unique isogeny $\phi^* : E_2 \rightarrow E_1$, called its dual, such that*

$$\phi^* \circ \phi = [n]_{E_1}, \quad \phi \circ \phi^* = [n]_{E_2}.$$

Proof. The kernel $\ker(\phi)$ is a finite group scheme of order equal to $\deg(\phi)$. By Deligne's Theorem (Theorem 4.10), $\ker(\phi)$ is n -torsion. This means $\ker(\phi) \subseteq E[n]$. Proposition 14.6 implies that there exists an isogeny ϕ^* with $\phi^* \circ \phi = [n]_{E_1}$. Then

$$\phi \circ \phi^* \circ \phi = \phi \circ [n]_{E_1} \stackrel{(\star)}{=} [n]_{E_2} \circ \phi$$

where $(\star)$ holds because $[n]$ commutes with every group homomorphism. The cancellation law from Lemma 14.4 implies that also $\phi \circ \phi^* = [n]_{E_2}$. $\square$

It is clear that $(\phi \circ \psi)^* = \psi^* \circ \phi^*$ and that $[n]^* = [n]$. In this way, taking the dual extends multiplicatively to a well-defined map

$$\mathrm{Hom}^0(E_1, E_2) \longrightarrow \mathrm{Hom}^0(E_2, E_1), \quad \frac{\phi_0}{n} \longmapsto \frac{\phi_0^*}{n}. \quad (14.3)$$

Corollary 14.8. *Let E be an elliptic curve over some field k . Then $\mathrm{End}^0(E)$ is a skew-field of characteristic 0.*

In general, every non-zero quasi-homomorphism $\phi \in \mathrm{Hom}^0(E_1, E_2)$ has a (unique) inverse $\phi^{-1} \in \mathrm{Hom}^0(E_2, E_1)$.

Proof. Skew-field means that every quasi-isogeny $\phi : E \rightarrow E$ has a quasi-isogeny inverse. Both for $\mathrm{End}^0(E)$ and $\mathrm{Hom}^0(E_1, E_2)$ this follows from Corollary 14.7: the inverse to ϕ is $(\deg(\phi))^{-1} \cdot \phi^*$. Being of characteristic 0 just says that $n\phi \neq 0$ for all $n \in \mathbb{Z}_{\geq 1}$ and $\phi \neq 0$, which was already explained before. $\square$

14.3. The structure of $\mathrm{End}^0(E)$. For every pair of elliptic curves E_1 and E_2 , we have just constructed a bijection

$$* : \mathrm{Hom}(E_1, E_2) \longrightarrow \mathrm{Hom}(E_2, E_1)$$

with the properties

$$*^2 = \mathrm{id}, \quad (\phi \circ \psi)^* = \psi^* \circ \phi^*, \quad \phi^* \circ \phi = [\deg(\phi)].$$

It is called the *Rosati involution*. We will next cite an important result stating that $*$ is also additive. This is not obvious from the definition and represents a non-trivial fact about the structure of elliptic curves.

Theorem 14.9 ([12, Theorem 10.8]). *The Rosati involution is a group homomorphism, meaning*

$$(\phi + \psi)^* = \phi^* + \psi^*.$$

The existence of $*$ has several consequences, which we next discuss in detail. Everything in the following will be purely arithmetic (no algebraic geometry involved). Let E be an elliptic curve over a field k . Recall that $\text{End}(E)$ has no zero-divisors and is $\mathbb{Z}$ -torsion free (Corollary 14.8). By Theorem 14.9, the Rosati involution on $\text{End}(E) = \text{Hom}(E, E)$ is an involution in the sense that

$$*^2 = \text{id}, \quad (x + y)^* = x^* + y^*, \quad (xy)^* = y^*x^*.$$

These three properties give the usual definition of an involution on a (not necessarily commutative) ring.

Proposition 14.10. *The following properties hold.*

(1) *For every $x \in \text{End}(E)$, the endomorphism $\text{tr}(x) = x + x^*$ lies in $\mathbb{Z}$. It is called the trace of x . In particular, the subring $\mathbb{Z}[x] \subseteq \text{End}(E)$ generated by x is stable under $*$.*

(2) *If $x \notin \mathbb{Z}$, then $\mathbb{Z}[x]$ is an order in an imaginary-quadratic extension of $\mathbb{Q}$. In particular, $\text{End}(E)^{*= \text{id}} = \mathbb{Z}$.*

(3) *More precisely, x is a zero of the following quadratic equation with coefficients in $\mathbb{Z}$:*

$$(T - x)(T - x^*) = T^2 - \text{tr}(x)T + \deg(x). \quad (14.4)$$

If $x \notin \mathbb{Z}$, then $\text{tr}(x)^2 - 4\deg(x) < 0$, so $\mathbb{Q}(x)$ is an imaginary-quadratic extension.

Proof. Let $x, y \in \text{End}(E)$ be two endomorphisms. By Theorem 14.9,

$$\begin{aligned} \deg(x + y) &= (x + y)^*(x + y) \\ &= x^*x + (x^*y + y^*x) + y^*y \\ &= \deg(x) + \text{tr}(xy^*) + \deg(y). \end{aligned} \quad (14.5)$$

Since $\deg(x + y)$, $\deg(x)$ and $\deg(y)$ all lie in $\mathbb{Z}$, we obtain that $\text{tr}(y^*x) \in \mathbb{Z}$. Apply this with $y = 1$ (meaning $y = \text{id}_E$) to obtain that $\text{tr}(x) \in \mathbb{Z}$. Then $x^* = \text{tr}(x) - x \in \mathbb{Z}[x]$. This implies that $\mathbb{Z}[x]$ is $*$ -stable and we have proved (1).

Next, it is clear that x is a zero of (14.4). Namely,

$$x^2 - (x + x^*)x + x^*x = x^2 - x^2 - x^*x + x^*x = 0.$$

For every $p/q \in \mathbb{Q}$, $q \neq 0$, we have

$$\left(\frac{p}{q}\right)^2 - \text{tr}(x) \left(\frac{p}{q}\right) + \deg(x) = \frac{\deg(p - qx)}{q^2} \geq 0.$$

So $\lambda^2 - \text{tr}(x)\lambda + \deg(x) \geq 0$ for every $\lambda \in \mathbb{R}$, which means that either $T^2 - \text{tr}(x)T + \deg(x)$ has a double zero in $\mathbb{R}$ or defines an imaginary-quadratic extension of $\mathbb{Q}$. The first case happens if and only if $x = x^*$. Then $2x = \text{tr}(x) \in \mathbb{Z}$ and hence $x \in \mathbb{Z}$. But x is also integral over $\mathbb{Z}$ because it is a zero of $T^2 - \text{tr}(x)T + \deg(x)$, and so even $x \in \mathbb{Z}$. This proves (2) and (3). $\square$

Corollary 14.11. *Assume that $\text{End}(E)$ is commutative. Then $K := \text{End}^0(E)$ equals $\mathbb{Q}$ or an imaginary-quadratic extension of $\mathbb{Q}$.*

Proof. If $\text{End}(E)$ is commutative, then K is a field by Corollary 14.8. Then $*$ $\in \text{Aut}(K)$ is a field automorphism. Moreover, Proposition 14.10 established that $K^{*= \text{id}} = \mathbb{Q}$. So K is either $\mathbb{Q}$ or a quadratic extension. By Proposition 14.10 again, such a quadratic extension has to be imaginary. $\square$

Corollary 14.12. *Assume that $\text{End}(E)$ is non-commutative. Then $B := \text{End}^0(E)$ is a quaternion algebra over $\mathbb{Q}$ and the Rosati involution equals the main involution of B . By definition this means that there exist $a, b \in \mathbb{Q}^\times$ as well as $i, j, k \in B$ such that*

$$B = \mathbb{Q} \oplus \mathbb{Q}i \oplus \mathbb{Q}j \oplus \mathbb{Q}k \quad (14.6)$$

with

$$i^2 = a, \quad j^2 = b, \quad ij = -ji = k$$

and

$$i^* = -i, \quad j^* = -j, \quad k^* = -k.$$

Moreover, B is definite in the sense that

$$\mathbb{R} \otimes_{\mathbb{Q}} B \cong \mathbb{H} \quad (\text{Hamilton quaternions}).$$

Proof. Since B is non-commutative, $B \neq \mathbb{Q}$. Let $z \in B \setminus \mathbb{Q}$ and set $K = \mathbb{Q}(z)$. This is a quadratic extension of $\mathbb{Q}$. The Rosati involution preserves K by Proposition 14.10 and hence agrees with the Galois conjugation of $K/\mathbb{Q}$. Choose $i \in K^\times$ with $i^* = -i$ and set $a = i^2 \in \mathbb{Q}^\times$.

Next, B can be viewed as K -vector space via left-multiplication or via right-multiplication. In this way, B becomes a $K \otimes_{\mathbb{Q}} K$ -module,

$$(x \otimes y) \cdot \alpha := x\alpha y.$$

As ring, $K \otimes_{\mathbb{Q}} K \cong K \times K$, so we obtain a decomposition $B = B_0 \times B_1$ with

$$B_0 = \{\alpha \in B \mid x\alpha = \alpha x \ \forall x \in K\}, \quad B_1 = \{\alpha \in B \mid x\alpha = \alpha x^* \ \forall x \in K\}. \quad (14.7)$$

For every $\alpha \in B_0$, the ring $K[\alpha]$ is a finite commutative $\mathbb{Q}$ -algebra without zero-divisors, meaning a finite field extension. Since $*$ preserves $K[\alpha]$ (Proposition 14.10) and satisfies $K[\alpha]^{*=id} = \mathbb{Q}$, it has to be a quadratic extension of $\mathbb{Q}$. So we see that $K[\alpha] = K$, and hence that $B_0 = K$.

Our assumption is that B is non-commutative. So $K \subsetneq B$, meaning $B_1 \neq 0$. For any two non-zero $j_1, j_2 \in B_1$, the definition of B_1 by (14.7) implies that $j_1 j_2 \in K$. This means that B_1 is a 1-dimensional K -vector space via left-multiplication. Let $j \in B_1$ be any generator and set $k = ij$. Then

$$ji \stackrel{\text{Def. of } B_1}{=} (i^*)j = -ij$$

as required. Consider next the Rosati involution. For $x \in K$, we have

$$(j^*)x = ((x^*)j)^* = (jx)^* = (x^*)(j^*)$$

which means that $j^* \in B_1$. Write $j^* = cj$ with $c \in K$. Then

$$j = (j^*)^* = (cj)^* = j^* c^* = c j c^* \stackrel{j \in B_1}{=} c^2 j$$

implies that $c^2 = 1$. Since $B^{*=id} = \mathbb{Q}$ by Proposition 14.10, the case $c = 1$ is excluded. This means $c = -1$. It is clear that also $k^* = -k$ because

$$(ij)^* = j^* i^* = -j i^* = -ij.$$

Moreover, $j^* = -j$ implies that

$$b := j^2 = -j^* j \in \mathbb{Q}^\times.$$

At this point, we have shown that B is a quaternion division algebra with generators and relations as in (14.6). It is only left to prove that $\mathbb{R} \otimes_{\mathbb{Q}} B \cong \mathbb{H}$. Observe for this that $a, b < 0$ because $\mathbb{Q}(i)$ and $\mathbb{Q}(j)$ have to be imaginary quadratic extensions of $\mathbb{Q}$ (Proposition 14.10). Then

$$i' = \sqrt{-a} \otimes i, \quad j' = \sqrt{-b} \otimes j, \quad k' = i' j'$$

give a standard Hamilton quaternion basis for $\mathbb{R} \otimes_{\mathbb{Q}} B$. □

Corollary 14.13. (1) For any two elliptic curves E_1 and E_2 , $\text{Hom}^0(E_1, E_2)$ is a finite-dimensional $\mathbb{Q}$ -vector space of dimension 0, 1, 2 or 4.

(2) In general, $\text{Hom}(E_1, E_2)$ is a lattice in $\text{Hom}^0(E_1, E_2)$.

Proof. (1) We can consider $\mathrm{Hom}(E_1, E_2)$ as an $\mathrm{End}(E_1)$ -right module or $\mathrm{End}(E_2)$ -left module by composition. The same applies after $\mathbb{Q} \otimes_{\mathbb{Z}} -$. We choose one of the two possibilities and view $\mathrm{Hom}^0(E_1, E_2)$ as an $\mathrm{End}^0(E_2)$ -vector space in the following.

If $\phi_1, \phi_2 : E_1 \rightarrow E_2$ are two non-zero homomorphisms, then $\phi_2 \circ (\phi_1)^{-1}$ lies in $\mathrm{End}^0(E_2)$. This shows that $\mathrm{Hom}^0(E_1, E_2)$ is at most one-dimensional over $\mathrm{End}^0(E_2)$. We know from Corollaries 14.11 and 14.12 that

$$\dim_{\mathbb{Q}}(\mathrm{End}^0(E_2)) \in \{1, 2, 4\},$$

so we deduce that the $\mathbb{Q}$ -dimension of $\mathrm{Hom}^0(E_1, E_2)$ is 0, 1, 2 or 4 as claimed.

(2) Consider the function

$$\begin{aligned} \mathrm{Hom}^0(E_1, E_2) &\longrightarrow \mathbb{Q}_{\geq 0} \\ \phi &\longmapsto \deg(\phi). \end{aligned} \tag{14.8}$$

We obviously have $\phi(qx) = q^2\phi(x)$. But Theorem 14.9 implies something stronger, namely that $\deg$ is a quadratic form: Let $x_1, \dots, x_r$ be a $\mathbb{Q}$ -basis of $\mathrm{Hom}^0(E_1, E_2)$ and define

$$b_{ii} = \deg(x_i) \quad \text{and} \quad b_{ij} = x_i x_j^* + x_j^* x_i.$$

Then $B = (b_{ij})$ is a symmetric $(r \times r)$ -matrix with entries in $\mathbb{Z}$ by the argument from (14.5), and for every tuple of $q_i \in \mathbb{Q}$,

$$\deg(q_1 x_1 + \dots + q_r x_r) = (q_1, \dots, q_r) \cdot B \cdot (q_1, \dots, q_r)^t$$

by Theorem 14.9. Since $\deg(\phi)$ only takes positive values, B is a positive definite symmetric bilinear form. The ball

$$\{x \in \mathbb{R} \otimes_{\mathbb{Z}} \mathrm{Hom}(E_1, E_2) \mid B(x, x) \leq 1/2\}$$

is a neighborhood of $\{0\}$ which does not contain any other points from $\mathrm{Hom}(E_1, E_2)$. This shows that $\mathrm{Hom}(E_1, E_2)$ is discrete, hence a lattice in $\mathrm{Hom}^0(E_1, E_2)$. $\square$

14.4. The Tate module. Our final goal in this section is to get more information about the non-commutative case.

Definition 14.14. Let E be an elliptic curve over a field k and let ℓ be a prime different from $\mathrm{char}(k)$. The ℓ -adic Tate module of E is defined as

$$T_{\ell}(E) := \varprojlim_{n \geq 1} E[\ell^n](\bar{k}).$$

The transition maps here are given by multiplication by ℓ ,

$$[\ell] : E[\ell^{n+1}] \longrightarrow E[\ell^n].$$

By Theorems 6.7 and 4.10 (2), $T_{\ell}(E)$ is free of rank 2 as $\mathbb{Z}_{\ell}$ -module. We also set $V_{\ell}(E) = \mathbb{Q}_{\ell} \otimes_{\mathbb{Z}_{\ell}} T_{\ell}(E)$ which is a 2-dimensional $\mathbb{Q}_{\ell}$ -vector space. It is called the *rational Tate module* of E .

Any homomorphism $\phi : E_1 \rightarrow E_2$ restricts to a compatible family of homomorphisms $\phi[\ell^n] : E_1[\ell^n] \rightarrow E_2[\ell^n]$ between ℓ^n -torsion group schemes, and so defines a $\mathbb{Z}_{\ell}$ -linear map $T_{\ell}(\phi) : T_{\ell}(E_1) \rightarrow T_{\ell}(E_2)$. In other words, $T_{\ell}(-)$ is a covariant functor from elliptic curves over k to $\mathbb{Z}_{\ell}$ -modules.

Theorem 14.15. *Let E_1 and E_2 be elliptic curves over k . Then the natural map*

$$\mathbb{Z}_{\ell} \otimes_{\mathbb{Z}} \mathrm{Hom}(E_1, E_2) \longrightarrow \mathrm{Hom}(T_{\ell}(E_1), T_{\ell}(E_2))$$

is injective.

Remark 14.16. Note that this is stronger than just saying that the map from $\mathrm{Hom}(E_1, E_2)$ to $\mathrm{Hom}(T_{\ell}(E_1), T_{\ell}(E_2))$ is injective. For example, $V = \mathbb{Q}(\sqrt{2}) \subset \mathbb{R}$ is a 2-dimensional $\mathbb{Q}$ -subvector space, but the induced map $\mathbb{R} \otimes_{\mathbb{Q}} V \rightarrow \mathbb{R}$ is not injective. The element $\sqrt{2} \otimes 1 - 1 \otimes \sqrt{2}$ lies in the kernel.

Proof of Theorem 14.15. First observe that there is an isomorphism (natural in E)

$$\begin{aligned} T_\ell(E)/\ell^n T_\ell(E) &\xrightarrow{\sim} E[\ell^n](\bar{k}) \\ (\dots, x_3, x_2, x_1) &\longmapsto x_n. \end{aligned} \quad (14.9)$$

Also note that because $L = \text{Hom}(E_1, E_2)$ is finitely generated by Corollary 14.13, i.e. abstractly isomorphic to $\mathbb{Z}^r$ for some $r \geq 1$, the tensor product $\mathbb{Z}_\ell \otimes_{\mathbb{Z}} L$ coincides with the ℓ -adic completion L_ℓ of L . Consider some $\tilde{\phi}$ that lies in the kernel of $L_\ell \rightarrow \text{Hom}(T_\ell(E_1), T_\ell(E_2))$. We can assume that $\tilde{\phi}$ is not divisible by ℓ in L_ℓ . Let $\phi \in L$ be an approximation in the sense that $\phi - \tilde{\phi} \in \ell L_\ell$. But then both $\tilde{\phi}$ and $\phi - \tilde{\phi}$ restrict to zero on $E_1[\ell]$. Proposition 14.6 then implies that ϕ is divisible by ℓ in L . Then $\tilde{\phi}$ is divisible by ℓ in L_ℓ — contradiction! $\square$

We can now complete our classification of $\text{End}^0(E)$.

Corollary 14.17. *Assume that $\text{End}(E)$ is non-commutative as in Corollary 14.12; set $B = \text{End}^0(E)$. Then for every $\ell \neq \text{char}(k)$,*

$$\mathbb{Q}_\ell \otimes_{\mathbb{Q}} B \cong M_2(\mathbb{Q}_\ell).$$

In particular, the characteristic $p = \text{char}(k)$ is necessarily positive and B is the unique (up to isomorphism) quaternion algebra over $\mathbb{Q}$ that is non-split at p and ∞ .

Proof. For every prime $\ell \neq \text{char}(k)$, consider the action of $B_\ell = \mathbb{Q}_\ell \otimes_{\mathbb{Q}} B$ on the rational Tate module $V_\ell(E)$. By Theorem 14.15, this action is faithful, meaning induced from an injective map of $\mathbb{Q}_\ell$ -algebras

$$B_\ell \longrightarrow \text{End}_{\mathbb{Q}_\ell}(V_\ell(E)) \cong M_2(\mathbb{Q}_\ell). \quad (14.10)$$

Both sides here are 4-dimensional $\mathbb{Q}_\ell$ -vector spaces, so this map is an isomorphism. We moreover know from Corollary 14.12 that $\mathbb{R} \otimes_{\mathbb{Q}} B$ is isomorphic to the Hamilton quaternions. In particular, $\mathbb{R} \otimes_{\mathbb{Q}} B \not\cong M_2(\mathbb{R})$.

At this point, we use the classification of quaternion algebras over a number field F . It states that

$$\begin{aligned} \left\{ \begin{array}{c} \text{Isom. classes of} \\ \text{quaternion algebras over } F \end{array} \right\} &\xrightarrow{\sim} \{ \text{finite } \Sigma \subset \{ \text{places of } F \} \text{ s.t. } |\Sigma| \text{ even} \} \\ B &\longmapsto \{ v \mid F_v \otimes_F B \not\cong M_2(F_v) \}. \end{aligned} \quad (14.11)$$

Applied to $B = \text{End}^0(E)$ as before, we already know that $\mathbb{R} \otimes_{\mathbb{Q}} B \not\cong M_2(\mathbb{R})$, so there has to be at least one prime number p such that $\mathbb{Q}_p \otimes_{\mathbb{Q}} B \not\cong M_2(\mathbb{Q}_p)$. Since we have (14.10) for all $\ell \neq \text{char}(k)$, the only possibility is $p = \text{char}(k) > 0$. Then (14.11) and (14.10) pin down B as the unique quaternion algebra (up to isomorphism) that is non-split precisely at $\{p, \infty\}$. $\square$

15. THE HONDA–TATE THEOREM

We next discuss the Honda–Tate Theorem, which classifies isogeny classes of elliptic curves over finite fields.

15.1. The Frobenius. Let $\phi : E_1 \rightarrow E_2$ be an isogeny (or quasi-isogeny) of elliptic curves. Given a quasi-endomorphism $x \in \text{End}^0(E_1)$, we can transport it to E_2 as $\phi x \phi^{-1}$. The transported element is the unique quasi-endomorphism of E_2 that makes the following

square commute.

$$\begin{array}{ccc} E_1 & \xrightarrow{\phi} & E_2 \\ x \downarrow & & \downarrow \phi x \phi^{-1} \\ E_1 & \xrightarrow{\phi} & E_2. \end{array} \quad (15.1)$$

Note that even if ϕ and x are actual homomorphisms, $\phi x \phi^{-1}$ is often just a quasi-endomorphism because $\phi^{-1} = \phi^* / \deg(\phi)$ introduces a denominator. We are interested in the following, very special endomorphism.

Lemma 15.1. *Let $\phi : X \rightarrow Y$ be a morphism of $\mathbb{F}_q$ -schemes. Let F_X and F_Y denote the q -Frobenii of X and Y . Then*

$$F_Y \circ \phi = \phi \circ F_X.$$

Proof. Recall from Definition 13.8 that the Frobenius is the identity on underlying topological spaces. So obviously $|F_Y \circ \phi| = |\phi \circ F_X|$. For every open $U \subseteq Y$ and every function $h \in \mathcal{O}_Y(U)$, we have

$$\phi^*(F_Y^*(h)) = \phi^*(h^q) = \phi^*(h)^q = F_X^*(\phi^*(h)).$$

□

The point here was that raising to the q -th power commutes with every ring homomorphism and hence defines an endomorphism of the whole category of $\mathbb{F}_q$ -algebras.

Corollary 15.2. (1) *Let E be an elliptic curve over $\mathbb{F}_q$. The q -Frobenius of E , denoted by π_E , defines an endomorphism $\pi_E \in \text{End}(E)$.*

(2) *Assume that $\phi : E_1 \rightarrow E_2$ is a quasi-isogeny of elliptic curves over $\mathbb{F}_q$. Then $\phi \pi_{E_1} \phi^{-1} = \pi_{E_2}$.*

Proof. For statement (1), we need to check that π_E is a group scheme endomorphism. Recall that this means $\pi_E \circ m_E = m_E \circ (\pi_E \times \pi_E)$, where m_E denotes the multiplication map $m_E : E \times_{\text{Spec } \mathbb{F}_q} E \rightarrow E$. This commutativity is a special case of Lemma 15.1. Claim (2) follows in a similar way from that lemma. □

Definition 15.3. Let E be an elliptic curve over a field k and let $x \in \text{End}(E)$ be an endomorphism. The *characteristic polynomial* of x is defined as the polynomial from (14.4):

$$\text{char}(x; T) := T^2 - \text{tr}(x)T + \deg(x).$$

Recall that $\text{tr}(x) = x + x^*$ lies in $\mathbb{Z}$. The definition also applies to quasi-endomorphisms; the coefficients then lie in $\mathbb{Q}$.

Lemma 15.4. *Let $\phi : E_1 \rightarrow E_2$ be an isogeny and let $x \in \text{End}^0(E)$ a quasi-endomorphism. Then*

$$\text{char}(\phi x \phi^{-1}; T) = \text{char}(x; T).$$

Proof. We need to show that $\phi x \phi^{-1}$ has the same trace and degree as x . This is easy to see from definitions:

$$\deg(\phi x \phi^{-1}) = \deg(\phi) \deg(x) \deg(\phi)^{-1} = \deg(x)$$

$$\begin{aligned} \text{tr}(\phi x \phi^{-1}) &= \phi x \phi^{-1} + (\phi x \phi^{-1})^* \\ &= \phi x \phi^{-1} + (\phi x \phi^* / \deg(\phi))^* \\ &= \phi x \phi^{-1} + \phi x^* \phi^* / \deg(\phi) \\ &= \phi(x + x^*) \phi^{-1} \\ &= x + x^* \end{aligned}$$

where the last equality holds because $x + x^*$ lies in $\mathbb{Q}$ and hence commutes with all quasi-homomorphisms. $\square$

Let us also calculate the degree of the Frobenius endomorphism.

Lemma 15.5. *Let $E/\mathbb{F}_q$ be an elliptic curve and let π_E denote the q -Frobenius of E . Then $\deg(\pi_E) = q$.*

Proof. Elliptic curves are smooth and one-dimensional, so the local ring $R = \mathcal{O}_{E,e}$ at the identity element is a discrete valuation ring. Its residue field is $\mathbb{F}_q$. Let $t \in R$ denote a uniformizer. The kernel of π_E is then calculated as the spectrum of

$$R \otimes_{r^q \leftarrow r, R} R/(t) \cong R/(t^q)$$

which has dimension q as $\mathbb{F}_q$ -vector space. $\square$

Let us sum up the whole discussion. Every elliptic curve $E/\mathbb{F}_q$ has a naturally defined Frobenius endomorphism π_E . Its degree is q , so its characteristic polynomial has the form

$$T^2 - \text{tr}(\pi_E)T + q.$$

This polynomial only depends on E up to isogeny. That is, if there exists an isogeny $E_1 \rightarrow E_2$, then

$$\text{tr}(\pi_{E_1}) = \text{tr}(\pi_{E_2}).$$

Moreover, by Proposition 14.10 (3), the characteristic polynomial is positive semi-definite, meaning that the discriminant $\text{tr}(\pi_E)^2 - 4\deg(\pi_E)$ is ≤ 0 . That is,

$$-2\sqrt{q} \leq \text{tr}(\pi_E) \leq 2\sqrt{q}. \quad (15.2)$$

In this way, we have constructed a map

$$\left\{ \begin{array}{l} \text{Isogeny classes of} \\ \text{ellipt. curves over } \mathbb{F}_q \end{array} \right\} \longrightarrow \left\{ \begin{array}{l} \text{Quadratic integral polynomials} \\ T^2 - aT + q \text{ with } |a| \leq 2\sqrt{q} \end{array} \right\}. \quad (15.3)$$

Theorem 15.6 (Honda–Tate classification). *The map (15.3) is injective. Its image is precisely those polynomials $T^2 - aT + q$ that satisfy one of the following two conditions:*

- (1) $p \nmid a$. In this case, E is ordinary.
- (2) $p \mid a$ but p is non-split in $\mathbb{Q}[T]/(T^2 - aT + q)$. In this case, E is supersingular.

Remark 15.7. Let $\overline{\mathbb{Q}} \subset \mathbb{C}$ denote the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. An algebraic integer $\alpha \in \overline{\mathbb{Q}}$ is called a *Weil q -number* if, for every $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, $|\sigma(\alpha)| = q^{1/2}$. The zeroes of a polynomial $T^2 - aT + q$ with $|a| \leq 2\sqrt{q}$ are Weil q -numbers. Theorem 15.6 generalizes to a bijection

$$\left\{ \begin{array}{l} \text{Isogeny classes of simple} \\ \text{abelian varieties over } \mathbb{F}_q \end{array} \right\} \xrightarrow{\sim} \left\{ \begin{array}{l} \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})\text{-orbits of} \\ \text{Weil } q\text{-numbers} \end{array} \right\} \quad (15.4)$$

$$A \longmapsto \text{zeroes of } \text{char}(\pi_A; T).$$

Requiring α that is the zero of a polynomial of the form $T^2 - aT + q$ with $|a| \leq 2\sqrt{q}$, and adding condition (1) or (2) from Theorem 15.6 singles out those quadratic Weil q -numbers which correspond to elliptic curves.

Proposition 15.8. *Let E be an elliptic curve over $\mathbb{F}_q$ with Frobenius π_E . Then*

$$|E(\mathbb{F}_{q^n})| = 1 - \text{tr}(\pi_E^n) + q^n.$$

Proof. The $\mathbb{F}_{q^n}$ -points $E(\mathbb{F}_{q^n})$ are precisely the q^n -Frobenius fixed points in $E(\overline{\mathbb{F}_q})$. These are nothing but the $\overline{\mathbb{F}_q}$ -points of $\ker(1 - \pi_E^n)$. The Frobenius map is zero on $\text{Lie}(E)$, so

$1 - \pi_E^n$ is the identity on $\text{Lie}(E)$. This means that $\ker(1 - \pi_E^n)$ is a reduced scheme, which yields

$$\begin{aligned} |E(\mathbb{F}_{q^n})| &= \deg(1 - \pi_E^n) \\ &= (1 - \pi_E^n)(1 - \pi_E^n)^* \\ &= 1 - \text{tr}(\pi_E^n) + \deg(\pi_E)^n. \end{aligned}$$

Now use that $\deg(\pi_E) = q$ (Lemma 15.5). $\square$

Corollary 15.9. *Let E , E_1 and E_2 be elliptic curves over $\mathbb{F}_q$.*

(1) *The number of $\mathbb{F}_q$ -points satisfies the Hasse bound*

$$|E(\mathbb{F}_q) - q - 1| \leq 2\sqrt{q}.$$

(2) *The cardinality of $E(\mathbb{F}_q)$ determines the cardinalities of $E(\mathbb{F}_{q^n})$ for all $n \geq 1$.*

(3) *The two curves E_1 and E_2 are isogeneous if and only if $|E_1(\mathbb{F}_q)| = |E_2(\mathbb{F}_q)|$.*

Proof. (1) Use Proposition 15.8 and the bound $|\text{tr}(\pi_E)| \leq 2\sqrt{q}$ from (15.2).

(2) In general, the characteristic polynomial of an endomorphism f will uniquely determine the characteristic polynomials of all powers f^n . In the case at hand, we have the recursive relation (with $\pi = \pi_E$)

$$\begin{aligned} \text{tr}(\pi^{n-1})\text{tr}(\pi) &= \text{tr}(\pi^n) + \pi^{n-1}\pi^* + (\pi^*)^{n-1}\pi \\ &= \text{tr}(\pi^n) + \text{tr}(\pi^{n-2}) \cdot q. \end{aligned} \tag{15.5}$$

By Proposition 15.8, knowing $|E(\mathbb{F}_q)|$ is equivalent to knowing $\text{tr}(\pi)$, which by (15.5) determines all $\text{tr}(\pi^n)$ and then all $|E(\mathbb{F}_{q^n})|$.

(3) By the Honda–Tate classification (Theorem 15.6), E_1 and E_2 are isogeneous if and only if $\text{tr}(\pi_{E_1}) = \text{tr}(\pi_{E_2})$, which by Proposition 15.8 is equivalent to $|E_1(\mathbb{F}_q)| = |E_2(\mathbb{F}_q)|$. $\square$

15.2. Properties of π_E . Proving Theorem 15.6 is not so easy and beyond our course. But we can at least understand where the two conditions on $\text{tr}(\pi_E)$ in Theorem 15.6 come from.

Lemma 15.10. *Let E be an elliptic curve over a field k of characteristic 0. Then either $E[p](\bar{k}) = \{0\}$ or $E[p](\bar{k}) \cong \mathbb{Z}/(p)$.*

Proof. We know that $E[p]$ is a finite k -group scheme which is p -torsion and of order p^2 . So, up to isomorphism,

$$E[p](\bar{k}) \in \{0, \mathbb{Z}/(p), (\mathbb{Z}/(p))^2\}.$$

However, $[p] : E \rightarrow E$ induces multiplication by p on $\text{Lie}(E)$. This is a general principle: n -multiplication $[n] : G \rightarrow G$ on a commutative group scheme G induces multiplication by n on $\text{Lie}(G)$. So $E[p]$ cannot be reduced, which implies $|E[p](\bar{k})| < \deg(E[p])$. $\square$

Definition 15.11. E is called *ordinary* if $|E[p](\bar{k})| = p$. Otherwise, we have $E[p](\bar{k}) = \{0\}$ and E is called *supersingular*.

Note that for every power $q = p^n$, the q -torsion $E[q]$ is a successive extension of copies of $E[p]$. It follows that

$$E[q](\bar{k}) \cong \begin{cases} \mathbb{Z}/(q) & \text{if } E \text{ ordinary} \\ \{0\} & \text{if } E \text{ supersingular.} \end{cases} \tag{15.6}$$

An equivalent way to describe the situation is as follows. We can consider the local ring $\mathcal{O}_{E[p],e}$ of $E[p]$ at the identity element. It is a quotient of the DVR $\mathcal{O}_{E,e}$ and hence isomorphic to $k[\varepsilon]/\varepsilon^m$ for some integer m . We also have

$$\deg(E[p]) = \dim_k(\mathcal{O}_{E[p],e}) \cdot |E[p](\bar{k})|$$

because the local rings of $E[p]_{\bar{k}}$ at the various closed points $E[p](\bar{k})$ are all isomorphic by a translation argument using the group structure. So we deduce

$$\mathcal{O}_{E[q],e} \cong \begin{cases} k[\varepsilon]/(\varepsilon^q) & \text{if } E \text{ ordinary} \\ k[\varepsilon]/(\varepsilon^{q^2}) & \text{if } E \text{ supersingular.} \end{cases} \quad (15.7)$$

We now come back to the case $k = \mathbb{F}_q$ and study how $\text{tr}(\pi_E)$ encodes E being ordinary or supersingular.

Proposition 15.12. *Let $E/\mathbb{F}_q$ be an elliptic curve. Then E is ordinary if and only if $\text{tr}(\pi_E)$ is prime to p . In this case, $\pi_E \notin \mathbb{Q}$ and p is split in the quadratic extension $\mathbb{Q}(\pi_E)$.*

Proof. Let $R = \mathcal{O}_{E,e}$ and let $t \in R$ be a uniformizer. The kernel of π_E is equal to $\text{Spec } R/(t^q)$. In particular, π_E is an isomorphism on $\mathbb{F}_q$ -points. This is also clear from its definition (it is the q -Frobenius on $E(\mathbb{F}_q)$.) Recall that the dual isogeny π_E^* satisfies $\pi_E \circ \pi_E^* = [q]$. So we see that

$$E[q](\mathbb{F}_q) = \ker(\pi_E^*)(\mathbb{F}_q). \quad (15.8)$$

First assume that E is ordinary. We have $|E[q](\mathbb{F}_q)| = q$ by (15.6), so (15.8) shows that

$$|\ker(\pi_E^*)(\mathbb{F}_q)| = \deg(\pi_E^*).$$

It follows that $\ker(\pi_E^*)$ is reduced. Comparing with the definition of the Lie algebra (10.1), this means

$$0 = \text{Lie}(\ker(\pi_E^*)) = \ker(\pi_E^* : \text{Lie}(E) \rightarrow \text{Lie}(E))$$

which shows that π_E^* induces an automorphism of $\text{Lie}(E)$. Since π_E induces the zero map on $\text{Lie}(E)$, it follows that $\pi_E + \pi_E^*$ induces an automorphism of $\text{Lie}(E)$. Hence, this integer cannot be divisible by p .

Conversely, assume that $\text{tr}(\pi_E)$ is not divisible by p . Then it induces an automorphism of $\text{Lie}(E)$. So $\pi_E^* = \text{tr}(\pi_E) - \pi_E$ induces an automorphism of $\text{Lie}(E)$. Hence $\ker(\pi_E^*)$ is reduced, which implies $|\ker(\pi_E^*)(\mathbb{F}_q)| = q$. By (15.8), this shows that E is ordinary.

Assume E is ordinary. The only possibility for $\pi_E \in \mathbb{Z}$ would be $\pi_E = \pm\sqrt{q}$. In this case, $\text{tr}(\pi_E) = \pm 2\sqrt{q}$ would not be prime to p , contradicting what we already proved. So $F = \mathbb{Q}(\pi_E)$ is a quadratic extension of $\mathbb{Q}$.

Assume p is inert or ramified in F , and let $\mathfrak{p}$ be the unique prime ideal above p . Then $\pi_E \in \mathfrak{p}$ implies $\text{tr}(\pi_E) \in \mathfrak{p}$, which would again imply that $\text{tr}(\pi_E)$ is divisible by p , again contradicting what we already showed. This shows that p is split in F , finishing the proof. $\square$

Proposition 15.13. *Let $E/\mathbb{F}_q$ be an elliptic curve. Then E is supersingular if and only if $\pi_E^m \in \mathbb{Z}$ for some $m \geq 1$. Moreover, if $\pi_E \notin \mathbb{Z}$, then p is non-split in $\mathbb{Q}(\pi_E)$.*

Proof. First assume that E is supersingular. Then, by (15.7), $E[q] = \ker(\pi_E^2)$. By Proposition 14.6, there exists a (unique) isogeny $\zeta \in \text{End}(E)$ with $\pi_E^2 = q\zeta$. Then $\deg(\zeta) = 1$, so ζ is an automorphism of E . The only integral elements of norm 1 in $\mathbb{Q}$ or imaginary-quadratic extensions $F/\mathbb{Q}$ are 4-th or 6-th roots of unity. So $(\pi_E^2)^{12} = q^{12}$ lies in $\mathbb{Z}$.

Conversely, assume that $\pi_E^m \in \mathbb{Z}$ for some m . For degree reasons, the only possibility is $\pi_E^m = \pm q^{m/2}$. This means that q and π_E^2 differ by a root of unity as before, showing that $E[q] = \ker(\pi_E^2)$. By (15.7), this means E is supersingular.

It is left to show that if E is supersingular and $\pi_E \notin \mathbb{Q}$, then p is non-split in $\mathbb{Q}(\pi_E)$. This requires the following additional piece of information:

Proposition 15.14 ([22, Theorem 3.1]¹²). *Let E be an elliptic curve over a field k of characteristic p . Then E is supersingular if and only if $\text{End}^0(\bar{k} \otimes_k E)$ is a quaternion algebra.*

For our given supersingular E , we observe that $\text{End}^0(E) \hookrightarrow \text{End}^0(\bar{k} \otimes_k E)$ which is a quaternion algebra by Proposition 15.14. In general, if a quadratic extension $F/\mathbb{Q}$ embeds into a quaternion algebra $B/\mathbb{Q}$, then

$$F \text{ split at } p \implies B \text{ split at } p.$$

It follows that $\mathbb{Q}(\pi_E)$ cannot be split at p because $\text{End}^0(\bar{k} \otimes_k E)$ is non-split at p by Corollary 14.17. $\square$

16. LATTICES AND ISOGENIES

Having classified isogeny classes of elliptic curves over $\mathbb{F}_q$ (Honda–Tate Theorem 15.6), our next problem is as follows: Given an elliptic curve E_0 over $\mathbb{F}_q$, describe the set

$$\left\{ \begin{array}{l} \text{Isomorphism classes of elliptic curves } E/\mathbb{F}_q \\ \text{s.t. there exists a quasi-isogeny } E \rightarrow E_0 \end{array} \right\}. \quad (16.1)$$

The basic idea is to overparametrize this set by making the quasi-isogeny $E \rightarrow E_0$ part of the datum which will be similar to what we did around (8.8). We develop these ideas over a general field k .

Fix an elliptic curve E_0 . Consider pairs (E, ρ) where E/k is an elliptic curve and $\rho \in \text{Hom}^0(E, E_0)$ a quasi-isogeny. Two such pairs (E, ρ) and (E', ρ') are said to be *isomorphic* if there exists an isomorphism $\alpha : E \rightarrow E'$ such that $\rho = \rho' \circ \alpha$. The group $\text{End}^0(E_0)^\times$ of self quasi-isogenies of E_0 acts on such pairs by

$$\gamma \cdot (E, \rho) := (E, \gamma \circ \rho).$$

Assume that $\rho, \rho' : E \rightarrow E_0$ are two quasi-isogenies from the same elliptic curve E . Then $\gamma := \rho' \circ \rho^{-1}$ lies in $\text{End}^0(E_0)^\times$ and $\gamma \cdot (E, \rho) = (E, \rho')$. This shows that forgetting about ρ is the same as taking the quotient by $\text{End}^0(E_0)^\times$. In this way, we have found a bijection

$$\text{End}^0(E_0)^\times \backslash \left\{ \begin{array}{l} \text{Isom. classes of} \\ \text{pairs } (E, \rho) \end{array} \right\} \xrightarrow{\sim} \left\{ \begin{array}{l} \text{Isom. classes of } E \text{ s.t.} \\ \text{there exists a } \rho : E \rightarrow E_0 \end{array} \right\}. \quad (16.2)$$

Our task thus becomes solving the following two subproblems:

Problem 16.1. (1) Describe the set of isomorphism classes of pairs (E, ρ) .

(2) Determine the ring $\text{End}^0(E_0)$ in dependence of E_0 , as well as the action of its units on the set of all pairs (E, ρ) .

In this section, we consider problem (1), and we will see that it is closely related to lattices in 2-dimensional vector spaces.

16.1. The case $k = \mathbb{C}$. We first discuss the situation over $\mathbb{C}$. Let $E_0 = \mathbb{C}/\Lambda_0$ be a fixed complex elliptic curve and define $V = \mathbb{Q} \otimes_{\mathbb{Z}} \Lambda_0$. The following proposition explains the basic mechanism.

Proposition 16.2. *There is a natural bijection between the following three sets.*

- (1) *The set of lattices $\Lambda \subset V$ containing Λ_0 .*
- (2) *The set of finite subgroups $K \subset E_0$.*
- (3) *The set of isomorphism classes of isogenies $\theta : E_0 \rightarrow E$.*

Extending to quasi-isogenies, there is a natural bijection between

¹²Note that Silverman's notation and our notation are related by $\text{End}^{\text{Silverman}}(E) = \text{End}(\bar{k} \otimes_k E)$ and $\text{End}_k^{\text{Silverman}}(E) = \text{End}(E)$.

(1') The set of all lattices $\Lambda \subset V$.

(3') The set of isomorphism classes of quasi-isogenies $\theta : E_0 \rightarrow E$.

Proof. The quotient map $\pi : V \rightarrow E_0$ defines an isomorphism

$$V/\Lambda_0 \xrightarrow{\sim} E_{\text{tors}}$$

between V/Λ_0 and the torsion points on E . Given a lattice $\Lambda_0 \subseteq \Lambda$ as in (1), the image $\pi(\Lambda) = \Lambda/\Lambda_0$ is a finite subgroup of E_{tors} . Conversely, a finite subgroup $K \subset E_{\text{tors}}$ as in (2) defines the lattice $\pi^{-1}(K)$. The finite subgroup K also defines a quotient isogeny $\theta : E_0 \rightarrow E_0/K$ as in (3). Conversely, an isogeny θ defines the finite subgroup $\ker(\theta)$. This explains the passage between (1), (2) and (3).

Consider an isogeny $\theta : E_0 \rightarrow E$ corresponding to a lattice $\Lambda_0 \subseteq \Lambda$. Then $n\theta = \theta \circ [n]$ corresponds to the lattice $n^{-1}\Lambda$ because multiplication by n on E_0 induces multiplication by n on V . This allows to extend (1) $\leftrightarrow$ (3) to a bijection (1') $\leftrightarrow$ (3') as follows.

Given a lattice $\Lambda \subset V$ as in (1'), choose $n \geq 1$ with $\Lambda_0 \subseteq n^{-1}\Lambda$. Let θ' be the isogeny with kernel $(n^{-1}\Lambda)/\Lambda_0$. Then $\theta := n^{-1}\theta'$ is the quasi-isogeny corresponding to Λ .

Conversely, given a quasi-isogeny $\theta : E_0 \rightarrow E$ as in (3'), choose $n \geq 1$ such that $\theta' = n\theta$ is an isogeny. Let $\Lambda_0 \subseteq \Lambda'$ be the corresponding lattice. Then $\Lambda = n\Lambda'$ is the lattice corresponding to θ . $\square$

Proposition 16.2 is very intuitive to prove. However, it feels unnatural that multiplication by n on isogenies in (3) by *division* by n on the lattices in (1). For this reason, we swap the arrow direction in (3) via $\phi \leftrightarrow \phi^{-1}$ and use the following bijection (consistent with (16.2)):

$$\left\{ \begin{array}{c} \text{Isom. classes of} \\ \text{quasi-isogenies } (E, \rho : E \rightarrow E_0) \end{array} \right\} \xrightarrow{\sim} \{\text{Lattices } \Lambda \subset V\} \quad (16.3)$$

$$(E, \rho) \mapsto n^{-1} \cdot (\widetilde{n\rho})(L)$$

where $E = \mathbb{C}/L$, where n is any choice of integer such that the multiple $n\rho$ is an isogeny, and where $\widetilde{n\rho} : \mathbb{C} \rightarrow \mathbb{C}$ is the lifting of $n\rho$ to universal covers. The definition is independent of the choice of n . Moreover, we can restrict to actual isogenies on the left hand side. For these, we may choose $n = 1$ and obtain the bijection

$$\left\{ \begin{array}{c} \text{Isom. classes of} \\ \text{isogenies } (E, \rho : E \rightarrow E_0) \end{array} \right\} \xrightarrow{\sim} \{\text{Sublattices } \Lambda \subseteq \Lambda_0\} \quad (16.4)$$

$$(E, \rho) \mapsto \widetilde{\rho}(L).$$

We now go about describing $\text{End}^0(E_0)$. Every endomorphism $x : \mathbb{C}/\Lambda_0 \rightarrow \mathbb{C}/\Lambda_0$ lifts to a $\mathbb{C}$ -linear endomorphism of $\mathbb{C}$, and any such endomorphism is given by multiplication with a complex number. In this way, we find

$$\text{End}(\mathbb{C}/\Lambda_0) = \{x \in \mathbb{C} \mid x\Lambda_0 \subseteq \Lambda_0\}.$$

This is either $\mathbb{Z}$ or an order in an imaginary-quadratic field. So $\text{End}^0(E_0)$ is either $\mathbb{Q}$ or an imaginary-quadratic field K . In this way, we see that

$$\left\{ \begin{array}{c} \text{Isom. classes of } E \\ \text{isogeneous to } E_0 \end{array} \right\} \xrightarrow{\sim} \begin{cases} \mathbb{Q}^\times \backslash \mathcal{Latt}(V) & \text{or} \\ K^\times \backslash \mathcal{Latt}(V) \end{cases} \quad (16.5)$$

where $\mathcal{Latt}(V)$ denotes the set of lattices in V . In either case, we see that there are infinitely many isomorphism classes of elliptic curves E that are isogeneous to E_0 , and we have described them in terms of lattices.

16.2. ℓ -Isogenies: The case k algebraically closed. Let k be algebraically closed. Fix an elliptic curve E_0/k and a prime $\ell \neq \text{char}(k)$. Let $V_\ell = V_\ell(E_0)$ be the rational Tate module of E_0 . Our goal is to establish an analog of (16.5) for E_0 and V_ℓ .

Observe that multiplication by n induces an automorphism of $T_\ell(E_0)$ when n is coprime to ℓ . For this reason, lattices in V_ℓ can only ever detect ℓ -quasi-isogenies:

Definition 16.3. An ℓ -isogeny is an isogeny $\phi : E_1 \rightarrow E_2$ such that $\ker(\phi)$ has order ℓ^r for some $r \geq 0$. An ℓ -quasi-isogeny is a quasi-isogeny ϕ such that $\ell^m \phi$ is an ℓ -isogeny for some $m \geq 1$. These notions are stable under composition and dual.

We now have a direct analog of Proposition 16.2. The two differences are that we have to restrict to ℓ -isogenies when working with V_ℓ , and that we need to use our abstract result Proposition 14.6 to construct quotients.

Proposition 16.4. *There is a natural bijection between the following three sets.*

- (1) *The set of $\mathbb{Z}_\ell$ -lattices $\Lambda_\ell \subset V_\ell$ containing $T_\ell(E_0)$.*
- (2) *The set of finite subgroups $K \subset E_0$ of ℓ -power order.*
- (3) *The set of isomorphism classes of ℓ -isogenies $\theta : E_0 \rightarrow E$.*

Extending to quasi-isogenies, there is a natural bijection between

- (1') *The set of all $\mathbb{Z}_\ell$ -lattices $\Lambda_\ell \subset V_\ell$.*
- (3') *The set of isomorphism classes of ℓ -quasi-isogenies $\theta : E_0 \rightarrow E$.*

Proof. We begin with an observation that is similar to (14.9) but without fixing a specific power ℓ^n . Namely, for every field k and every elliptic curve E/k , there is a canonical isomorphism

$$\begin{aligned} V_\ell(E)/T_\ell(E) &\xrightarrow{\sim} E(\bar{k})[\ell^\infty] \\ \ell^{-r}(\dots, x_3, x_2, x_1) &\longmapsto x_r. \end{aligned} \tag{16.6}$$

Indeed, an equivalent fraction representative of the form

$$\ell^{-r-k}(\dots, \ell^k x_3, \ell^k x_2, \ell^k x_1)$$

gets sent to $\ell^k x_{r+k}$, which equals x_r because the transition map $E[\ell^{k+r}] \rightarrow E[\ell^r]$ in the definition of $T_\ell(E)$ is multiplication by ℓ^k . So (16.6) is well-defined. In this way, taking image or preimage, we obtain a bijection

$$\{\mathbb{Z}_\ell\text{-Lattices } T_\ell(E) \subseteq \Lambda_\ell \subset V_\ell(E)\} \xrightarrow{\sim} \left\{ \begin{array}{c} \text{finite subgroups } K \subset E(\bar{k}) \\ \text{of } \ell\text{-power order} \end{array} \right\}. \tag{16.7}$$

Coming back to the situation $k = \bar{k}$ and our fixed elliptic curve E_0 , we note that the finite subgroups $K \subset E_0(\bar{k})[\ell^\infty]$ are essentially the same as the ℓ -power order k -subgroup schemes of E_0 . (Since $k = \bar{k}$, any such K is a union of rational points, which may also be viewed as a closed subscheme.) This constructs the bijection between (1) and (2).

The bijection between (2) and (3) comes from Proposition 14.6. Namely, every finite ℓ -power order subgroup scheme $K \subset E_0$ is the kernel of the ℓ -isogeny $\theta : E_0 \rightarrow E_0/K$. Conversely, every ℓ -isogeny has such a kernel. Moreover, if $\theta_1 : E_0 \rightarrow E_1$ and $\theta_2 : E_0 \rightarrow E_2$ are two isogenies with the same kernel, then Proposition 14.6 (2) ensures that there exists $\alpha : E_1 \xrightarrow{\sim} E_2$ with $\theta_2 = \alpha \circ \theta_1$.

The extension to a bijection (1') $\leftrightarrow$ (3') is done in the same way as during the proof of Proposition 16.2. $\square$

Just like over $\mathbb{C}$, we are in the situation that the isogeny $[\ell^r] : E_0 \rightarrow E_0$ corresponds to the lattice $\ell^{-r}T_\ell(E_0)$. So we again renormalize our bijections by switching $\phi \leftrightarrow \phi^{-1}$. We obtain:

$$\left\{ \begin{array}{c} \text{Isom. classes of} \\ \ell\text{-quasi-isogenies } (E, \rho : E \rightarrow E_0) \end{array} \right\} \xrightarrow{\sim} \{ \mathbb{Z}_\ell\text{-Lattices } \Lambda_\ell \subset V_\ell \} \quad (16.8)$$

$$(E, \rho) \mapsto \ell^{-r} \cdot (\ell^r \rho)(T_\ell(E)),$$

where $r \geq 1$ is chosen such that $\ell^r \rho$ is an isogeny. Moreover, we can again restrict to actual isogenies on the left hand side and obtain the bijection

$$\left\{ \begin{array}{c} \text{Isom. classes of} \\ \ell\text{-isogenies } (E, \rho : E \rightarrow E_0) \end{array} \right\} \xrightarrow{\sim} \{ \mathbb{Z}_\ell\text{-Sublattices } \Lambda_\ell \subseteq T_\ell(E_0) \} \quad (16.9)$$

$$(E, \rho) \mapsto \rho(T_\ell(E)).$$

16.3. ℓ -Isogenies: The general case. Let us write k^{sep} for a separable closure of k and $G_k = \text{Gal}(k^{\text{sep}}/k)$ for the absolute Galois group. Recall that G_k is nothing but the group of field automorphisms of k^{sep} over k . It can be written as the inverse limit $G_k = \lim \text{Gal}(K/k)$ where $K \subseteq k^{\text{sep}}$ runs through the finite Galois extensions of k .

Let X be a scheme over k . Then G_k acts on the set of k^{sep} -valued points of X . Namely, given $\sigma \in G_k$ and $x \in X(k^{\text{sep}})$, we may define $\sigma(x)$ as the composition

$$\text{Spec}(k^{\text{sep}}) \xrightarrow{\text{Spec}(\sigma)} \text{Spec}(k^{\text{sep}}) \xrightarrow{x} X.$$

If we locally have $X = \text{Spec } k[T_i, i \in I]/J$ for some indexing set I and some ideal J , then x corresponds to a tuple $(x_i) \in (k^{\text{sep}})^I$ and $\sigma(x)$ is the tuple $(\sigma(x_i))_{i \in I}$. This makes sense because every $f \in J$ has coefficients in k , and hence

$$f(\sigma(x_i)_{i \in I}) = \sigma(f((x_i)_{i \in I})) = 0.$$

Clearly, if $X \rightarrow Y$ is a morphism of k -schemes, then the induced map $X(k^{\text{sep}}) \rightarrow Y(k^{\text{sep}})$ is G_k -equivariant.

Construction 16.5. Let E/k be an elliptic curve. As just remarked, the G_k -action on $E(k^{\text{sep}})$ commutes with the group law $m : E \times_{\text{Spec}(k)} E \rightarrow E$, meaning it defines a group homomorphism

$$G_k \longrightarrow \text{GL}_{\mathbb{Z}}(E(k^{\text{sep}})).$$

Restricting to ℓ^n -torsion, we obtain the torsion representations

$$G_k \longrightarrow \text{GL}_{(\mathbb{Z}/\ell^n\mathbb{Z})}(E[\ell^n](k^{\text{sep}})).$$

Passing to the inverse limit under the multiplication maps $[\ell] : E[\ell^{n+1}] \rightarrow E[\ell^n]$, we obtain the ℓ -adic Galois representation defined by E ,

$$G_k \longrightarrow \text{GL}_{\mathbb{Z}_\ell}(T_\ell(E)), \quad G_k \longrightarrow \text{GL}_{\mathbb{Q}_\ell}(V_\ell(E)).$$

Proposition 16.6. *Let E_0/k be an elliptic curve and let $\ell \neq \text{char}(k)$ be a prime. There is a natural bijection between the following three sets.*

- (1) *The set of G_k -stable $\mathbb{Z}_\ell$ -lattices $\Lambda_\ell \subset V_\ell(E_0)$ containing $T_\ell(E_0)$.*
- (2) *The set of finite k -subgroup schemes $K \subset E_0$ of ℓ -power order.*
- (3) *The set of isomorphism classes of ℓ -isogenies $\theta : E_0 \rightarrow E$.*

Extending to quasi-isogenies, there is a natural bijection between

- (1') *The set of all G_k -stable $\mathbb{Z}_\ell$ -lattices $\Lambda_\ell \subset V_\ell$.*
- (3') *The set of isomorphism classes of ℓ -quasi-isogenies $\theta : E_0 \rightarrow E$.*

Proof. The bijection between (2) and (3) is the same as before. To $K \subset E_0$, we associate the isogeny $E_0 \rightarrow E_0/K$. To an isogeny $\theta : E_0 \rightarrow E$, we associate $\ker(\theta)$. Next, we have the following lemma which is an example of Galois descent.

Lemma 16.7. *Let X be a finite type k -scheme. There is a bijection*

$$\begin{aligned} \left\{ \begin{array}{c} \text{Finite étale} \\ \text{subschemes } Z \subseteq X \end{array} \right\} &\xrightarrow{\sim} \left\{ \begin{array}{c} \text{Finite } G_k\text{-stable} \\ \text{subsets of } X(k^{\text{sep}}) \end{array} \right\} \\ Z &\longmapsto Z(k^{\text{sep}}). \end{aligned}$$

Recall that finite group schemes of ℓ -power order over k are necessarily étale (Theorem 4.10). So applying Lemma 16.7 to elliptic curves (exercise) provides the bijection

$$\begin{aligned} \left\{ \begin{array}{c} \text{Finite } k\text{-subgroup schemes} \\ K \subset E_0 \text{ of } \ell\text{-power order} \end{array} \right\} &\xrightarrow{\sim} \left\{ \begin{array}{c} \text{Finite } G_k\text{-stable subgroups} \\ \text{of } E_0(k^{\text{sep}}) \text{ of } \ell\text{-power order} \end{array} \right\} \\ K &\longmapsto K(k^{\text{sep}}). \end{aligned}$$

Moreover, $E_0[\ell^n](\bar{k}) = E_0[\ell^n](k^{\text{sep}})$ by étaleness of $E_0[\ell^n]$, and the bijection (16.6) is G_k -equivariant (clear by definition). Thus, the preimages of the G_k -stable finite subgroups of $E_0(k^{\text{sep}})[\ell^\infty]$ are precisely the G_k -stable lattices $\Lambda_\ell \subset V_\ell(E_0)$ that contain $T_\ell(E_0)$.

The extension to quasi-isogenies is done as before. $\square$

As in (16.8) and (16.9) before, we will usually invert arrows and consider the bijection

$$\begin{aligned} \left\{ \begin{array}{c} \text{Isom. classes of} \\ \ell\text{-quasi-isogenies } (E, \rho : E \rightarrow E_0) \end{array} \right\} &\xrightarrow{\sim} \{G_k\text{-stable } \mathbb{Z}_\ell\text{-Lattices } \Lambda_\ell \subset V_\ell(E_0)\} \\ (E, \rho) &\longmapsto \ell^{-r} \cdot (\ell^r \rho)(T_\ell(E)), \quad r \gg 0. \end{aligned} \quad (16.10)$$

Explanation 16.8. What does it mean concretely for a lattice $\Lambda_\ell \subset V_\ell(E_0)$ to be G_k -stable? First, choose integers a and b such that $\ell^a T_\ell(E_0) \subseteq \Lambda_\ell \subseteq \ell^b T_\ell(E_0)$. The multiples $\ell^k T_\ell(E_0)$ are all G_k -stable, so G_k acts on the quotient

$$\ell^b T_\ell(E_0) / \ell^a T_\ell(E_0) \xrightarrow{\sim} E[\ell^{a-b}](k^{\text{sep}}). \quad (16.11)$$

Since $E[\ell^{a-b}]$ is finite, there exists a finite Galois extension $K \subseteq k^{\text{sep}}$ such that $E[\ell^{a-b}](k^{\text{sep}}) = E[\ell^{a-b}](K)$. Then the action of G_k on (16.11) factors through $\text{Gal}(K/k)$ and the question becomes whether the finite group $\Lambda_\ell / \ell^a T_\ell(E_0)$ is $\text{Gal}(K/k)$ -stable.

Example 16.9. (1) Let us consider the case $k = \mathbb{Q}$. In this case, we have Serre's Open Image Theorem for elliptic curves without complex multiplication.

Theorem 16.10 (Serre Open Image Theorem). *Let E be an elliptic curve over $\mathbb{Q}$ such that $\text{End}(\bar{\mathbb{Q}} \otimes_{\mathbb{Q}} E) = \mathbb{Z}$. Then, for every prime number ℓ , the homomorphism $G_{\mathbb{Q}} \rightarrow \text{GL}_{\mathbb{Z}_\ell}(T_\ell(E))$ has open image. It is surjective for almost all primes.*

If $H \subset \text{GL}_2(\mathbb{Z}_\ell)$ is an open subgroup, then there are only finitely many $\mathbb{Z}_\ell$ -lattices $\Lambda \subset \mathbb{Q}_\ell^2$ up to multiplication by $\mathbb{Z}^\ell$ which are H -stable (Exercise)¹³.

So, if E_0 satisfies the “no CM” condition from Theorem 16.10, then there are only finitely many ℓ -isogenies $\rho_1 : E_1 \rightarrow E_0, \dots, \rho_r : E_r \rightarrow E_0$ of elliptic curves over $\mathbb{Q}$, such that every ℓ -quasi-isogeny $\rho : E \rightarrow E_0$ is of the form $\ell^a \rho_i$ for some i . For the primes ℓ such that $G_{\mathbb{Q}} \twoheadrightarrow \text{GL}_{\mathbb{Z}_\ell}(T_\ell(E_0))$, the multiplication maps $[\ell^a] : E_0 \rightarrow E_0$ are the only ℓ -quasi-isogenies.

(2) Let us now consider the case $k = \mathbb{F}_q$. The Galois group $G_{\mathbb{F}_q}$ is isomorphic to $\widehat{\mathbb{Z}}$ and topologically generated by the q -Frobenius F_q . For an elliptic curve $E/\mathbb{F}_q$, the representation

$$G_{\mathbb{F}_q} \longrightarrow \text{GL}_{\mathbb{Z}_\ell}(T_\ell(E))$$

is the unique continuous homomorphism that takes F_q to $T_\ell(\pi_E)$. That is, the Galois action is realized by an endomorphism of E ! A lattice $\Lambda_\ell \subset V_\ell(E)$ is Galois-stable if and only if it is π_E -stable.

¹³You may draw inspiration from the proof of Proposition 8.7.

16.4. p -Isogenies in characteristic p . Consider now a field k of characteristic p and p -(quasi-)isogenies. We do not have the Tate module as before anymore, but there is a simple classification of all possibilities.

Proposition 16.11. *Let E be an elliptic curve over k . The finite subgroup schemes of E of p -power order have the following description.*

(1) *Assume E is supersingular. Then they are exactly the subschemes of the form $\mathrm{Spec} \mathcal{O}_{E,e}/\mathfrak{m}_e^{p^h}$. In particular, for every $h \in \mathbb{Z}$, there is a unique quasi-isogeny $\theta : E \rightarrow E'$ of degree p^h (up to isomorphism in E').*

(2) *Assume that k is perfect and E ordinary. Then they are exactly the subschemes of the form*

$$\mathrm{Spec} (\mathcal{O}_{E,e}/\mathfrak{m}_e^{p^a}) \times E[p^b]_{\mathrm{red}}$$

where $a, b \geq 0$, and where $(-)\mathrm{red}$ denotes the underlying reduced subscheme. In particular, the quasi-isogenies $\theta : E \rightarrow E'$ (up to isomorphism in E') are in bijection with the set of pairs $(a, b) \in \mathbb{Z}^2$.

We proceed the proof with a lemma.

Lemma 16.12. *Let E be an elliptic curve over a field k of characteristic p . For every $m \geq 0$, the subscheme $\mathrm{Spec} \mathcal{O}_{E,e}/\mathfrak{m}_e^{p^m}$ is a subgroup scheme. It coincides with the kernel of the relative p^m -Frobenius $F_{E/k}^m : E \rightarrow E^{(p^m)}$.*

Proof. We explain the construction of the relative Frobenius. Consider the p -Frobenius $F : \mathrm{Spec}(k) \rightarrow \mathrm{Spec}(k)$ and define $E^{(p^m)} = (F^m)^*(E)$ as the base change of E along F^m . Consider the *absolute* p^m -Frobenius $F_E^m : E \rightarrow E$ as in Definition 13.8. It is not a $\mathrm{Spec}(k)$ -morphism; instead, it fits into the following commutative diagram making the outer square commute:

$$\begin{array}{ccc} E & \xrightarrow{F_E^m} & E \\ \downarrow F_{E/k}^m & \searrow & \downarrow \\ E^{(p^m)} & \xrightarrow{\quad} & E \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \xrightarrow{F^m} & \mathrm{Spec}(k). \end{array} \quad (16.12)$$

By the universal property of fiber products, this defines the morphism $F_{E/k}^m$ which now is a $\mathrm{Spec}(k)$ -morphism (consider the left triangle). Moreover, $F_{E/k}^m$ takes the identity element to the identity and is hence a group scheme homomorphism ([12, Proposition 3.6]). It is called the *relative p^m -Frobenius* of E over k , and we leave it as an exercise to show that it is an isogeny of degree p^m with kernel $\mathrm{Spec} \mathcal{O}_{E,e}/\mathfrak{m}_e^{p^m}$ (exercise). $\square$

Exercise 16.13. Show that the relative p^m -Frobenius can also be defined as the m -fold composition of relative p -Frobenii $F_{E^{(p^{m-1})}/k} : E^{(p^{m-1})} \rightarrow E^{(p^m)}$:

$$F_{E/k}^m = F_{E^{(p^m)}/k} \circ F_{E^{(p^{m-1})}/k} \circ \cdots \circ F_{E^{(p)}/k} \circ F_{E/k}. \quad (16.13)$$

Proof of Proposition 16.11. (1) First assume E is supersingular. Then $E[p^n] = \mathrm{Spec} \mathcal{O}_{E,e}/\mathfrak{m}_e^{p^{2n}}$ for every $n \geq 0$, see (15.7). Every p -power order subgroup of E is contained in some $E[p^n]$ by Deligne's Theorem 4.10 (1) and hence of the form $\mathrm{Spec} \mathcal{O}_{E,e}/\mathfrak{m}_e^{p^m}$ for some $m \geq 0$. Conversely, all these subschemes are subgroup schemes by Lemma 16.12.

(2) Now assume that E is ordinary. Consider the p^n -torsion group scheme $E[p^n]$. By (15.7), its identity connected component is given by

$$(E[p^n])^\circ = \mathrm{Spec} \mathcal{O}_{E,e}/\mathfrak{m}_e^{p^n}.$$

Let $E[p^n]_{\text{red}} \subset E[p^n]$ be the maximal closed reduced subscheme. It is a finite, reduced, 0-dimensional k -scheme; equivalently, it is a finite disjoint union $\bigsqcup_{i \in I} \text{Spec}(k_i)$ for certain finite field extensions k_i/k . Since k is perfect by assumption, the tensor products $k_i \otimes_k k_j$ are again reduced. In other words, $E[p^n]_{\text{red}} \times_{\text{Spec}(k)} E[p^n]_{\text{red}}$ is again reduced, and hence the restriction of the multiplication map to it

$$\begin{array}{ccc} E[p^n]_{\text{red}} \times_{\text{Spec}(k)} E[p^n]_{\text{red}} & \xrightarrow{m} & E[p^n] \\ & \searrow & \uparrow \\ & & E[p^n]_{\text{red}} \end{array}$$

factors through $E[p^n]_{\text{red}}$. This shows that m induces a group scheme structure on $E[p^n]_{\text{red}}$.

Given $a, b \geq 0$, we may now consider the product map

$$(E[p^a])^\circ \times_{\text{Spec}(k)} E[p^b]_{\text{red}} \longrightarrow E. \quad (16.14)$$

It is not difficult to show that this is a closed immersion (exercise). Its image is characterized as the unique closed subgroup scheme $K \subset E$ with

$$K^\circ = \text{Spec } \mathcal{O}_{E,e}/\mathfrak{m}_e^{p^a} \quad \text{and} \quad K(\bar{k}) = E[p^b](\bar{k}).$$

□

16.5. Putting everything together. We come back to studying isogenies of arbitrary degree. These can always be factored into a sequence of ℓ -isogenies for varying primes ℓ .

Proposition 16.14. *Let K be a commutative finite k -group scheme of degree n . Let $p_1^{e_1} \dots p_r^{e_r}$ be the prime factorization of n . Then $K[p_i^{e_i}]$ has degree $p_i^{e_i}$, and multiplication defines an isomorphism*

$$K[p_1^{e_1}] \times \dots \times K[p_r^{e_r}] \xrightarrow{\sim} K. \quad (16.15)$$

We begin with a lemma which is a kind of converse to Theorem 4.10 (1).

Lemma 16.15. *Let K be a commutative finite k -group scheme which is p^k -torsion for some prime p . Then the degree of K is a power of p .*

Proof. We may assume that $k = \bar{k}$. First assume that $p \neq \text{char}(k)$. Then Theorem 4.10 (2) states that K is étale. Since we have assumed $k = \bar{k}$, this means K is the constant group scheme $\underline{K(k)}$ over $\text{Spec}(k)$. In this way, we have reduced to the statement

$$K(k) \text{ is } p^k\text{-torsion} \implies |K(k)| \text{ is a power of } p$$

This is a statement about a finite abelian group which is clear.

Assume now that $p = \text{char}(k)$. By the existence of quotients as explained in §9.1, there exists an exact sequence

$$1 \longrightarrow K^\circ \longrightarrow K \longrightarrow K/K^\circ \longrightarrow 1.$$

Both K° and K/K° are again p -power torsion. The quotient is étale and the same argument as before applies. For the identity connected component, which is the most non-trivial part, we have to refer to [19, Theorem 17.1]. □

Proof of Proposition 16.14. By Deligne's result (Theorem 4.10 (1)), K is n -torsion. In this way, it is endowed with an action by $\mathbb{Z}/n\mathbb{Z}$. This ring decomposes as the product

$$\mathbb{Z}/p_1^{e_1}\mathbb{Z} \times \dots \times \mathbb{Z}/p_r^{e_r}\mathbb{Z}$$

and it is a general principle that every object on which $\mathbb{Z}/n\mathbb{Z}$ acts decomposes accordingly. We recall this principle:

Let $\pi_i \in \mathbb{Z}/n\mathbb{Z}$ be the idempotent corresponding to the p_i -factor. It satisfies $p_i^{e_i}\pi_i = 0$, so multiplication with π_i defines a homomorphism

$$K \longrightarrow K[p_i^{e_i}].$$

Since $\pi_i^2 = \pi_i$, $\pi_i\pi_j = 0$ if $i \neq j$, and $\pi_1 + \dots + \pi_r = 1$, the tuple $(\pi_1, \dots, \pi_r)$ defines a right-inverse to the natural map (16.15). It is also surjective because for every S -valued $p_i^{e_i}$ -torsion point $x \in K(S)$, we have $\pi_i(x) = x$. Thus (16.15) is an isomorphism.

By Lemma 16.15, the order of $K[p_i^{e_i}]$ is a power of p_i . Since both sides of (16.15) have the same order, we deduce that $|K[p_i^{e_i}]| = p_i^{e_i}$. $\square$

Corollary 16.16. *Let $\rho : E \rightarrow E_0$ be a (quasi-)isogeny of elliptic curves over k . Then there exist finitely many primes $\ell_1, \dots, \ell_r$ as well as a factorization of ρ as a chain of ℓ_i -(quasi-)isogenies*

$$E \xrightarrow{\rho_1} E_1 \xrightarrow{\rho_2} E_2 \xrightarrow{\rho_3} \dots \xrightarrow{\rho_r} E_0.$$

Proof. Assume that ρ is an isogeny with kernel K . Let $K = \prod_{i=1}^r K[p_i^{e_i}]$ be its factorization as in Proposition 16.14. Then we set $\ell_1 = p_1$ and define ρ_1 as the quotient isogeny

$$\rho_1 : E \longrightarrow E_1 := E/K[p_1^{e_1}].$$

Since $\ker(\rho_1) \subseteq \ker(\rho)$, we obtain a factorization $\rho = \rho' \circ \rho_1$ for $\rho' : E_1 \rightarrow E_0$. The kernel of ρ' is $K/K[p_1^{e_1}]$, and we conclude by induction on $\deg(\rho)$. Extending to quasi-isogenies is formal. $\square$

If $\text{char}(k) = 0$, then we can define the *complete Tate module* of E/k as

$$\widehat{T}(E) := \prod_{\ell \text{ prime}} T_\ell(E), \quad \widehat{V}(E) := \prod_{\ell \text{ prime}} V_\ell(E). \quad (16.16)$$

These are $\widehat{\mathbb{Z}}$ - and $\mathbb{A}_f$ -modules, respectively. Combining Corollary 16.16 with (16.10), we obtain a bijection

$$\left\{ \begin{array}{c} \text{Isom. classes of} \\ \text{quasi-isogenies } (E, \rho : E \rightarrow E_0) \end{array} \right\} \xrightarrow{\sim} \{G_k\text{-stable } \widehat{\mathbb{Z}}\text{-Lattices } \widehat{\Lambda} \subset \widehat{V}(E_0)\} \quad (16.17)$$

$$(E, \rho) \longmapsto n^{-1} \cdot (n\rho)(\widehat{T}(E)),$$

where n is chosen such that $n\rho$ is an isogeny. The notion of $\widehat{\mathbb{Z}}$ -lattice is the same as in Definition 8.8. If we restrict to actual isogenies, then the description simplifies to

$$\left\{ \begin{array}{c} \text{Isom. classes of} \\ \text{isogenies } (E, \rho : E \rightarrow E_0) \end{array} \right\} \xrightarrow{\sim} \{G_k\text{-stable } \widehat{\mathbb{Z}}\text{-Sublattices } \widehat{\Lambda} \subset \widehat{T}(E_0)\} \quad (16.18)$$

$$(E, \rho) \longmapsto \rho(\widehat{T}(E)).$$

If $\text{char}(k) = p$ is positive, then we may define the complete Tate module *away from p* ,

$$\widehat{T}^p(E) := \prod_{\ell \neq p} T_\ell(E), \quad \widehat{V}^p(E) := \prod_{\ell \neq p} V_\ell(E). \quad (16.19)$$

We can add an artificial factor

$$\mathbb{I}_p := \begin{cases} \mathbb{Z} & \text{if } E_0 \text{ supersingular} \\ \mathbb{Z} \times \mathbb{Z} & \text{if } E_0 \text{ ordinary} \end{cases} \quad (16.20)$$

that parametrizes the p -quasi-isogenies into E_0 as in Proposition 16.11. At least for perfect k , we obtain

$$\left\{ \begin{array}{c} \text{Isom. classes of} \\ \text{quasi-isogenies } (E, \rho : E \rightarrow E_0) \end{array} \right\} \xrightarrow{\sim} \{G_k\text{-stable } \widehat{\mathbb{Z}}^p\text{-Lattices } \widehat{\Lambda}^p \subset \widehat{V}^p(E_0)\} \times \mathbb{I}_p$$

$$(E, \rho) \longmapsto [n^{-1} \cdot (n\rho)(\widehat{T}^p(E)), \delta(\rho)] \quad (16.21)$$

where $\delta(\rho)$ is defined as follows. Let v_p be the p -adic valuation normalized by $v_p(p) = 1$. If ρ is an isogeny, then

$$\delta(\rho) = \begin{cases} v_p(\rho) & \text{if } E_0 \text{ supersingular} \\ (v_p(\deg \ker(\rho)^\circ), v_p(\deg \ker(\rho)_{\text{red}})) & \text{if } E_0 \text{ ordinary.} \end{cases} \quad (16.22)$$

We extend this definition to the general case by

$$\delta(\rho/n) = \begin{cases} \delta(\rho) - 2v_p(n) & \text{if } E_0 \text{ supersingular} \\ \delta(\rho) - (v_p(n), v_p(n)) & \text{if } E_0 \text{ ordinary.} \end{cases}$$

17. ORBITAL INTEGRALS

17.1. Lattice counting. Let $x \in \text{GL}_n(\mathbb{Q})$ be an endomorphism of $\mathbb{Q}^n$, and let us ask the following simple question:

What is the number of lattices $\Lambda \subset \mathbb{Q}^n$ that satisfy $x\Lambda \subseteq \Lambda$?

This is too naive of course. If $x\Lambda \subseteq \Lambda$, then also $x(q\Lambda) \subseteq q\Lambda$ for every $q \in \mathbb{Q}^\times$, so this number is always 0 or infinite. We first make two observations that allow us to formulate a proper mathematical problem.

(1) If x is degenerate, for example $x = 1_n$ (unit matrix), then there will be too many x -stable lattices for the counting to make sense. So we assume that x is *regular semi-simple*. Recall that this means that the characteristic polynomial $\alpha(T)$ of x is separable. Equivalently, it means that the $\mathbb{Q}$ -algebra $F := \mathbb{Q}[x] \subset \text{M}_n(\mathbb{Q})$ generated by x is a product $F = F_1 \times \dots \times F_r$ of field extensions of $\mathbb{Q}$ such that $\sum_{i=1}^r [F_i : \mathbb{Q}] = n$.

(2) Every element $t \in F^\times$ can be written as a polynomial in x and hence commutes with x . Hence, if Λ satisfies $x\Lambda \subseteq \Lambda$, then

$$x(t\Lambda) = t(x\Lambda) \subseteq t\Lambda.$$

In this way, $F^\times$ acts on the set of x -stable lattices. The orbit of some Λ will always be infinite because, for example, it contains all the multiples $q\Lambda$, $q \in \mathbb{Q}^\times$.

So our precise problem is the following:

Let $x \in \text{GL}_n(\mathbb{Q})$ be regular semi-simple and let $F = \mathbb{Q}[x]$.

What is the number of x -stable lattices $\Lambda \subseteq \mathbb{Q}^n$ up to multiplication by $F^\times$?

Lemma 17.1. *This number is finite.*

Proof. If there exists a lattice Λ with $x\Lambda \subseteq \Lambda$, then the characteristic polynomial α of x lies in $\mathbb{Z}[T]$. Then $\mathbb{Z}[x]$ is an order in F . (This means it is both a subring and a full-rank lattice.) There exists an integer $N \geq 1$ such that $N \cdot O_F \subseteq \mathbb{Z}[x] \subseteq O_F$. If Λ is x -stable, then the lattice

$$O_F \cdot \Lambda := \text{span}\{a\lambda \mid a \in O_F, \lambda \in \Lambda\}$$

is O_F -stable and satisfies

$$N(O_F \cdot \Lambda) \subseteq \Lambda \subseteq O_F \cdot \Lambda. \quad (17.1)$$

There only exist finitely many O_F -stable lattices in $\mathbb{Q}^n$ up to $F^\times$ -scaling. More precisely, if $F = F_1 \times \dots \times F_r$ is written as a product of fields as before, then these classes are in bijection with the product of class groups

$$\mathcal{C}\ell_{F_1} \times \dots \times \mathcal{C}\ell_{F_r}.$$

For every O_F -stable lattice L , there only exist finitely many lattices Λ which are sandwiched as $NL \subseteq \Lambda \subseteq L$. Together with (17.1), this finishes the proof. $\square$

Proposition 17.2. *It only depends on the characteristic polynomial $\alpha(T)$ of x .*

Proof. Recall that two regular semi-simple elements of $\mathrm{GL}_n(\mathbb{Q})$ are conjugate if and only if they have the same characteristic polynomial. For a conjugate element gxg^{-1} , we have bijections

$$\Lambda \longmapsto g\Lambda, \quad t \longmapsto gtg^{-1}$$

from x -stable lattices to gxg^{-1} -stable lattices, and from $\mathbb{Q}[x]$ to $\mathbb{Q}[gxg^{-1}]$. These bijections identify the two lattice counting problems. $\square$

17.2. Adelic lattice counting. Recall that $\mathrm{GL}_n(\mathbb{Q})$ is a subgroup of $\mathrm{GL}_n(\mathbb{A}_f)$ by diagonal embedding. In this way, we may view x as an element of $\mathrm{GL}_n(\mathbb{A}_f)$.

Also recall that we have a notion of $\widehat{\mathbb{Z}}$ -lattice in $\mathbb{A}_f^n$ (Definition 8.8). According to (2) of that definition, these are the $\widehat{\mathbb{Z}}$ -submodules $\widehat{\Lambda}$ such that there exists an integer $c \geq 1$ with

$$c \cdot \widehat{\mathbb{Z}}^n \subseteq \widehat{\Lambda} \subseteq c^{-1} \widehat{\mathbb{Z}}^n. \quad (17.2)$$

Proposition 17.3. *There is a bijection*

$$\begin{aligned} \{\mathbb{Z}\text{-Lattices } \Lambda \subset \mathbb{Q}^n\} &\xrightarrow{\sim} \{\widehat{\mathbb{Z}}\text{-lattices } \widehat{\Lambda} \subset \mathbb{A}_f^n\} \\ \Lambda &\longmapsto \widehat{\mathbb{Z}} \cdot \Lambda \\ \widehat{\Lambda} \cap \mathbb{Q}^n &\longleftarrow \widehat{\Lambda}. \end{aligned} \quad (17.3)$$

This bijection identifies the x -stable lattices on both sides.

Proof. The given map sends the standard lattice $\mathbb{Z}^n$ to $\widehat{\mathbb{Z}}^n$. From this we recover $\mathbb{Z}^n$ as $\widehat{\mathbb{Z}}^n \cap \mathbb{Q}^n$. For a general $\mathbb{Z}$ -lattice Λ , there exists $g \in \mathrm{GL}_n(\mathbb{Q})$ with $\Lambda = g\mathbb{Z}^n$, and hence

$$\begin{aligned} (\widehat{\mathbb{Z}} \cdot \Lambda) \cap \mathbb{Q}^n &= g(\widehat{\mathbb{Z}} \cdot \mathbb{Z}^n) \cap \mathbb{Q}^n \\ &= g(\widehat{\mathbb{Z}}^n \cap \mathbb{Q}^n) \\ &= \Lambda. \end{aligned}$$

This proves the injectivity of $\Lambda \mapsto \widehat{\mathbb{Z}} \cdot \Lambda$. For the surjectivity, given some $\widehat{\mathbb{Z}}$ -lattice $\widehat{\Lambda}$, we choose an integer c as in (17.2). There is a bijection

$$(c^{-1}\mathbb{Z}^n)/(c\mathbb{Z}^n) \xrightarrow{\sim} (c^{-1}\widehat{\mathbb{Z}}^n)/(c\widehat{\mathbb{Z}}^n)$$

under which the finite subgroup $\widehat{\Lambda}/(c\widehat{\mathbb{Z}}^n)$ is the image of a subgroup $\Lambda/(c\mathbb{Z}^n)$. This proves the bijectivity of (17.3). The compatibility with x -stability is essentially clear. $\square$

Assume that there exists an x -stable lattice $\Lambda \subset \mathbb{Q}^n$. Then the characteristic polynomial α of x has to be integral. In particular, $|\det(x)|$ is an integer ≥ 1 and determined by

$$|\det(x)| = [\Lambda : x\Lambda].$$

We can capture the property of being contained of index $m \geq 1$ by a *test function* $\Phi^{(m)} \in \mathcal{C}_c^\infty(\mathrm{GL}_n(\mathbb{A}_f))$. Here, the notation means locally constant functions with compact support. Every such function is a finite linear combination of characteristic functions 1_{gH} , where $H \subset \mathrm{GL}_n(\mathbb{A}_f)$ is an open compact subgroup and $g \in \mathrm{GL}_n(\mathbb{A}_f)$ an element. Concretely,

one may assume that all occurring H are principal congruence subgroups (3.8). The specific function $\Phi^{(m)}$ that we now define will simply be the characteristic function of an open compact subset $\mu_m \subset \mathrm{GL}_n(\mathbb{A}_f)$.

Definition 17.4. For any integer $m \geq 1$, consider the open compact subset

$$\mu^{(m)} = \{y \in \mathrm{M}_n(\widehat{\mathbb{Z}}) \mid v_p(\det(y)) = v_p(m) \text{ for all primes } p\}.$$

Let $\Phi^{(m)} = 1_{\mu_m}$ be its characteristic function.

In the following, we let $K = \mathrm{GL}_n(\widehat{\mathbb{Z}})$ denote the standard maximal compact subgroup of $\mathrm{GL}_n(\mathbb{A}_f)$.

Lemma 17.5. *Let $x \in \mathrm{GL}_n(\mathbb{Q})$ be regular semi-simple with determinant $m = |\det(x)|$ in $\mathbb{Z}$. Let $F = \mathbb{Q}[x]$ be the subalgebra of $\mathrm{M}_n(\mathbb{Q})$ generated by x . There is a bijection*

$$\begin{aligned} \{\bar{g} \in F^\times \backslash \mathrm{GL}_n(\mathbb{A}_f)/K \mid \Phi^{(m)}(g^{-1}xg) = 1\} &\xrightarrow{\sim} F^\times \backslash \{x\text{-stable lattices } \Lambda \subset \mathbb{Q}^n\} \\ \bar{g} &\longmapsto (g \cdot \widehat{\mathbb{Z}}^n) \cap \mathbb{Q}^n. \end{aligned} \quad (17.4)$$

Let us first explain why the expression $\Phi^{(m)}(g^{-1}xg)$ is independent of the choice of g in the double coset $\bar{g} = F^\times \cdot g \cdot K$. Namely, first, $F^\times$ commutes with x and, second, μ_m is invariant under conjugation by K . So for $t \in F^\times$ and $k \in K$, we have

$$\Phi(k^{-1}g^{-1}t^{-1}xtgk) = \Phi(g^{-1}xg).$$

Proof. The subgroup K is precisely the stabilizer of $\widehat{\mathbb{Z}}^n$ in $\mathrm{GL}_n(\mathbb{A}_f)$. So we have a bijection

$$\begin{aligned} \mathrm{GL}_n(\mathbb{A}_f)/K &\xrightarrow{\sim} \{\widehat{\mathbb{Z}}\text{-lattices } \widehat{\Lambda} \subset \mathbb{A}_f^n\} \\ g &\longmapsto g \cdot \widehat{\mathbb{Z}}^n. \end{aligned}$$

Then

$$x(g \cdot \widehat{\mathbb{Z}}^n) \subseteq g \cdot \widehat{\mathbb{Z}}^n \iff g^{-1}xg \in \mathrm{M}_n(\widehat{\mathbb{Z}}).$$

Since $\det(x)$ is fixed and invariant under conjugation, this is if and only if $g^{-1}xg \in \mu_m$, i.e. $\Phi^{(m)}(g^{-1}xg) = 1$. Taking the quotient by the $F^\times$ -action on both sides and applying Proposition 17.3 finishes the proof. $\square$

17.3. Orbital integrals. We want to recast Lemma 17.5 in terms of orbital integrals on $\mathrm{GL}_n(\mathbb{A}_f)$ and, later, on the factors $\mathrm{GL}_n(\mathbb{Q}_p)$. There is one subtlety however. Namely, the natural way to count isomorphism classes of objects in a groupoid is to weigh every object Λ by $|\mathrm{Aut}(\Lambda)|^{-1}$. In our situation, the automorphism group of an x -stable lattice Λ is defined as

$$\mathrm{Aut}(\Lambda) := \{t \in F^\times \mid t\Lambda = \Lambda\}.$$

Note that for every x -stable lattice Λ ,

$$\mathbb{Z}[x]^\times \subseteq \mathrm{Aut}(\Lambda) \subseteq O_F^\times. \quad (17.5)$$

Let $F = F_1 \times \dots \times F_r$ be a decomposition of F as product of fields. Let (r_i, s_i) be the number of real (resp. complex) places of F_i . By Dirichlet's unit theorem, the three groups in (17.5) are finitely generated of rank

$$\mathrm{rk}_{\mathbb{Z}}(\mathrm{Aut}(\Lambda)) = \sum_{i=1}^r r_i + s_i - 1.$$

If this rank is ≥ 1 , then the quantity $|\mathrm{Aut}(\Lambda)|^{-1}$ does not make sense anymore. For this reason, we will henceforth assume that each F_i is either $\mathbb{Q}$ or imaginary-quadratic.

Example 17.6. Let $n = 2$ and let $\alpha(T) = T^2 + 1$. In this case, $\mathbb{Z}[T]/(\alpha(T))$ is the Gaussian integers $\mathbb{Z}[i]$ which have class number 1. Moreover, $\mathbb{Z}[i]^\times$ has four elements. Thus

$$\sum_{\mathbb{Q}(i)^\times \setminus \{i\Lambda \subseteq \Lambda\}} \frac{1}{|\text{Aut}(\Lambda)|} = \frac{1}{4}.$$

Now let $\alpha(T) = T^2 + 4$, meaning x generates the order $\mathbb{Z}[2i]$. Let L be the unique $\mathbb{Z}[i]$ -lattice in $\mathbb{Q}(i)$ up to $\mathbb{Q}(i)^\times$ -scaling (use class number 1). It is obviously also $\mathbb{Z}[2i]$ -stable. Looking at (17.1), we moreover need to find all Λ with $2L \subset \Lambda \subset L$. These are in bijection with $\mathbb{P}^1(\mathbb{F}_2)$ which has three elements. One of them is the lattice $(1+i)L$ (2 is ramified in $\mathbb{Z}[i]$.) So there are two non- $\mathbb{Z}[i]$ -stable Λ . These have to be interchanged by i . Their automorphism group is $\{\pm 1\}$. So we see

$$\sum_{\mathbb{Q}(i)^\times \setminus \{2i\Lambda \subseteq \Lambda\}} \frac{1}{|\text{Aut}(\Lambda)|} = \frac{1}{4} + \frac{1}{2}.$$

Lemma 17.7. *Assume each F_i is either $\mathbb{Q}$ or imaginary-quadratic. Then $F^\times$ is discrete in $(\mathbb{A}_f \otimes_{\mathbb{Q}} F)^\times$.*

Proof. The situation is a product of the elementary cases that $F = \mathbb{Q}$ or imaginary-quadratic. In this situation, we take the intersection with the open subgroup $\widehat{O}_F^\times \subset \mathbb{A}_{F,f}^\times$,

$$F^\times \cap \widehat{O}_F^\times = O_F^\times,$$

which is finite. The claim follows. $\square$

Remark 17.8. The subgroup $F^\times \subset (\mathbb{A} \otimes_{\mathbb{Q}} F)^\times$ is always discrete. This was proved in Corollary 3.6. The difference is that we left out the archimedean place in Lemma 17.7.

Let G be a nice topological group such as $\mathbb{G}(\mathbb{Q})$, $\mathbb{G}(\mathbb{Q}_p)$, $\mathbb{G}(\mathbb{A}_f)$, $\mathbb{G}(\mathbb{R})$, or $\mathbb{G}(\mathbb{A})$ for a reductive algebraic group $\mathbb{G}$. Then there exists a measure μ_G on G that is left and right translation invariant. It is unique up to scaling¹⁴ and called the Haar measure of G . We will use the following examples of this measure.

Example 17.9. (1) If G is of the form $\mathbb{G}(\mathbb{Q})$, then G is discrete and we endow it with the counting measure.

(2) If G is of the form $\mathbb{G}(\mathbb{Q}_p)$ or $\mathbb{G}(\mathbb{A}_f)$, then we fix an open compact subgroup $K \subseteq G$ to normalize the measure as $\mu_G(K) = 1$. A smaller open subgroup $K' \subseteq K$ then has to have measure $\mu_G(K') = [K : K']^{-1}$ because

$$\begin{aligned} \mu_G(K) &= \mu_G(\bigsqcup_{gK' \in K/K'} gK') \\ &= \sum_{gK' \in K/K'} \mu_G(gK') \\ &= [K : K'] \cdot \mu_G(K'). \end{aligned}$$

Here, the crucial point was that translation invariance ensures $\mu_G(gK') = \mu_G(K')$ for all $g \in G$. In general, every open compact subset $U \subseteq G$ is a finite disjoint union of translates $g_j K_j$ for open subgroups $K_j \subset K$ and elements $g_j \in G$. By translation invariance of the measure,

$$\mu_G\left(\bigsqcup_{j=1}^d g_j K_j\right) = \sum_{j=1}^d [K : K_j]^{-1}.$$

¹⁴One needs to impose a mild regularity assumption on μ_G to characterize it in this way.

Definition 17.10 (Quotient measure). Let $H \subseteq G$ be a closed subgroup. Assume that both H and G are of the nice type considered before, and let μ_G and μ_H be Haar measures on them. The *quotient measure* on $H \backslash G$ is the unique measure $\mu_H \backslash \mu_G$ such that for every measurable function f on G ,

$$\int_{H \backslash G} \left[\int_H f(hg) \mu_H(h) \right] \frac{\mu_G}{\mu_H}(Hg) = \int_G f(g) \mu_G(g). \quad (17.6)$$

Example 17.11. Again consider a regular semi-simple element $x \in \mathrm{GL}_n(\mathbb{Q})$. Its centralizer is a closed subgroup scheme $\mathrm{GL}_{n,x} \subset \mathrm{GL}_n$. We can take $\mathbb{Q}$ or $\mathbb{A}_f$ -points and obtain groups

$$F^\times \subset (\mathbb{A}_f \otimes_{\mathbb{Q}} F)^\times \subset \mathrm{GL}_n(\mathbb{A}_f).$$

Here, $F = \mathbb{Q}[x]$ is as before. Let us assume that the conditions of Lemma 17.7 hold. Then the first inclusion is that of a discrete subgroup. The second inclusion is a closed immersion because it comes from a closed immersion of algebraic varieties. In particular, $F^\times \subset \mathrm{GL}_n(\mathbb{A}_f)$ is a discrete subgroup.

We endow $F^\times$ with the counting measure, and $\mathrm{GL}_n(\mathbb{A}_f)$ with the measure such that $\mathrm{Vol}(K) = 1$. Definition 17.10 provides a measure on the quotient

$$F^\times \backslash \mathrm{GL}_n(\mathbb{A}_f).$$

Proposition 17.12. Let $x \in \mathrm{GL}_n(\mathbb{Q})$ have determinant in $\mathbb{Z}$ and set $m = |\det(x)|$. Let $\Phi^{(m)}$ be the test function from Definition 17.4. Assume that the conditions of Lemma 17.7 hold for $F = \mathbb{Q}[x]$. Then

$$\int_{F^\times \backslash \mathrm{GL}_n(\mathbb{A}_f)} \Phi^{(m)}(g^{-1}xg) dg = \sum_{F^\times \backslash \{x\Lambda \subseteq \Lambda\}} \frac{1}{|\mathrm{Aut}(\Lambda)|}. \quad (17.7)$$

Here, dg denotes the quotient measure on $F^\times \backslash \mathrm{GL}_n(\mathbb{A}_f)$.

Proof. As explained after Lemma 17.5, the function $\Phi^{(m)}$ is conjugation invariant under K . So $g \mapsto \Phi^{(m)}(g^{-1}xg)$ takes constant value 0 or 1 on each double coset $F^\times \cdot h \cdot K$. Decomposing $\mathrm{GL}_n(\mathbb{A}_f)$ as a disjoint union of such double cosets, each (obviously) being $F^\times$ -stable, we find

$$\int_{F^\times \backslash \mathrm{GL}_n(\mathbb{A}_f)} \Phi^{(m)}(g^{-1}xg) dg = \sum_{F^\times gK \in F^\times \backslash \mathrm{GL}_n(\mathbb{A}_f)/K} \Phi^{(m)}(g) \cdot \mathrm{Vol}(F^\times \backslash (F^\times gK)). \quad (17.8)$$

Lemma 17.13. The volume of $F^\times \backslash (F^\times gK)$ is $|F^\times \cap gKg^{-1}|^{-1}$.

Proof. We start from the definition of the quotient measure in Definition 17.10. Let f be the characteristic function of gK . Then, the integral on the right hand side of (17.6) is simply

$$\int_{\mathrm{GL}_n(\mathbb{A}_f)} f(h) dh = 1. \quad (17.9)$$

On the left hand side, we determine the inner function

$$\begin{aligned} \varphi : F^\times \backslash \mathrm{GL}_n(\mathbb{A}_f) &\longrightarrow \mathbb{C} \\ h &\longmapsto \sum_{t \in F^\times} f(th). \end{aligned}$$

We see that $\varphi(F^\times h) \neq 0$ only if $h \in F^\times gK$, in which case $\varphi(F^\times h)$ equals the cardinality

$$|\{t \in F^\times \mid th \in gK\}|. \quad (17.10)$$

Given $t_1, t_2 \in F^\times$ with $t_1 h, t_2 h \in gK$, we have $t_1 t_2^{-1} \in gKg^{-1}$. So (17.10) is always the same and equal to $|F^\times \cap gKg^{-1}|$ as claimed. In this way, we have shown that φ is given by

$$\varphi = |F^\times \cap gKg^{-1}| \cdot 1_{F^\times \backslash (F^\times gK)}.$$

So

$$\begin{aligned} |F^\times \cap gKg^{-1}| \cdot \text{Vol}(F^\times \backslash (F^\times gK)) &= \int_{F^\times \backslash \text{GL}_n(\mathbb{A}_f)} \varphi(h) dh \\ &\stackrel{\text{Def. 17.10}}{=} \int_{\text{GL}_n(\mathbb{A}_f)} f(h) dh \\ &\stackrel{(17.9)}{=} 1. \end{aligned}$$

□

By Lemma 17.5, the double cosets $F^\times gK$ with $\Phi^{(m)}(g) = 1$ are precisely those such that $\widehat{\Lambda} = g \cdot \widehat{\mathbb{Z}}^n$ is x -stable. Moreover, this sets up a bijection between these double cosets and the $F^\times$ -orbits of x -stable lattices Λ . For such g ,

$$F^\times \cap gKg^{-1} = \{t \in F^\times \mid t\Lambda = \Lambda\} = \text{Aut}(\Lambda).$$

We obtain with Lemma 17.13 that the two right hand sides of (17.7) and (17.8) are equal as claimed. □

17.4. Factorization into local orbital integrals. Let $x \in \text{GL}_n(\mathbb{Q})$ be regular semi-simple as before. Assume that $F = \mathbb{Q}[x]$ satisfies the condition of Lemma 17.7. Write $\mathbb{A}_{F,f} = (\mathbb{A}_f \otimes_{\mathbb{Q}} F)$ and choose any Haar measure on $\mathbb{A}_{F,f}^\times$. Then Definition 17.10 defines measures on both

$$F^\times \backslash \mathbb{A}_{F,f}^\times \quad \text{and} \quad \mathbb{A}_{F,f}^\times \backslash \text{GL}_n(\mathbb{A}_f).$$

For every function $\Phi \in \mathcal{C}_c^\infty(\text{GL}_n(\mathbb{A}_f))$, we obtain

$$\int_{F^\times \backslash \text{GL}_n(\mathbb{A}_f)} \Phi(g^{-1}xg) dg = \text{Vol}(F^\times \backslash \mathbb{A}_{F,f}^\times) \cdot \int_{\mathbb{A}_{F,f}^\times \backslash \text{GL}_n(\mathbb{A}_f)} \Phi(g^{-1}xg) dg. \quad (17.11)$$

Note that one measure on the right hand side is proportional, one inverse proportional to the choice of measure on $\mathbb{A}_{F,f}^\times$. This is why their product is independent of any choices.

By convention, we always choose the measure on $\mathbb{A}_{F,f}^\times$ such that $\text{Vol}(\widehat{O}_F^\times) = 1$.

Lemma 17.14. *With this choice of measure,*

$$\text{Vol}(F^\times \backslash \mathbb{A}_{F,f}^\times) = \frac{|\mathcal{C}\ell_F|}{|O_F^\times|}.$$

Proof. Exercise. □

Definition 17.15 ($\mathbb{A}_f$ -orbital integral). The orbital integral of $\Phi \in \mathcal{C}_c^\infty(\text{GL}_n(\mathbb{A}_f))$ along the orbit defined by the regular semi-simple element x is

$$\text{Orb}(x, \Phi) = \int_{\mathbb{A}_{F,f}^\times \backslash \text{GL}_n(\mathbb{A}_f)} \Phi(g^{-1}xg) dg.$$

Let $y \in \text{GL}_n(\mathbb{Q}_p)$ be regular semi-simple and let $F_p = \mathbb{Q}_p[y]$ be the subalgebra of $M_n(\mathbb{Q}_p)$ generated by y . It is a product of field extensions of $\mathbb{Q}_p$ whose degrees sum up to n , just as in the global setting. If $y = x$ comes from a $\mathbb{Q}$ -point, then F_p equals $\mathbb{Q}_p \otimes_{\mathbb{Q}} \mathbb{Q}[x]$.

In general, we normalize the Haar measures on $\text{GL}_n(\mathbb{Q}_p)$ and $F_p^\times$ by

$$\text{Vol}(\text{GL}_n(\mathbb{Z}_p)) = \text{Vol}(O_{F,p}^\times) = 1.$$

Note that this matches our global conventions prime by prime.

Definition 17.16 (*p*-adic orbital integral). The quotient $F_p^\times \backslash \mathrm{GL}_n(\mathbb{Q}_p)$ is endowed with the measure from Definition 17.10. For any function $\Phi_p \in \mathcal{C}_c^\infty(\mathrm{GL}_n(\mathbb{Q}_p))$, the orbital integral along the orbit of y is defined by

$$\mathrm{Orb}(y, \Phi_p) := \int_{F_p^\times \backslash \mathrm{GL}_n(\mathbb{Q}_p)} \Phi_p(g^{-1}yg) dg.$$

Definition 17.17 (Factorizable functions). A function $\Phi \in \mathcal{C}_c^\infty(\mathrm{GL}_n(\mathbb{A}_f))$ is said to be *factorizable* if there exist functions $\Phi_p \in \mathcal{C}_c^\infty(\mathrm{GL}_n(\mathbb{Q}_p))$, almost all of which are $1_{\mathrm{GL}_n(\mathbb{Z}_p)}$, such that

$$\Phi(x_2, x_3, x_5, \dots) = \prod_{p < \infty} \Phi_p(x_p).$$

We note that every Φ is a finite sum of factorizable functions. For $\Phi = \bigotimes_{p < \infty} \Phi_p$ factorizable and $x \in \mathrm{GL}_n(\mathbb{Q})$ regular semi-simple, we have

$$\mathrm{Orb}(x, \Phi) = \prod_{p < \infty} \mathrm{Orb}(x, \Phi_p). \quad (17.12)$$

We leave it as an exercise to show that almost all of the factors $\mathrm{Orb}(x, \Phi_p)$ are equal to 1, making the product well-defined.

Example 17.18. The function $\Phi^{(m)}$ from Definition 17.4 is factorizable. It equals $\bigotimes_{p < \infty} \Phi_p^{(m)}$, where $\Phi_p^{(m)}$ is the characteristic function of

$$\{g \in \mathrm{M}_n(\mathbb{Z}_p) \mid v_p(\det(g)) = v_p(m)\}.$$

Exercise 17.19. Prove the following local analog of Proposition 17.12. Let $y \in \mathrm{GL}_n(\mathbb{Q}_p)$ be regular semi-simple and let $F_p = \mathbb{Q}_p[y]$ as before. Let Φ be the characteristic function of all $g \in \mathrm{M}_n(\mathbb{Z}_p)$ with $v_p(\det(g)) = v_p(\det(y))$. Show that

$$\mathrm{Orb}(y, \Phi) = \sum_{F_p^\times \backslash \{y\Lambda \subseteq \Lambda\}} [O_{F_p}^\times : \mathrm{Aut}(\Lambda)], \quad (17.13)$$

where $\mathrm{Aut}(\Lambda) = \{t \in F_p^\times \mid t\Lambda = \Lambda\}$.

Example 17.20. Assume that $n = 2$ and that $y \in \mathrm{GL}_2(\mathbb{Q}_p)$ is regular semi-simple with integral characteristic polynomial. Let $F_p = \mathbb{Q}_p[y]$. There exists a unique integer $c \geq 0$ such that

$$\mathbb{Z}_p[y] = \mathbb{Z}_p + p^c O_{F_p}$$

which is called the *conductor* of y . The local orbital integral (17.13) is given by

$$\mathrm{Orb}(y, \Phi) = \begin{cases} 2 + 2p + \dots + 2p^{c-1} + p^c & \text{if } F_p/\mathbb{Q}_p \text{ is inert} \\ 1 + p + \dots + p^c & \text{if } F_p/\mathbb{Q}_p \text{ is ramified} \\ p^c & \text{if } F_p \cong \mathbb{Q}_p \times \mathbb{Q}_p. \end{cases} \quad (17.14)$$

These quantities arise as follows. For each $0 \leq t \leq c$, there exists a unique $F_p^\times$ -orbit in (17.13) such that

$$\{a \in O_{F_p} \mid a\Lambda \subseteq \Lambda\} = \mathbb{Z}_p + p^t O_{F_p}.$$

The sum in (17.13) becomes the sum of group indices

$$\sum_{t=0}^c \left[O_{F_p}^\times : (\mathbb{Z}_p + p^t O_{F_p})^\times \right]$$

which equals (17.14).

17.5. A summarizing example. In conclusion, we have obtained the following result. Let $x \in \mathrm{GL}_n(\mathbb{Q})$ be regular semi-simple and assume that $F = \mathbb{Q}[x]$ satisfies the condition of Lemma 17.7. This assumption is equivalent to $O_F^\times$ being finite. Assume that the characteristic polynomial of x is integral, for otherwise there cannot be any x -stable lattices. Set $m = |\det(x)|$ and let $\Phi^{(m)} \in \mathcal{C}_c^\infty(\mathrm{GL}_n(\mathbb{A}_f))$ be the function from Definition 17.4. It is factorizable, equal to $\bigotimes_{p < \infty} \Phi_p^{(m)}$ as stated in Example 17.18. Then

$$\sum_{F^\times \setminus \{x\Lambda \subseteq \Lambda\}} \frac{1}{|\mathrm{Aut}(\Lambda)|} = \frac{|\mathcal{C}\ell_F|}{|O_F^\times|} \prod_{p < \infty} \mathrm{Orb}(x, \Phi_p^{(m)}). \quad (17.15)$$

Moreover, if $n = 2$, then we have the simple formula (17.14) for the local orbital integrals. They only depend on the ramification behaviour of $F/\mathbb{Q}$ at p and the conductor of the order $\mathbb{Z}_p[x]$.

Example 17.21. Consider the (2×2) -matrix $x = 2^3 \cdot 5^2 \cdot 7 \cdot \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$. Identifying x with $2^3 \cdot 5^2 \cdot 7 \cdot i \in \mathbb{Q}(i)$, our counting problem is asking about $\mathbb{Z}[2^3 \cdot 5^2 \cdot 7 \cdot i]$ -stable $\mathbb{Z}$ -lattices in $\mathbb{Q}(i)$. The order $\mathbb{Z}_p[x]$ in $\mathbb{Q}_p[x]$ is maximal unless $p \in \{2, 5, 7\}$; the local orbital integrals at such primes equal 1. So identity (17.15) specializes to

$$\sum_{\mathbb{Q}(i)^\times \setminus \{x\Lambda \subseteq \Lambda\}} \frac{1}{|\mathrm{Aut}(\Lambda)|} = \frac{|\mathcal{C}\ell_{\mathbb{Q}(i)}|}{|\mathbb{Z}[i]^\times|} \mathrm{Orb}(x, \Phi_2^{(m)}) \mathrm{Orb}(x, \Phi_5^{(m)}) \mathrm{Orb}(x, \Phi_7^{(m)}). \quad (17.16)$$

The primes 2, 5 and 7 are ramified, split, and inert, respectively. The conductor of $\mathbb{Z}[x]$ at these primes is equal to 3, 2, and 1, respectively. (These are simply the exponents in $2^3 \cdot 5^2 \cdot 7$.) Substituting in (17.16), we find that

$$\sum_{\mathbb{Q}(i)^\times \setminus \{x\Lambda \subseteq \Lambda\}} \frac{1}{|\mathrm{Aut}(\Lambda)|} = \frac{(1 + 2 + 4 + 8) \cdot 5^2 \cdot (2 + 7)}{4}.$$

18. COUNTING POINTS MOD p ON $\mathcal{M}_n$

We have two final goals in this course. The first is to apply our new adelic techniques to obtain a local-global decomposition as in (17.15) for the number of isomorphism classes of elliptic curves (possibly with level structure) in a given isogeny class. The second is to recast this counting for all isogeny classes simultaneously in a single group-theoretic expression for GL_2 .

18.1. Counting in a given isogeny class. Fix a prime power $q = p^d$ and an elliptic curve E_0 over $\mathbb{F}_q$. The counting problem asks for an expression for

$$\sum_{\{E \text{ isogeneous to } E_0\}} \frac{1}{|\mathrm{Aut}(E)|} \quad (18.1)$$

where the sum runs over all isomorphism classes of elliptic curves $E/\mathbb{F}_q$ for which there exists an isogeny $E \rightarrow E_0$. We can also fix an integer $n \geq 1$ and consider isomorphism classes of elliptic curves with level- n -structure $(E, \alpha)/\mathbb{F}_q$ such that E is isogeneous to E_0 . Recall that always $\mathrm{Aut}(E, \alpha) = \{1\}$ when $n \geq 3$ (Proposition 7.11). The counting quantity is

$$\sum_{\{(E, \alpha) \text{ with } E \text{ isogeneous to } E_0\}} \frac{1}{|\mathrm{Aut}(E, \alpha)|}. \quad (18.2)$$

Let $\pi \in \mathrm{End}(E_0)$ denote the q -Frobenius endomorphism of E_0 .

Proposition 18.1. (1) If $\pi \notin \mathbb{Z}$, then $\text{End}^0(E_0) = \mathbb{Q}(\pi)$ is an imaginary-quadratic extension of $\mathbb{Q}$.

(2) If instead $\pi \in \mathbb{Z}$, meaning d even and $\pi = \pm q^{1/2}$, then $\text{End}^0(E_0)$ is a quaternion algebra as in Corollary 14.17.

Proof. Statement (1) is easy to show. Choose a prime $\ell \neq p$ and consider the injective map (Theorem 14.15)

$$\mathbb{Q}_\ell \otimes_{\mathbb{Z}} \text{End}(E_0) \longrightarrow \text{End}_{\mathbb{Q}_\ell}(V_\ell(E_0)). \quad (18.3)$$

Its image is contained in the $\mathbb{Q}_\ell$ -subvector space of endomorphisms *commuting with* π (Lemma 15.1. If π does not lie in $\mathbb{Q}$, then its centralizer in $\text{End}_{\mathbb{Q}_\ell}(V_\ell(E_0))$ is 2-dimensional and equal to $\mathbb{Q}_\ell[\pi]$. For dimension reasons, the map

$$\mathbb{Q}_\ell \otimes_{\mathbb{Q}} \mathbb{Q}[\pi] \xrightarrow{\sim} \mathbb{Q}_\ell[\pi]$$

has to be an isomorphism, so $\text{End}^0(E_0) = \mathbb{Q}[\pi]$ as claimed.

Statement (2) is much harder and a theorem of Tate. It states that (18.3) in fact defines a *bijection* with the centralizer of π on the right hand side. If $\pi \in \mathbb{Q}$, then this centralizer is all of $\text{End}_{\mathbb{Q}_\ell}(V_\ell(E_0))$, so $\text{End}^0(E_0)$ has to be four-dimensional. $\square$

Example 18.2. Consider a Weil p^2 -number of the form $\pi = p \cdot i$. Then $\mathbb{Q}(\pi) = \mathbb{Q}(i)$. Recall that p is non-split in $\mathbb{Q}(i)$ if and only if $p = 2$ or $p \equiv 3 \pmod{4}$. In these cases, Theorem 15.6 states that there exists an elliptic curve $E_0/\mathbb{F}_{p^2}$ with Weil number π . By Proposition 18.1, we have $\text{End}^0(E_0) \cong \mathbb{Q}(i)$.

Consider now the bases change $\mathbb{F}_{p^4} \otimes_{\mathbb{F}_{p^2}} E_0$. Its p^4 -Frobenius is $\pi^2 = -p^2$, which lies in $\mathbb{Q}$. By Proposition 18.1, $\text{End}^0(\mathbb{F}_{p^4} \otimes E_0)$ is a quaternion algebra.

18.2. The non-commutative case. We need to discuss the two cases from Proposition 18.1 separately. Let us first consider the case that $B = \text{End}^0(E_0)$ is a quaternion algebra. Thus, we assume that d is even and consider the situation $\pi = \pm p^{d/2}$. Note that such π is not regular semi-simple (its characteristic polynomial is $(T \mp p^{d/2})^2$) which makes it fall out of the framework developed in §17.

Proposition 18.3. The ring $O_B = \text{End}(E_0)$ is a maximal order in B . That is, if an order $O \subset B$ satisfies $O_B \subseteq O$, then $O = O_B$.

An equivalent characterization is as follows. For every $\ell \neq p$,

$$\mathbb{Z}_\ell \otimes_{\mathbb{Z}} O_B \cong M_2(\mathbb{Z}_\ell),$$

and $\mathbb{Z}_p \otimes_{\mathbb{Z}} O_B$ is the unique maximal order in $\mathbb{Q}_p \otimes_{\mathbb{Q}} B$.

Proof. One can extend Theorem 14.15 a little bit and prove that the image of

$$\mathbb{Z}_\ell \otimes_{\mathbb{Z}} \text{Hom}(E_1, E_2) \longrightarrow \text{Hom}(T_\ell(E_1), T_\ell(E_2))$$

is always *saturated*. That is, if $T_\ell(\varphi)$ is divisible by ℓ , then also φ is divisible by ℓ . For our case where $\text{End}(E_0)$ is an order in a quaternion algebra, we obtain that

$$\mathbb{Z}_\ell \otimes_{\mathbb{Z}} \text{End}(E_0) \xrightarrow{\sim} \text{End}_{\mathbb{Z}_\ell}(T_\ell(E_0)) \cong M_2(\mathbb{Z}_\ell).$$

This is a maximal order. A similar argument but using the p -divisible group instead of the ℓ -adic Tate module applies at p .

Finally, $\mathbb{Z}_\ell \otimes_{\mathbb{Z}} O_B$ being a maximal order in $\mathbb{Q}_\ell \otimes_{\mathbb{Q}} B$ for every prime ℓ (including $\ell = p$) is equivalent to O_B being a maximal order, which follows from Proposition 17.3. $\square$

Let us from now on write $B_\ell = \mathbb{Q}_\ell \otimes_{\mathbb{Q}} B$ and $O_{B,\ell} = \mathbb{Z}_\ell \otimes_{\mathbb{Z}} O_B$. We can identify

$$\mathbb{A}_f \otimes_{\mathbb{Q}} B = M_2(\mathbb{A}_f^p) \times B_p \quad \text{and} \quad \widehat{\mathbb{Z}} \otimes_{\mathbb{Z}} O_B = M_2(\widehat{\mathbb{Z}}^p) \times O_{B,p}.$$

The division algebra $B_p/\mathbb{Q}_p$ behaves like a discrete valuation ring: There is a non-archimedean valuation $\omega : B_p \rightarrow \mathbb{Z} \cup \{\infty\}$ for which $O_{B,p}$ is the ring of elements with valuation ≥ 0 . On quasi-endomorphisms $\varphi : E_0 \rightarrow E_0$, it is given by $\omega(\varphi) = v_p(\deg(\varphi))$ (p -adic valuation). Then

$$\omega : B_p^\times / O_{B,p}^\times \xrightarrow{\sim} \mathbb{Z}$$

gives a description of the set $\mathbb{I}_p$ from (16.20) in terms of B . Since every $\widehat{\mathbb{Z}}^p$ -lattice in $\widehat{V}^p(E_0)$ is π -stable because $\pi \in \mathbb{Z}$, (16.21) specializes to

$$\begin{aligned} \left\{ \begin{array}{c} \text{Isom. classes of} \\ \text{quasi-isogenies } (E, \rho : E \rightarrow E_0) \end{array} \right\} &\xrightarrow{\sim} \{ \widehat{\mathbb{Z}}^p\text{-Lattices } \widehat{\Lambda}^p \subset \widehat{V}^p(E_0) \} \times \mathbb{Z} \\ &\xrightarrow{\sim} \text{GL}_{\mathbb{A}_f^p}(\widehat{V}^p(E_0)) / \text{GL}_{\widehat{\mathbb{Z}}^p}(\widehat{T}^p(E_0)) \times B_p^\times / O_{B,p}^\times \\ &\xrightarrow{\sim} (\mathbb{A}_f \otimes_{\mathbb{Q}} B)^\times / \widehat{O}_B^\times. \end{aligned} \quad (18.4)$$

Giving volume 1 to $\widehat{O}_B^\times$ and dividing out the action of $\text{End}^0(E_0)^\times$ as in (16.2), we obtain that

$$\sum_{\{E \text{ isogenous to } E_0\}} \frac{1}{|\text{Aut}(E)|} = \text{Vol}(B^\times \backslash (\mathbb{A}_f \otimes_{\mathbb{Q}} B)^\times). \quad (18.5)$$

The right hand side is complete analogous to the quantity in Lemma 17.14. It can be understood as a class number of O_B weighted by automorphism groups. The volume in (18.5) is known, in fact, and equal to $(p-1)/24$. This is the *Deuring–Eichler Mass Formula*, see [12, §15].¹⁵

How to add level structure into the picture? Let $n \geq 1$ be prime to p . For every E isogenous to E_0 , the Frobenius action on $E[n](\overline{\mathbb{F}}_q)$ is multiplication by π . If $\pi \equiv 1 \pmod{n}$ then this action is trivial, and E has $|\text{GL}_2(\mathbb{Z}/n\mathbb{Z})|$ many level structures. If, on the other hand, $\pi \not\equiv 1 \pmod{n}$, then there are no level structures defined over $\mathbb{F}_q$. Hence, our final result for (18.2) is

$$\sum_{\{(E, \alpha) \text{ with } E \text{ isogenous to } E_0\}} \frac{1}{|\text{Aut}(E, \alpha)|} = \frac{p-1}{24} \cdot \begin{cases} 0 & \text{if } \pi \not\equiv 1 \pmod{n} \\ |\text{GL}_2(\mathbb{Z}/n\mathbb{Z})| & \text{if } \pi \equiv 1 \pmod{n}. \end{cases} \quad (18.6)$$

18.3. The commutative case. We now assume that $\pi \notin \mathbb{Q}$. Note that although this includes both ordinary and supersingular examples, we always have $\text{End}^0(E_0) = \mathbb{Q}(\pi)$. This was illustrated in Example 18.2.

We set $F = \mathbb{Q}(\pi)$. Combining (16.2) and (16.21), the set of isomorphism classes of elliptic curves $E/\mathbb{F}_q$ isogenous to E_0 is described by

$$F^\times \backslash \left[\{ \pi\text{-stable } \widehat{\mathbb{Z}}^p\text{-Lattices } \widehat{\Lambda}^p \subset \widehat{V}^p(E_0) \} \times \mathbb{I}_p \right]. \quad (18.7)$$

Choosing a basis $(\mathbb{A}_f^p)^2 \xrightarrow{\sim} \widehat{V}^p(E_0)$ allows to view π as an element of $\text{GL}_2(\mathbb{A}_f^p)$ and to identify $\widehat{\mathbb{Z}}^p$ -lattices in $\widehat{V}^p(E_0)$ with $\text{GL}_2(\mathbb{A}_f^p) / \text{GL}_2(\widehat{\mathbb{Z}}^p)$. In this way, (18.7) can be rewritten as

$$F^\times \backslash \left[\{ g \in \text{GL}_2(\mathbb{A}_f^p) \mid g^{-1} \pi g \in \text{M}_2(\widehat{\mathbb{Z}}^p) \} \times \mathbb{I}_p \right]. \quad (18.8)$$

Our goal is to count this set with elements weighted by $|\text{Aut}(E)|^{-1}$. The essential point is that this problem can be localized in the same way as we did in §17.4 before. We endow

¹⁵The cited formula counts isomorphism classes of supersingular elliptic curves over $\overline{\mathbb{F}}_q$. This number also agrees with the volume in (18.5) by the same arguments.

$\mathbb{A}_{F,f}^\times$ with the Haar measure such that $\text{Vol}(\widehat{O}_F^\times) = 1$. Performing the manipulations in (17.11) and (17.12), the analog of the final expression (17.15) is

$$\sum_{E \text{ isogenous to } E_0} \frac{1}{|\text{Aut}(E)|} = \frac{|\mathcal{C}_F|}{O_F^\times} \cdot \text{Vol}(F_p^\times \backslash \mathbb{I}_p) \cdot \prod_{\ell \neq p} \text{Orb}(\pi, 1_{M_2(\mathbb{Z}_\ell)}). \quad (18.9)$$

We still need to explain the volume term at p , which is simply the number of elements of $F_p^\times \backslash \mathbb{I}_p$. Looking at (16.22), we distinguish two cases:

- If F_p is a field, then $t \in F_p^\times$ acts on $\mathbb{I}_p = \mathbb{Z}$ by translation with $v_p(N_{F_p/\mathbb{Q}_p}(t))$.
- If $F_p \cong \mathbb{Q}_p \times \mathbb{Q}_p$, then $(t_1, t_2) \in F_p$ acts on $\mathbb{Z} \times \mathbb{Z}$ by translation with $(v_p(t_1), v_p(t_2))$.

Hence, we have

$$\text{Vol}(F_p^\times \backslash \mathbb{I}_p) = \begin{cases} 2 & \text{if } p \text{ is inert in } F \\ 1 & \text{if } p \text{ is ramified in } F \\ 1 & \text{if } p \text{ is split in } F. \end{cases} \quad (18.10)$$

How can we add level structure to the picture? Let $n \geq 1$ be prime to p . Let E be an elliptic curve isogenous to E_0 represented by a point $(\widehat{\Lambda}^p, \delta)$ on the right hand side of (18.7). Then

$$E[n](\bar{\mathbb{F}}_q) = (n^{-1} \cdot \widehat{\Lambda}^p) / (\widehat{\Lambda}^p)$$

as in (16.6). The Frobenius action on the left hand side is given by the action by π on the right. (Note that $\widehat{\Lambda}^p$ is π -stable!) So the n -torsion points of E are $\mathbb{F}_q$ -rational if and only if $\pi \equiv \text{id} \pmod{n \text{End}(\widehat{\Lambda}^p)}$. If this congruence holds, then there are $|\text{GL}_2(\mathbb{Z}/n\mathbb{Z})|$ many level structures, otherwise none. This condition corresponds to replacing the characteristic functions of $M_2(\mathbb{Z}_\ell)$ in (18.9) by the functions

$$\Psi'_{n,\ell} := |\text{GL}_2(\mathbb{Z}_\ell/n\mathbb{Z}_\ell)| \cdot 1_{\{1+nM_2(\mathbb{Z}_\ell)\}}.$$

Note that if $\ell \nmid n$, then simply $\Psi'_{n,\ell} = 1_{M_2(\mathbb{Z}_\ell)}$ as before. Our final expression is

$$\begin{aligned} \sum_{(E, \alpha) \text{ with } E \text{ isogenous to } E_0} \frac{1}{|\text{Aut}(E, \alpha)|} \\ = \frac{|\mathcal{C}_F|}{O_F^\times} \cdot \prod_{\ell \neq p} \text{Orb}(\pi, \Psi'_{\ell,n}) \cdot \begin{cases} 2 & \text{if } p \text{ is inert in } F \\ 1 & \text{if } p \text{ is ramified in } F \\ 1 & \text{if } p \text{ is split in } F. \end{cases} \end{aligned} \quad (18.11)$$

18.4. The final expression for $|\mathcal{M}_n(\mathbb{F}_q)|$. In (18.11), we are still *starting* from a Weil q -number π that satisfies one of the conditions of the Honda–Tate classification for elliptic curves (Theorem 15.6). For this, we improve our choices of functions:

Definition 18.4 (Test function for counting $\mathcal{M}_n(\mathbb{F}_q)$). (1) For all $\ell \neq p, \infty$, we define

$$\Psi_{n,\ell} := |\text{GL}_2(\mathbb{Z}_\ell/n\mathbb{Z}_\ell)| \cdot 1_{\ker(\text{GL}_2(\mathbb{Z}_\ell) \rightarrow \text{GL}_2(\mathbb{Z}_\ell/n\mathbb{Z}_\ell))}.$$

That is, $\Psi_{n,\ell}$ is a multiple of the characteristic function of the ℓ -factor of the level subgroup that defines $\mathcal{M}_n$.

(2) At p , we choose the following special test function $\Psi_p \in \mathcal{C}_c^\infty(\text{GL}_2(\mathbb{Q}_p))$. Let $\mathcal{S}_q$ be the set

$$\mathcal{S}_q := \{x \in M_2(\mathbb{Z}_p) \mid v_p(\det(x)) = v_p(q)\}. \quad (18.12)$$

It is open and compact. Also consider the set

$$\mathcal{T}_q := \{x \in p \cdot M_2(\mathbb{Z}_p) \mid v_p(\det(x)) = v_p(q)\}. \quad (18.13)$$

Put in words, $\mathcal{T}_q \subset \mathcal{S}_q$ is the subset of those elements that are divisible by p in $M_2(\mathbb{Z}_p)$. We define

$$\Psi_p := 1_{\mathcal{S}_q} - p \cdot 1_{\mathcal{T}_q}.$$

(3) Finally, the theory of discrete series representations ensures that there exists a compactly supported smooth function $\Psi_\infty \in \mathcal{C}_c^\infty(\mathrm{GL}_2(\mathbb{R}))$ such that all regular semi-simple $\pi \in \mathrm{GL}_2(\mathbb{R})$,

$$\mathrm{Orb}(\pi, \Psi_\infty) = \begin{cases} 1 & \text{if } \mathbb{R}[\pi] \cong \mathbb{C} \\ 0 & \text{if } \mathbb{R}[\pi] \cong \mathbb{R} \times \mathbb{R}. \end{cases} \quad (18.14)$$

Lemma 18.5. *For integral, regular semi-simple $\pi \in \mathrm{GL}_2(\mathbb{Q}_p)$ with $v_p(\det(x)) = v_p(q)$, the orbital integral of Ψ_p at π is given by*

$$\mathrm{Orb}(\pi, \Psi_p) = \begin{cases} 2 & \text{if } \mathbb{Q}_p[\pi]/\mathbb{Q}_p \text{ is an unramified field extension} \\ 1 & \text{if } \mathbb{Q}_p[\pi]/\mathbb{Q}_p \text{ is a ramified field extension} \\ 1 & \text{if } \mathbb{Z}_p[\pi] \cong \mathbb{Z}_p \times \mathbb{Z}_p \\ 0 & \text{if } \mathbb{Q}_p[\pi] \cong \mathbb{Q}_p \times \mathbb{Q}_p \text{ but } \mathbb{Z}_p[\pi] \text{ not maximal.} \end{cases} \quad (18.15)$$

Proof. We have

$$\mathrm{Orb}(\pi, 1_{\mathcal{T}_q}) = \mathrm{Orb}(\pi/p, 1_{\mathcal{S}_{q/p^2}}) \quad (18.16)$$

because $\pi \in pM_2(\mathbb{Z}_p)$ if and only if $\pi/p \in M_2(\mathbb{Z}_p)$. Moreover, the orbital integrals $\mathrm{Orb}(\pi, 1_{\mathcal{S}_q})$ and $\mathrm{Orb}(\pi/p, 1_{\mathcal{S}_{q/p^2}})$ are described by (17.14). We can now evaluate this expression case by case.

(1) Assume that $\mathbb{Q}_p[\pi]/\mathbb{Q}_p$ is an unramified field extension and that the conductor of $\mathbb{Z}_p[\pi]$ is c . The element π/p is still integral and $\mathbb{Z}_p[\pi/p]$ has conductor $c-1$. So $\mathrm{Orb}(\pi, \Psi_p)$ becomes

$$\begin{aligned} \mathrm{Orb}(\pi, 1_{\mathcal{S}_q}) - p \mathrm{Orb}(\pi, 1_{\mathcal{T}_q}) &= \\ [2 + 2p + \dots + 2p^{c-1} + p^c] - [p \cdot (2 + 2p + \dots + 2p^{c-2} + p^{c-1})] \end{aligned} \quad (18.17)$$

which equals 2 as claimed.

(2) The ramified case is similar. A minor difference is that π/p is not integral if $q = p$. Then we have

$$\mathrm{Orb}(\pi, 1_{\mathcal{S}_q}) - p \mathrm{Orb}(\pi, 1_{\mathcal{T}_q}) = 1 - 0$$

which is still 1 as desired for (18.15).

(3) Finally, assume $\mathbb{Q}_p[\pi] \cong \mathbb{Q}_p \times \mathbb{Q}_p$ and fix such an isomorphism. Write $\pi = (u, v)$. We have $\mathbb{Z}_p[\pi] = \mathbb{Z}_p \times \mathbb{Z}_p$ if and only if one out of u, v lies in $\mathbb{Z}_p^\times$. In this case, π/p is not integral anymore, so

$$\mathrm{Orb}(\pi, 1_{\mathcal{S}_q}) - p \mathrm{Orb}(\pi, 1_{\mathcal{T}_q}) = 1 - 0$$

equals 1 as claimed in (18.15). If both $u, v \in p\mathbb{Z}_p$, then π/p is still integral and the conductors of π and π/p are c and $c-1$ for some $c \geq 1$. Looking at (17.14), we get

$$\mathrm{Orb}(\pi, 1_{\mathcal{S}_q}) - p \mathrm{Orb}(\pi, 1_{\mathcal{T}_q}) = p^c - p \cdot p^{c-1}$$

which vanishes as claimed in (18.15). $\square$

Theorem 18.6. *Given $q = p^d$ and $n \geq 1$ prime to p , let $\Psi_{n,\ell}$, Ψ_p and Ψ_∞ be as in Definition 18.4. Let*

$$\Psi = \bigotimes_{\ell \neq p} \Psi_{n,\ell} \otimes \Psi_p \otimes \Psi_\infty \in \mathcal{C}_c^\infty(\mathrm{GL}_2(\mathbb{A}))$$

by their tensor product. Then

$$\sum_{(E, \alpha) \in \mathcal{M}_n(\mathbb{F}_q)} \frac{1}{|\text{Aut}(E, \alpha)|} = (18.6) + \sum_{\pi \in \left\{ \begin{array}{c} \text{reg. ss. conj.} \\ \text{classes of } \text{GL}_2(\mathbb{Q}) \end{array} \right\}} \text{Orb}(\pi, \Psi) \quad (18.18)$$

where the extra term (18.6) is understood as zero if d is odd. The occurring orbital integrals are defined by¹⁶

$$\begin{aligned} \text{Orb}(\pi, \Psi) &= \int_{\mathbb{Q}(\pi)^\times \cdot \mathbb{R}_{>0} \setminus \text{GL}_2(\mathbb{A})} \Psi(g^{-1}\pi g) dg \\ &= \frac{\mathcal{C}_{\mathbb{Q}(\pi)}}{|O_{\mathbb{Q}(\pi)}^\times|} \cdot \prod_{\ell \neq p} \text{Orb}(\pi, \Psi_{\ell, p}) \cdot \text{Orb}(\pi, \Psi_p) \cdot \text{Orb}(\pi, \Psi_\infty). \end{aligned}$$

Proof. We can subdivide the left hand side of (18.18) as

$$\sum_{(E, \alpha) \in \mathcal{M}_n(\mathbb{F}_q)} \frac{1}{|\text{Aut}(E, \alpha)|} = \sum_{\substack{E_0/\mathbb{F}_q \text{ up} \\ \text{to isogeny}}} \sum_{\substack{(E, \alpha) \text{ with } E \\ \text{isogeneous to } E_0}} \frac{1}{|\text{Aut}(E, \alpha)|}. \quad (18.19)$$

Taking the characteristic polynomial defines a bijection

$$\left\{ \begin{array}{c} \text{reg. ss. conj.} \\ \text{classes of } \text{GL}_2(\mathbb{Q}) \end{array} \right\} \xrightarrow{\sim} \left\{ \begin{array}{c} \text{Monic quadratic} \\ \text{separable } p(T) \in \mathbb{Q}[T] \end{array} \right\}. \quad (18.20)$$

By the Honda–Tate classification (Theorem 15.6), taking characteristic polynomial of Frobenius defines an injection

$$\{E_0/\mathbb{F}_q \text{ up to isogeny}\} \hookrightarrow \left\{ \begin{array}{c} \text{Monic quadratic} \\ p(T) \in \mathbb{Q}[T] \end{array} \right\}.$$

Given E_0 with Frobenius π , $\text{char}(\pi; T)$ not being separable means that $\sqrt{q} \in \mathbb{Q}$ and $\text{char}(\pi; T) = (T \mp \sqrt{q})^2$. This summand of (18.19) is captured by the term (18.6) in (18.18).

If $\text{char}(\pi; T)$ is separable, then (18.20) allows to view it as a conjugacy class of $\text{GL}_2(\mathbb{Q})$. In this case, (18.11) with (18.15) and (18.14) shows

$$\sum_{\substack{(E, \alpha) \text{ with } E \\ \text{isogeneous to } E_0}} \frac{1}{|\text{Aut}(E, \alpha)|} = \text{Orb}(\pi, \Psi).$$

In this way, we have explained that the LHS of (18.18) agrees with the *subsum* of the RHS of all classes π coming from elliptic curves. It is left to show that all remaining π satisfy $\text{Orb}(\pi, \Psi) = 0$.

It might be for several reasons that a regular semi-simple conjugacy class $\pi \in \text{GL}_2(\mathbb{Q})$ does not come from an elliptic. We go through the cases step by step following (15.3) and Theorem 15.6. Note throughout that the characteristic polynomial of π is

$$T^2 - \text{tr}(\pi)T + \det(\pi).$$

(1) The characteristic polynomial might not be integral. Then π cannot be conjugate to an element from $M_2(\widehat{\mathbb{Z}})$. Since the support of $\bigotimes_{\ell \neq p} \Psi_{n, \ell} \otimes \Psi_p$ is contained in $M_2(\widehat{\mathbb{Z}})$, it follows that $\text{Orb}(\pi, \Psi) = 0$.

¹⁶If $O_{\mathbb{Q}(\pi)}^\times$ is infinite, then $\text{Orb}(\pi, \Psi_\infty) = 0$ and the product is understood to be 0.

(2) Even if the characteristic polynomial is integral, it might happen that $|\det(\pi)| \neq q$. Then π can never be conjugate to an element $\prod_{\ell \neq p} \mathrm{GL}_2(\mathbb{Z}_\ell) \times \mathcal{S}_q$. Again arguing with the support of Ψ as before, we see $\mathrm{Orb}(\pi, \Psi) = 0$.

(3) Even if $|\det(\pi)| = q$, it might still happen that $\det(\pi) = -q$. In this case, the discriminant $\mathrm{tr}(\pi)^2 - 4\det(\pi)$ will be > 0 . Then $\mathbb{R}[\pi] \cong \mathbb{R} \times \mathbb{R}$ and hence $\mathrm{Orb}(\pi, \Psi_\infty) = 0$ by (18.14).

(4) Even if the characteristic polynomial is $T^2 - aT + q$ with $a \in \mathbb{Z}$, it might still happen that the discriminant $a^2 - 4q$ is > 0 . In this case, we again have $\mathrm{Orb}(\pi, \Psi_\infty) = 0$ as before.

Assume that π passed quality gates (1)–(4). In other words, assume that π has an integral characteristic polynomial of the form $T^2 - aT + q$ with $|a| \leq 2\sqrt{q}$ as in (15.3). It is left to consider the conditions in Theorem 15.6.

(5) None of the two conditions holding means that $p \mid a$ and p split in $F = \mathbb{Q}(\pi)$. If we identify $F_p \cong \mathbb{Q}_p \times \mathbb{Q}_p$, we can identify π with some $(u, v) \in \mathbb{Z}_p \times \mathbb{Z}_p$. The condition says $u + v \in p\mathbb{Z}_p$. Since also $uv \in p\mathbb{Z}_p$ because $\det(\pi) = q$, it follows that $u, v \in p\mathbb{Z}_p$. This implies that $\mathbb{Z}_p[\pi]$ is not the maximal order in F_p . By (18.15), we have $\mathrm{Orb}(\pi, \Psi_p) = 0$. This finishes the proof of the theorem. $\square$

Example 18.7. If $q = p$, then Ψ_p can simply be chosen as the characteristic function of

$$\mathcal{S}_p = \{x \in \mathrm{M}_2(\mathbb{Z}_p) \mid v_p(\det(x)) = 1\}.$$

Indeed, if $\pi \in \mathcal{S}_p$ is regular semi-simple, then $F := \mathbb{Q}_p[\pi]$ can only be a ramified quadratic extension or isomorphic to $\mathbb{Q}_p \times \mathbb{Q}_p$. (There are no elements a with $v_p(N_{F/\mathbb{Q}_p}(a)) = 1$ in an unramified quadratic extension $F/\mathbb{Q}_p$.)

Moreover, $\mathbb{Z}_p[\pi]$ will always equal $\mathcal{O}_F$: If $F/\mathbb{Q}_p$ as a ramified extension, then π will be a uniformizer. If $F \cong \mathbb{Q}_p \times \mathbb{Q}_p$, then π will be of the form (u, v) with one out of $\{u, v\}$ in $\mathbb{Z}_p^\times$ and the other in $p\mathbb{Z}_p$.

After these observations, Example 17.20 shows that

$$\mathrm{Orb}(\pi, \Psi_p) = 1$$

for all $\pi \in \mathcal{S}_p$ and that this aligns with the conditions in (18.15).

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